ALGEBRA I · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2100 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • HSN-RN.A.1 • NUMBER & QUANTITY

Explain the meaning of rational exponents from the properties of integer exponents

MAIN	Algebra I main question	**HSN-RN.A.1**
BACKWARD 1	Apply the properties of integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Express the solution of an exponential equation as a logarithm	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship between exponents and logarithms	**HSF-BF.B.5**

SET 1 • HSN-RN.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

For the power-of-a-power property to hold, $(7^{1/4})^4$ must equal $7^{(1/4)\cdot 4} = 7$.

What number must $7^{1/4}$ be? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt[4]{7}$, the fourth root of 7 (the positive number whose fourth power is 7).

GRADING NOTE Any equivalent form, such as the fourth root of 7. The explanation must use the fact that $(7^{1/4})^4 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q1/5

Use the properties of exponents to explain why $8^{2/3} = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

$8^{2/3} = (8^{1/3})^2 = 2^2 = 4.$

GRADING NOTE Any equivalent reasoning, such as $8^{2/3} = (8^2)^{1/3} = \sqrt[3]{64} = 4$. The explanation must use a property of exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q2/5

For $x > 0$, which expression is equal to $x^{3/4}$?

A. $x^3 \div 4$

B. the fourth root of x, cubed

C. the cube root of x, to the fourth power

D. $3x \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the fourth root of x, cubed

GRADING NOTE $x^{3/4} = (x^{1/4})^3$: the denominator 4 is the root and the numerator 3 is the power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q3/5

True or false?

$16^{1/2} = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $16^{1/2}$ is $\sqrt{16} = 4$. An exponent of 1/2 is a square root, not a division by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q4/5

True or false?

For $x > 0$, $x^{1/2} \cdot x^{1/2} = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The product property gives $x^{1/2+1/2} = x^1 = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 1

Apply the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$2^3 \cdot 2^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2^8

GRADING NOTE Form required: a single power. Add exponents when multiplying powers with the same base: 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q1/5

Write as a single power.

$(3^2)^4$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3^8

GRADING NOTE Form required: a single power. Multiply exponents for a power of a power: 2 · 4 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q2/5

Write as a single power.

$5^6 \div 5^2$

A. 1^4

B. 5^4

C. 5^3

D. 5^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5^4

GRADING NOTE Subtract exponents when dividing powers with the same base: $6 - 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$(7^2)^3 = 7^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiply the exponents: $2 \cdot 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$4^0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero number to the power 0 is 1, so $4^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

$\sqrt{81}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $9^2 = 81$, and $\sqrt{}$ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

Find the value.

$\sqrt[3]{27}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

$\sqrt[3]{64}$

A. 32

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE $4^3 = 64$. (8 is $\sqrt{64}$, the square root.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-2)^3 = -8$, and a cube root of a negative number is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\sqrt{25} = \pm 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The symbol √ means the nonnegative square root, so √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 1

Express the solution of an exponential equation as a logarithm • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $10^x = 500$. Write the solution as a logarithm, then use a calculator to round it to the nearest hundredth.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

27

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \log 500 \approx 2.70$

GRADING NOTE Form required: the logarithm and the rounded value. log means log base 10; log 500 = 2.69897...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q1/5

A colony of 400 bacteria grows by a factor of 10 every hour: $N = 400 \cdot 10^t$. Solve $400 \cdot 10^t = 3{,}000$. Give the exact logarithm and round to the nearest hundredth. (log means base 10.)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

28

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

log 7.5 ≈ 0.88 hour

GRADING NOTE Form required: the exact logarithm and the rounded value. $10^t = 7.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q2/5

Which expression is the solution of $3 \cdot 10^x = 60$? (log means base 10.)

A. x = log 60

B. x = 20

C. x = log 57

D. x = log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

29

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = log 20

GRADING NOTE Divide both sides by 3 first: $10^x = 20$, so x = log 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q3/5

True or false?

The solution of $2 \cdot 10^t = 1{,}000$ is $t = \log 500 \approx 2.70$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

30

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^t = 500$, and $\log 500 \approx 2.699$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = -10$ is $x = -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

31

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10^x is positive for every real x, so $10^x = -10$ has no real solution. ($10^{-1} = 0.1$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-RN.A.1

FORWARD 2

Use the inverse relationship between exponents and logarithms • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$f(x) = 10^x$ and $g(x) = \log x$ (log means base 10).

Find f(g(100)) and g(f(2)), and say what they show.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

33

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

f(g(100)) = 100 and g(f(2)) = 2: f and g undo each other, so they are inverse functions.

GRADING NOTE Form required: both values and the conclusion. g(100) = 2 and f(2) = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q1/5

A sound level is $L = 10 \log(I \div I_0)$ decibels, where log means base 10.

Solve for the intensity I in terms of L.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

34

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$I_0 \cdot 10^{L/10}$

GRADING NOTE Any equivalent form. Divide by 10, then use 10 to the power of each side: $I \div I_0 = 10^{L/10}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q2/5

Which function is the inverse of $f(x) = 5^x$?

A. $f^{-1}(x) = x \div 5$

B. $f^{-1}(x) = x^5$

C. $f^{-1}(x) = \log_5 x$

D. $f^{-1}(x) = 5^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

35

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = \log_5 x$

GRADING NOTE $\log_5 x$ undoes 5^x: $\log_5(5^x) = x$ and $5^{\log_5 x} = x$ for $x > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q3/5

True or false?

$10^{\log_{10} 7} = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

36

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10^x and log x are inverse functions, so each undoes the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\log(10^x) = 10x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

37

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(10^x) = x$, because log undoes 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 1 • FORWARD 2 • Q5/5

SET 2 • HSN-RN.A.2 • NUMBER & QUANTITY

Rewrite expressions with radicals and rational exponents

MAIN	Algebra I main question	**HSN-RN.A.2**
BACKWARD 1	Solve equations of the form $x^2 = p$ and $x^3 = p$	**8.EE.A.2**
BACKWARD 2	Use the meaning of rational exponents	**HSN-RN.A.1**
FORWARD 1	Solve radical equations and identify extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Add, subtract, and multiply complex numbers	**HSN-CN.A.2**

SET 2 • HSN-RN.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Assume $x \geq 0$. Write $x^{5/2}$ using a radical.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

40

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{(x^5)}$

GRADING NOTE Form required: radical form. $(\sqrt{x})^5$ is also correct. $x^{5/2} = (x^5)^{1/2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $\sqrt[3]{(y^2)}$ using a rational exponent.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

41

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y^{2/3}$

GRADING NOTE Form required: a rational exponent. The cube root gives the denominator 3, and the power 2 gives the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q2/5

Simplify.

$\sqrt{50} \cdot \sqrt{2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

42

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $\sqrt{50} \cdot \sqrt{2} = \sqrt{100} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-M-Q3**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q3/5

Which expression is equal to $(27x^6)^{1/3}$ for every real number x?

A. $27x^2$

B. $9x^2$

C. $3x^3$

D. $3x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

43

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $3x^2$

GRADING NOTE $27^{1/3} = 3$ and $(x^6)^{1/3} = x^2$ (a cube root is defined for every real number).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q4/5

True or false?

$\sqrt{(x^2)} = x$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

44

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$. For $x = -3$, $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 1

Solve equations of the form $x^2 = p$ and $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x.

$x^3 = 125$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

46

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$, and $x^3 = 125$ has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$x^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

47

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 6 and x = −6

GRADING NOTE Any equivalent form (x = ±6). Both solutions are required: $6^2 = (-6)^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q2/5

Solve for x.

$x^3 = -8$

A. ±2

B. −2

C. −4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

48

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2

GRADING NOTE $(-2)^3 = -8$. A cube equation has one real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$x^2 = 7$ has the solutions $\sqrt{7}$ and $-\sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

49

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both $(\sqrt{7})^2$ and $(-\sqrt{7})^2$ equal 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$x^3 = 27$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

50

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only x = 3 works; $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

BACKWARD 2

Use the meaning of rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$25^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

52

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $25^{1/2} = \sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q1/5

Find the value.

$27^{2/3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

53

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q2/5

Find the value.

$32^{1/5}$

A. 160

B. 2

C. 6.4

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

54

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE $2^5 = 32$, so the fifth root of 32 is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$9^{3/2} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

55

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$8^{1/3} = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

56

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $8^{1/3}$ is the cube root of 8, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 1

Solve radical equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 3) = 5. Check your solution.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

58

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. Square both sides: x + 3 = 25, so x = 22. Check: √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q1/5

Solve √(2x + 1) = x − 1. Check for extraneous solutions.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

59

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 (x = 0 is extraneous)

GRADING NOTE Form required: the solution, with x = 0 identified as extraneous. Squaring gives $x^2 - 4x = 0$, so x = 0 or 4. x = 0 gives √1 = 1 but 0 − 1 = −1, so it fails; x = 4 gives 3 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q2/5

What are the solutions of $\sqrt{x} = -3$?

A. $x = 3$

B. $x = 9$

C. $x = -9$

D. There is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

60

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. There is no solution.

GRADING NOTE $\sqrt{x}$ is never negative, so no x works. Squaring would give $x = 9$, but $\sqrt{9} = 3$, not -3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q3/5

True or false?

Squaring both sides of an equation can introduce extraneous solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

61

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Squaring can make a false equation true, for example $-3 = 3$ becomes $9 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = -4$ is a solution of $\sqrt{(x + 8)} = x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

62

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{4} = 2$, but $-4 + 2 = -2$, so the equation is false at $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-RN.A.2

FORWARD 2

Add, subtract, and multiply complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

(3 + 2i) + (1 − 5i)

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

64

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 − 3i

GRADING NOTE Form required: a + bi. Add real parts (3 + 1) and imaginary parts (2 − 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q1/5

Multiply. Write the answer in the form a + bi.

$(2 + i)(3 - i)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

65

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7 + i

GRADING NOTE Form required: a + bi. $6 - 2i + 3i - i^2 = 6 + i + 1 = 7 + i$, because $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q2/5

Simplify.

$i^2 \cdot i$

A. 1

B. $-i$

C. i

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

66

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-i$

GRADING NOTE $i^2 = -1$, so $i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

$(1 + i)^2 = 2i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

67

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

$i \cdot i = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

68

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i \cdot i = i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the sum of a rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Find exact lengths and perimeters on the coordinate plane	**HSG-GPE.B.7**
FORWARD 2	Write numbers in the complex number system as a + bi	**HSN-CN.A.1**

SET 3 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why 5 + √2 is irrational. (Suppose 5 + √2 = r for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

71

QUESTION ID

S3-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

5 + √2 is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if 5 + √2 = r with r rational, then √2 = r − 5, a difference of rational numbers, which is rational. But √2 is irrational, so 5 + √2 cannot be rational.

Any equivalent form of the argument. The key step is that rational numbers are closed under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q1/5

Explain why √3 − 4 is irrational. (Suppose √3 − 4 = r for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

72

QUESTION ID

S3-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

√3 − 4 is irrational.

GRADING NOTE

REQUIRED WORK: An argument like this: if √3 − 4 = r with r rational, then √3 = r + 4, a sum of rational numbers, which is rational. But √3 is irrational, so √3 − 4 cannot be rational.

Form required: the argument by contradiction. It must use the fact that a sum of rational numbers is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q2/5

Prove that if a is rational and x is irrational, then a + x is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

73

QUESTION ID

S3-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

a + x is irrational.

GRADING NOTE REQUIRED WORK: Suppose a + x = r with r rational. Then x = r − a, a difference of rational numbers, which is rational. That contradicts x being irrational, so a + x is irrational.

Form required: the general argument by contradiction, using closure of the rational numbers under subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q3/5

Which sum is irrational?

A. $\sqrt{2} + (-\sqrt{2})$

B. $0.25 + \frac{3}{4}$

C. $\sqrt{16} + 1.5$

D. $\sqrt{5} + 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

74

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{5} + 1.5$

GRADING NOTE $\sqrt{5}$ is irrational and 1.5 is rational, so their sum is irrational. $\sqrt{16} + 1.5 = 5.5$, $0.25 + 0.75 = 1$, and $\sqrt{2} + (-\sqrt{2}) = 0$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q4/5

True or false?

The sum of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

75

QUESTION ID
S3-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 3 • MAIN • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √49 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

77

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Rational;

√49 = 7 = 7/1.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q1/5

The number 0.1010010001... continues with one more 0 before each 1. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

78

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Irrational; its decimal never ends and never repeats.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q2/5

Which number is rational?

A. √2

B. √10

C. π

D. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

79

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2727... (the digits 27 repeat)

GRADING NOTE A repeating decimal is rational: 0.2727... = 27/99 = 3/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

Every integer is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

80

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any integer n can be written as n/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

√20 is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

81

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20 is not a perfect square, so √20 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write 1.25 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

83

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/4

GRADING NOTE Any equivalent form, such as 125/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q1/5

Write $-3\frac{1}{3}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

84

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10/3

GRADING NOTE Any equivalent form, such as 10/(−3). $-3\frac{1}{3} = -(9/3 + 1/3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q2/5

Which quotient is equal to 0.6?

A. $\frac{6}{100}$

B. $\frac{5}{3}$

C. $\frac{3}{5}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

85

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/5

GRADING NOTE $3 \div 5 = 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-7 = \frac{-7}{1}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

86

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every integer is a quotient of itself and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

87

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined, so 2/0 is not a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 1

Find exact lengths and perimeters on the coordinate plane • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (3, 0), and (0, 3).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

89

QUESTION ID
S3-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6 + 3√2 units

GRADING NOTE Form required: exact form. The legs are 3 and 3, and the hypotenuse is √18 = 3√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q1/5

A square has vertices (0, 0), (1, 1), (0, 2), and (−1, 1). Find its exact perimeter, and say whether it is rational or irrational.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

90

QUESTION ID
S3-F1-Q2

TYPE
Short answer

CORRECT ANSWER
4√2 units; irrational

GRADING NOTE Form required: exact form. Each side is $\sqrt{(1^2 + 1^2)} = \sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q2/5

What is the area of the triangle with vertices (0, 0), (4, 0), and (4, 3)?

A. 6

B. 12

C. 7

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

91

QUESTION ID
S3-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. 6

GRADING NOTE It is a right triangle with legs 4 and 3: $\frac{1}{2} \cdot 4 \cdot 3 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (1, 0), and (0, 1) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

92

QUESTION ID
S3-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The perimeter is 2 + √2, a rational number plus an irrational number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q4/5

True or false?

The triangle with vertices (0, 0), (6, 0), and (0, 8) has an irrational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

93

QUESTION ID
S3-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The sides are 6, 8, and 10, so the perimeter is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-RN.B.3

FORWARD 2

Write numbers in the complex number system as a + bi • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{-49}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

95

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7i

GRADING NOTE Form required: bi. $\sqrt{-49} = \sqrt{49} \cdot \sqrt{-1} = 7i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q1/5

Name the real part and the imaginary part of $4 - 3i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

96

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Real part 4; imaginary part −3

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q2/5

Which of these is a real number?

A. i

B. i^2

C. 2i

D. 1 + i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

97

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. i^2

GRADING NOTE $i^2 = -1$, which is real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q3/5

True or false?

Every real number is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

98

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real number a is a + 0i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$i^2 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

99

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE i is defined so that $i^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F2-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • FORWARD 2 • Q5/5

SET 69 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that the product of a nonzero rational number and an irrational number is irrational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
BACKWARD 2	Multiply and divide rational numbers	**7.NS.A.2.c**
FORWARD 1	Find exact lengths, perimeters, and areas with irrational side lengths	**HSG-GPE.B.7**
FORWARD 2	Factor sums of squares over the complex numbers	**HSN-CN.C.8**

SET 69 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why $3\sqrt{7}$ is irrational. (Suppose $3\sqrt{7} = r$ for some rational number r.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

102

QUESTION ID

S69-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$3\sqrt{7}$ is irrational.

GRADING NOTE REQUIRED WORK: An argument like this: if $3\sqrt{7} = r$, then $\sqrt{7} = r \div 3$, a quotient of a rational number by a nonzero rational number, which is rational. But $\sqrt{7}$ is irrational, so $3\sqrt{7}$ is irrational.

Form required: the argument by contradiction. It must divide by the nonzero rational number 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q1/5

Prove that if a is a nonzero rational number and x is irrational, then ax is irrational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

103

QUESTION ID

S69-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

ax is irrational.

GRADING NOTE REQUIRED WORK: Suppose ax = r with r rational. Since a ≠ 0, x = r ÷ a, a quotient of rational numbers with a nonzero divisor, which is rational. That contradicts x being irrational.

Form required: the general argument by contradiction, including where a ≠ 0 is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q2/5

Why does the statement "a rational number times an irrational number is irrational" need the rational number to be nonzero?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

104

QUESTION ID

S69-M-Q3

TYPE

Short answer

CORRECT ANSWER

If a = 0, then ax = 0, which is rational. The proof also divides by a, which needs a ≠ 0.

GRADING NOTE Any equivalent explanation that gives the counterexample 0 · x = 0 or the division by a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q3/5

Which product is irrational?

A. $\sqrt{5} \cdot \sqrt{5}$

B. $\sqrt{2} \cdot \sqrt{8}$

C. $0 \cdot \sqrt{5}$

D. $\frac{2}{3}\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

105

QUESTION ID

S69-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2/3√5

GRADING NOTE $\frac{2}{3}$ is a nonzero rational number and $\sqrt{5}$ is irrational. $0 \cdot \sqrt{5} = 0$, $\sqrt{5} \cdot \sqrt{5} = 5$, and $\sqrt{2} \cdot \sqrt{8} = 4$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

106

QUESTION ID
S69-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $\sqrt{2} \cdot \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 69 • MAIN • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 1

Classify numbers as rational or irrational • Grade 8 • The Number System • 8.NS.A.1

Is √64 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

108

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Rational: √64 = 8.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q1/5

The decimal 0.1010010001... has one more 0 between the 1s each time. Is it rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

109

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Irrational: it never ends and never repeats.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q2/5

Which of these numbers is irrational?

A. $\sqrt{10}$

B. $-\frac{4}{7}$

C. $\sqrt{100}$

D. 0.333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

110

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{10}$

GRADING NOTE 10 is not a perfect square. $\sqrt{100} = 10$, $0.333... = \frac{1}{3}$, and $-\frac{4}{7}$ are rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q3/5

True or false?

Every terminating decimal is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

111

QUESTION ID

S69-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A terminating decimal is a fraction with a power of 10 in the denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\pi = \frac{22}{7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

112

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE π is irrational; $\frac{22}{7} \approx 3.142857$ is only an approximation of $\pi \approx 3.141593$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

BACKWARD 2

Multiply and divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find $(-\frac{2}{3}) \cdot \frac{9}{4}$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

114

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−3/2

GRADING NOTE Any equivalent form, such as −1.5. −18/12 = −3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q1/5

Find $\frac{5}{6} \div \left(-\frac{10}{3}\right)$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

115

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/4

GRADING NOTE Any equivalent form. $\frac{5}{6} \cdot \left(-\frac{3}{10}\right) = -\frac{15}{60}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q2/5

Which product is positive?

A. $(0.5)(-8)$

B. $0 \cdot 8$

C. $(-0.5)(-8)$

D. $(-0.5)(8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

116

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(-0.5)(-8)$

GRADING NOTE A negative times a negative is positive: 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q3/5

True or false?

The quotient of two nonzero rational numbers is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

117

QUESTION ID

S69-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (a/b) ÷ (c/d) = ad/(bc), a quotient of integers with bc ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{4} \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

118

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 1

Find exact lengths, perimeters, and areas with irrational side lengths • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A square has vertices (0, 0), (2, 1), (1, 3), and (−1, 2). Find its exact side length and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

120

QUESTION ID
S69-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Side √5; area 5 square units

GRADING NOTE Form required: the exact side and the area. Each side is $\sqrt{(2^2 + 1^2)} = \sqrt{5}$, and $(\sqrt{5})^2 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has vertices (0, 0), (3, 3), (1, 5), and (−2, 2). Find its exact perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

121

QUESTION ID
S69-F1-Q2

TYPE
Short answer

CORRECT ANSWER
10√2 units

GRADING NOTE Form required: exact. The sides are √18 = 3√2 and √8 = 2√2, so the perimeter is 2(3√2 + 2√2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q2/5

What is the exact perimeter of the triangle with vertices (0, 0), (4, 0), and (0, 4)?

A. 4√2

B. 16

C. 8 + 2√2

D. 8 + 4√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

122

QUESTION ID
S69-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 8 + 4√2

GRADING NOTE The legs are 4 and 4, and the hypotenuse is √32 = 4√2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q3/5

True or false?

A square with side $\sqrt{5}$ has a rational area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

123

QUESTION ID
S69-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its area is $(\sqrt{5})^2 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q4/5

True or false?

A square with side √5 has a rational perimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

124

QUESTION ID
S69-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its perimeter is 4√5, a nonzero rational number times an irrational number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSN-RN.B.3

FORWARD 2

Factor sums of squares over the complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 9$ over the complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

126

QUESTION ID
S69-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(x + 3i)(x − 3i)

GRADING NOTE Form required: factored with complex numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand (x + 2i)(x − 2i).

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

127

QUESTION ID

S69-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 4$

GRADING NOTE Form required: expanded. $-(2i)^2 = -4i^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q2/5

Which is a factorization of $x^2 + 25$?

A. $(x + 5i)(x - 5i)$

B. $(x + 5)(x - 5)$

C. $(x + 5i)^2$

D. $(x - 5i)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

128

QUESTION ID

S69-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x + 5i)(x - 5i)$

GRADING NOTE $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q3/5

True or false?

$x^2 + 16 = (x + 4i)(x - 4i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q4/5

129

QUESTION ID

S69-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-(4i)^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q4/5

True or false?

$x^2 + 4$ cannot be factored, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

130

QUESTION ID

S69-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 4 = (x + 2i)(x - 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSN-RN.B.3 • NUMBER & QUANTITY

Prove that sums and products of two rational numbers are rational

MAIN	Algebra I main question	**HSN-RN.B.3**
BACKWARD 1	Write quotients of integers as rational numbers	**7.NS.A.2.b**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Show that sums and products of complex numbers are complex n…	**HSN-CN.A.2**
FORWARD 2	Show that rational expressions are closed under addition and mult…	**HSA-APR.D.7**

SET 70 • HSN-RN.B.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3

Explain why the sum of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1
High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

QUESTION ID
S70-M-Q1

TYPE
Short answer

CORRECT ANSWER
a/b + c/d = (ad + bc)/(bd). ad + bc and bd are integers and bd ≠ 0, so the sum is a quotient of integers.

GRADING NOTE Any equivalent explanation that writes the sum as a quotient of integers with a nonzero denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1
High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q1/5

Prove that the product of two rational numbers a/b and c/d (with integers a, b, c, d, and b ≠ 0, d ≠ 0) is rational.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

134

QUESTION ID

S70-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The product is rational.

GRADING NOTE

REQUIRED WORK: (a/b)(c/d) = (ac)/(bd); ac and bd are integers, and bd ≠ 0 because b ≠ 0 and d ≠ 0, so the product is a quotient of integers.

Form required: the product written as a quotient of integers, with the reason the denominator is not 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q2/5

In the proof that (a/b)(c/d) is rational, why is bd ≠ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

135

QUESTION ID

S70-M-Q3

TYPE

Short answer

CORRECT ANSWER

b ≠ 0 and d ≠ 0, and a product of two nonzero integers is not 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q3/5

Which statement shows that $(\frac{2}{5})(-\frac{15}{4})$ is rational?

A. It can be written as a decimal with a calculator.

B. It equals −30/20 = −3/2, a quotient of integers with a nonzero denominator.

C. It is negative.

D. Its first factor has an even numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

136

QUESTION ID

S70-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. It equals −30/20 = −3/2, a quotient of integers with a nonzero denominator.

GRADING NOTE Rational means a quotient of integers with a nonzero denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q4/5

True or false?

The difference of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

137

QUESTION ID

S70-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE a/b − c/d = (ad − bc)/(bd), a quotient of integers with bd ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 70 • MAIN • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 1

Write quotients of integers as rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-\frac{3}{4}$ as a quotient of integers in two different ways.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

139

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−3/4 and 3/(−4)

GRADING NOTE Any two correct quotients, such as −6/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is −12 ÷ 5 a rational number? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

140

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. It is a quotient of integers with a nonzero divisor.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q2/5

Which quotient is undefined?

A. $-7 \div 2$

B. $0 \div 7$

C. $7 \div 0$

D. $7 \div (-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

141

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \div 0$

GRADING NOTE Division by 0 is undefined; $0 \div 7 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q3/5

True or false?

For integers p and q with q ≠ 0,
−(p/q) = (−p)/q = p/(−q).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

142

QUESTION ID

S70-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each is the same negative number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

Some integers are not rational numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

143

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every integer n equals n/1, so every integer is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

BACKWARD 2

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Find $\frac{3}{4} \div \frac{1}{8}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

145

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. $\frac{3}{4} \cdot 8 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q1/5

Find $\frac{2}{3} \div \frac{5}{6}$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

146

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form. $\frac{2}{3} \cdot \frac{6}{5} = \frac{12}{15}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q2/5

Which number equals $\frac{5}{2} \div \frac{1}{4}$?

A. $\frac{5}{8}$

B. $\frac{1}{10}$

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

147

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10

GRADING NOTE $\frac{5}{2} \cdot 4 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

148

QUESTION ID

S70-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE There are two quarters in a half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{5} \div 3 = \frac{9}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

149

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{3}{5} \div 3 = \frac{1}{5}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 1

Show that sums and products of complex numbers are complex numbers • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2

Add. Write the answer in the form a + bi.

(3 − 2i) + (−1 + 6i)

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

151

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 + 4i

GRADING NOTE Form required: a + bi. Add the real parts and the imaginary parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q1**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q1/5

Multiply. Write the answer in the form a + bi.

(2 + i)(3 + 2i)

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

152

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4 + 7i

GRADING NOTE Form required: a + bi. 6 + 4i + 3i + 2i² = 6 + 7i − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q2/5

Which shows that the product of two complex numbers is a complex number?

A. $i^2 = 1$

B. $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

C. $(a + bi)(c + di) = ac + bdi$

D. The product of two complex numbers is always real

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

153

QUESTION ID
S70-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B.
$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, and $ac - bd$ and $ad + bc$ are real numbers

GRADING NOTE Expanding and using $i^2 = -1$ gives a real part and an imaginary part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q3/5

True or false?

The sum of two complex numbers is a complex number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

154

QUESTION ID

S70-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (a + bi) + (c + di) = (a + c) + (b + d)i.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q4**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q4/5

True or false?

(1 + i) + (1 − i) is not a complex number because it has no i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

155

QUESTION ID

S70-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is 2 = 2 + 0i, a complex number (a real number).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSN-RN.B.3

FORWARD 2

Show that rational expressions are closed under addition and multiplication • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Assume $x \neq 0$ and $x \neq -1$. Write $1 \div x + 2 \div (x + 1)$ as one fraction.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

157

QUESTION ID
S70-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$(3x + 1) \div (x(x + 1))$

GRADING NOTE Any equivalent form, such as $(3x + 1)/(x^2 + x)$. The common denominator is $x(x + 1)$: $(x + 1 + 2x) \div (x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q1/5

Assume $x \neq 2$ and $x \neq -3$. Multiply and simplify:
$(x \div (x - 2)) \cdot ((x - 2) \div (x + 3))$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

158

QUESTION ID
S70-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x \div (x + 3)$

GRADING NOTE Form required: simplified. The factor $x - 2$ cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q2/5

Why is the sum of two rational expressions a rational expression?

A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

B. The denominators always cancel.

C. Rational expressions are always integers.

D. Any sum of fractions equals 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

159

QUESTION ID
S70-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Over a common denominator, the sum is a polynomial divided by a nonzero polynomial.

GRADING NOTE This is the same argument as for rational numbers, with polynomials in place of integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q3/5

True or false?

The quotient of two rational expressions, when the divisor is not 0, is a rational expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

160

QUESTION ID
S70-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Dividing multiplies by the reciprocal, which is also a rational expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q4/5

True or false?

$1 \div x + 1 \div y = 2 \div (x + y)$ for all nonzero x and y with $x + y \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

161

QUESTION ID
S70-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For x = y = 1, the left side is 2 and the right side is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 70 • FORWARD 2 • Q5/5

SET 4 • HSN-Q.A.1 • NUMBER & QUANTITY

Use units to guide multistep solutions and to choose scales for graphs

MAIN	Algebra I main question	**HSN-Q.A.1**
BACKWARD 1	Convert measurement units with ratio reasoning	**6.RP.A.3.d**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Use density in a model	**HSG-MG.A.2**
FORWARD 2	Solve problems with velocity vectors — nearest related	**HSN-VM.A.3**

SET 4 • HSN-Q.A.1

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

A car goes 32 miles on each gallon of gas, and gas costs $3.60 per gallon.

Use units to find the cost of gas for a 400-mile trip. Show how the units cancel.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

164

QUESTION ID

S4-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$45

GRADING NOTE REQUIRED WORK: A product of rates with units, such as 400 miles × (1 gallon ÷ 32 miles) × ($3.60 ÷ 1 gallon), with miles and gallons canceling to leave dollars.

Form required: the cost with the unit work shown. 400 ÷ 32 = 12.5 gallons, and 12.5 × $3.60 = $45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q1/5

Convert 72 kilometers per hour to meters per second.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

165

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 meters per second

GRADING NOTE Any equivalent form. 72 km/h × 1,000 m/km ÷ 3,600 s/h = 20 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q2/5

The formula d = rt gives distance. A rate r is in meters per second and a time t is in minutes.

What must be done before multiplying?

A. Divide r by t.

B. Convert t to seconds (or r to meters per minute).

C. Nothing; the units already match.

D. Convert d to minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

166

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Convert t to seconds (or r to meters per minute).

GRADING NOTE The time units must match: (m/s) × s = m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q3/5

A graph will show a town's population each year from 1990 to 2020. The populations range from 12,000 to 31,000.

Choose a scale for the vertical axis, and explain your choice.

Scale: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

167

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

For example, 0 to 35,000 in steps of 5,000, labeled "Population"; every value fits and the steps are equal.

GRADING NOTE Any equivalent form: any labeled scale with equal steps that includes 12,000 to 31,000 (starting at 10,000 instead of 0 is also correct if the break is shown or stated).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q4/5

True or false?

A speed of 5 miles per hour is about 7.3 feet per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

168

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 × 5,280 ÷ 3,600 = 7.33... feet per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 4 • MAIN • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 1

Convert measurement units with ratio reasoning • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

How many inches are in 3.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

170

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

42

GRADING NOTE Any equivalent form. 3.5 × 12 = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q1/5

How many kilograms are in 2,500 grams?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

171

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form. 2,500 ÷ 1,000 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q2/5

How many feet are in 4 yards?

A. 48

B. 36

C. $\frac{4}{3}$

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

172

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE 1 yard = 3 feet, so 4 yards = 12 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q3/5

True or false?

1 hour = 3,600 seconds

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

173

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 minutes × 60 seconds = 3,600 seconds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q4/5

True or false?

5 km = 500 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

174

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 km = 1,000 m, so 5 km = 5,000 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A walker goes $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

176

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (3/4) ÷ (1/4) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q1/5

Half a pound of cheese costs $3.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

177

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$6

GRADING NOTE Any equivalent form. $3 ÷ (1/2) = $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\frac{2}{3}$ gallon of paint covers $\frac{1}{3}$ of a fence.

How many gallons cover the whole fence?

A. $\frac{1}{2}$

B. $\frac{2}{9}$

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

178

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE (2/3) ÷ (1/3) = 2 gallons per fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{6}$ hour is a speed of 3 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

179

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1/2) ÷ (1/6) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q4/5

True or false?

Painting $\frac{1}{4}$ of a wall in $\frac{1}{2}$ hour is a rate of $\frac{1}{8}$ wall per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

180

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (1/4) ÷ (1/2) = 1/2 wall per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 1

Use density in a model • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

Copper has a density of 8.96 grams per cubic centimeter.

What is the mass of a 5 cubic-centimeter piece of copper?

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

182

QUESTION ID
S4-F1-Q1

TYPE
Short answer

CORRECT ANSWER
44.8 grams

GRADING NOTE Any equivalent form. Mass = density × volume = 8.96 × 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q1/5

A county has 240,000 people living in 600 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

183

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

GRADING NOTE Any equivalent form, with units. 240,000 ÷ 600 = 400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q2/5

A 2 cm by 3 cm by 4 cm block has a mass of 60 grams.

What is its density?

A. 0.4 gram per cubic centimeter

B. 60 grams

C. 2.5 grams per cubic centimeter

D. 24 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

184

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5 grams per cubic centimeter

GRADING NOTE The volume is 24 cubic centimeters, and 60 ÷ 24 = 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Population density is the number of people divided by the area they live on.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

185

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Density is a quantity per unit of area (or volume).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

An object with a volume of 10 cubic centimeters and a density of 3 grams per cubic centimeter has a mass of 3.3 grams.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

186

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Mass = 3 × 10 = 30 grams.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-Q.A.1

FORWARD 2

Solve problems with velocity vectors • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.A.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Precalculus standard. Connection: Velocity vectors are rate quantities with units, like the unit rates of the MAIN set, with a direction added.

A boat heads east at 12 km/h across a river whose current flows north at 5 km/h.

What is the boat's resulting speed?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

188

QUESTION ID
S4-F2-Q1

TYPE
Short answer

CORRECT ANSWER
13 km/h

GRADING NOTE Any equivalent form. The velocities are perpendicular, so the speed is $\sqrt{(12^2 + 5^2)} = 13$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q1/5 • NEAREST RELATED

A plane points north at 300 km/h, and the wind blows west at 40 km/h.

Write its velocity as (east part, north part) in km/h.

Velocity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

189

QUESTION ID
S4-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(−40, 300)

GRADING NOTE Form required: (east part, north part). West is the negative east direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q2/5 • NEAREST RELATED

Two velocities, (3, 4) and (5, −1), are added.

What is the result?

A. (8, 5)

B. (8, 3)

C. (15, −4)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

190

QUESTION ID
S4-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (8, 3)

GRADING NOTE Add the parts: (3 + 5, 4 + (−1)) = (8, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An object with velocity (6, 8) meters per second has a speed of 10 meters per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

191

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(6^2 + 8^2)} = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S4-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Adding the velocities (3, 0) and (0, 4) gives a speed of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

192

QUESTION ID
S4-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The sum is (3, 4), whose speed is √(9 + 16) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5
High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities
HSN-VM.A.3 • Set 4 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 5 • HSN-Q.A.2 • NUMBER & QUANTITY

Define appropriate quantities for descriptive modeling

MAIN	Algebra I main question	**HSN-Q.A.2**
BACKWARD 1	Use variables to represent quantities in a situation	**6.EE.B.6**
BACKWARD 2	Write and solve px + q = r from a word problem	**7.EE.B.4.a**
FORWARD 1	Use geometric quantities to make design decisions	**HSG-MG.A.3**
FORWARD 2	Recognize surveys, experiments, and observational studies and th…	**HSS-IC.B.3**

SET 5 • HSN-Q.A.2

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2

A school wants to model the total cost of a field trip by bus.

Name the quantities the model needs, with a unit for each.

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

195

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

The number of students (students), the students each bus holds (students per bus), the cost of each bus (dollars per bus), and the admission price (dollars per student).

GRADING NOTE Any equivalent form: any complete, sensible set of quantities, each with a unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q1/5

A club rents a room for $300 and pays $40 for each member who attends. Its cost model is C = 40n + 300.

Define C and n, with units.

Definitions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

196

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

C is the total cost in dollars; n is the number of members who attend.

GRADING NOTE Any equivalent wording, with both units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q2/5

Which quantity best describes how fast a phone battery runs down?

A. Number of apps on the phone

B. Percent of charge used per hour

C. Width of the screen in inches

D. Mass of the phone in grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

197

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Percent of charge used per hour

GRADING NOTE A rate of drain needs an amount of charge per unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q3/5

Two towns have different populations and different areas.

Define one quantity, with units, that compares how crowded they are.

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

198

QUESTION ID

S5-M-Q4

TYPE

Short answer

CORRECT ANSWER

Population density: people per square mile (or per square kilometer).

GRADING NOTE Any equivalent quantity that is people per unit of area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q4/5

True or false?

Kilowatt-hours per month is a sensible quantity for a model of a household's electricity use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

199

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It measures energy used per unit of time, which is what an electricity bill is based on.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 1

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost $8 each.

Write an expression for the cost of n tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

201

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A worker earns $15 per hour plus a $20 bonus.

Write an expression for the pay for h hours.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

202

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15h + 20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q2/5

A typist types 30 words per minute.

Which expression gives the number of words typed in m minutes?

A. $\frac{m}{30}$

B. $m + 30$

C. $30m$

D. $\frac{30}{m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

203

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 30m

GRADING NOTE Words = rate × time = 30m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q3/5

True or false?

If w is a number of weeks, 7w is the number of days.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

204

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each week has 7 days.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q4/5

True or false?

If d is a number of dollars, d + 5 is 5 times as many dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

205

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5 times as many is 5d; d + 5 is 5 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

BACKWARD 2

Write and solve px + q = r from a word problem • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a $25 fee plus $10 per class. Tom paid $95.

How many classes did he take?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

207

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 25 + 10c = 95 gives c = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q1/5

A phone costs $50 plus $15 per month. The total so far is $140.

For how many months has it been paid?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

208

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 50 + 15m = 140 gives m = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q2/5

A plumber charges a $12 fee plus r dollars per hour. A 3-hour job costs $57.

Which equation fits?

A. 3r + 12 = 57

B. 3(r + 12) = 57

C. 12r + 3 = 57

D. r + 3 + 12 = 57

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

209

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3r + 12 = 57

GRADING NOTE 3 hours at r dollars is 3r, plus the $12 fee.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q3/5

True or false?

x = 6 is the solution of 4x + 6 = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

210

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 \cdot 6 + 6 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

211

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \cdot 3 - 5 = 1$; the solution is $x = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 1

Use geometric quantities to make design decisions • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden needs an area of at least 200 square feet and can use at most 60 feet of fence.

Does a 10 ft by 20 ft garden meet both requirements? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

213

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Its area is 200 square feet and its perimeter is 60 feet.

GRADING NOTE Any equivalent wording. Both checks are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q1/5

A box with a square base 10 cm on each side must hold 1,000 cubic centimeters.

How tall must it be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

214

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 cm

GRADING NOTE Any equivalent form. 10 × 10 × h = 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q2/5

A rectangular pen must have an area of 100 square meters.

Which dimensions use the least fence?

A. 5 m by 20 m

B. 2 m by 50 m

C. 10 m by 10 m

D. 4 m by 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

215

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 m by 10 m

GRADING NOTE The perimeters are 40, 50, 58, and 104 meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q3/5

True or false?

A 6 by 6 square has more area than a 4 by 8 rectangle, even though their perimeters are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

216

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 36 > 32, and both perimeters are 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with a perimeter of 20, the 1 by 9 rectangle has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

217

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is 9; the 5 by 5 square has area 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-Q.A.2

FORWARD 2

Recognize surveys, experiments, and observational studies and the variables they measure • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Defining the quantities a model needs leads to deciding which variables a study measures and how they are collected.

A researcher randomly assigns 40 plants to two fertilizers and measures each plant's growth in centimeters.

What kind of study is this, and what is the response variable (with units)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

219

QUESTION ID
S5-F2-Q1

TYPE
Short answer

CORRECT ANSWER
An experiment; the response is growth in centimeters.

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q1/5 • NEAREST RELATED

A survey asks 200 randomly chosen students how many hours they sleep on a school night.

What kind of study is this, and what variable does it measure?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

220

QUESTION ID
S5-F2-Q2

TYPE
Short answer

CORRECT ANSWER
A survey (sample survey); hours of sleep per school night.

GRADING NOTE Any equivalent wording. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which kind of study can show cause and effect?

A. An observational study

B. A poll of volunteers

C. A randomized experiment

D. A survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

221

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A randomized experiment

GRADING NOTE Random assignment tends to balance other variables between the groups, so a difference can be credited to the treatment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

In an observational study, the researchers do not assign treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

222

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They observe groups that already exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A survey of volunteers who answer an online poll is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

223

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 5 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 6 • HSN-Q.A.3 • NUMBER & QUANTITY

Choose a level of accuracy that fits the limits of measurement

MAIN	Algebra I main question	**HSN-Q.A.3**
BACKWARD 1	Estimate to check whether an answer is reasonable	**7.EE.B.3**
BACKWARD 2	Round decimals to any place	**5.NBT.A.4**
FORWARD 1	Use volume formulas and report results to a sensible accuracy	**HSG-GMD.A.3**
FORWARD 2	Use the margin of error of a sample survey — nearest related	**HSS-IC.B.4**

SET 6 • HSN-Q.A.3

MAIN

Algebra I main question • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A ruler measures to the nearest 0.1 cm. A rectangle measures 4.3 cm by 2.7 cm, and a calculator gives its area as 11.61 square cm.

Which report of the area best fits the accuracy of the measurements?

A. 11.610 square cm

B. 11.61 square cm

C. About 10 square cm

D. About 12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

226

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. About 12 square cm

GRADING NOTE Each length could be off by 0.05 cm, so the true area is between about 11.3 and 12.0 square cm; the hundredths digit is not known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q1/5

A stopwatch reads to the nearest 0.1 second. A runner covers 100 m in 12.4 seconds, and a calculator gives 8.0645161 meters per second.

Report the speed to a sensible accuracy.

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

227

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 8.1 meters per second

GRADING NOTE Form required: a rounded value. 8.1 or 8.06 m/s is accepted; the true speed is between about 8.03 and 8.10 m/s, so more digits are not meaningful.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q2/5

A tape measure is marked in centimeters.

Which is the most sensible report of a person's height measured with it?

A. 167.3482 cm

B. 170.00 cm

C. 2 m

D. 167 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

228

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 167 cm

GRADING NOTE The tape measures to the nearest centimeter, so report whole centimeters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q3/5

A bridge is 1,250 m long, measured to the nearest 10 m.

Between which two lengths is its true length?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

229

QUESTION ID

S6-M-Q4

TYPE

Short answer

CORRECT ANSWER

Between 1,245 m and 1,255 m

GRADING NOTE Any equivalent form. Measuring to the nearest 10 m allows an error of up to 5 m either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q4/5

True or false?

Giving more decimal places always makes a measured answer more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

230

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A result cannot be more accurate than the measurements it comes from.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 1

Estimate to check whether an answer is reasonable • Grade 7 • Expressions & Equations • 7.EE.B.3

Estimate.

4.9×21.2

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

232

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

About 100

GRADING NOTE Any equivalent estimate near 100: $5 \times 20 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q1/5

Is 38.4 ÷ 4.8 = 80 reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

233

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 40 ÷ 5 = 8, so the answer should be about 8.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q2/5

Which is the best estimate of 19.8% of 503?

A. 100

B. 50

C. 1,000

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

234

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100

GRADING NOTE About 20% of 500 is 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$29.95 × 4 is about $120.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

235

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $30 × 4 = $120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q4/5

True or false?

0.48 × 61 is about 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

236

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About 0.5 × 60 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

BACKWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 8.0645 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

238

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.06

GRADING NOTE Form required: hundredths. The thousandths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q1/5

Round 11.61 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

239

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Form required: a whole number. The tenths digit is 6, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q2/5

Round 0.0357 to the nearest thousandth.

A. 0.035

B. 0.0357

C. 0.036

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

240

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.036

GRADING NOTE The ten-thousandths digit is 7, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q3/5

True or false?

3.149 rounded to the nearest tenth is 3.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

241

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The hundredths digit is 4, so round down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

6.95 rounded to the nearest tenth is 6.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

242

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hundredths digit is 5, so round up to 7.0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 1

Use volume formulas and report results to a sensible accuracy • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has a radius of 3.0 cm and a height of 10.0 cm.

Find its volume to the nearest cubic centimeter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

244

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

283 cubic centimeters

GRADING NOTE Form required: rounded to the nearest whole number. $V = \pi \cdot 3^2 \cdot 10 = 90\pi \approx 282.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q1/5

A sphere has a radius of 5 m.

Find its volume exactly in terms of π, then to the nearest cubic meter.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

245

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{500}{3}\pi$ cubic meters, about 524 cubic meters

GRADING NOTE Form required: both the exact and the rounded value. V = (4/3)π · 125.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q2/5

A cone has a radius of 6 and a height of 4.

What is its volume?

A. 144π

B. 96π

C. 48π

D. 16π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

246

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 48π

GRADING NOTE $V = (1/3)\pi \cdot 6^2 \cdot 4 = 48\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q3/5

True or false?

A cylinder with radius 2 cm and height 2 cm has a volume of 8π cm³, which is about 25 cm³ to the nearest cubic centimeter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

247

QUESTION ID
S6-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $V = \pi \cdot 2^2 \cdot 2 = 8\pi \approx 25.1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a sphere doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

248

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volume depends on r^3, so it is multiplied by 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-Q.A.3

FORWARD 2

Use the margin of error of a sample survey • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: The accuracy a measurement supports leads to the accuracy of an estimate from a sample, given by its margin of error.

A poll reports that 54% of voters support a plan, with a margin of error of 3 percentage points.

What interval of values for the true percent does this give?

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

250

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

51% to 57%

GRADING NOTE Any equivalent form. 54% − 3% to 54% + 3%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q1/5 • NEAREST RELATED

A poll reports 54% support with a margin of error of 3 percentage points.

Do the results suggest that a majority supports the plan? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

251

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Every value in the interval, 51% to 57%, is above 50%.

GRADING NOTE Any equivalent explanation that uses the interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q2/5 • NEAREST RELATED

A poll reports 48% support with a margin of error of 4 percentage points.

Which statement fits?

A. The poll must be wrong.

B. Fewer than half support the plan, for certain.

C. The true percent could be above or below 50%.

D. The true percent is exactly 48%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

252

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The true percent could be above or below 50%.

GRADING NOTE The interval 44% to 52% includes values on both sides of 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Larger random samples tend to have smaller margins of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

253

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More data gives less variation from sample to sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A margin of error of 3 percentage points means the poll is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

254

QUESTION ID
S6-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It gives a range of plausible values, not the exact error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 6 • FORWARD 2 • Q5/5 • NEAREST RELATED

Interpret the terms, factors, and coefficients of an expression in context

MAIN	Algebra I main question	**HSA-SSE.A.1.a**
BACKWARD 1	Identify the parts of an expression	**6.EE.A.2.b**
BACKWARD 2	Rewrite an expression to show how quantities are related	**7.EE.A.2**
FORWARD 1	Rewrite a rational expression and interpret its parts	**HSA-APR.D.6**
FORWARD 2	Compose functions and interpret the composite in context	**HSF-BF.A.1.c**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A gym charges C = 35 + 20m dollars for m months of membership.

What do 35 and 20 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

257

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

35 is a one-time joining fee in dollars; 20 is the cost in dollars for each month.

GRADING NOTE Any equivalent wording. Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The expression 1.08p gives the total cost, including sales tax, of an item with price p.

What does the factor 1.08 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

258

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

The total is 108% of the price: the price plus 8% tax (1.08p = p + 0.08p).

GRADING NOTE Any equivalent wording that names the 8% tax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

The perimeter of a rectangle is 2(l + w).

What does the factor (l + w) represent?

A. The area

B. The length of one side

C. Half the perimeter: one length plus one width

D. Twice the perimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

259

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Half the perimeter: one length plus one width

GRADING NOTE 2(l + w) is two copies of (l + w), so (l + w) is half of the perimeter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

For the expression $3x^2 - 5x + 7$, name its terms, the coefficient of x, and the constant term.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

260

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

Terms: $3x^2$, $-5x$, and 7; the coefficient of x is -5; the constant term is 7.

GRADING NOTE Any equivalent form. All three parts are required. The sign belongs to the term: the x-term is $-5x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

True or false?

A car's value is $500(0.9)^t$ dollars after t years. The factor 0.9 means the car loses 90% of its value each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

261

QUESTION ID
S7-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Multiplying by 0.9 keeps 90% of the value each year, so the car loses 10% each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Identify the parts of an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.b

How many terms does 4a + 3b − 2 have?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

263

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. The terms are 4a, 3b, and −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of y in 7y − 2?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

264

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. The coefficient is the number multiplying y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q2/5

Which description fits 5(x + 2)?

A. A product of two factors, 5 and (x + 2)

B. A sum of three terms

C. A single term with coefficient 2

D. A quotient of two numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

265

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A product of two factors, 5 and (x + 2)

GRADING NOTE 5(x + 2) means 5 times (x + 2), a product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 8 + 2n, the term 2n has a coefficient of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

266

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2n = 2 \cdot n$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q4/5

True or false?

The terms of $6(m - 1)$ are 6 and m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

267

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6(m - 1)$ is a product with the factors 6 and $(m - 1)$; expanded, its terms are $6m$ and -6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Rewrite an expression to show how quantities are related • Grade 7 • Expressions & Equations • 7.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite a + 0.05a as a single term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

269

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.05a

GRADING NOTE Form required: a single term. a + 0.05a = (1 + 0.05)a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A price p is lowered by 20%.

Write the new price as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

270

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8p

GRADING NOTE Form required: a product. $p - 0.2p = 0.8p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q2/5

Which expression shows a 15% increase of n?

A. 1.15n

B. 15n

C. n + 15

D. 0.15n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

271

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1.15n

GRADING NOTE n + 0.15n = 1.15n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$p - 0.25p = 0.75p$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

272

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $p - 0.25p = (1 - 0.25)p$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$x + 0.3x = 0.3x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

273

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 0.3x = 1.3x$; adding like terms does not multiply them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Rewrite a rational expression and interpret its parts • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

A company's average cost per item for n items is (500 + 3n) ÷ n dollars.

Rewrite it as a sum of two terms, and say what each term means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

275

QUESTION ID
S7-F1-Q1

TYPE
Short answer

CORRECT ANSWER
3 + 500/n: $3 is the cost of making each item, and 500/n is each item's share of a $500 fixed cost.

GRADING NOTE Form required: 3 + 500/n (in either order). Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume x ≠ 0. Rewrite (2x + 5) ÷ x in the form a + b/x.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

276

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 5/x

GRADING NOTE Form required: a + b/x. (2x + 5)/x = 2x/x + 5/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression is equal to $(x^2 + 3x) \div x$?

A. $x^2 + 3$

B. $x + 3x$

C. $x + 3$

D. $4x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

277

QUESTION ID
S7-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. x + 3

GRADING NOTE Divide each term by x: $x^2/x + 3x/x = x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q3/5

True or false?

$(6x + 4) \div 2 = 3x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

278

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide each term by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q4/5

True or false?

Assume $x \neq 0$. $(x + 6) \div x = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

279

QUESTION ID
S7-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 6)/x = 1 + 6/x$, which depends on x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Compose functions and interpret the composite in context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

T(h) = 20 + 5h is a temperature in °C after h hours, and h(t) = t ÷ 60 converts t minutes to hours.

Write T(h(t)) in simplest form, and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

281

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

T(h(t)) = 20 + t/12: the temperature after t minutes.

GRADING NOTE Form required: simplest form, 20 + t/12 (or t/12 + 20); 20 + 5t/60 is equivalent but not simplified. The meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 3$ and $g(x) = 2x$.

Find $f(g(4))$.

$f(g(4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

282

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form. g(4) = 8, and f(8) = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x - 1$.

Which expression is f(g(x))?

A. $x - 1^2$

B. $x^2 - 1$

C. $x^2 - x$

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

283

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(x - 1)^2$

GRADING NOTE Put g(x) into f: $f(x - 1) = (x - 1)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q3/5

True or false?

If f(x) = 3x and g(x) = x + 2, then g(f(1)) = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

284

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(1) = 3, and g(3) = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q4/5

True or false?

f(g(x)) is always equal to g(f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

285

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For $f(x) = x^2$ and $g(x) = x - 1$, $f(g(2)) = 1$ but $g(f(2)) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S7-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 7 • FORWARD 2 • Q5/5

Interpret complicated expressions by viewing one or more of their parts as a single quantity

MAIN	Algebra I main question	**HSA-SSE.A.1.b**
BACKWARD 1	Apply the distributive property to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Evaluate expressions for given values	**6.EE.A.2.c**
FORWARD 1	Use the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Solve exponential growth models with logarithms	**HSF-LE.A.4**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

An investment of P dollars is worth $P(1 + r)^n$ dollars after n years, where $r > 0$.

Treating $(1 + r)^n$ as one quantity, explain why the investment grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

288

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

Since $r > 0$, $1 + r > 1$, so $(1 + r)^n > 1$ for $n > 0$. P times a number greater than 1 is greater than P.

GRADING NOTE Any equivalent explanation that treats $(1 + r)^n$ as one factor greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

For which value of x is $10 - (x - 3)^2$ greatest, and what is that greatest value? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

289

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

At x = 3 the value is 10. $(x - 3)^2$ is never negative and is 0 only at x = 3, so subtracting it gives at most 10.

GRADING NOTE Form required: the x-value, the greatest value, and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

Which expression is never greater than 4, for every real number x?

A. $(x + 1)^2 - 4$

B. $4 - (x + 1)^2$

C. $4(x + 1)^2$

D. $4 + (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

290

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4 - (x + 1)^2$

GRADING NOTE $(x + 1)^2 \geq 0$, so $4 - (x + 1)^2 \leq 4$. The other expressions can be larger than 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A square picture with side s is surrounded by a frame 2 units wide on every side. The frame's area is $(s + 4)^2 - s^2$.

What do $(s + 4)^2$ and s^2 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

291

QUESTION ID
S8-M-Q4

TYPE
Short answer

CORRECT ANSWER
$(s + 4)^2$ is the area of the whole framed square (side s + 2 + 2); s^2 is the area of the picture. Their difference is the frame.

GRADING NOTE Any equivalent wording. Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

True or false?

For every real number x, $5 + 3(x - 2)^2 \geq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

292

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $3(x - 2)^2$ is never negative, so adding it to 5 gives at least 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Apply the distributive property to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

3(2x + 5)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

294

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6x + 15

GRADING NOTE Form required: expanded. Multiply each term by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q1/5

Factor using the greatest common factor.

12y + 18

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

295

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6(2y + 3)

GRADING NOTE Form required: factored with the greatest common factor 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is equal to 4(a − 3) + 2a?

A. 6a + 12

B. 6a − 12

C. 6a − 3

D. 4a − 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

296

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6a − 12

GRADING NOTE 4a − 12 + 2a = 6a − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q3/5

True or false?

$2(x + y) = 2x + 2y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

297

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The 2 multiplies both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$3(x + 4) = 3x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

298

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(x + 4) = 3x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(x - 3)^2$ when $x = 7$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

300

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. $(7 - 3)^2 = 4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q1/5

Evaluate $100(1.05)^2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

301

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

110.25

GRADING NOTE Any equivalent form. $1.05^2 = 1.1025$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $10 - (x - 3)^2$ when $x = 1$.

A. 8

B. -6

C. 14

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

302

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE $(1 - 3)^2 = 4$, and $10 - 4 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

When $x = 3$, $(x - 3)^2 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

303

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 - 3 = 0$, and $0^2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

When $x = -2$, $x^2 = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

304

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(-2)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Use the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3 + 6 + 12 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

306

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

189

GRADING NOTE Any equivalent form. $S = 3(2^6 - 1)/(2 - 1) = 3 \cdot 63 = 189$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find $2 + 1 + 0.5 + 0.25 + 0.125$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

307

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.875

GRADING NOTE Any equivalent form. a = 2, r = 0.5, n = 5: 2(1 − 0.03125)/0.5 = 3.875.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q2/5

A geometric series has first term 5, ratio 3, and 4 terms.

What is its sum?

A. 405

B. 80

C. 200

D. 121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

308

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 200

GRADING NOTE $5(3^4 - 1)/(3 - 1) = 5 \cdot 80/2 = 200$ (5 + 15 + 45 + 135).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q3/5

True or false?

$1 + 2 + 4 + \ldots + 2^9 = 2^{10} - 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

309

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sum of 10 terms is $(2^{10} - 1)/(2 - 1) = 1{,}023$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ gives the sum of a geometric series with r = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

310

QUESTION ID
S8-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE With r = 1 the formula divides by 0; the sum is simply na.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Solve exponential growth models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

An investment is worth $1{,}000(1.05)^n$ dollars after n years.

Write the number of years to reach $2,000 as an expression with logarithms, then round it to the nearest tenth.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

312

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

n = log 2 ÷ log 1.05

≈ 14.2 years

GRADING NOTE Form required: the logarithm expression and the rounded value. $1.05^n = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q1/5

A town of 5,000 people grows 3% per year:
$P = 5{,}000(1.03)^t$. When will it reach 8,000? Round to the nearest tenth of a year. (log means base 10.)

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

313

QUESTION ID
S8-F2-Q2

TYPE
Short answer

CORRECT ANSWER
15.9 years

GRADING NOTE Form required: rounded to the nearest tenth. $1.03^t = 1.6$, so $t = \log 1.6 \div \log 1.03 \approx 15.90$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^t = 60$. (log means base 10.)

A. t = log 60 ÷ log 10 ≈ 1.78

B. t = 3

C. t = log 12 ÷ log 2 ≈ 3.58

D. t = 12 ÷ 2 = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

314

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = log 12 ÷ log 2 ≈ 3.58

GRADING NOTE $2^t = 12$, so t = log 12 ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q3/5

True or false?

$\log(1.05^n) = n \cdot \log 1.05$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q4/5

315

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The power property of logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q4/5

True or false?

The solution of $2^x = 10$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

316

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^5 = 32$. The solution is $x = \log 10 \div \log 2 \approx 3.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S8-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 8 • FORWARD 2 • Q5/5

Add and subtract polynomials and explain why the result is a polynomial

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Add and subtract linear expressions	**7.EE.A.1**
BACKWARD 2	Combine like terms to write equivalent expressions	**6.EE.A.4**
FORWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
FORWARD 2	Rewrite a quotient of polynomials as q(x) + r(x)/b(x)	**HSA-APR.D.6**

SET 9 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add. Write the answer in standard form.

$(4x^2 - 3x + 5) + (2x^2 + 7x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

319

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$6x^2 + 4x - 4$

GRADING NOTE Form required: simplified, in standard form (terms in descending powers of x). Add like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-M-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q1/5

Subtract. Write the answer in standard form.

$(5y^3 + 2y - 1) - (y^3 - 4y^2 + 6y - 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

320

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$4y^3 + 4y^2 - 4y + 2$

GRADING NOTE Form required: simplified, in standard form (descending powers of y). Subtracting changes the sign of every term of the second polynomial: $5y^3 - y^3 + 4y^2 + 2y - 6y - 1 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q2/5

Which expression is equal to $(3a^2 - 2ab) - (a^2 + 5ab)$?

A. $2a^2 - 7ab$

B. $2a^2 - 3ab$

C. $2a^2 + 3ab$

D. $4a^2 + 3ab$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

321

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $2a^2 - 7ab$

GRADING NOTE $3a^2 - a^2 = 2a^2$, and $-2ab - 5ab = -7ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q3/5

Explain why the sum of two polynomials is always a polynomial.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

322

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

Adding combines like terms by adding their coefficients. The coefficients stay real numbers and the exponents stay whole numbers, so the result is a polynomial.

GRADING NOTE Any equivalent explanation (closure: the result has the same form).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q4/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

323

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtracting adds the opposite of each term, which is still a sum of terms with whole-number exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • MAIN • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 1

Add and subtract linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$(3x + 4) + (5x - 9)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

325

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x − 5

GRADING NOTE Form required: simplified. Combine like terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q1/5

Subtract.

$(7a - 2) - (3a - 6)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

326

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4a + 4

GRADING NOTE Form required: simplified. $-(3a - 6) = -3a + 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q2/5

Subtract.

$(2y + 1) - (y + 5)$

A. $y - 6$

B. $3y - 4$

C. $y + 6$

D. $y - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

327

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y - 4$

GRADING NOTE $2y - y = y$, and $1 - 5 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$-(x - 3) = -x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

328

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The minus sign changes the sign of each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$(4n + 2) - (n + 2) = 3n + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

329

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4n + 2) - (n + 2) = 3n$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

BACKWARD 2

Combine like terms to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.4

Simplify.

4m + 2 + m

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

331

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5m + 2

GRADING NOTE Form required: simplified. 4m + m = 5m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q1/5

Are 2(3 + x) and 6 + 2x equivalent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

332

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(3 + x) = 6 + 2x for every x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q2/5

Which expression is equal to b + b + b + 4?

A. 7b

B. 3b + 4

C. 4b + 3

D. $b^3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

333

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3b + 4

GRADING NOTE b + b + b = 3b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q3/5

True or false?

$x + x = 2x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

334

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two copies of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q4/5

True or false?

$5y - y = 4y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

335

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $5y - 1y = 4y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 1

Use the Remainder Theorem • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^2 + 3x - 4$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

337

QUESTION ID
S9-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE Any equivalent form. By the Remainder Theorem, the remainder is $p(2) = 4 + 6 - 4 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q1/5

Is $x + 1$ a factor of $x^3 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

338

QUESTION ID
S9-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes.
$p(-1) = -1 + 1 = 0$, so the remainder is 0.

GRADING NOTE Any equivalent explanation using $p(-1)$ or division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q2/5

What is the remainder when $p(x) = x^3 - 2x + 5$ is divided by $x - 1$?

A. 8

B. 4

C. -4

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

339

QUESTION ID
S9-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 4

GRADING NOTE $p(1) = 1 - 2 + 5 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q3/5

True or false?

If p(3) = 0, then x − 3 is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

340

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A remainder of 0 means x − 3 divides p(x) evenly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q4/5

True or false?

The remainder when p(x) is divided by x + 2 is p(2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

341

QUESTION ID
S9-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE x + 2 = x − (−2), so the remainder is p(−2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-APR.A.1

FORWARD 2

Rewrite a quotient of polynomials as q(x) + r(x)/b(x) • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq -2$. Write $(x^2 + 5x + 7) \div (x + 2)$ in the form $q(x) + r/(x + 2)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

343

QUESTION ID
S9-F2-Q1

TYPE
Short answer

CORRECT ANSWER
x + 3 + 1/(x + 2)

GRADING NOTE Form required: $q(x) + r/(x + 2)$. $(x + 2)(x + 3) = x^2 + 5x + 6$, which leaves a remainder of 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume x ≠ 0. Write (3x + 7) ÷ x in the form a + b/x.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

344

QUESTION ID
S9-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3 + 7/x

GRADING NOTE Form required: a + b/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q2/5

Assume $x \neq 1$. Which expression is equal to $(x^2 - 1) \div (x - 1)$?

A. $x + 1$

B. x

C. $x^2 + 1$

D. $x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

345

QUESTION ID
S9-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. x + 1

GRADING NOTE $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q3/5

True or false?

Assume $x \neq 0$. $(2x + 1) \div x = 2 + 1/x$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

346

QUESTION ID
S9-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $2x/x + 1/x = 2 + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q4/5

True or false?

Assume $x \neq -2$. $(x^2 + 4) \div (x + 2) = x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q5/5

347

QUESTION ID
S9-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 2)^2 = x^2 + 4x + 4$, not $x^2 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 9 • FORWARD 2 • Q5/5

Multiply polynomials

MAIN	Algebra I main question	**HSA-APR.A.1**
BACKWARD 1	Apply the distributive property to linear expressions	**6.EE.A.3**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Expand powers of binomials with the binomial theorem	**HSA-APR.C.5**

SET 10 • HSA-APR.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply. Write the answer in standard form.

$(2x + 3)(x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

350

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$2x^2 - 7x - 15$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $2x^2 - 10x + 3x - 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q1/5

Multiply. Write the answer in standard form.

$(x - 4)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

351

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 8x + 16$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $(x - 4)(x - 4) = x^2 - 4x - 4x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q2/5

Multiply. Write the answer in standard form.

$(y + 2)(y^2 - 3y + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

352

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

$y^3 - y^2 - 5y + 2$

GRADING NOTE Form required: expanded, simplified, in standard form (descending powers). $y^3 - 3y^2 + y + 2y^2 - 6y + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-M-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q3/5

Multiply.

$(3a - 1)(3a + 1)$

A. $9a^2 - 1$

B. $9a^2 - 6a - 1$

C. $6a - 1$

D. $9a^2 + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

353

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $9a^2 - 1$

GRADING NOTE The middle terms cancel: $9a^2 + 3a - 3a - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q4/5

True or false?

$(x + 3)^2 = x^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

354

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$; the middle term 6x comes from 3x + 3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 10 • MAIN • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 1

Apply the distributive property to linear expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

$4(3x - 2)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

356

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$12x - 8$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand.

$-2(5 - 3a)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

357

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−10 + 6a

GRADING NOTE Form required: expanded (6a − 10 is the same). −2 times −3a is +6a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q2/5

Expand.

$7(2b + 1)$

A. $14b + 8$

B. $14b + 7$

C. $9b + 8$

D. $14b + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

358

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 14b + 7

GRADING NOTE $7 \cdot 2b = 14b$ and $7 \cdot 1 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$5(m - 2) = 5m - 10$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

359

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 multiplies both terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3(2x + 4) = 6x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

360

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3(2x + 4) = 6x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$7^2 \cdot 7^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

362

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7^5

GRADING NOTE Form required: a single power. 2 + 3 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$10^4 \cdot 10$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

363

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^5

GRADING NOTE Form required: a single power. $10 = 10^1$, and $4 + 1 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write as a single power.

$6^3 \cdot 6^3$

A. 6^6

B. 6^9

C. 12^6

D. 36^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

364

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6^6

GRADING NOTE Add the exponents: 3 + 3 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$9 \cdot 9^4 = 9^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

365

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9^1 \cdot 9^4 = 9^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$2^3 \cdot 2^3 = 2^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

366

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 \cdot 2^3 = 2^6$; exponents are added, not multiplied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(a + b)^2 - (a - b)^2 = 4ab$ for all numbers a and b.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

368

QUESTION ID
S10-F1-Q1

TYPE
Answer with required work

CORRECT ANSWER
$(a + b)^2 - (a - b)^2 = 4ab$

GRADING NOTE REQUIRED WORK: Expand both squares: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. The steps must be shown.

Any correct algebraic proof is accepted; checking numbers alone is not a proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q1/5

Use the identity $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ to factor $x^3 - 8$.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

369

QUESTION ID
S10-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)(x^2 + 2x + 4)$

GRADING NOTE Form required: factored. Here y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q2/5

Which identity is true for all numbers a and b?

A. $(a - b)^2 = a^2 - b^2$

B. $(a + b)(a - b) = a^2 - b^2$

C. $(a + b)^2 = a^2 + b^2$

D. $a^2 + b^2 = (a + b)(a + b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

370

QUESTION ID
S10-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. $(a + b)(a - b) = a^2 - b^2$

GRADING NOTE $(a + b)(a - b) = a^2 - ab + ab - b^2 = a^2 - b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ for all x and y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

371

QUESTION ID
S10-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both sides expand to $x^4 + 2x^2y^2 + y^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

372

QUESTION ID
S10-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the binomial theorem • High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand. Write the answer in standard form.

$(2x - 1)^3$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

374

QUESTION ID
S10-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$8x^3 - 12x^2 + 6x - 1$

GRADING NOTE Form required: expanded, in standard form (descending powers). Coefficients 1, 3, 3, 1 with $(2x)^k$ and $(-1)^{3-k}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q1/5

What is the coefficient of the x^3y^2 term in the expansion of $(x + y)^5$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

375

QUESTION ID
S10-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. It is 5 choose 2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q2/5

How many terms does the expansion of $(a + b)^7$ have?

A. 128

B. 8

C. 14

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

376

QUESTION ID
S10-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 8

GRADING NOTE $(a + b)^n$ has n + 1 terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q3/5

True or false?

The number 5 choose 2 equals 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

377

QUESTION ID
S10-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $5 \cdot 4 \div 2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q4/5

True or false?

Every coefficient in the expansion of $(a - b)^4$ is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

QUESTION ID
S10-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(a - b)^4 = a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5
High School — Precalculus (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 10 • FORWARD 2 • Q5/5

Use structure to factor with a greatest common factor and the difference of squares

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Use the greatest common factor and the distributive property	**6.NS.B.4**
BACKWARD 2	Multiply a sum and a difference	**HSA-APR.A.1**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Factor sums of squares with complex numbers	**HSN-CN.C.8**

SET 11 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor completely.

$6x^3 - 15x^2$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

381

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^2(2x - 5)$

GRADING NOTE Form required: factored completely. The greatest common factor is $3x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor completely.

$49y^2 - 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

382

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

(7y − 5)(7y + 5)

GRADING NOTE Form required: factored completely (factors in either order). It is a difference of squares: $(7y)^2 - 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q2/5

Factor completely.

$x^4 - 16$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

383

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x^2 + 4)(x + 2)(x - 2)$

GRADING NOTE Form required: factored completely over the integers. $x^4 - 16 = (x^2)^2 - 4^2 = (x^2 + 4)(x^2 - 4)$, and $x^2 - 4$ factors again; $x^2 + 4$ does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q3/5

Which is the complete factorization of $2x^2 - 18$?

A. $(2x - 6)(x + 3)$

B. $2(x - 3)(x + 3)$

C. $(x - 3)(x + 3)$

D. $2(x^2 - 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

384

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $2(x - 3)(x + 3)$

GRADING NOTE Take out the common factor 2, then factor $x^2 - 9$. $(2x - 6)(x + 3)$ and $2(x^2 - 9)$ are equal to $2x^2 - 18$ but not completely factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q4/5

True or false?

$x^2 + 9$ can be factored as $(x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

385

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$. A sum of squares like $x^2 + 9$ does not factor over the real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 1

Use the greatest common factor and the distributive property • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

387

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the greatest common factor to write 42 + 28 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

388

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14(3 + 2)

GRADING NOTE Form required: the GCF times a sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q2/5

What is the greatest common factor of 15 and 40?

A. 5

B. 8

C. 120

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

389

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE $15 = 3 \cdot 5$ and $40 = 8 \cdot 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q3/5

True or false?

$18 + 12 = 6(3 + 2)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

390

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6 \cdot 3 = 18$ and $6 \cdot 2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q4/5

True or false?

The greatest common factor of 9 and 16 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

391

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 and 16 share no factor greater than 1, so their GCF is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

BACKWARD 2

Multiply a sum and a difference • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x - 7)(x + 7)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

393

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - 49$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(2y + 3)(2y - 3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

394

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$4y^2 - 9$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q2/5

Multiply.

$(x + 4)(x - 4)$

A. $x^2 - 8$

B. $x^2 + 16$

C. $x^2 - 8x - 16$

D. $x^2 - 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

395

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 16$

GRADING NOTE The middle terms 4x and −4x cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$(a - 5)(a + 5) = a^2 - 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

396

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $a^2 + 5a - 5a - 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$(x + 3)(x - 3) = x^2 - 6x - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

397

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

What are the zeros of p(x) = x(x − 2)(x + 5)?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

399

QUESTION ID
S11-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0, 2, and −5

GRADING NOTE Any order. All three are required. Each factor equals 0 at one zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q1/5

Factor $x^3 - 9x$ completely, then give all of its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

400

QUESTION ID
S11-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$x(x - 3)(x + 3)$; zeros 0, 3, and −3

GRADING NOTE Form required: factored completely, and all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q2/5

Which polynomial has the zeros −1, 0, and 4?

A. (x + 1)(x − 4)

B. x(x + 1)(x + 4)

C. x(x + 1)(x − 4)

D. x(x − 1)(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

401

QUESTION ID
S11-F1-Q3

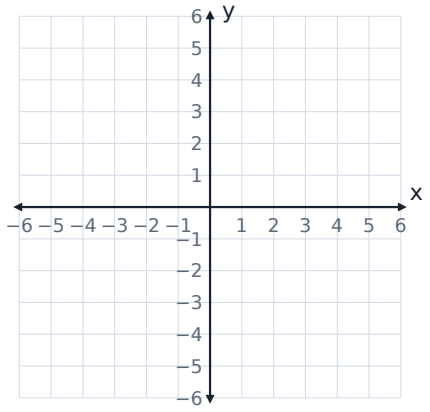
TYPE
Multiple choice

CORRECT ANSWER
C. x(x + 1)(x − 4)

GRADING NOTE A zero r comes from the factor (x − r): x, (x + 1), and (x − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = x(x - 2)(x + 5)$. Mark the zeros and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

402

QUESTION ID

S11-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −5, 0, and 2, falls to the left and rises to the right, with one turn between −5 and 0 and one between 0 and 2

GRADING NOTE Any correct sketch: crossings at −5, 0, and 2, and end behavior of a cubic with positive leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q4/5

True or false?

$p(x) = (x - 3)^2(x + 1)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

403

QUESTION ID
S11-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its zeros are 3 (twice) and −1, only two different values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.A.2

FORWARD 2

Factor sums of squares with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

405

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5i)(x - 5i)$

GRADING NOTE Form required: two linear factors. $(x + 5i)(x - 5i) = x^2 - 25i^2 = x^2 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 16 = 0$. Give all solutions.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

406

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4i and x = −4i

GRADING NOTE Any equivalent form (±4i). Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q2/5

Which is the complete factorization of $x^4 - 16$ into linear factors with complex numbers?

A. $(x - 4)(x + 4)$

B. $(x^2 - 4)(x^2 + 4)$

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

D. $(x - 2)^2(x + 2)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

407

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 2)(x + 2)(x - 2i)(x + 2i)$

GRADING NOTE $x^2 + 4 = (x - 2i)(x + 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q3/5

True or false?

$(x + i)(x - i) = x^2 + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q4/5

408

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - i^2 = x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S11-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q4/5

True or false?

$x^2 + 4$ has no linear factors, even with complex numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

409

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 4 = (x + 2i)(x - 2i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 11 • FORWARD 2 • Q5/5

Factor quadratic trinomials

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find factor pairs of a whole number	**4.OA.B.4**
BACKWARD 2	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Use factored form to describe the graph of a polynomial function	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its zeros and find zeros by factoring	**HSA-APR.B.3**

SET 12 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 7x + 12$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

412

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x + 4)

GRADING NOTE Form required: factored (factors in either order). 3 · 4 = 12 and 3 + 4 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q1**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q1/5

Factor.

$x^2 - 2x - 15$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

413

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 5)(x + 3)$

GRADING NOTE Form required: factored. $-5 \cdot 3 = -15$ and $-5 + 3 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-M-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q2/5

Factor completely.

$2x^2 + 7x + 3$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

414

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2x + 1)(x + 3)

GRADING NOTE Form required: factored. Check: $2x^2 + 6x + x + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q3/5

Which is a factor of $3x^2 - 10x - 8$?

A. $3x - 2$

B. $x + 4$

C. $x - 4$

D. $x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

415

QUESTION ID

S12-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x - 4$

GRADING NOTE $3x^2 - 10x - 8 = (3x + 2)(x - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q4/5

True or false?

$x^2 + 5x + 6 = (x + 1)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

416

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 1)(x + 6) = x^2 + 7x + 6$. The correct factoring is $(x + 2)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 1

Find factor pairs of a whole number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 12.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

418

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 and 12, 2 and 6, 3 and 4

GRADING NOTE Any order. All three pairs are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q1/5

Two whole numbers multiply to 15 and add to 8.

What are they?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

419

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 and 5

GRADING NOTE Any order. 3 · 5 = 15 and 3 + 5 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q2/5

Which factor pair of 24 has a sum of 10?

A. 4 and 6

B. 3 and 8

C. 1 and 24

D. 2 and 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

420

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 and 6

GRADING NOTE $4 + 6 = 10$, and $4 \cdot 6 = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q3/5

True or false?

3 and 7 are a factor pair of 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

421

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 7 = 21$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q4/5

True or false?

4 and 5 are a factor pair of 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

422

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \cdot 5 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

BACKWARD 2

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x + 2)(x + 9)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

424

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 11x + 18$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q1/5

Multiply.

$(x - 6)(x + 1)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

425

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 5x - 6$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q2/5

Multiply.

$(3x + 1)(x + 2)$

A. $3x^2 + 6x + 2$

B. $3x^2 + 7x + 2$

C. $4x^2 + 7x + 2$

D. $3x^2 + 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

426

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3x^2 + 7x + 2$

GRADING NOTE $3x^2 + 6x + x + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q3/5

True or false?

$(x - 2)(x - 6) = x^2 - 8x + 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

427

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - 6x - 2x + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$(x + 1)(x - 4) = x^2 - 3x + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q5/5

428

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The constant is $1 \cdot (-4) = -4$: $x^2 - 3x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 1

Use factored form to describe the graph of a polynomial function • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$p(x) = (x + 2)(x - 1)(x - 3)$.

What are its zeros and its y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

430

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Zeros −2, 1, and 3; y-intercept (0, 6)

GRADING NOTE Any equivalent form. All parts are required. $p(0) = 2 \cdot (-1) \cdot (-3) = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q1/5

Describe the end behavior of $y = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

431

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

As x increases without bound, y increases without bound; as x decreases without bound, y decreases without bound.

GRADING NOTE Any equivalent wording (up to the right, down to the left). The leading term x^3 decides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q2/5

Which describes the end behavior of $y = -x^3$?

A. It rises to the left and falls to the right.

B. It rises on both ends.

C. It falls to the left and rises to the right.

D. It falls on both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

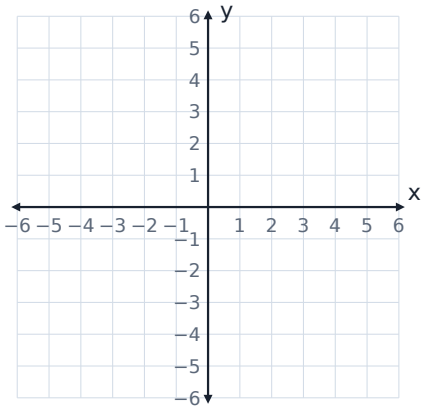
A. It rises to the left and falls to the right.

GRADING NOTE For large positive x, $-x^3$ is very negative; for large negative x, it is very positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q3/5

Graph
$p(x) = (x + 2)(x - 1)(x - 3)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

433

QUESTION ID
S12-F1-Q4

TYPE
Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 1, and 3, through (0, 6), falling to the left and rising to the right

GRADING NOTE Any correct graph. $p(0) = (2)(-1)(-3) = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q4/5

True or false?

The graph of $y = (x - 1)^2$ crosses the x-axis at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

434

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It touches the x-axis at $x = 1$ and turns back; $(x - 1)^2$ is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.A.2

FORWARD 2

Write a polynomial from its zeros and find zeros by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write a polynomial function of degree 3 whose only zeros are 2, −3, and 5.

p(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

436

QUESTION ID
S12-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(x − 2)(x + 3)(x − 5)

GRADING NOTE Any equivalent form, or any nonzero constant multiple, such as 2(x − 2)(x + 3)(x − 5). A degree-3 polynomial with these three zeros has exactly these factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q1/5

Factor $x^3 - x^2 - 6x$ completely, then give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

437

QUESTION ID
S12-F2-Q2

TYPE
Short answer

CORRECT ANSWER
x(x − 3)(x + 2);
zeros 0, 3, and −2

GRADING NOTE Form required: factored completely, and all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q2/5

What are the zeros of p(x) = (2x − 1)(x + 4)?

A. 2 and −4

B. 1 and 4

C. $\frac{1}{2}$ and −4

D. $-\frac{1}{2}$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

438

QUESTION ID
S12-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 1/2 and −4

GRADING NOTE 2x − 1 = 0 gives x = 1/2, and x + 4 = 0 gives x = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q3/5

True or false?

If $(x - 7)$ is a factor of $p(x)$, then 7 is a zero of p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

439

QUESTION ID
S12-F2-Q4

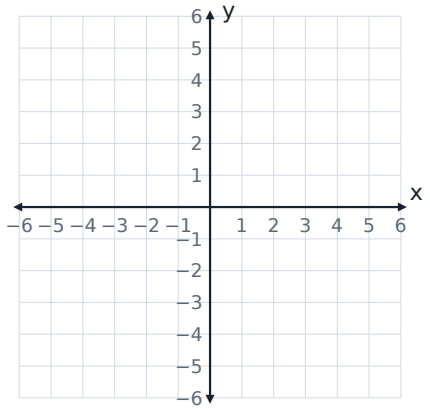
TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $p(7)$ has the factor $7 - 7 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q4/5

Sketch a rough graph of $p(x) = (x - 1)^2(x + 2)(x - 3)$. Show what the graph does at each zero and its end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

440

QUESTION ID
S12-F2-Q5

TYPE
Drawing / plotting

CORRECT ANSWER
A curve rising on both ends, crossing the x-axis at −2 and 3 and touching it at 1 (double zero)

GRADING NOTE Any correct sketch: degree 4 with positive leading coefficient, so it rises on both ends; even multiplicity at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 12 • FORWARD 2 • Q5/5

Use structure to factor perfect squares and expressions in quadratic form

MAIN	Algebra I main question	**HSA-SSE.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Square a binomial	**HSA-APR.A.1**
FORWARD 1	Find the zeros of polynomial functions in quadratic form	**HSF-IF.C.7.c**
FORWARD 2	Factor quadratic expressions with complex numbers	**HSN-CN.C.8**

SET 13 • HSA-SSE.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor.

$x^2 + 10x + 25$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

443

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 5)^2$

GRADING NOTE Form required: factored; $(x + 5)(x + 5)$ is also correct. It is a perfect square: $x^2 + 2 \cdot 5 \cdot x + 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q1/5

Factor completely. (Think of x^2 as one quantity.)

$x^4 - 5x^2 + 4$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

444

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 1)(x + 1)(x - 2)(x + 2)$

GRADING NOTE Form required: factored completely. With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = (x^2 - 1)(x^2 - 4)$, and each factors again.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q2/5

Factor, and simplify each factor. (Think of x + 1 as one quantity.)

$(x + 1)^2 - 9$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

445

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x + 4)

GRADING NOTE Form required: factored and simplified. It is a difference of squares: ((x + 1) − 3)((x + 1) + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q3/5

Which expression is a perfect square trinomial?

A. $x^2 + 9$

B. $4x^2 - 12x + 9$

C. $4x^2 + 6x + 9$

D. $x^2 - 6x - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

446

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 12x + 9$

GRADING NOTE $4x^2 - 12x + 9 = (2x - 3)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q4/5

True or false?

$9x^2 + 30x + 25 = (3x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

447

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(3x + 5)^2 = 9x^2 + 2 \cdot 3x \cdot 5 + 25 = 9x^2 + 30x + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 13 • MAIN • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

449

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What whole number squared is 169?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

450

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 13 · 13 = 169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

2.5^2

A. 5

B. 4.25

C. 62.5

D. 6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

451

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6.25

GRADING NOTE $2.5 \cdot 2.5 = 6.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$11^2 = 121$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

452

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $11 \cdot 11 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$6^2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

453

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $6^2 = 6 \cdot 6 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

BACKWARD 2

Square a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x + 5)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

455

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 10x + 25$

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q1/5

Multiply.

$(3y - 2)^2$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

456

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$9y^2 - 12y + 4$

GRADING NOTE Form required: expanded. $(3y)^2 - 2 \cdot 3y \cdot 2 + 2^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q2/5

Multiply.

$(x - 6)^2$

A. $x^2 - 6x + 36$

B. $x^2 - 36$

C. $x^2 + 36$

D. $x^2 - 12x + 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

457

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 - 12x + 36$

GRADING NOTE $x^2 - 6x - 6x + 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-B2-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$(a + 1)^2 = a^2 + 2a + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q4/5

458

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a + 1)(a + 1) = a^2 + a + a + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q4/5

True or false?

$(x - 3)^2 = x^2 - 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

459

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 3)^2 = x^2 - 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 1

Find the zeros of polynomial functions in quadratic form • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Find all real zeros of $f(x) = x^4 - 5x^2 + 4$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

461

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2, −1, 1, and 2

GRADING NOTE Any order. All four are required. $f(x) = (x^2 - 1)(x^2 - 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q1/5

Find all real zeros of $y = x^4 - 9x^2$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

462

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 3, and −3

GRADING NOTE Any order. All three are required. $x^4 - 9x^2 = x^2(x - 3)(x + 3)$; 0 is a double zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q2/5

How many real zeros does $y = x^4 - 1$ have?

A. 2

B. 1

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

463

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

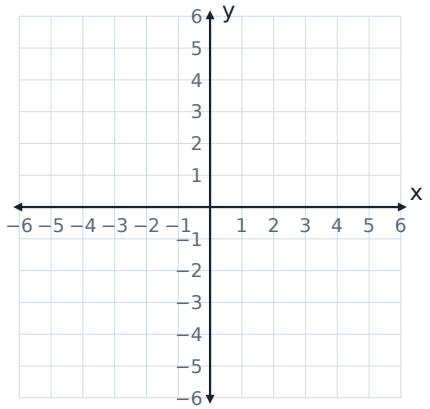
A. 2

GRADING NOTE $x^4 - 1 = (x^2 + 1)(x - 1)(x + 1)$: only 1 and −1 are real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q3/5

Graph $f(x) = x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$. Mark the zeros and the y-intercept, and show the end behavior.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

464

QUESTION ID

S13-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A W-shaped curve crossing the x-axis at −2, −1, 1, and 2, with a high point at (0, 4), low points near (±1.6, −2.25), rising on both ends

GRADING NOTE Any correct graph showing all four zeros, the y-intercept 4, and both ends rising.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$y = x^4 + 1$ has a real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

465

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^4 + 1 \geq 1$ for every real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-SSE.A.2

FORWARD 2

Factor quadratic expressions with complex numbers • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 49$ using complex numbers.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

467

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 7i)(x − 7i)

GRADING NOTE Form required: two linear factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q1/5

The solutions of $x^2 + 2x + 5 = 0$ are $-1 + 2i$ and $-1 - 2i$.

Write $x^2 + 2x + 5$ as a product of two linear factors.

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

468

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1 - 2i)(x + 1 + 2i)$

GRADING NOTE Form required: two linear factors. A zero r gives the factor $(x - r)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q2/5

Multiply.

$(x - 3i)(x + 3i)$

A. $x^2 - 9$

B. $x^2 + 6ix - 9$

C. $x^2 + 9$

D. $x^2 - 9i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

469

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x^2 + 9$

GRADING NOTE $x^2 - 9i^2 = x^2 + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q3/5

True or false?

$x^4 - 1 = (x - 1)(x + 1)(x - i)(x + i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q4/5

470

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^4 - 1 = (x^2 - 1)(x^2 + 1)$, and $x^2 + 1 = (x - i)(x + i)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q4/5

True or false?

$(x + 2i)^2 = x^2 - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q5/5

471

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 2i)^2 = x^2 + 4ix + 4i^2 = x^2 + 4ix - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 13 • FORWARD 2 • Q5/5

Factor a quadratic expression to reveal the zeros of the function it defines

MAIN	Algebra I main question	**HSA-SSE.B.3.a**
BACKWARD 1	Factor trinomials with a leading coefficient of 1	**HSA-SSE.A.2**
BACKWARD 2	Use substitution to decide which values make an equation true	**6.EE.B.5**
FORWARD 1	Find zeros of cubic functions by factoring	**HSA-APR.B.3**
FORWARD 2	Sketch polynomial graphs from their zeros	**HSF-IF.C.7.c**

SET 14 • HSA-SSE.B.3.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

$f(x) = x^2 - x - 12$.

Factor f(x), then give the zeros of f.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

474

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 4)(x + 3); zeros 4 and −3

GRADING NOTE Form required: the factored form and both zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q1/5

A ball's height is $h(t) = -16t^2 + 48t$ feet after t seconds.

Factor h(t), and use the factors to say when the ball is at ground level.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

475

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

h(t) = −16t(t − 3); the height is 0 at t = 0 seconds (when it is thrown) and t = 3 seconds (when it lands).

GRADING NOTE Form required: the factored form and both times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q2/5

What are the zeros of $g(x) = 2x^2 + 5x - 3$?

A. $\frac{1}{2}$ and -3

B. $-\frac{1}{2}$ and 3

C. 2 and -3

D. 1 and -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

476

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/2 and −3

GRADING NOTE $2x^2 + 5x - 3 = (2x - 1)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q3/5

Write a quadratic function, in factored form with a leading coefficient of 1, whose zeros are 2 and −5.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

477

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x + 5)

GRADING NOTE Form required: factored form. Factors in either order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q4/5

True or false?

The zeros of f(x) = (x − 6)(x + 1) are 6 and −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

478

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each factor is 0 at one zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 14 • MAIN • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 1

Factor trinomials with a leading coefficient of 1 • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor.

$x^2 + 9x + 20$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

480

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 4)(x + 5)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q1/5

Factor.

$x^2 - 9x + 14$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

481

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x − 7)

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q2**

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q2/5

Factor.

$x^2 + x - 6$

A. $(x - 6)(x + 1)$

B. $(x - 3)(x + 2)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

482

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (x + 3)(x − 2)

GRADING NOTE $3 \cdot (-2) = -6$ and $3 + (-2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

$x^2 - 4x - 21 = (x - 7)(x + 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

483

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-7 \cdot 3 = -21$ and $-7 + 3 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 8x + 15 = (x + 3)(x + 6)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

484

QUESTION ID
S14-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 3)(x + 6) = x^2 + 9x + 18$; the correct factors are $(x + 3)(x + 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

BACKWARD 2

Use substitution to decide which values make an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Is $x = 4$ a solution of $x^2 = 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

486

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $4^2 = 16$.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 make $x^2 - 5x + 6 = 0$ true?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

487

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 and 3

GRADING NOTE Any equivalent form. Both are required. $4 - 10 + 6 = 0$ and $9 - 15 + 6 = 0$; $x = 1$ gives 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q2/5

Which value makes $3x - 6 = 0$ true?

A. 3

B. -2

C. 2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

488

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $3 \cdot 2 - 6 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q3/5

True or false?

$x = 5$ makes $x(x - 5) = 0$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

489

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $5 \cdot 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q4/5

True or false?

$x = -2$ makes $x + 2 = 4$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

490

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-2 + 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 1

Find zeros of cubic functions by factoring • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Find the zeros of $f(x) = x^3 + 2x^2 - 15x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

492

QUESTION ID
S14-F1-Q1

TYPE
Short answer

CORRECT ANSWER
0, 3, and −5

GRADING NOTE Any order. All three are required. $x(x^2 + 2x - 15) = x(x + 5)(x - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q1/5

$g(x) = (x - 1)(x^2 - 4)$.

List all the zeros of g.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

493

QUESTION ID
S14-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1, 2, and −2

GRADING NOTE Any order. All three are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q2**
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q2/5

Which function has a double zero at x = 2?

A. $f(x) = (x - 2)^2(x + 1)$

B. $f(x) = 2x(x - 1)$

C. $f(x) = (x + 2)^2(x - 1)$

D. $f(x) = (x - 2)(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

494

QUESTION ID
S14-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A.
$f(x) = (x - 2)^2(x + 1)$

GRADING NOTE The factor $(x - 2)$ appears twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q3/5

True or false?

$f(x) = x^3 - x$ has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

495

QUESTION ID
S14-F1-Q4

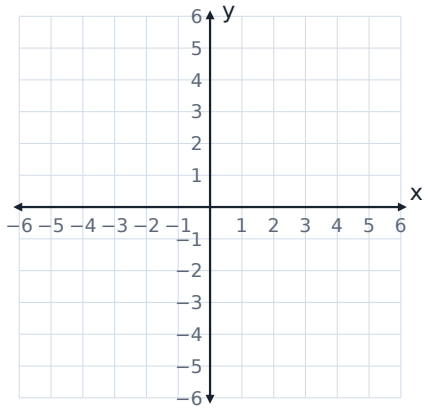
TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x(x - 1)(x + 1)$ has zeros -1, 0, and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q4/5

Sketch a rough graph of $f(x) = (x - 2)^2(x + 1)$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

496

QUESTION ID

S14-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −1, touches it at 2 and turns back up (double zero), passes through (0, 4), falls to the left and rises to the right

GRADING NOTE Any correct sketch: a crossing at −1, a touch at 2 (even multiplicity), y-intercept 4, and cubic end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S14-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-SSE.B.3.a

FORWARD 2

Sketch polynomial graphs from their zeros • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Describe where f(x) = (x + 1)(x − 2)(x − 4) is above the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

498

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

For −1 < x < 2 and for x > 4.

GRADING NOTE Any equivalent form. Test a value in each interval: f(0) = 8 > 0, f(3) = −4 < 0, f(5) = 12 > 0, f(−2) = −24 < 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q1/5

A cubic function has zeros −3, 0, and 3 and a positive leading coefficient.

Describe its end behavior.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

499

QUESTION ID
S14-F2-Q2

TYPE
Short answer

CORRECT ANSWER
It falls to the left and rises to the right.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many turning points can the graph of a cubic function have, at most?

A. 2

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

500

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

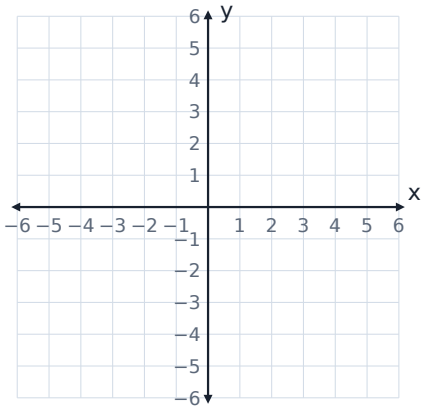
A. 2

GRADING NOTE A polynomial of degree n has at most n − 1 turning points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q3/5

Graph
$f(x) = (x + 1)(x - 2)(x - 4)$.
Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

501

QUESTION ID
S14-F2-Q4

TYPE
Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −1, 2, and 4, through (0, 8), falling to the left and rising to the right; above the axis for $-1 < x < 2$ and $x > 4$

GRADING NOTE Any correct graph. $f(0) = (1)(-2)(-4) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q4/5

True or false?

A polynomial function with a negative leading coefficient always falls on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

502

QUESTION ID
S14-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $y = -x^3$ rises on the left and falls on the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 14 • FORWARD 2 • Q5/5

Complete the square to reveal the maximum or minimum value of a quadratic function

MAIN	Algebra I main question	**HSA-SSE.B.3.b**
BACKWARD 1	Recognize and complete perfect square trinomials	**HSA-SSE.A.2**
BACKWARD 2	Describe where a graph rises, falls, and reaches its highest point	**8.F.B.5**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Derive the equations of ellipses and hyperbolas from their foci	**HSG-GPE.A.3**

SET 15 • HSA-SSE.B.3.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Write f(x) = x² + 6x + 5 in vertex form by completing the square. Show the steps.

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

505

QUESTION ID

S15-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(x + 3)² − 4

GRADING NOTE

REQUIRED WORK: The completed square shown: x² + 6x + 9 − 9 + 5 (add and subtract (6 ÷ 2)² = 9), then (x + 3)² − 4.

Form required: vertex form a(x − h)² + k, with the completing-the-square steps. x² + 6x + 9 − 9 + 5 = (x + 3)² − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q1/5

$g(x) = x^2 - 10x + 30$.

Complete the square to write g in vertex form, then give the minimum value of g and where it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

506

QUESTION ID

S15-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$g(x) = (x - 5)^2 + 5$; the minimum is 5, at $x = 5$.

GRADING NOTE REQUIRED WORK: The completed square shown: $x^2 - 10x + 25 - 25 + 30 = (x - 5)^2 + 5$, and the minimum read from the vertex form.

Form required: the vertex form with its steps, the minimum value, and the x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q2/5

A ball's height is $h(t) = -t^2 + 8t + 2$ meters after t seconds.

Complete the square (write h in vertex form) to find the greatest height and when it happens. Show the steps.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

507

QUESTION ID

S15-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$h(t) = -(t - 4)^2 + 18$; the greatest height is 18 meters, at t = 4 seconds.

GRADING NOTE

REQUIRED WORK: The completed square shown: $-(t^2 - 8t) + 2 = -(t^2 - 8t + 16) + 16 + 2 = -(t - 4)^2 + 18$, and the maximum read from it.

Form required: the vertex form with its steps, the height, and the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q3/5

Which is $x^2 - 4x + 7$ in vertex form?

A. $(x - 4)^2 + 7$

B. $(x - 2)^2 + 7$

C. $(x - 2)^2 + 3$

D. $(x + 2)^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

508

QUESTION ID

S15-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 2)^2 + 3$

GRADING NOTE $x^2 - 4x + 4 + 3 = (x - 2)^2 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q4/5

True or false?

The minimum value of $(x + 1)^2 - 6$ is -6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

509

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1)^2 \geq 0$, so the value is at least -6, reached at $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 15 • MAIN • Q5/5

SET 15 • HSA-SSE.B.3.b

BACKWARD 1

Recognize and complete perfect square trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor.

$x^2 + 14x + 49$

Factored: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

511

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 7)^2$

GRADING NOTE Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q1/5

What number c makes $x^2 - 8x + c$ a perfect square trinomial?

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

512

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form. $(-8/2)^2 = 16$, giving $(x - 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q2/5

Which number completes $x^2 + 12x + ___$ to a perfect square trinomial?

A. 36

B. 6

C. 24

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

513

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 36

GRADING NOTE $(12/2)^2 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q3/5

True or false?

$x^2 - 2x + 1 = (x - 1)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

514

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 1)^2 = x^2 - 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 6x + 6$ is a perfect square trinomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

515

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$, not + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 15 • BACKWARD 1 • Q5/5

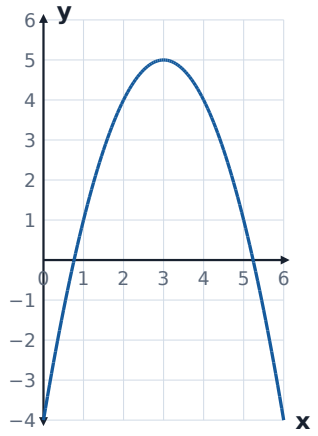
SET 15 • HSA-SSE.B.3.b

BACKWARD 2

Describe where a graph rises, falls, and reaches its highest point • Grade 8 • Functions • 8.F.B.5

Where is the highest point of the graph?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

517

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 5)

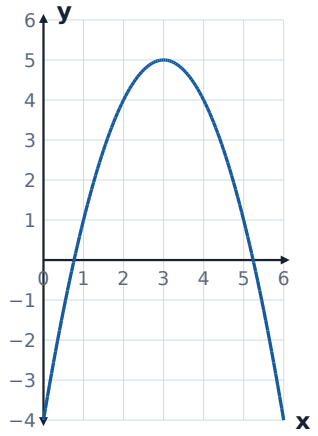
GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q1/5

For which x-values shown is the graph rising?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

518

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

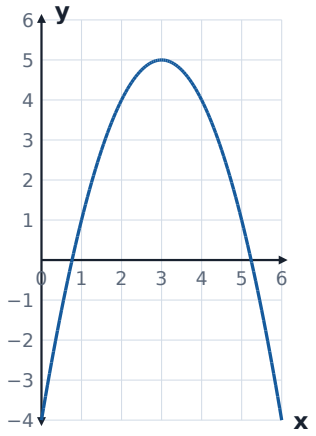
From x = 0 to x = 3

GRADING NOTE Any equivalent form (0 < x < 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q2/5

For which x-values shown is the graph falling?



A. From x = 3 to x = 6

B. It never falls.

C. From x = 0 to x = 3

D. From x = 0 to x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

519

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. From x = 3 to x = 6

GRADING NOTE After its highest point at x = 3, the graph goes down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

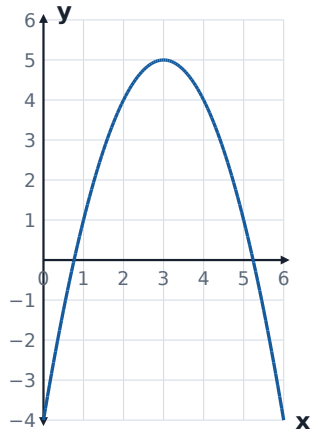
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q3/5

True or false?

The graph has the same height at x = 2 and at x = 4.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

520

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is symmetric about x = 3, and 2 and 4 are each 1 unit from 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

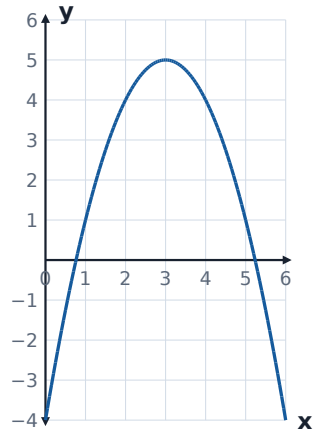
Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q4/5

True or false?

The graph is a straight line.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

521

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It curves: it rises and then falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 3 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

523

QUESTION ID
S15-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (3, −2), radius 4

GRADING NOTE Any equivalent form. Both are required. $(x - 3)^2 + (y + 2)^2 = 3 + 9 + 4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the circle with center (−1, 4) and radius 3 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

524

QUESTION ID
S15-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x + 1)^2 + (y - 4)^2 = 9$

GRADING NOTE Form required: standard form $(x - h)^2 + (y - k)^2 = r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q2/5

Find the center and radius of $x^2 + y^2 + 8x = 9$.

A. Center (4, 0), radius 5

B. Center (0, 0), radius 9

C. Center (−4, 0), radius 3

D. Center (−4, 0), radius 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

525

QUESTION ID
S15-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Center (−4, 0), radius 5

GRADING NOTE $(x + 4)^2 + y^2 = 9 + 16 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + (y + 1)^2 = 4$ has a radius of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

526

QUESTION ID
S15-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $r^2 = 4$, so $r = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + y^2 = 25$ has its center at (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

527

QUESTION ID
S15-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its center is (0, 0), and its radius is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-SSE.B.3.b

FORWARD 2

Derive the equations of ellipses and hyperbolas from their foci • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

An ellipse has foci (4, 0) and (−4, 0), and the sum of the distances from any point on it to the foci is 10.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

529

QUESTION ID
S15-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$x^2/25 + y^2/9 = 1$

GRADING NOTE Form required: standard form. $a = 5$, $c = 4$, and $b^2 = a^2 - c^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q1/5

A hyperbola has foci (5, 0) and (−5, 0), and the absolute value of the difference of the distances from any point on it to the foci is 8.

Write its equation in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

530

QUESTION ID
S15-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2/16 - y^2/9 = 1$

GRADING NOTE Form required: standard form. $a = 4$, $c = 5$, and $b^2 = c^2 - a^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q2/5

The ellipse $x^2 \div 25 + y^2 \div 9 = 1$ has foci (4, 0) and (−4, 0).

Show that (3, 2.4) is on the ellipse and that its distances to the two foci add to 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

531

QUESTION ID
S15-F2-Q3

TYPE
Answer with required work

CORRECT ANSWER
It satisfies the equation, and 2.6 + 7.4 = 10.

GRADING NOTE REQUIRED WORK: $9 \div 25 + 5.76 \div 9 = 0.36 + 0.64 = 1$ shown; the distances $\sqrt{(1 + 5.76)} = 2.6$ and $\sqrt{(49 + 5.76)} = 7.4$ computed and added.

Form required: the substitution and both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q3/5

True or false?

$x^2/25 + y^2/9 = 1$ is the equation of an ellipse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

532

QUESTION ID
S15-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A sum of two squared terms with different denominators, equal to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q4/5

True or false?

For every point on an ellipse, the difference of the distances to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

533

QUESTION ID
S15-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For an ellipse the sum is constant; for a hyperbola the difference is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 15 • FORWARD 2 • Q5/5

Use the properties of exponents to transform exponential expressions

MAIN	Algebra I main question	**HSA-SSE.B.3.c**
BACKWARD 1	Use the meaning of rational exponents and the power-of-a-power…	**HSN-RN.A.1**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Use the sum of a geometric series in a growth model	**HSA-SSE.B.4**

SET 16 • HSA-SSE.B.3.c

MAIN

Algebra I main question • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

An account grows by 6% each year: $A = 500(1.06)^t$ after t years.

Rewrite the expression to show the monthly growth factor, rounded to four decimal places.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

536

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$500(1.0049)^{12t}$

GRADING NOTE Form required: 500 times (monthly factor)^{12t}. The factor is $1.06^{1/12} \approx 1.0049$, so $A = 500(1.06^{1/12})^{12t}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q1/5

Use the properties of exponents to explain why $1.21^t = 1.1^{2t}$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

537

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

$1.21 = 1.1^2$, so $1.21^t = (1.1^2)^t = 1.1^{2t}$ by the power-of-a-power property.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q2/5

A population doubles every 5 years: $P = 300 \cdot 2^{t/5}$ after t years.

Find the yearly growth factor to three decimal places, and the yearly percent growth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

538

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

$2^{1/5} \approx 1.149$, so about 14.9% growth per year.

GRADING NOTE Form required: the factor to three decimal places and the percent. $2^{t/5} = (2^{1/5})^t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is equal to 8^t?

A. 2^{t+3}

B. 3^{2t}

C. 4^{2t}

D. 2^{3t}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

539

QUESTION ID

S16-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2^{3t}

GRADING NOTE $8 = 2^3$, so $8^t = (2^3)^t = 2^{3t}$. ($4^{2t} = 16^t$.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q4/5

True or false?

In $y = 100(0.5)^{t/2}$, the amount is cut in half every 2 time units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

540

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE When t increases by 2, the exponent t/2 increases by 1, so y is multiplied by 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 16 • MAIN • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 1

Use the meaning of rational exponents and the power-of-a-power property • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Find the value.

$4^{1/2}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

542

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $4^{1/2} = \sqrt{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $(1.1^2)^t$ as a single power of 1.1.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

543

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.1^{2t}

GRADING NOTE Form required: a single power. Multiply the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B1-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q2/5

Find the value.

$9^{1/2}$

A. $\frac{1}{9}$

B. 4.5

C. 3

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

544

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE $9^{1/2} = \sqrt{9} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$(2^3)^t = 2^{3t}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

545

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power: multiply the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$5^{1/2} = 2.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q5/5

546

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5^{1/2} = \sqrt{5} \approx 2.24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Increase $500 by 6%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

548

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$530

GRADING NOTE Any equivalent form. 500 · 1.06.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q1/5

Decrease 80 by 25%.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

549

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form. 80 · 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What do you multiply by to increase an amount by 4%?

A. 0.04

B. 1.4

C. 1.04

D. 0.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

550

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.04

GRADING NOTE 100% + 4% = 104% = 1.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Decreasing an amount by 10% is the same as multiplying it by 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

551

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 100% − 10% = 90%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q4/5

True or false?

An increase of 5% followed by a decrease of 5% gives back the original amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

552

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.05 \cdot 0.95 = 0.9975$, slightly less than the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $500(1.06)^t = 800$. Round t to the nearest tenth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

554

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.1

GRADING NOTE Form required: rounded to the nearest tenth. $1.06^t = 1.6$, so $t = \log 1.6 \div \log 1.06 \approx 8.07$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q1/5

A population is $P = 300 \cdot 2^{t/5}$ after t years. When is $P = 1{,}000$? Round to the nearest tenth. (log means base 10.)

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

555

QUESTION ID
S16-F1-Q2

TYPE
Short answer

CORRECT ANSWER
8.7 years

GRADING NOTE Form required: rounded to the nearest tenth. $2^{t/5} = 10/3$, so $t = 5 \cdot \log(10/3) \div \log 2 \approx 8.68$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q2/5

Solve $10^{2x} = 50$. (log means base 10.)

A. x = 2.5

B. x = (log 50) ÷ 2 ≈ 0.85

C. x = 25

D. x = log 25 ≈ 1.40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q3/5

556

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = (log 50) ÷ 2 ≈ 0.85

GRADING NOTE 2x = log 50 ≈ 1.699.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

If $3^x = 3^5$, then x = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

557

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal powers of the same base (other than 1) have equal exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The solution of $1.06^t = 2$ is $t = 2 \div 1.06$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

558

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = \log 2 \div \log 1.06 \approx 11.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S16-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-SSE.B.3.c

FORWARD 2

Use the sum of a geometric series in a growth model • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Deposits of \$100, \$100(1.005), \$100(1.005)^2, ..., \$100(1.005)^{11} are added. Use $S = a(1 - r^n) \div (1 - r)$ to find the total, to the nearest dollar.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

560

QUESTION ID
S16-F2-Q1

TYPE
Short answer

CORRECT ANSWER
\$1,234

GRADING NOTE Form required: to the nearest dollar. a = 100, r = 1.005, n = 12: $100(1 - 1.005^{12}) \div (1 - 1.005) \approx 1{,}233.56$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum 5 + 10 + 20 + ... + 640.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

561

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1,275

GRADING NOTE Any equivalent form. There are 8 terms ($640 = 5 \cdot 2^7$): $5(2^8 - 1) = 1{,}275$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q2/5

What is 1 + 3 + 9 + 27 + 81?

A. 81

B. 121

C. 243

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

562

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 121

GRADING NOTE $(3^5 - 1) \div (3 - 1) = 242 \div 2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q3/5

True or false?

The series 2 + 6 + 18 + 54 has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q4/5

563

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each term is 3 times the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q4/5

True or false?

The sum of the first n terms of a geometric series with first term a and ratio $r \neq 1$ is $a \cdot r^n$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $a \cdot r^n$ is not the sum; the sum is $a(1 - r^n) \div (1 - r)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 16 • FORWARD 2 • Q5/5

Explain each step in solving an equation and justify the solution method

MAIN	Algebra I main question	**HSA-REI.A.1**
BACKWARD 1	Use substitution to decide whether a number makes an equation…	**6.EE.B.5**
BACKWARD 2	Solve linear equations by expanding and collecting like terms	**8.EE.C.7.b**
FORWARD 1	Justify the steps of a proof about lines and angles	**HSG-CO.C.9**
FORWARD 2	Solve radical and rational equations and identify extraneous solut..	**HSA-REI.A.2**

SET 17 • HSA-REI.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve 3(x − 4) + 5 = 2x + 7. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

567

QUESTION ID

S17-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 14

GRADING NOTE REQUIRED WORK: Each step written as an equation with its reason: distributive property (3x − 12 + 5), combining like terms (3x − 7 = 2x + 7), subtraction property of equality (x − 7 = 7), addition property of equality (x = 14).

Form required: the solution with a reason for every step. Check: 3(10) + 5 = 35 and 2(14) + 7 = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q1/5

A student solves 5x − 8 = 12.

Step 1: 5x = 20

Step 2: x = 4

Which property justifies Step 1?

A. Distributive property

B. Addition property of equality: add 8 to both sides

C. Subtraction property of equality: subtract 8 from both sides

D. Multiplication property of equality: multiply both sides by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

568

QUESTION ID

S17-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Addition property of equality: add 8 to both sides

GRADING NOTE Adding 8 to both sides of 5x − 8 = 12 gives 5x = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q2/5

A student wrote:

4(x + 2) = 20

4x + 2 = 20

4x = 18

x = 4.5

Which step has the error? What is the correct solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

569

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

The step from 4(x + 2) = 20 to 4x + 2 = 20: the 4 must multiply both terms, giving 4x + 8 = 20. The solution is x = 3.

GRADING NOTE Form required: the step with the error and the correct solution. 4x + 8 = 20, so 4x = 12 and x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q3/5

True or false?

If a = b, then a ÷ c = b ÷ c for every number c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

570

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined, so the property holds only when c ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q4/5

True or false?

Subtracting 2x from both sides of $2(x + 1) = 2x + 2$ gives $2 = 2$, so every real number is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

571

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2(x + 1) = 2x + 2$ is true for every x. The step leaves a true statement with no variable, so the solution set is all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • MAIN • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 1

Use substitution to decide whether a number makes an equation true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 3 a solution of 4x − 5 = 7? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

573

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

4(3) − 5 = 12 − 5 = 7.

GRADING NOTE Any equivalent explanation that substitutes 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q1/5

Which number from 1, 2, 3, 4 makes 2x + 1 = 9 true?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

574

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 2(4) + 1 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q2/5

Which value of y makes 15 − y = 9 true?

A. 6

B. −6

C. 9

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

575

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE 15 − 6 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$x = 5$ is a solution of $3x = 15$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

576

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 5 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

n = 2 is a solution of n + 7 = 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

577

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 + 7 = 9, not 5. The solution is n = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

BACKWARD 2

Solve linear equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve 2(x + 3) = 14.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

579

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 2x + 6 = 14, so 2x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5y − 2y + 4 = 19.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

580

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 3y + 4 = 19, so 3y = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q2/5

Solve 3(n − 2) = n + 8.

A. n = 7

B. n = 5

C. n = 3

D. n = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

581

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. n = 7

GRADING NOTE 3n − 6 = n + 8, so 2n = 14 and n = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q3/5

True or false?

The solution of $0.5x + 1 = 4$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

582

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5x = 3$, so $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

The solution of $4(x - 1) = 12$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

583

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4x - 4 = 12$, so $4x = 16$ and $x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 1

Justify the steps of a proof about lines and angles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9

Angles 1 and 2 are vertical angles. Angle 1 and angle 3 form a linear pair, and so do angle 2 and angle 3.

Explain why m∠1 = m∠2.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

585

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

m∠1 + m∠3 = 180° and m∠2 + m∠3 = 180°, so m∠1 + m∠3 = m∠2 + m∠3. Subtracting m∠3 from both sides gives m∠1 = m∠2.

GRADING NOTE Any equivalent explanation that uses both linear pairs and the subtraction property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

In a proof, the line m∠A + 40° = 90° is followed by m∠A = 50°.

Which reason justifies that step?

A. Definition of a linear pair

B. Vertical angles are congruent

C. Addition property of equality

D. Subtraction property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

586

QUESTION ID

S17-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Subtraction property of equality

GRADING NOTE 40° is subtracted from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two angles form a linear pair. One measures (3x + 10)° and the other measures (2x + 20)°.

Find x and both angle measures.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

587

QUESTION ID

S17-F1-Q3

TYPE

Short answer

CORRECT ANSWER

x = 30; the angles are 100° and 80°.

GRADING NOTE Form required: x and both angle measures. 5x + 30 = 180, so x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

588

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the vertical angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

True or false?

If two angles have equal measures, they must be vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

589

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The converse is false: for example, two angles in different parts of a figure can both measure 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-REI.A.1

FORWARD 2

Solve radical and rational equations and identify extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 3) = 5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

591

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. Square both sides: x + 3 = 25. Check: √25 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q1/5

Solve $\sqrt{(2x + 3)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

592

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3; x = −1 is extraneous.

GRADING NOTE Form required: the solution and the extraneous value. Squaring gives $x^2 - 2x - 3 = 0$, so $x = 3$ or $x = -1$. For $x = -1$, $\sqrt{1} = 1 \neq -1$, so -1 is not a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q2/5

Assume $x \neq 0$. Solve $6 \div x = 3$.

A. $x = \frac{1}{2}$

B. $x = 3$

C. $x = 18$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q3/5

593

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2

GRADING NOTE Multiply both sides by x: 6 = 3x, so x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q3/5

True or false?

$x = 4$ is a solution of $\sqrt{x} = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q4/5

594

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{4} = 2$, and a principal square root is never negative, so the equation has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q4/5

True or false?

Squaring both sides of an equation can produce a value that does not satisfy the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

595

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, $x = 2$ has one solution, but $x^2 = 4$ also has $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 17 • FORWARD 2 • Q5/5

Solve linear equations in one variable, including equations with letter coefficients

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Name the property that justifies a step	**HSA-REI.A.1**
BACKWARD 2	Solve two-step equations from word problems	**7.EE.B.4.a**
FORWARD 1	Prove and apply theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Solve rational equations that lead to linear equations	**HSA-REI.A.2**

SET 18 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{2}{3}x - 5 = \frac{1}{4}x + 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

598

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. Multiply both sides by 12: 8x − 60 = 3x + 60, so 5x = 120. Check: 16 − 5 = 11 and 6 + 5 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q1/5

Solve 0.4(3x − 5) = 2.2x + 1.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

599

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form. 1.2x − 2 = 2.2x + 1, so −x = 3. Check: 0.4(−14) = −5.6 and 2.2(−3) + 1 = −5.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does $5x - 2(x + 4) = 3x + 1$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

600

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE The left side is $3x - 8$, so $3x - 8 = 3x + 1$ gives $-8 = 1$, which is false for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ax + b = c for x, where a ≠ 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

601

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = (c − b) ÷ a

GRADING NOTE Any equivalent form, such as c/a − b/a. Subtract b, then divide by a (allowed because a ≠ 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q4/5

True or false?

The equation $2(x - 3) = 2x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

602

QUESTION ID
S18-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both sides equal $2x - 6$ for every x, so every real number is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 1

Name the property that justifies a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Name the property used to go from 3x + 4 = 19 to 3x = 15.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

604

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Subtraction property of equality (subtract 4 from both sides)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q1/5

Name the property used to go from $\frac{x}{5} = 3$ to $x = 15$.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

605

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Multiplication property of equality (multiply both sides by 5)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q2/5

Which step keeps the same solutions as 2x = 10?

A. Divide both sides by 2

B. Square the right side only

C. Add 2 to the left side only

D. Multiply both sides by 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

606

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Divide both sides by 2

GRADING NOTE Doing the same nonzero operation to both sides keeps the solutions. The other steps change the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q3/5

True or false?

Adding the same number to both sides of an equation keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

607

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Addition property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

608

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It turns every equation into 0 = 0, which every number satisfies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

BACKWARD 2

Solve two-step equations from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride cost $19.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

610

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 3 + 2m = 19, so 2m = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q1/5

Solve −4x + 9 = 1.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

611

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. −4x = −8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{1}{2}(x + 6) = 10$.

A. x = 14

B. x = 17

C. x = 2

D. x = 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

612

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 14

GRADING NOTE x + 6 = 20, so x = 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q3/5

True or false?

The solution of $6x - 4 = 20$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

613

GRADING NOTE $6x = 24$.

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q4/5

True or false?

The solution of $2x + 5 = -1$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

614

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2x = -6$, so $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 1

Prove and apply theorems about triangles • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of any triangle ABC add to 180°.
Start by drawing the line through B parallel to AC.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

616

QUESTION ID
S18-F1-Q1

TYPE
Answer with required work

CORRECT ANSWER
m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: The parallel line through B makes two angles with BA and BC; each equals ∠A or ∠C because they are alternate interior angles; the three angles at B form a straight angle (180°).

Form required: the proof with reasons (parallel postulate, alternate interior angles, straight angle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q1/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

617

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The exterior angle and the adjacent interior angle form a linear pair, so they add to 180°. The three interior angles also add to 180°. Subtracting the adjacent angle from both shows the exterior angle equals the other two interior angles together.

GRADING NOTE Any equivalent explanation that uses the linear pair and the triangle angle sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

618

QUESTION ID

S18-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE REQUIRED WORK: Triangles ABD and ACD are congruent by SAS (AB = AC, ∠BAD = ∠CAD, AD = AD), so the corresponding angles ∠B and ∠C are congruent.

Form required: the congruence with its reasons and the corresponding-parts conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q3/5

D and E are the midpoints of AB and AC in triangle ABC, and BC = 18.

What does the midsegment theorem let you conclude?

A. DE ⊥ BC and DE = 9

B. DE ∥ BC and DE = 18

C. DE = 9, but DE need not be parallel to BC

D. DE ∥ BC and DE = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

619

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. DE ∥ BC and DE = 9

GRADING NOTE The segment joining the midpoints of two sides is parallel to the third side and half as long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q4/5

True or false?

The three medians of a triangle meet at one point, and that point is $\frac{2}{3}$ of the way from each vertex to the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

620

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the centroid theorem, one of the triangle theorems of HSG-CO.C.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-REI.B.3

FORWARD 2

Solve rational equations that lead to linear equations • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 0$. Solve $3 \div x + 2 = 5 \div x$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

622

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. Multiply by x: 3 + 2x = 5, so x = 1. Check: 3 + 2 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 2$. Solve $(x + 1) \div (x - 2) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

623

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $x + 1 = 4x - 8$, so $x = 3$. Check: $4 \div 1 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q2/5

Assume $x \neq 3$. Solve $x \div (x - 3) = 3 \div (x - 3)$.

A. $x = 0$

B. No solution: $x = 3$ is extraneous

C. $x = 3$

D. Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

624

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No solution: x = 3 is extraneous

GRADING NOTE Multiplying by $x - 3$ gives $x = 3$, but 3 makes the denominator 0, so it is not a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x = 1$ is a solution of $2x \div (x + 1) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q4/5

625

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2 \div 2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

The solution of $6 \div (x + 1) = 2$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

626

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 1 = 3$, so $x = 2$. For $x = 3$, $6 \div 4 = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S18-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 18 • FORWARD 2 • Q5/5

Solve linear inequalities and compound inequalities in one variable

MAIN	Algebra I main question	**HSA-REI.B.3**
BACKWARD 1	Solve two-step inequalities from word problems	**7.EE.B.4.b**
BACKWARD 2	Write and graph simple inequalities	**6.EE.B.8**
FORWARD 1	Use inequalities to meet constraints in a design problem	**HSG-MG.A.3**
FORWARD 2	Write a margin-of-error interval as a compound inequality — neare…	**HSS-IC.B.4**

SET 19 • HSA-REI.B.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $5 - 3x \leq 17$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

629

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq -4$

GRADING NOTE Form required: an inequality in x ($x \geq -4$ or $-4 \leq x$). $-3x \leq 12$; dividing by -3 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-2 < 3x + 4 \leq 13$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

630

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

$-2 < x \leq 3$

GRADING NOTE Form required: a compound inequality. Subtract 4 from all three parts ($-6 < 3x \leq 9$), then divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q2/5

The number line shows the solution set of which inequality?



A. $x + 5 > 7$

B. $4x + 3 < 2x + 7$

C. $3x \leq 6$

D. $-2x < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

631

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x + 3 < 2x + 7$

GRADING NOTE The graph is $x < 2$. $4x + 3 < 2x + 7$ gives $2x < 4$, so $x < 2$. $x + 5 > 7$ and $-2x < -4$ give $x > 2$, and $3x \leq 6$ gives $x \leq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\frac{x}{4} - 1 > \frac{x}{2} + 2$.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

632

QUESTION ID

S19-M-Q4

TYPE

Short answer

CORRECT ANSWER

x < −12

GRADING NOTE Form required: an inequality in x. Multiply by 4: x − 4 > 2x + 8, so −x > 12 and x < −12. Check x = −16: −5 > −6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q4/5

True or false?

The solution of $-4x > 8$ is $x > -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

633

QUESTION ID
S19-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Dividing by -4 reverses the sign: $x < -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 19 • MAIN • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 1

Solve two-step inequalities from word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x + 2 > 14.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

635

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x > 4

GRADING NOTE Form required: an inequality in x. 3x > 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q1/5

A phone plan costs $20 plus $5 per gigabyte g. Lena can spend at most $45.

Write and solve an inequality for g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

636

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20 + 5g ≤ 45; g ≤ 5, so at most 5 gigabytes.

GRADING NOTE Form required: the inequality and its solution. 5g ≤ 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q2/5

Solve $-2x + 1 \geq 9$.

A. $x \leq -4$

B. $x \geq 5$

C. $x \leq 4$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

637

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x \leq -4$

GRADING NOTE $-2x \geq 8$; dividing by -2 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q3/5

True or false?

$x = 6$ is a solution of $2x - 3 > 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

638

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2(6) - 3 = 9$, and $9 > 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q4/5

True or false?

Dividing both sides of an inequality by a positive number reverses the inequality sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

639

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only multiplying or dividing by a negative number reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

BACKWARD 2

Write and graph simple inequalities • Grade 6 • Expressions & Equations • 6.EE.B.8

A rider must be at least 48 inches tall.

Write an inequality for the rider's height h.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

641

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

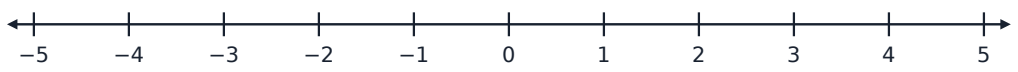
h ≥ 48

GRADING NOTE Form required: an inequality. "At least" includes 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q1/5

Graph x < 3 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

642

QUESTION ID

S19-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An open circle at 3 and a ray shaded to the left

GRADING NOTE Any correct graph. The open circle shows 3 is not included.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q2/5

Which number is a solution of $x > -2$?

A. -2

B. -3

C. -5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

643

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE 0 is greater than −2; −2 itself is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

The inequality x > 5 has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

644

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every number greater than 5 is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

On a number line, $x \leq 2$ is shown with an open circle at 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

645

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ≤ includes 2, so the circle is closed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 1

Use inequalities to meet constraints in a design problem • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular garden is 12 ft long. Its perimeter can be at most 50 ft.

Write and solve an inequality for the width w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

647

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$24 + 2w \leq 50$;

$0 < w \leq 13$ feet

GRADING NOTE Form required: the inequality and its solution. A width must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q1/5

A box with a 10 in by 10 in base must hold at least 600 in^3.

Write and solve an inequality for the height h.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

100h ≥ 600; h ≥ 6 inches

GRADING NOTE Form required: the inequality and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q2/5

A poster 15 in wide must have an area of at least 300 in².

Which lengths L work?

A. L ≥ 285 inches

B. L ≥ 20 inches

C. L ≥ 4.5 inches

D. L ≤ 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

649

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. L ≥ 20 inches

GRADING NOTE 15L ≥ 300, so L ≥ 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q3/5

True or false?

A rectangle 8 cm long and w cm wide has an area of at most 40 cm² exactly when $0 < w \leq 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

650

QUESTION ID
S19-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $8w \leq 40$ gives $w \leq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

A square with side s cm has a perimeter less than 20 cm exactly when $s < 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

651

QUESTION ID
S19-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4s < 20$ gives $s < 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-REI.B.3

FORWARD 2

Write a margin-of-error interval as a compound inequality • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A margin-of-error interval is written as a compound inequality like those solved in the MAIN set.

A survey estimates that 54% of voters support a plan, with a margin of error of 3 percentage points.

Write the interval for the true proportion p as a compound inequality.

Interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

653

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$0.51 \leq p \leq 0.57$

GRADING NOTE Any equivalent form, such as 51% ≤ p ≤ 57%. 54 − 3 = 51 and 54 + 3 = 57.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A poll reports 62% with a margin of error of 4 percentage points.

Is 60% inside the interval? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

654

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The interval is 58% ≤ p ≤ 66%, and 60% is in it.

GRADING NOTE Any equivalent explanation that states the interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q2/5 • NEAREST RELATED

The interval $0.40 \leq p \leq 0.48$ is an estimate plus or minus a margin of error.

What are the estimate and the margin of error?

A. 0.44 and 0.08

B. 0.40 and 0.08

C. 0.44 and 0.04

D. 0.48 and 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

655

QUESTION ID
S19-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 0.44 and 0.04

GRADING NOTE The estimate is the midpoint, 0.44; the margin is half the width, 0.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An estimate of 70% with a margin of error of 5 percentage points gives $65\% \le p \le 75\%$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

656

QUESTION ID
S19-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $70 - 5 = 65$ and $70 + 5 = 75$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If the interval is $0.30 \leq p \leq 0.36$, the margin of error is 0.06.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

657

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The margin is half the width: $0.06 \div 2 = 0.03$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.4 • Set 19 • FORWARD 2 • Q5/5 • NEAREST RELATED

Create linear equations and inequalities in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve linear equations and inequalities	**HSA-REI.B.3**
BACKWARD 2	Write and solve one-step equations	**6.EE.B.7**
FORWARD 1	Write and solve equations from similar triangles	**HSG-SRT.B.5**
FORWARD 2	Write and solve an equation for an expected value — nearest relat.	**HSS-MD.A.2**

SET 20 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges a $65 call fee plus $48 per hour. A bill is $257.

Write an equation for the number of hours h, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

660

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

65 + 48h = 257;
h = 4 hours

GRADING NOTE Form required: an equation and its solution. 48h = 192.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q1/5

The sum of three consecutive integers is 141.

Write an equation and find the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

661

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

n + (n + 1) + (n + 2) = 141; the integers are 46, 47, and 48.

GRADING NOTE Form required: an equation and all three integers. 3n + 3 = 141, so n = 46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q2/5

Maya has $180 and saves $15 each week. She wants at least $400.

Write and solve an inequality for the number of weeks w. How many whole weeks must she save?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

662

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

180 + 15w ≥ 400; w ≥ $14\frac{2}{3}$, so at least 15 weeks.

GRADING NOTE Form required: the inequality, its solution, and the whole number of weeks. 15w ≥ 220. After 14 weeks she has only $390.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q3/5

A rental costs $40 per day plus a one-time $25 fee.

Which inequality gives the number of days d you can rent for at most $265?

A. 40(d + 25) ≤ 265

B. 40d + 25 ≥ 265

C. 40d + 25 ≤ 265

D. 25d + 40 ≤ 265

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

663

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 40d + 25 ≤ 265

GRADING NOTE The cost 40d + 25 must be at most (≤) 265.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q4/5

True or false?

Plan A costs 30 + 0.10m dollars and Plan B costs 0.25m dollars for m minutes. Plan A costs less when m < 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

664

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30 + 0.10m < 0.25m gives 30 < 0.15m, so m > 200. For fewer than 200 minutes, Plan B costs less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • MAIN • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 1

Solve linear equations and inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 7x − 4 = 3x + 20.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

666

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 4x = 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B1-Q1**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(5 − x) ≥ 16.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

667

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x ≤ −3

GRADING NOTE Form required: an inequality in x. 10 − 2x ≥ 16, so −2x ≥ 6 and x ≤ −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.25t + 3 = 7.

A. t = 1

B. t = 40

C. t = 2.5

D. t = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

668

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. t = 16

GRADING NOTE 0.25t = 4, so t = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$x = -3$ is a solution of $4 - x > 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

669

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4 - (-3) = 7$, and $7 > 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q4/5

True or false?

The solution of $12 - 3x < 0$ is $x < 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

670

QUESTION ID
S20-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $-3x < -12$; dividing by -3 reverses the sign, so $x > 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

BACKWARD 2

Write and solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve x + 17 = 42.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

672

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form. Subtract 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tickets cost $9 each. Sam spent $63.

Write an equation and find the number of tickets t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

673

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9t = 63; t = 7

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q2/5

Solve 4.5 = 1.5y.

A. y = 6.75

B. y = 6

C. y = 3

D. y = 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

674

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3

GRADING NOTE 4.5 ÷ 1.5 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q3/5

True or false?

If n − 8 = 15, then n = 23.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

675

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

If 6p = 3, then p = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

676

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $p = 3 \div 6 = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 1

Write and solve equations from similar triangles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-ft person casts a 4-ft shadow at the same time a tree casts a 22-ft shadow.

Write a proportion and find the tree's height.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

678

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/4 = h/22; h = 33 feet

GRADING NOTE Form required: a proportion and the height. The triangles formed by the sun's rays are similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q1/5

Triangle ABC ~ triangle DEF with AB = 8, BC = 12, and DE = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

679

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 8/10 = 12/EF, so EF = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q2/5

Triangle PQR ~ triangle STU, with PQ = 5, ST = 15, QR = x + 2, and TU = 2x + 9.

What is x?

A. x = 1

B. x = 4.5

C. x = 3

D. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

680

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3

GRADING NOTE The scale factor is 3, so 3(x + 2) = 2x + 9 and x = 3 (QR = 5, TU = 15).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

If two triangles are similar, their corresponding side lengths are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

681

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is what similarity means for triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

In similar triangles with a ratio of 2 : 3, a side of 10 corresponds to a side of 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

682

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot \frac{3}{2} = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-CED.A.1

FORWARD 2

Write and solve an equation for an expected value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Writing and solving a linear equation from a situation is applied to an expected-value condition.

ANSWER KEY Reduced copy of the question page, with its answer

A game pays $10 with probability 0.2 and $0 otherwise.

What is the expected payout?

Expected payout: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

684

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2

GRADING NOTE Any equivalent form. 10(0.2) + 0(0.8) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q1/5 • NEAREST RELATED

A game pays x dollars with probability 0.25 and $0 otherwise.

Write and solve an equation for x so that the expected payout is $3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

685

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.25x = 3; x = 12

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q2/5 • NEAREST RELATED

A random variable is 1 with probability 0.5, 2 with probability 0.3, and 4 with probability 0.2.

What is its expected value?

A. 7

B. 1.9

C. 2

D. 2.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

686

QUESTION ID
S20-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 1.9

GRADING NOTE $1(0.5) + 2(0.3) + 4(0.2) = 0.5 + 0.6 + 0.8 = 1.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The expected value of one roll of a fair number cube numbered 1 to 6 is 3.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

687

QUESTION ID
S20-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(1 + 2 + 3 + 4 + 5 + 6) \div 6 = 3.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A game costs $5 to play. It pays $20 with probability 0.2 and $0 otherwise. The expected net gain is $4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

688

QUESTION ID
S20-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The expected payout is 20(0.2) + 0(0.8) = $4, so the expected net gain is 4 − 5 = −$1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 20 • FORWARD 2 • Q5/5 • NEAREST RELATED

Rearrange formulas to highlight a quantity of interest

MAIN	Algebra I main question	**HSA-CED.A.4**
BACKWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
BACKWARD 2	Evaluate formulas for given values	**6.EE.A.2.c**
FORWARD 1	Rearrange the arc-length formula $s = r\theta$	**HSG-C.B.5**
FORWARD 2	Solve exponential formulas for the exponent with logarithms	**HSF-LE.A.4**

SET 21 • HSA-CED.A.4

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $V = \pi r^2 h$ for h. ($r > 0$)

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

691

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

$h = V \div (\pi r^2)$

GRADING NOTE Any equivalent form. Divide both sides by πr^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $C = \frac{5}{9}(F - 32)$ for F.

F = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

692

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

$F = \frac{9}{5}C + 32$

GRADING NOTE Any equivalent form, such as F = 1.8C + 32. Multiply by $\frac{9}{5}$, then add 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve A = P(1 + rt) for t. (P ≠ 0, r ≠ 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

693

QUESTION ID

S21-M-Q3

TYPE

Short answer

CORRECT ANSWER

t = (A − P) ÷ (Pr)

GRADING NOTE Any equivalent form, such as (A/P − 1)/r. Divide by P: A/P = 1 + rt; subtract 1, then divide by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q3/5

Ohm's law is V = IR. Solve for R. (I ≠ 0)

A. $R = \frac{I}{V}$

B. $R = V - I$

C. $R = VI$

D. $R = \frac{V}{I}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

694

QUESTION ID

S21-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. R = V/I

GRADING NOTE Divide both sides by I.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q4/5

True or false?

Solving $y = mx + b$ for x ($m \neq 0$) gives $x = y \div m - b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

695

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtract b first, then divide: $x = (y - b) \div m$. The given form divides only y by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 21 • MAIN • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 1

Solve linear equations with letter coefficients • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3x + k = 10 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

697

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = (10 − k) ÷ 3

GRADING NOTE Any equivalent form. Subtract k, then divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ax = bx + c for x, where a ≠ b.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

698

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = c ÷ (a − b)

GRADING NOTE Any equivalent form. (a − b)x = c, and a − b ≠ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x \div m = n$ for x. ($m \neq 0$)

A. $x = m + n$

B. $x = mn$

C. $x = n \div m$

D. $x = m \div n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

699

QUESTION ID
S21-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. x = mn

GRADING NOTE Multiply both sides by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 2x − p = q, then x = (p + q) ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q4/5

700

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2x = p + q.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $kx = 5$ and $k \neq 0$, then $x = 5k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q5/5

701

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 5 \div k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

BACKWARD 2

Evaluate formulas for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Use A = lw to find A when l = 9 and w = 4.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

703

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form. 9 · 4 = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q1/5

Use d = rt to find d when r = 55 and t = 3.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

704

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

165

GRADING NOTE Any equivalent form. 55 · 3 = 165.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q2/5

Use P = 2l + 2w to find P when l = 7 and w = 3.

A. 10

B. 21

C. 20

D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

705

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 20

GRADING NOTE 14 + 6 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q3/5

True or false?

$V = s^3$ gives $V = 27$ when $s = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

706

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q4/5

True or false?

$A = \frac{1}{2}bh$ gives $A = 40$ when $b = 10$ and $h = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

707

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\frac{1}{2} \cdot 10 \cdot 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 1

Rearrange the arc-length formula s = rθ • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 6 cm. An arc has length 9 cm.

Use $s = r\theta$ to find the central angle θ in radians.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

709

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 radians

GRADING NOTE Any equivalent form. $\theta = s \div r = 9 \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q1/5

Solve s = rθ for r. Then find r when s = 10π and θ = π ÷ 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

710

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

r = s ÷ θ; r = 20

GRADING NOTE Form required: the formula for r and the value. 10π ÷ (π ÷ 2) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q2/5

The area of a sector is $A = \frac{1}{2}r^2\theta$. Solve for θ. ($r > 0$)

A. $\theta = A - \frac{1}{2}r^2$

B. $\theta = 2Ar^2$

C. $\theta = A \div (2r^2)$

D. $\theta = 2A \div r^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

711

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\theta = 2A \div r^2$

GRADING NOTE Multiply by 2, then divide by r^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q3/5

True or false?

An arc of length 2πr has a central angle of 2π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

712

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE θ = 2πr ÷ r = 2π, the whole circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q4/5

True or false?

If $s = r\theta$, then $\theta = r \div s$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

713

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\theta = s \div r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-CED.A.4

FORWARD 2

Solve exponential formulas for the exponent with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $A = Pe^{rt}$ for t. (A, P, r > 0)

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

715

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = ln(A ÷ P) ÷ r

GRADING NOTE Any equivalent form, such as (ln A − ln P) ÷ r. Divide by P, take ln of both sides, then divide by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q1/5

$500 grows to $1,000 with continuous interest at r = 0.05.

Find t to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

716

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13.9 years

GRADING NOTE Form required: rounded to the nearest tenth. t = ln 2 ÷ 0.05 ≈ 13.86.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q2/5

Solve $2^x = y$ for x. ($y > 0$)

A. $x = y^2$

B. $x = \sqrt{y}$

C. $x = y \div 2$

D. $x = \log_2 y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

717

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \log_2 y$

GRADING NOTE By definition, x is the exponent on 2 that gives y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q3/5

True or false?

If $10^k = 1{,}000$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

718

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S21-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q4/5

True or false?

Solving $N = 5 \cdot 3^t$ for t gives $t = \log_3 N - 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

719

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by 5 first: $3^t = N \div 5$, so $t = \log_3(N \div 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 21 • FORWARD 2 • Q5/5

Complete the square to solve quadratic equations and derive the quadratic formula

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square to write a quadratic in vertex form	**HSA-SSE.B.3.b**
BACKWARD 2	Solve $x^2 = p$ and $x^3 = p$ with square and cube roots	**8.EE.A.2**
FORWARD 1	Complete the square to find the center and radius of a circle	**HSG-GPE.A.1**
FORWARD 2	Complete the square to write the equation of an ellipse	**HSG-GPE.A.3**

SET 22 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 6x - 7 = 0$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

722

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 1 or x = −7

GRADING NOTE REQUIRED WORK: The equation rewritten as $(x + 3)^2 = 16$, then the square root of both sides ($x + 3 = 4$ or $x + 3 = -4$).

Form required: both solutions with the completed square shown. $x^2 + 6x = 7$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $x^2 - 10x + 18 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

723

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 5)^2 = 7$

GRADING NOTE Form required: $(x - p)^2 = q$. $x^2 - 10x = -18$; add 25 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q2/5

Rewrite $2x^2 + 12x + 5 = 0$ in the form $(x - p)^2 = q$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

724

QUESTION ID

S22-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(x + 3)^2 = 6.5$

GRADING NOTE Form required: $(x - p)^2 = q$, with q as any equivalent number (13/2 or 6.5). Divide by 2: $x^2 + 6x = -2.5$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-M-Q3**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q3/5

Which number completes the square?

$x^2 + 8x + ___$

A. 8

B. 16

C. 4

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

725

QUESTION ID

S22-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16

GRADING NOTE $(8 \div 2)^2 = 16$, and $x^2 + 8x + 16 = (x + 4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q4/5

True or false?

Completing the square on $ax^2 + bx + c = 0$ ($a \neq 0$) gives $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$, which leads to the quadratic formula.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

726

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by a, move c ÷ a, and add $(b \div (2a))^2$ to both sides. Taking square roots gives $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 22 • MAIN • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 1

Complete the square to write a quadratic in vertex form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $x^2 + 4x + 1$ in the form $(x - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

728

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 2)^2 - 3$

GRADING NOTE Form required: $(x - h)^2 + k$. $x^2 + 4x + 4 - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q1/5

Write $x^2 - 6x + 11$ in the form $(x - h)^2 + k$, and give its minimum value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

729

QUESTION ID
S22-B1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + 2$; the minimum value is 2.

GRADING NOTE Form required: the completed square and the minimum. $(x - 3)^2$ is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q2/5

Which expression is equal to $x^2 + 2x + 5$?

A. $(x + 2)^2 + 1$

B. $(x + 1)^2 + 5$

C. $(x + 1)^2 + 4$

D. $(x - 1)^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

730

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x + 1)^2 + 4$

GRADING NOTE $(x + 1)^2 = x^2 + 2x + 1$, and $1 + 4 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q3/5

True or false?

$(x - 4)^2 = x^2 - 8x + 16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

731

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 4)(x - 4) = x^2 - 4x - 4x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 10x = (x + 5)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

732

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 5)^2 = x^2 + 10x + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

BACKWARD 2

Solve $x^2 = p$ and $x^3 = p$ with square and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 = 49$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

734

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 or x = −7

GRADING NOTE Form required: both solutions. $7^2 = 49$ and $(-7)^2 = 49$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = -64$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

735

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form. $(-4)^3 = -64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

A. 9 and −9

B. 9 only

C. −9 only

D. 40.5 and −40.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

736

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 and −9

GRADING NOTE Both 9^2 and $(-9)^2$ equal 81.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{64} = 8$ and $\sqrt[3]{64} = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

737

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $8^2 = 64$ and $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The only solution of $x^2 = 25$ is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

738

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE −5 is also a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 1

Complete the square to find the center and radius of a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Find the center and radius of the circle
$x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

740

QUESTION ID
S22-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Center (3, −2), radius 5

GRADING NOTE Form required: the center and the radius. $(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q1/5

Write an equation of the circle with center (2, −5) and radius 6 in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

741

QUESTION ID
S22-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$(x - 2)^2 + (y + 5)^2 = 36$

GRADING NOTE Form required: standard form $(x - h)^2 + (y - k)^2 = r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q2/5

What is the center of the circle $x^2 + 8x + y^2 = 9$?

A. $(-8, 0)$

B. $(-4, 0)$

C. $(4, 0)$

D. $(0, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

742

QUESTION ID
S22-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (−4, 0)

GRADING NOTE $(x + 4)^2 + y^2 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 2)^2 + y^2 = 16$ has radius 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

743

QUESTION ID
S22-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $r^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q4/5

True or false?

The circle $x^2 + (y + 5)^2 = 9$ has center (0, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

744

QUESTION ID
S22-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $y + 5 = y - (-5)$, so the center is $(0, -5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-REI.B.4.a

FORWARD 2

Complete the square to write the equation of an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Write $x^2 + 4y^2 - 6x - 7 = 0$ in standard form, and give the center of the ellipse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

746

QUESTION ID
S22-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 \div 16 + y^2 \div 4 = 1$; center $(3, 0)$

GRADING NOTE Form required: standard form and the center. $(x - 3)^2 + 4y^2 = 16$; divide by 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q1/5

Derive the equation of the set of points whose distances to the foci (3, 0) and (−3, 0) add to 10.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

747

QUESTION ID
S22-F2-Q2

TYPE
Answer with required work

CORRECT ANSWER
$x^2 \div 25 + y^2 \div 16 = 1$

GRADING NOTE REQUIRED WORK: $\sqrt{((x-3)^2+y^2)} + \sqrt{((x+3)^2+y^2)} = 10$ written; one radical isolated and both sides squared twice; simplified to $16x^2 + 25y^2 = 400$.

Form required: the derivation and the standard form. $a = 5$, $c = 3$, and $b^2 = 25 - 9 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q2/5

The ellipse $(x - 1)^2 \div 25 + (y + 3)^2 \div 9 = 1$ has a horizontal major axis.

How long is the major axis?

A. 6

B. 5

C. 25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

748

QUESTION ID
S22-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 10

GRADING NOTE The semi-major axis is $\sqrt{25} = 5$, so the major axis is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

Completing the square on $x^2 - 8x$ gives $(x - 4)^2 - 16$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

749

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x - 4)^2 = x^2 - 8x + 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S22-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The ellipse $x^2 \div 4 + y^2 \div 9 = 1$ is wider than it is tall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

750

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It reaches 2 units left and right but 3 units up and down, so it is taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 22 • FORWARD 2 • Q5/5

Derive the quadratic formula by completing the square

MAIN	Algebra I main question	**HSA-REI.B.4.a**
BACKWARD 1	Complete the square in expressions	**HSA-SSE.B.3.b**
BACKWARD 2	Solve literal equations for a variable	**HSA-CED.A.4**
FORWARD 1	Use the quadratic formula when the discriminant is negative	**HSN-CN.C.7**
FORWARD 2	Count the complex solutions of quadratic equations	**HSN-CN.C.9**

SET 71 • HSA-REI.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

Derive the quadratic formula. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, and complete the square to solve for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

753

QUESTION ID

S71-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$

GRADING NOTE REQUIRED WORK: Every step shown: divide by a; subtract c ÷ a; add (b ÷ (2a))² to both sides; write $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$; take square roots, using ±; subtract b ÷ (2a).

Form required: the complete derivation with every step, ending in the quadratic formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q1/5

The equation $x^2 + (b \div a)x = -c \div a$ comes from $ax^2 + bx + c = 0$. What must be added to both sides to complete the square, and what does the left side become?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

754

QUESTION ID

S71-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

Add $b^2 \div (4a^2)$; the left side becomes $(x + b \div (2a))^2$.

GRADING NOTE REQUIRED WORK: Half of the x-coefficient, $b \div (2a)$, squared: $(b \div (2a))^2 = b^2 \div (4a^2)$; then the left side factored as a square.

Form required: the number added and the squared binomial.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q2/5

After completing the square, $(x + b \div (2a))^2 = (b^2 - 4ac) \div (4a^2)$. Explain how taking square roots of both sides gives the ± and the denominator 2a.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

755

QUESTION ID

S71-M-Q3

TYPE

Short answer

CORRECT ANSWER

Both the positive and the negative square roots of the right side work, so $x + b \div (2a) = \pm\sqrt{(b^2 - 4ac)} \div \sqrt{(4a^2)}$. $\sqrt{(4a^2)} = 2|a|$, and the ± already covers both signs, so the denominator can be written 2a.

GRADING NOTE Any equivalent explanation that mentions both square roots and $\sqrt{(4a^2)} = 2|a|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q3/5

In the derivation of the quadratic formula, why must a ≠ 0?

A. Because the discriminant must be positive.

B. Because b must be positive.

C. Because c must be 0.

D. The first step divides by a, and if a = 0 the equation is not quadratic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

756

QUESTION ID
S71-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. The first step divides by a, and if a = 0 the equation is not quadratic.

GRADING NOTE Dividing by 0 is not allowed; with a = 0 the equation is linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q4/5

True or false?

The derivation shows that when $b^2 - 4ac < 0$, the equation has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

757

QUESTION ID

S71-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The right side $(b^2 - 4ac) \div (4a^2)$ is negative, but a square of a real number is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 71 • MAIN • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 1

Complete the square in expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 + 12x + 7$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

759

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 6)^2 - 29$

GRADING NOTE Form required: $(x - h)^2 + k$. $x^2 + 12x + 36 - 36 + 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 - 5x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

760

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2.5)^2 - 6.25$

GRADING NOTE Form required: $(x - h)^2 + k$, with fractions or decimals: $(x - \frac{5}{2})^2 - \frac{25}{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What number completes the square for $x^2 + \frac{2}{3}x$?

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{4}{9}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

761

QUESTION ID

S71-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/9

GRADING NOTE Half of $\frac{2}{3}$ is $\frac{1}{3}$, and $(\frac{1}{3})^2 = \frac{1}{9}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q3/5

True or false?

$x^2 + bx + (b \div 2)^2 = (x + b \div 2)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

762

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + b \div 2)^2 = x^2 + bx + b^2 \div 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q4/5

True or false?

To complete the square on $2x^2 + 8x$, add 16 to the expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

763

QUESTION ID

S71-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Factor 2 first: $2(x^2 + 4x + 4) - 8 = 2(x + 2)^2 - 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

BACKWARD 2

Solve literal equations for a variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $ax + b = 0$ for x. ($a \neq 0$)

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

765

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-b \div a$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q1/5

Solve $(x + p)^2 = q$ for x. ($q \geq 0$)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

766

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-p + \sqrt{q}$ or $-p - \sqrt{q}$

GRADING NOTE Form required: both solutions ($x = -p \pm \sqrt{q}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q2/5

Solve $y = kx^2$ for x, where $k > 0$ and $y \geq 0$.

A. $x = \pm\sqrt{(ky)}$

B. $x = \sqrt{y} \div k$

C. $x = \pm\sqrt{(y \div k)}$

D. $x = y \div k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

767

QUESTION ID

S71-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $x = \pm\sqrt{(y \div k)}$

GRADING NOTE $x^2 = y \div k$, so $x = \pm\sqrt{(y \div k)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Solving $A = \pi r^2$ for r (with $r > 0$) gives $r = \sqrt{(A \div \pi)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

768

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $r^2 = A \div \pi$, and r is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q4/5

True or false?

Solving $2x - c = d$ for x gives $x = (d - c) \div 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

769

QUESTION ID

S71-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2x = c + d$, so $x = (c + d) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 1

Use the quadratic formula when the discriminant is negative • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Use the quadratic formula to solve $x^2 + 6x + 13 = 0$.
Write the solutions as a + bi.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

771

QUESTION ID
S71-F1-Q1

TYPE
Short answer

CORRECT ANSWER
x = −3 + 2i or
x = −3 − 2i

GRADING NOTE Form required: a + bi (both). $x = (-6 \pm \sqrt{(36 - 52)}) \div 2 = (-6 \pm 4i) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q1/5

Solve $2x^2 - 2x + 5 = 0$. Write the solutions as a + bi.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

772

QUESTION ID

S71-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0.5 + 1.5i or
x = 0.5 − 1.5i

GRADING NOTE Form required: a + bi (both), with fractions or decimals. x = (2 ± √(4 − 40)) ÷ 4 = (2 ± 6i) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q2/5

What is the discriminant of $3x^2 + x + 2$?

A. 25

B. −23

C. −25

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

773

QUESTION ID

S71-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −23

GRADING NOTE $1 - 4(3)(2) = 1 - 24 = -23$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q3/5

True or false?

When $b^2 - 4ac < 0$, the quadratic formula gives two complex solutions that are conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

774

QUESTION ID
S71-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The ± gives p + qi and p − qi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q4/5

True or false?

The solutions of $x^2 + 9 = 0$ are 3 and -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

775

QUESTION ID

S71-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = -9$, so $x = 3i$ or $x = -3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F1-Q5**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSA-REI.B.4.a

FORWARD 2

Count the complex solutions of quadratic equations • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

ANSWER KEY Reduced copy of the question page, with its answer

How many complex solutions, counted with multiplicity, does every quadratic equation have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

777

QUESTION ID

S71-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. A degree-2 polynomial has 2 complex zeros counted with multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q1/5

$x^2 - 6x + 9 = 0$ has one real solution. How is it counted by the Fundamental Theorem of Algebra?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

778

QUESTION ID

S71-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 is a double solution (multiplicity 2): $(x - 3)^2 = 0$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q2/5

A quadratic equation with real coefficients has the solution 2 + i. What is its other solution?

A. −2 − i

B. 2 − i

C. −2 + i

D. 2 + i again

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

779

QUESTION ID

S71-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 − i

GRADING NOTE Nonreal solutions of real quadratics come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q3/5

True or false?

Every quadratic equation with real coefficients has two solutions in the complex numbers, counted with multiplicity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

780

QUESTION ID
S71-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the Fundamental Theorem of Algebra for degree 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q4/5

True or false?

A quadratic equation can have three different complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

781

QUESTION ID
S71-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A degree-2 equation has at most 2 solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 71 • FORWARD 2 • Q5/5

Solve quadratic equations by inspection, square roots, and factoring

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Factor quadratic expressions	**HSA-SSE.A.2**
BACKWARD 2	Solve the linear equations that come from each factor	**HSA-REI.B.3**
FORWARD 1	Find the zeros of a polynomial from its factors	**HSA-APR.B.3**
FORWARD 2	Use the Fundamental Theorem of Algebra to count complex zeros	**HSN-CN.C.9**

SET 23 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve by factoring.

$x^2 - 5x - 24 = 0$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

784

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 8 or x = −3

GRADING NOTE REQUIRED WORK: The factored form (x − 8)(x + 3) = 0, and each factor set equal to 0.

Form required: both solutions, with the factoring shown. A product is 0 only when a factor is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3(x - 2)^2 = 75$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

785

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 7$ or $x = -3$

GRADING NOTE Form required: both solutions. $(x - 2)^2 = 25$, so $x - 2 = 5$ or $x - 2 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2x^2 = 7x$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

786

QUESTION ID

S23-M-Q3

TYPE

Short answer

CORRECT ANSWER

x = 0 or x = 3.5

GRADING NOTE Form required: both solutions (3.5 or 7/2). $2x^2 - 7x = 0$, so $x(2x - 7) = 0$. Dividing both sides by x would lose x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q3/5

Solve $6x^2 + x - 2 = 0$.

A. $x = 2$ or $x = -1$

B. $x = -\frac{2}{3}$ or $x = \frac{1}{2}$

C. $x = \frac{2}{3}$ or $x = -\frac{1}{2}$

D. $x = -2$ or $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

787

QUESTION ID

S23-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = −2/3 or x = 1/2

GRADING NOTE $6x^2 + x - 2 = (3x + 2)(2x - 1)$, so $3x + 2 = 0$ or $2x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q4/5

True or false?

A student divides both sides of $x^2 = 4x$ by x and gets $x = 4$. This finds every solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

788

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 4x = x(x - 4) = 0$, so $x = 0$ is also a solution. Dividing by x assumed $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 23 • MAIN • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 1

Factor quadratic expressions • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor completely.

$x^2 + 2x - 15$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

790

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 5)(x − 3)

GRADING NOTE Form required: factored completely. $5 \cdot (-3) = -15$ and $5 + (-3) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q1/5

Factor completely.

$4x^2 - 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

791

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2x + 3)(2x − 3)

GRADING NOTE Form required: factored completely. A difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q2/5

Factor $x^2 - 9x + 20$.

A. $(x - 4)(x + 5)$

B. $(x - 2)(x - 10)$

C. $(x - 4)(x - 5)$

D. $(x + 4)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

792

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(x - 4)(x - 5)$

GRADING NOTE $-4 \cdot -5 = 20$ and $-4 + -5 = -9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q3/5

True or false?

$3x^2 - 12x = 3x(x - 4)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

793

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3x \cdot x = 3x^2$ and $3x \cdot (-4) = -12x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

794

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 3)(x - 3) = x^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

BACKWARD 2

Solve the linear equations that come from each factor • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 7 = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

796

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−3.5

GRADING NOTE Any equivalent form, such as −7/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q1/5

Solve $3x - 4 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

797

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4/3

GRADING NOTE Any equivalent form, such as $1\frac{1}{3}$. A rounded decimal is not exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5 - 2x = 0$.

A. $x = 3$

B. $x = -2.5$

C. $x = 2.5$

D. $x = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

798

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 2.5

GRADING NOTE $-2x = -5$, so $x = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q3/5

True or false?

If $x - 6 = 0$, then $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

799

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 4x = 0, then x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5

800

QUESTION ID
S23-B2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Divide both sides by 4: 0 ÷ 4 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 1

Find the zeros of a polynomial from its factors • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Find all zeros of $p(x) = (x - 2)(x + 1)(x - 5)$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

802

QUESTION ID
S23-F1-Q1

TYPE
Short answer

CORRECT ANSWER
2, −1, and 5

GRADING NOTE Form required: all three zeros. Each factor is 0 at one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find all zeros of $p(x) = x^3 - 4x$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

803

QUESTION ID
S23-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0, 2, and −2

GRADING NOTE Form required: all three zeros. $x^3 - 4x = x(x - 2)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q2/5

Which polynomial has zeros −3, 0, and 4?

A. x(x + 3)(x − 4)

B. (x + 3)(x − 4)

C. x(x − 3)(x + 4)

D. x(x − 3)(x − 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

804

QUESTION ID
S23-F1-Q3

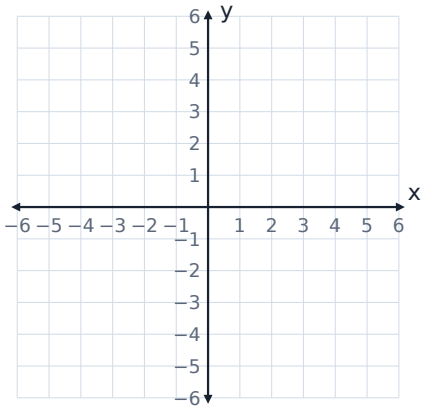
TYPE
Multiple choice

CORRECT ANSWER
A. x(x + 3)(x − 4)

GRADING NOTE A zero at −3 gives the factor x + 3; a zero at 4 gives x − 4; a zero at 0 gives x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q3/5

Sketch a rough graph of $p(x) = -(x + 3)(x - 1)(x - 4)$. Mark the zeros and the y-intercept, and show the end behavior. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

805

QUESTION ID

S23-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that crosses the x-axis at −3, 1, and 4, passes through (0, −12) (below the grid is fine), rises to the left and falls to the right

GRADING NOTE Any correct sketch: three crossings and the end behavior of a cubic with negative leading coefficient. $p(0) = -(3)(-1)(-4) = -12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S23-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q4/5

True or false?

$p(x) = x^2(x - 3)$ has three different zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

806

QUESTION ID
S23-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its zeros are 0 (twice) and 3: only two different values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-REI.B.4.b

FORWARD 2

Use the Fundamental Theorem of Algebra to count complex zeros • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does a polynomial of degree 5 have?

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

808

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. By the Fundamental Theorem of Algebra, a degree-n polynomial has n complex zeros counted with multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = x^3 - x^2 + 4x - 4 = (x - 1)(x^2 + 4)$.

Find all complex zeros.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

809

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2i, and −2i

GRADING NOTE Form required: all three zeros. $x^2 + 4 = 0$ gives $x^2 = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A polynomial with real coefficients has zeros 3 and 2i.

Which number must also be a zero?

A. 3i

B. 2

C. −3

D. −2i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

810

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −2i

GRADING NOTE Nonreal zeros of a polynomial with real coefficients come in conjugate pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q3/5

True or false?

$x^2 + 1 = (x + i)(x - i)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

811

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + i)(x - i) = x^2 - i^2 = x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q4/5

True or false?

Every polynomial of degree 3 has three real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

812

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^3 + x = x(x^2 + 1)$ has only one real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 23 • FORWARD 2 • Q5/5

Solve quadratic equations with the quadratic formula and use the discriminant

MAIN	Algebra I main question	**HSA-REI.B.4.b**
BACKWARD 1	Complete the square to solve a quadratic equation	**HSA-REI.B.4.a**
BACKWARD 2	Estimate square roots with rational approximations	**8.NS.A.2**
FORWARD 1	Solve right-triangle problems that lead to quadratic equations	**HSG-SRT.C.8**
FORWARD 2	Solve quadratic equations with complex solutions	**HSN-CN.C.7**

SET 24 • HSA-REI.B.4.b

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $2x^2 - 3x - 4 = 0$ with the quadratic formula.

Give exact answers in simplest radical form.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

815

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = (3 + \sqrt{41}) \div 4$ or
$x = (3 - \sqrt{41}) \div 4$

GRADING NOTE REQUIRED WORK: a = 2, b = −3, and c = −4 substituted into $x = (-b \pm \sqrt{(b^2 - 4ac)}) \div (2a)$, with the discriminant 9 + 32 = 41 shown.

Form required: exact, in simplest radical form (both solutions), with the formula work shown. Decimals are not exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S24-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q1/5

Solve $x^2 + 4x - 1 = 0$. Round each solution to the nearest hundredth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

816

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

$x \approx 0.24$ or $x \approx -4.24$

GRADING NOTE Form required: both solutions, rounded to the nearest hundredth. $x = -2 \pm \sqrt{5}$, and $\sqrt{5} \approx 2.236$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q2/5

Use the discriminant to find how many real solutions $3x^2 - 2x + 5 = 0$ has.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

817

QUESTION ID

S24-M-Q3

TYPE

Short answer

CORRECT ANSWER

None. The discriminant is −56.

GRADING NOTE Form required: the number of real solutions and the discriminant. $b^2 - 4ac = 4 - 60 = -56 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q3/5

Solve $x^2 + 2x + 7 = 0$ over the real numbers.

A. No real solution

B. $x = -1 + \sqrt{6}$ or $x = -1 - \sqrt{6}$

C. $x = -7$ or $x = 1$

D. $x = 1 + \sqrt{6}$ or $x = 1 - \sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

818

QUESTION ID

S24-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. No real solution

GRADING NOTE $b^2 - 4ac = 4 - 28 = -24 < 0$, so no real number is a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q4/5

True or false?

The equation $4x^2 - 12x + 9 = 0$ has exactly one real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

819

QUESTION ID
S24-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $b^2 - 4ac = 144 - 144 = 0$. The equation is $(2x - 3)^2 = 0$, so $x = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 24 • MAIN • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 1

Complete the square to solve a quadratic equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 4x = 5$ by completing the square.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

821

QUESTION ID

S24-B1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x = 5$ or $x = -1$

GRADING NOTE REQUIRED WORK: 4 added to both sides to get $(x - 2)^2 = 9$, then $x - 2 = 3$ or $x - 2 = -3$.

Form required: both solutions, with the completed square shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 2x - 2 = 0$ by completing the square. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

822

QUESTION ID

S24-B1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$x = -1 + \sqrt{3}$ or $x = -1 - \sqrt{3}$

GRADING NOTE REQUIRED WORK: $x^2 + 2x = 2$, then 1 added to both sides to get $(x + 1)^2 = 3$, then $x + 1 = \pm\sqrt{3}$.

Form required: exact (both solutions), with the completed square shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation results from completing the square on $x^2 - 6x + 4 = 0$?

A. $(x + 3)^2 = 5$

B. $(x - 3)^2 = 5$

C. $(x - 3)^2 = 13$

D. $(x - 6)^2 = 32$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

823

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(x - 3)^2 = 5$

GRADING NOTE $x^2 - 6x = -4$; add 9 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q3/5

True or false?

Adding 25 to both sides of $x^2 + 10x = 3$ gives $(x + 5)^2 = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

824

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 + 10x + 25 = (x + 5)^2$, and $3 + 25 = 28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q4/5

True or false?

Completing the square on $x^2 + 6x = 1$ gives $(x + 3)^2 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

825

QUESTION ID
S24-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Add 9 to both sides: $(x + 3)^2 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

BACKWARD 2

Estimate square roots with rational approximations • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is √41?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

827

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

GRADING NOTE Any equivalent form. 36 < 41 < 49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Round √5 to the nearest hundredth.

√5 ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

828

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.24

GRADING NOTE Form required: rounded to the nearest hundredth. √5 ≈ 2.236.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which is the best estimate of √30?

A. 6.2

B. 5.0

C. 15

D. 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

829

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5.5

GRADING NOTE √30 ≈ 5.48: $5.5^2 = 30.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{2} \approx 1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

830

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.41^2 = 1.9881$, close to 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

831

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 81 < 90 < 100, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 1

Solve right-triangle problems that lead to quadratic equations • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle's legs are x and x + 7, and its hypotenuse is 13.

Write an equation and find the lengths of the legs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

833

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + (x + 7)^2 = 169$; the legs are 5 and 12.

GRADING NOTE Form required: an equation and both legs. $x^2 + 7x - 60 = 0$, so $x = 5$ or $x = -12$; a length cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q1/5

A 10-ft ladder leans against a wall and makes a 70° angle with the ground.

How high up the wall does it reach, to the nearest tenth of a foot?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

834

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9.4 feet

GRADING NOTE Form required: rounded to the nearest tenth. h = 10 sin 70° ≈ 9.40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q2/5

A rectangle's diagonal is 10, and its length is 2 more than its width.

What is the width?

A. −8

B. 6

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

835

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE $w^2 + (w + 2)^2 = 100$ gives $w^2 + 2w - 48 = 0$, so $w = 6$ or $w = -8$. A width must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q3/5

True or false?

A ramp rises 1 ft over a horizontal run of 12 ft. Its angle with the ground is about 4.8°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

836

QUESTION ID
S24-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\tan\theta = 1 \div 12$, so $\theta = \tan^{-1}(1/12) \approx 4.76°$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs 1 and 1 has a hypotenuse of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

837

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The hypotenuse is √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-REI.B.4.b

FORWARD 2

Solve quadratic equations with complex solutions • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 - 4x + 13 = 0$. Write the solutions in the form $a + bi$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

839

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 2 + 3i or

x = 2 − 3i

GRADING NOTE Form required: a + bi (both solutions). $x = (4 \pm \sqrt{(-36)}) \div 2 = 2 \pm 3i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q1/5

Solve $x^2 + 25 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

840

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 5i$ or $x = -5i$

GRADING NOTE Form required: both solutions. $x^2 = -25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q2/5

What are the solutions of $x^2 + 2x + 5 = 0$?

A. $1 + 2i$ and $1 - 2i$

B. $-1 + 4i$ and $-1 - 4i$

C. $-2 + 4i$ and $-2 - 4i$

D. $-1 + 2i$ and $-1 - 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

841

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −1 + 2i and −1 − 2i

GRADING NOTE $x = (-2 \pm \sqrt{(4 - 20)}) \div 2 = (-2 \pm 4i) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q3/5

True or false?

If a quadratic equation with real coefficients has a negative discriminant, its two solutions are complex conjugates.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

842

QUESTION ID
S24-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The ± in the quadratic formula gives a + bi and a − bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q4/5

True or false?

The solutions of $x^2 + 2x + 2 = 0$ are $1 + i$ and $1 - i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q5/5

843

QUESTION ID
S24-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x = (-2 \pm \sqrt{(-4)}) \div 2 = -1 \pm i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 24 • FORWARD 2 • Q5/5

Create quadratic and exponential equations in one variable and use them to solve problems

MAIN	Algebra I main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations	**HSA-REI.B.4.b**
BACKWARD 2	Use properties of integer exponents	**8.EE.A.1**
FORWARD 1	Use volume formulas to write and solve equations for a dimension	**HSG-GMD.A.3**
FORWARD 2	Solve exponential models with logarithms	**HSF-LE.A.4**

SET 25 • HSA-CED.A.1

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 48t + 4$ feet after t seconds.

Write and solve an equation to find when the ball is 36 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

846

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-16t^2 + 48t + 4 = 36$; t = 1 second and t = 2 seconds

GRADING NOTE Form required: an equation and both times. $-16t^2 + 48t - 32 = 0$, so $t^2 - 3t + 2 = 0$ and $(t - 1)(t - 2) = 0$. The ball passes 36 ft going up and coming down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 5 cm more than its width. Its area is 84 cm².

Write an equation and find the dimensions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

847

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

w(w + 5) = 84; 7 cm by 12 cm

GRADING NOTE Form required: an equation and both dimensions. $w^2 + 5w - 84 = (w + 12)(w - 7) = 0$; a width cannot be −12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q2/5

A culture of 200 bacteria doubles every hour.

Write an equation for the number of hours t until there are 6,400 bacteria, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

848

QUESTION ID

S25-M-Q3

TYPE

Short answer

CORRECT ANSWER

$200 \cdot 2^t = 6{,}400$;

t = 5 hours

GRADING NOTE Form required: an equation and its solution. $2^t = 32 = 2^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q3/5

A $2,000 investment earns 5% interest, added once at the end of each year.

Which inequality gives the whole numbers of years n after which it is worth at least $3,000?

A. $2{,}000(1.5)^n \geq 3{,}000$

B. $2{,}000(0.05)^n \geq 3{,}000$

C. $2{,}000 + 0.05n \geq 3{,}000$

D. $2{,}000(1.05)^n \geq 3{,}000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

849

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D.

$2{,}000(1.05)^n \geq 3{,}000$

GRADING NOTE Each year the value is multiplied by 1 + 0.05 = 1.05. The first whole year it reaches $3,000 is n = 9: $2{,}000(1.05)^8 \approx \$2{,}954.91$ and $2{,}000(1.05)^9 \approx \$3{,}102.66$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q4/5

True or false?

A square garden of side x ft has a 2-ft path around it, and the garden and path cover 144 ft². Then $(x + 4)^2 = 144$, so $x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

850

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The path adds 2 ft on each side, so the outer side is x + 4 = 12. The other root, x = −16, is not a length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 25 • MAIN • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $t^2 - 3t + 2 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

852

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t = 1 or t = 2

GRADING NOTE Form required: both solutions. $(t - 1)(t - 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $w^2 + 5w = 84$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

853

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

w = 7 or w = −12

GRADING NOTE Form required: both solutions. (w + 12)(w − 7) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q2/5

Solve $(x + 4)^2 = 144$.

A. $x = 12$ or $x = -12$

B. $x = 140$ or $x = -148$

C. $x = 8$ only

D. $x = 8$ or $x = -16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

854

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 8$ or $x = -16$

GRADING NOTE $x + 4 = 12$ or $x + 4 = -12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

$x = -2$ is a solution of $x^2 + 5x + 6 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

855

QUESTION ID
S25-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $4 - 10 + 6 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The only solution of $x^2 = 3x$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

856

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x(x - 3) = 0$, so $x = 0$ is also a solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

BACKWARD 2

Use properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 32 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

858

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2^5

GRADING NOTE Form required: a power of 2. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $2^3 \cdot 2^4$. Write the answer as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

859

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2^7

GRADING NOTE Form required: a power of 2. Add the exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which power of 2 equals 64?

A. 2^{32}

B. 2^{8}

C. 2^{5}

D. 2^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

860

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2^{6}

GRADING NOTE $2^{6} = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q3/5

True or false?

$3^4 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

861

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 3 \cdot 3 \cdot 3 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$2^{-3} = \frac{1}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

862

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^{-3} = 1 \div 2^3 = 1 \div 8$. A negative exponent does not make the value negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 1

Use volume formulas to write and solve equations for a dimension • High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cylinder has height 10 cm and volume 250π cm^3.

Write an equation and find its radius.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

864

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\pi r^2(10) = 250\pi$; r = 5 cm

GRADING NOTE Form required: an equation and the radius. $r^2 = 25$, and a radius is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q1/5

A square pyramid has height 9 in and volume 300 in³.

Find the side length of its base.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

865

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 inches

GRADING NOTE Any equivalent form. $\frac{1}{3}s^2(9) = 300$, so $3s^2 = 300$ and $s^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q2/5

A cone has volume 12π and height 4.

What is its radius?

A. 9

B. 6

C. $\sqrt{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

866

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE $\frac{1}{3}\pi r^2(4) = 12\pi$, so $r^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q3/5

True or false?

A sphere of radius 3 has volume 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

867

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\frac{4}{3}\pi(27) = 36\pi$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

868

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The volume depends on r^2, so it is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-CED.A.1

FORWARD 2

Solve exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $200 \cdot 2^t = 5{,}000$. Give the exact answer using logarithms, then round it to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

870

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$t = \log 25 \div \log 2 \approx 4.64$

GRADING NOTE Form required: the logarithm expression and the rounded value. $2^t = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q1/5

Solve $2{,}000(1.05)^n = 3{,}000$. Round to the nearest tenth.

n ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

871

QUESTION ID
S25-F2-Q2

TYPE
Short answer

CORRECT ANSWER
8.3

GRADING NOTE Form required: rounded to the nearest tenth. n = log 1.5 ÷ log 1.05 ≈ 8.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q2/5

Solve $10^x = 500$. (log means base 10.)

A. x = 2.5

B. x = 5,000

C. x = 50

D. x = log 500 ≈ 2.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

872

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = log 500 ≈ 2.70

GRADING NOTE x = log 500 because log 500 is the power of 10 that gives 500; log 500 ≈ 2.699.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q3/5

True or false?

The solution of $e^x = 7$ is $x = \ln 7$. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

873

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ln is the logarithm base e.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

The solution of $5^x = 30$ is $x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

874

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5^x = 30$ gives $x = \log 30 \div \log 5 \approx 2.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 25 • FORWARD 2 • Q5/5

Show that replacing an equation by its sum with a multiple of the other keeps the solutions of a system

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Justify steps with the properties of equality	**HSA-REI.A.1**
BACKWARD 2	Recognize the solution of a system as a point on both lines	**8.EE.C.8.a**
FORWARD 1	Write a system as a matrix equation AX = B and use row operatio..	**HSA-REI.C.8**
FORWARD 2	Use determinants and inverse matrices	**HSN-VM.C.10**

SET 26 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

System: x + y = 7 and x − y = 1.

Replace the second equation by the sum of the two equations. Write the new system, solve it, and show that the solution also satisfies the original system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

877

QUESTION ID

S26-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(4, 3)

GRADING NOTE REQUIRED WORK: The new system x + y = 7 and 2x = 8, its solution, and a check of the solution in both original equations.

Form required: the new system, the solution, and the check. 2x = 8 gives x = 4, so y = 3. Check: 4 + 3 = 7 and 4 − 3 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Suppose (a, b) is a solution of the system $2x + y = 5$ and $x - 3y = 6$.

Explain why (a, b) is also a solution of $(2x + y) + 2(x - 3y) = 5 + 2(6)$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

878

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

Since 2a + b = 5 and a − 3b = 6, 2(a − 3b) = 12. Adding equal amounts to both sides of 2a + b = 5 gives (2a + b) + 2(a − 3b) = 5 + 12, a true equation.

GRADING NOTE Any equivalent explanation that substitutes (a, b) and adds equal quantities to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q2/5

System: 3x + 2y = 12 and x − y = −1.

Which system has the same solution?

A. 3x + 2y = 12 and 4x + y = 13

B. 3x + 2y = 12 and 2x + 3y = 11

C. 3x + 2y = 12 and 5x = 14

D. 3x + 2y = 12 and 5x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

879

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3x + 2y = 12 and 5x = 10

GRADING NOTE Adding the first equation and 2 times the second gives 5x = 10. The solution of both systems is (2, 3); (2, 3) does not satisfy 5x = 14, 4x + y = 13, or 2x + 3y = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q3/5

True or false?

Replacing one equation of a system by the sum of that equation and 3 times the other equation can change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

880

QUESTION ID
S26-M-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Any solution of the old system satisfies the new equation, and the old equation can be recovered by subtracting 3 times the other, so the solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q4/5

True or false?

In the system 2x + 3y = 7 and 4x − 3y = 5, adding the equations eliminates y and gives 6x = 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

881

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3y + (−3y) = 0 and 7 + 5 = 12. Then x = 2 and y = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 26 • MAIN • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 1

Justify steps with the properties of equality • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If a = b and c = d, explain why a + c = b + d.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

883

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Adding c to both sides of a = b gives a + c = b + c. Since c = d, replace c by d on the right: a + c = b + d.

GRADING NOTE Any equivalent explanation using the addition property and substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q1/5

Name the property used to go from x = 5 to 2x = 10.

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

884

QUESTION ID
S26-B1-Q2

TYPE
Short answer

CORRECT ANSWER
Multiplication property of equality (multiply both sides by 2)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

If p = q, which equation must also be true?

A. 4p = q

B. p = q + 4

C. p + 4 = q + 4

D. p + 4 = q

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

885

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. p + 4 = q + 4

GRADING NOTE Add 4 to both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If a = b, then 5a = 5b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

886

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplication property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If a = b and c = d, then a − c = b + d.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

887

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtracting equal amounts gives a − c = b − d.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

BACKWARD 2

Recognize the solution of a system as a point on both lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

Is (2, 5) a solution of the system y = 2x + 1 and y = x + 3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

889

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(2) + 1 = 5 and 2 + 3 = 5.

GRADING NOTE Any equivalent explanation that checks both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The lines y = x + 1 and y = −x + 5 meet at (2, 3).

What is the solution of the system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

890

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3)

GRADING NOTE Any equivalent form, such as x = 2, y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q2/5

Which point is a solution of both $x + y = 6$ and $x - y = 2$?

A. (2, 4)

B. (4, 2)

C. (3, 3)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

891

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 2)

GRADING NOTE $4 + 2 = 6$ and $4 - 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q3/5

True or false?

The solution of a system of two linear equations is the point where their graphs intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

892

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That point lies on both lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q4/5

True or false?

(1, 1) is a solution of the system y = 2x and y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

893

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2(1) = 2, not 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 1

Write a system as a matrix equation AX = B and use row operations • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

Write the system 2x + y = 7 and x − 3y = 0 as a matrix equation AX = B. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

895

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A has rows [2 1] and [1 −3]; X is the column [x, y]; B is the column [7, 0]; and AX = B.

GRADING NOTE Form required: A, X, and B, in this order of rows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the matrix equation AX = B, A has rows [1 2] and [3 −1], X is the column [x, y], and B is the column [5, 1].

Write the system it represents.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

896

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x + 2y = 5 and 3x − y = 1

GRADING NOTE Any equivalent form. Each row of A times X equals the matching entry of B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

On the rows [1 2 | 5] and [3 −1 | 1], replace row 2 by (row 2) − 3 × (row 1).

What is the new row 2?

A. [0 −7 | 14]

B. [2 −3 | −4]

C. [0 −7 | −14]

D. [0 5 | 16]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

897

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. [0 −7 | −14]

GRADING NOTE 3 − 3 = 0, −1 − 6 = −7, and 1 − 15 = −14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

True or false?

Adding a multiple of one row of an augmented matrix to another row does the same thing as replacing that equation by its sum with a multiple of the other equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

898

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each row is an equation, so this row operation is the elimination step written in matrix form. (Swapping rows or multiplying a row by a nonzero number are other row operations.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

True or false?

The coefficient matrix of x + y = 3 and 2x − y = 0 has the rows [1 1] and [2 1].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

899

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second row is [2 −1].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 2

Use determinants and inverse matrices • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10

Find the determinant of the matrix with rows [3 1] and [5 2].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

901

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. 3(2) − 1(5) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

Show that the matrix with rows [2 −1] and [−5 3] is the inverse of the matrix with rows [3 1] and [5 2].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

902

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The product has rows [3(2) + 1(−5) 3(−1) + 1(3)] = [1 0] and [5(2) + 2(−5) 5(−1) + 2(3)] = [0 1], the identity matrix.

GRADING NOTE Any equivalent work that multiplies the matrices and gets the identity matrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

Which matrix has no inverse?

A. Rows [2 4] and [1 2]

B. Rows [1 0] and [0 1]

C. Rows [2 1] and [1 1]

D. Rows [3 1] and [5 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

903

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [2 4] and [1 2]

GRADING NOTE 2(2) − 4(1) = 0. A 2-by-2 matrix has an inverse only when its determinant is not 0; each of the other three matrices has determinant 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system of two linear equations has an inverse, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

904

QUESTION ID
S26-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The solution is $X = A^{-1}B$, and it is the only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

True or false?

The inverse of the matrix with rows [2 0] and [0 4] has rows [−2 0] and [0 −4].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

905

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The inverse has rows $[\frac{1}{2}\ 0]$ and $[0\ \frac{1}{4}]$: each diagonal entry is replaced by its reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

Prove that replacing an equation by its sum with a multiple of the other gives an equivalent system, in both directions

MAIN	Algebra I main question	**HSA-REI.C.5**
BACKWARD 1	Use the properties of equality to justify a step	**HSA-REI.A.1**
BACKWARD 2	Solve simple systems by adding equations	**8.EE.C.8.b**
FORWARD 1	Use row operations on an augmented matrix as elimination steps	**HSA-REI.C.8**
FORWARD 2	Solve the same system with an inverse matrix	**HSA-REI.C.9**

SET 72 • HSA-REI.C.5

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

System A: $x + 2y = 7$ and $3x - y = 7$. System B: $x + 2y = 7$ and $7x = 21$, where $7x = 21$ is the first equation plus 2 times the second.

Show that every solution of A solves B and every solution of B solves A. Then give the solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

908

QUESTION ID

S72-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(3, 2)

GRADING NOTE REQUIRED WORK: A to B: a solution of A makes both equations true, so adding $x + 2y = 7$ and $2(3x - y) = 14$ gives $7x = 21$. B to A: from $7x = 21$ and $x + 2y = 7$, subtracting gives $6x - 2y = 14$, so $3x - y = 7$. The kept equation $x + 2y = 7$ is in both systems.

Form required: both directions and the solution. Check: $3 + 4 = 7$ and $9 - 2 = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q1/5

Let P = c and Q = d be the two equations of a system (P and Q are expressions in x and y). Prove that the system P = c and Q + kP = d + kc has exactly the same solutions, for any number k.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

909

QUESTION ID

S72-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

The two systems have the same solutions.

GRADING NOTE REQUIRED WORK: If (x, y) makes P = c and Q = d true, then Q + kP = d + kc (adding equal amounts), so it solves the new system. Conversely, if P = c and Q + kP = d + kc, subtracting kP = kc gives Q = d. So the solution sets are equal.

Form required: both directions, each with its reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q2/5

Why must the new system keep one of the original equations unchanged? Give an example.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

910

QUESTION ID

S72-M-Q3

TYPE

Short answer

CORRECT ANSWER

The kept equation is needed to recover the replaced one. For example, replacing both x + y = 2 and x − y = 0 with their sum 2x = 2 alone loses information: 2x = 2 has infinitely many solutions (x = 1, any y), but the system has only (1, 1).

GRADING NOTE Any equivalent explanation with an example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q3/5

System: 2x + y = 5 and x − y = 1. Which step can produce a system with DIFFERENT solutions?

A. Replacing both equations by their sum, 3x = 6, alone

B. Multiplying the first equation by 3

C. Writing the equations in the other order

D. Replacing the second equation by the sum of the two equations

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

911

QUESTION ID
S72-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. Replacing both equations by their sum, 3x = 6, alone

GRADING NOTE 3x = 6 alone allows any y. The other steps keep the solution (2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q4/5

True or false?

If (2, 1) solves x + y = 3 and x − y = 1, then (2, 1) also solves (x + y) + 4(x − y) = 3 + 4(1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

912

QUESTION ID
S72-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 3 + 4(1) = 7 on both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 72 • MAIN • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 1

Use the properties of equality to justify a step • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

If P = c and Q = d are true, explain why P + Q = c + d is true.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

914

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Add Q to both sides of P = c to get P + Q = c + Q; since Q = d, replace Q by d on the right.

GRADING NOTE Any equivalent explanation using the addition property and substitution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q1/5

If P + 3Q = c + 3d and Q = d, explain why P = c.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

915

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Since Q = d, 3Q = 3d. Subtracting equal amounts from both sides of P + 3Q = c + 3d gives P = c.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q2/5

Which property justifies going from $x - y = 1$ to $4x - 4y = 4$?

A. Addition property of equality

B. Commutative property

C. Substitution of y

D. Multiplication property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

916

QUESTION ID

S72-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Multiplication property of equality

GRADING NOTE Both sides are multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q3/5

True or false?

Subtracting equal quantities from both sides of a true equation gives a true equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

917

QUESTION ID
S72-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Subtraction property of equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q4/5

True or false?

If a + b = 10 and a = 4, then b = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

QUESTION ID

S72-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE b = 10 − 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

BACKWARD 2

Solve simple systems by adding equations • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve by adding the equations.

$x + y = 9$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

920

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 3)

GRADING NOTE Any equivalent form. 2x = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 11$

$y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

921

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Any equivalent form. 2x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q2/5

Solve.

$x + 2y = 8$

$x = 2$

A. (2, 5)

B. (8, 0)

C. (2, 3)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

922

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 3)

GRADING NOTE 2 + 2y = 8, so y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Adding $x + y = 5$ and $x - y = 1$ gives $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

923

QUESTION ID

S72-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y and −y cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q4/5

True or false?

The system x + y = 4 and 2x + 2y = 8 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

924

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 2 times the first: infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 1

Use row operations on an augmented matrix as elimination steps • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Write the augmented matrix of the system x + 2y = 7 and 3x − y = 7.

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

926

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

[1 2 | 7] and [3 −1 | 7]

GRADING NOTE Form required: both rows in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q1/5

Apply R2 → 2R2 + R1 to the rows [1 2 | 7] and [3 −1 | 7]. What is the new second row, and which equation does it represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

927

QUESTION ID
S72-F1-Q2

TYPE
Short answer

CORRECT ANSWER
[7 0 | 21], which is 7x = 21

GRADING NOTE Form required: the row and its equation. 2[3 −1 | 7] + [1 2 | 7].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q2/5

Which row operation matches "replace equation 2 by itself minus 3 times equation 1"?

A. R2 → R1

B. R2 → R2 − 3R1

C. R2 → 3R2 − R1

D. R1 → R1 − 3R2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

928

QUESTION ID

S72-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. R2 → R2 − 3R1

GRADING NOTE Row 2 changes; row 1 is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q3/5

True or false?

A row operation R2 → R2 + kR1 does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

929

QUESTION ID
S72-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is the elimination step proved in the MAIN section.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q4/5

True or false?

Replacing R2 by 0 · R2 + R1 keeps the solutions of every system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

930

QUESTION ID

S72-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It discards equation 2, so the solution set can grow.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSA-REI.C.5

FORWARD 2

Solve the same system with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For x + 2y = 7 and 3x − y = 7, the coefficient matrix A has rows [1 2] and [3 −1]. Find the determinant of A.

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

932

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−7

GRADING NOTE Any equivalent form. 1(−1) − 2(3) = −7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A^{-1} is $(1 \div (-7))$ times the matrix with rows $[-1\ -2]$ and $[-3\ 1]$. Use $X = A^{-1}B$, with B the column $[7, 7]$, to solve the system.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

933

QUESTION ID

S72-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

GRADING NOTE Form required: both values. $(1 \div (-7))$ times the column $[-7 - 14, -21 + 7] = [3, 2]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q2/5

If the determinant of the coefficient matrix of a 2-by-2 linear system is 0, which is true?

A. $X = A^{-1}B$ gives the solution.

B. The system has exactly one solution.

C. The system has no solution or infinitely many solutions.

D. The two equations must be identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

934

QUESTION ID

S72-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The system has no solution or infinitely many solutions.

GRADING NOTE A has no inverse, so the system does not have exactly one solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q3/5

True or false?

Solving $AX = B$ with $X = A^{-1}B$ gives the same solution as elimination.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

935

QUESTION ID

S72-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both methods find the one solution when A has an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, $A^{-1}B = BA^{-1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

936

QUESTION ID

S72-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Matrix multiplication is not commutative in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 72 • FORWARD 2 • Q5/5

Solve systems of linear equations exactly by substitution and elimination

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Eliminate a variable by adding a multiple of an equation	**HSA-REI.C.5**
BACKWARD 2	Solve simple systems of linear equations algebraically	**8.EE.C.8.b**
FORWARD 1	Use slopes and systems to find where parallel or perpendicular li..	**HSG-GPE.B.5**
FORWARD 2	Multiply a matrix by a column to check a solution	**HSN-VM.C.8**

SET 27 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve by substitution.

$y = 3x - 5$

$2x + y = 15$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

939

QUESTION ID

S27-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

(4, 7)

GRADING NOTE REQUIRED WORK: 3x − 5 substituted for y: 2x + (3x − 5) = 15 solved for x, then y found from y = 3x − 5.

Form required: the solution with the substitution shown; it may be written x = 4, y = 7. Check: 8 + 7 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

Solve by elimination.

$3x + 4y = 10$

$5x - 2y = 34$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

940

QUESTION ID

S27-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

(6, −2)

GRADING NOTE

REQUIRED WORK: The second equation multiplied by 2 and added to the first, so y cancels (13x = 78), then y found by substitution.

Form required: the solution with the elimination shown; it may be written x = 6, y = −2. Check: 30 + 4 = 34.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

Solve.

$\frac{1}{2}x + \frac{1}{3}y = 4$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

941

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

(6, 3)

GRADING NOTE Any equivalent form. Multiply the first equation by 6: 3x + 2y = 24. Substitute x = y + 3: 5y + 9 = 24, so y = 3. Check: 3 + 1 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

Solve.

$2x - 3y = -1$

$4x + y = 19$

A. $(-4, -3)$

B. $(4, 3)$

C. $(3, 4)$

D. $(5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

942

QUESTION ID

S27-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 3)

GRADING NOTE $y = 19 - 4x$, so $2x - 57 + 12x = -1$ and $x = 4$, $y = 3$. Check: $8 - 9 = -1$ and $16 + 3 = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

True or false?

The system 2x + y = 5 and 4x + 2y = 9 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

943

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Doubling the first equation gives 4x + 2y = 10, which cannot also equal 9. The lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 1

Eliminate a variable by adding a multiple of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Add the equations x + 2y = 9 and 3x − 2y = 3.

What equation results, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

945

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x = 12; x = 3

GRADING NOTE Form required: the new equation and x. The y-terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

System: 2x + y = 8 and 6x − 5y = 8.

What multiple of the first equation should you subtract from the second to eliminate x?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

946

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6x − 3(2x) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

System: x + y = 5 and 2x + 3y = 12.

Which equation can replace the second equation without changing the solution?

A. x = 2

B. y = 7

C. y = 2

D. 3x + 4y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

947

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2

GRADING NOTE (2x + 3y) − 2(x + y) = 12 − 10 gives y = 2. The solution is (3, 2), which does not satisfy x = 2, 3x + 4y = 12, or y = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

True or false?

Multiplying one equation of a system by a nonzero number does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

948

QUESTION ID
S27-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The new equation has exactly the same solutions as the old one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Adding the equations $x + y = 4$ and $x - y = 2$ gives $2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

949

QUESTION ID
S27-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 + 2 = 6$, so $2x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 2

Solve simple systems of linear equations algebraically • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = x + 2$

$y = 3x$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

951

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 3)

GRADING NOTE Any equivalent form. $3x = x + 2$, so $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

x + y = 10

y = 4

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

952

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(6, 4)

GRADING NOTE Any equivalent form. x + 4 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

Solve.

$y = 2x$

$x + y = 12$

A. (4, 8)

B. (6, 6)

C. (8, 4)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

953

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 8)

GRADING NOTE $x + 2x = 12$, so $x = 4$ and $y = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

The system $y = 2x + 3$ and $y = 2x - 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

954

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same slope, different y-intercepts: the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

The solution of $y = x$ and $y = -x + 6$ is $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

955

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = -x + 6$ gives $x = 3$, so the solution is $(3, 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 1

Use slopes and systems to find where parallel or perpendicular lines meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is $y = 2x + 1$. Write an equation of the line through $(5, 1)$ that is perpendicular to ℓ, and find where the two lines meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

957

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{1}{2}x + 3.5$; the lines meet at (1, 3).

GRADING NOTE Any equivalent equation; the point is required. The slope is $-\frac{1}{2}$. $2x + 1 = -\frac{1}{2}x + 3.5$ gives $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are the lines $2x + 3y = 6$ and $3x - 2y = 4$ perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

958

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Their slopes are $-\frac{2}{3}$ and $\frac{3}{2}$, and the product is -1.

GRADING NOTE Any equivalent explanation using slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line is parallel to $y = 3x - 2$ and passes through $(0, 5)$?

A. $y = 5x + 3$

B. $y = -3x + 5$

C. $y = 3x + 5$

D. $y = -\frac{1}{3}x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

959

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 3x + 5

GRADING NOTE Parallel lines have the same slope, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

True or false?

The lines $y = 4x + 1$ and $y = 4x - 7$ have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

960

QUESTION ID
S27-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are parallel and distinct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

True or false?

The lines y = 2x and y = −2x are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

961

QUESTION ID
S27-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 2(−2) = −4, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 2

Multiply a matrix by a column to check a solution • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8

Multiply the matrix with rows [2 1] and [1 −3] by the column [4, 1].

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

963

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The column [9, 1]

GRADING NOTE Any equivalent form. 2(4) + 1(1) = 9 and 1(4) − 3(1) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

The matrix A has rows [2 1] and [1 −3].

Use multiplication by A to decide whether x = 4, y = 1 solves 2x + y = 9 and x − 3y = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

964

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. A times the column [4, 1] is the column [9, 1], which matches the constants.

GRADING NOTE Any equivalent explanation using the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

What is the product of the matrix with rows [1 2] and [3 4] and the column [1, 1]?

A. The column [3, 7]

B. The column [2, 6]

C. The column [10]

D. The column [4, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

965

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The column [3, 7]

GRADING NOTE 1 + 2 = 3 and 3 + 4 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

True or false?

Multiplying a 2-by-2 matrix by a 2-by-1 column gives a 2-by-1 column.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

966

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2 \times 2)(2 \times 1)$ gives 2×1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, AB = BA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

967

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Matrix multiplication is not commutative in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

Solve systems of linear equations in context, including systems with no solution or infinitely many solutions

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Write and solve an equation for one quantity in context	**HSA-CED.A.1**
BACKWARD 2	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 1	Use matrices to organize data in context	**HSN-VM.C.6**
FORWARD 2	Solve a system in context with an inverse matrix	**HSA-REI.C.9**

SET 28 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A theater sold 340 tickets. Adult tickets cost $12 and student tickets cost $7. Sales totaled $3,255.

Write a system and find how many of each kind were sold.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

970

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

a + s = 340 and 12a + 7s = 3,255; 175 adult and 165 student tickets

GRADING NOTE Form required: a system and both numbers. 12a + 7(340 − a) = 3,255 gives 5a = 875. Check: 2,100 + 1,155 = 3,255.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

How many solutions does the system $6x - 4y = 10$ and $3x - 2y = 5$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

971

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

Infinitely many. The first equation is 2 times the second, so both describe the same line.

GRADING NOTE Form required: the number of solutions and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution.

How many liters of each are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

5 liters of the 10% solution and 15 liters of the 30% solution

GRADING NOTE Form required: both amounts, labeled. $x + y = 20$ and $0.10x + 0.30y = 5$. Then $6 - 0.2x = 5$, so $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

For which value of k does the system y = 3x + 2 and 6x − 2y = k have infinitely many solutions?

A. k = 2

B. k = −2

C. k = 4

D. k = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

973

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. k = −4

GRADING NOTE Substitute: 6x − 2(3x + 2) = −4 for every x, so the equations describe the same line only when k = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

True or false?

A system of two linear equations in two variables can have exactly two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

974

QUESTION ID
S28-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Two lines meet at no point, one point, or every point (the same line).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 1

Write and solve an equation for one quantity in context • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A gym has 340 members. There are 40 more adults than teens.

Write and solve an equation for the number of teens t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

976

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t + (t + 40) = 340;
t = 150

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pens cost $2 and notebooks cost $5. Ana buys 3 notebooks and some pens and spends $27.

Write and solve an equation for the number of pens p.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

977

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2p + 15 = 27; p = 6

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

A 20-liter mix has x liters of a 10% acid solution and the rest a 30% acid solution.

Which expression gives the liters of acid?

A. 0.40(20)

B. 0.10x + 0.30x

C. 0.10x + 0.30(20 − x)

D. 10x + 30(20 − x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

978

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

0.10x + 0.30(20 − x)

GRADING NOTE The rest of the mix is 20 − x liters; 10% = 0.10 and 30% = 0.30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

True or false?

25 coins are all dimes and quarters. If d coins are dimes, then 25 − d coins are quarters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

979

QUESTION ID
S28-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two counts add to 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

"3 more than 5 times a number is 48" gives 5n + 3 = 48, and its solution is n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

980

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5n = 45, so n = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

Two numbers have a sum of 30 and a difference of 8.

Find the numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

982

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 and 11

GRADING NOTE Any order. Adding the equations gives 2x = 38.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

Plan A costs $10 plus $2 per hour. Plan B costs $4 per hour.

After how many hours do they cost the same?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

983

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 10 + 2h = 4h, so h = 5 (both cost $20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

3 pencils and 2 erasers cost $1.30. 1 pencil and 2 erasers cost $0.70.

What does one pencil cost?

A. $0.60

B. $0.20

C. $0.35

D. $0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

984

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $0.30

GRADING NOTE Subtracting the equations gives 2 pencils = $0.60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

True or false?

Two cars leave the same place at the same time in the same direction, at 50 mi/h and 60 mi/h. After 2 hours the faster car is 20 miles ahead.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

985

QUESTION ID
S28-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $120 - 100 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

One company charges y = 2x + 5 dollars and another charges y = 2x + 8 dollars for x units. They charge the same for 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

986

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines are parallel, so the costs are never equal; the second always costs $3 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 1

Use matrices to organize data in context • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6

Friday: 120 adult and 90 student tickets. Saturday: 150 adult and 60 student tickets.

Write a 2-by-2 matrix with one row per day (adult first).

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

988

QUESTION ID
S28-F1-Q1

TYPE
Short answer

CORRECT ANSWER
[120 90] and [150 60]

GRADING NOTE Form required: Friday row first, adult column first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

Adult tickets cost $12 and student tickets $7.

Multiply the matrix with rows [120 90] and [150 60] by the column [12, 7]. What do the entries mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

989

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The column [2,070, 2,220]: Friday sales were $2,070 and Saturday sales were $2,220.

GRADING NOTE Form required: both entries and their meaning. 1,440 + 630 = 2,070 and 1,800 + 420 = 2,220.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

A matrix has rows [5 2] and [3 4]. Each row is a store and each column is a product.

What does the entry in row 2, column 1 represent?

A. Store 2's amount of product 1

B. Store 1's amount of product 2

C. Store 2's amount of product 2

D. The total of product 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

990

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Store 2's amount of product 1

GRADING NOTE Row 2 is store 2, and column 1 is product 1. The entry is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

True or false?

A matrix with 3 rows and 2 columns is a 3-by-2 matrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

991

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dimensions are written rows by columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

True or false?

Any two matrices can be added, whatever their dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

992

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only matrices with the same dimensions can be added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 2

Solve a system in context with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

Write the system a + s = 340 and 12a + 7s = 3,255 as a matrix equation AX = B. Give A, X, and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

994

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A has rows [1 1] and [12 7]; X is the column [a, s]; B is the column [340, 3,255]; and AX = B.

GRADING NOTE Form required: A, X, and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

The matrix with rows [1 1] and [12 7] has determinant −5.

Does the ticket system have exactly one solution? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

995

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The determinant is not 0, so A has an inverse and $X = A^{-1}B$ is the only solution.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

The inverse of the matrix with rows [1 1] and [12 7] is (1 ÷ (−5)) times which matrix?

A. Rows [7 −1] and [−12 1]

B. Rows [7 1] and [12 1]

C. Rows [−7 1] and [12 −1]

D. Rows [1 −1] and [−12 7]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

996

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [7 −1] and [−12 1]

GRADING NOTE For rows [a b] and [c d], the inverse is (1 ÷ (ad − bc)) times rows [d −b] and [−c a].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system has determinant 0, the system does not have exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

997

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The matrix has no inverse; the system has no solution or infinitely many.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

True or false?

The determinant of the matrix with rows [2 3] and [4 6] is 24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

998

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2(6) - 3(4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

Solve a system of a linear equation and a quadratic equation algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Solve systems of linear equations	**HSA-REI.C.6**
FORWARD 1	Decide whether points lie on a circle	**HSG-GPE.B.4**
FORWARD 2	Find complex solutions when a line and a parabola do not meet	**HSN-CN.C.7**

SET 29 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$y = x^2 - 3$

$y = 2x$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

1001

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6) and (−1, −2)

GRADING NOTE Form required: both ordered pairs. $x^2 - 3 = 2x$ gives $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$y = x^2 + 4x + 1$

$y = x - 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

1002

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, −2) and (−2, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + 3x + 2 = (x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many real solutions does the system $y = x^2 + 2$ and $y = x$ have?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

1003

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE $x^2 - x + 2 = 0$ has discriminant $1 - 8 = -7 < 0$, so the line and the parabola do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$x^2 + y^2 = 25$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

1004

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

(3, 4) and (−4, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + (x + 1)^2 = 25$ gives $x^2 + x - 12 = 0$, so $x = 3$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

True or false?

The line $y = 2x - 1$ meets the parabola $y = x^2$ at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

1005

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - 2x + 1 = (x - 1)^2 = 0$ has one solution, $x = 1$. The line touches the parabola at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 2x - 3 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

1007

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −1

GRADING NOTE Form required: both solutions. (x − 3)(x + 1) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + x - 12 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

1008

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −4

GRADING NOTE Form required: both solutions. (x + 4)(x − 3) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 3x + 2 = 0$.

A. $x = -1$ or $x = 2$

B. $x = 1$ or $x = 2$

C. $x = 1$ or $x = -2$

D. $x = -1$ or $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

1009

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = -1$ or $x = -2$

GRADING NOTE $(x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 - x + 2 = 0$ has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

1010

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The discriminant is $1 - 8 = -7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

1010

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solutions of $x^2 = 2x$ are $x = 2$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

1011

QUESTION ID
S29-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x(x - 2) = 0$, so $x = 0$ or $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 2

Solve systems of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = 2x$

$y = -x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

1013

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6)

GRADING NOTE Any equivalent form. $2x = -x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 1$

$y = x + 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

1014

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. $x + x + 1 = 1$, so $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 7$

$x - y = 2$

A. (4, −1)

B. (3, 1)

C. (1, 3)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

1015

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 1)

GRADING NOTE Adding gives 3x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

True or false?

Substitution and elimination give the same solution of a system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

1016

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both methods keep the solutions unchanged at every step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

True or false?

(2, 1) is a solution of the system $x + y = 3$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

1017

QUESTION ID
S29-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 - 1 = 1$, not 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 1

Decide whether points lie on a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and passes through (0, 5).

Does the point (3, −4) lie on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

1019

QUESTION ID
S29-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. The radius is 5, and the distance from (0, 0) to (3, −4) is √(9 + 16) = 5.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the points where the line $y = 3$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

1020

QUESTION ID
S29-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(4, 3) and (−4, 3)

GRADING NOTE Form required: both points. $x^2 + 9 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center (1, 2) and radius 5.

Which point is on the circle?

A. (1, 6)

B. (4, 6)

C. (3, 3)

D. (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

1021

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 6)

GRADING NOTE From (1, 2) to (4, 6): √(9 + 16) = 5. (3, 3) is √5 away, (6, 3) is √26 away, and (1, 6) is 4 away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

True or false?

The line $y = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

1022

QUESTION ID
S29-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The circle reaches only up to $y = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (2, 2) lies on the circle $x^2 + y^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

1023

QUESTION ID
S29-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 4 + 4 = 8, not 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 2

Find complex solutions when a line and a parabola do not meet • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Substituting $y = 2x$ into $y = x^2 + 2$ gives $x^2 - 2x + 2 = 0$.

Solve it over the complex numbers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

1025

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1 + i$ or $x = 1 - i$

GRADING NOTE Form required: a + bi (both solutions). $x = (2 \pm \sqrt{(-4)}) \div 2$. The solutions are not real, so the graphs do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 4 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

1026

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 2i$ or $x = -2i$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 6x + 10 = 0$.

A. $3 + i$ and $3 - i$

B. $6 + i$ and $6 - i$

C. $3 + 2i$ and $3 - 2i$

D. $-3 + i$ and $-3 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

1027

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 + i and 3 − i

GRADING NOTE $(x - 3)^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

True or false?

When a line and a parabola do not intersect, the equation from substitution has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

1028

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real solution x would give a common point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

True or false?

$i^3 = i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

1029

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^3 = i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

Understand the graph of an equation as the set of all its solutions

MAIN	Algebra I main question	**HSA-REI.D.10**
BACKWARD 1	Use tables and graphs to relate two variables	**6.EE.C.9**
BACKWARD 2	Understand a function's graph as its input-output pairs	**8.F.A.1**
FORWARD 1	Derive a parabola from its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use the unit circle as the set of points (cos θ, sin θ)	**HSF-TF.A.2**

SET 30 • HSA-REI.D.10

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, −2) on the graph of 2x + 5y = −4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

1032

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(3) + 5(−2) = 6 − 10 = −4, so (3, −2) is a solution.

GRADING NOTE Any equivalent explanation that substitutes the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

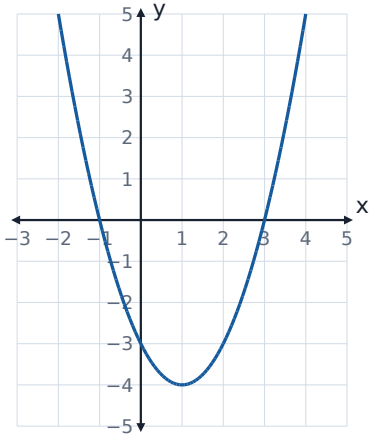
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

The graph of $y = x^2 - 2x - 3$ is shown.

Which point is NOT a solution of the equation?

A. (2, 3) **B.** (3, 0)

C. (0, −3) **D.** (1, −4)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

1033

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 3)

GRADING NOTE At $x = 2$, $y = 4 - 4 - 3 = -3$, not 3. The other points are on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

The point (k, 7) is on the graph of y = 3x − 2.

Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

1034

QUESTION ID
S30-M-Q3

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. 3k − 2 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

The graph of an equation in x and y passes through (4, 1) and (−2, 5).

What does that tell you about the equation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

1035

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 1 and x = −2, y = 5 both make the equation true: each is a solution.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

True or false?

A point that is not on the graph of $y = x^2$ can still be a solution of $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

1036

QUESTION ID
S30-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The graph is exactly the set of all solutions, so every solution is on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 1

Use tables and graphs to relate two variables • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

y = 3x. Find y for x = 0, 1, 2, and 3.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

1038

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, 6, 9

GRADING NOTE Any equivalent form, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A car travels at 40 miles per hour.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

1039

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

d = 40t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of y = x + 4?

A. (2, 6)

B. (1, 3)

C. (4, 0)

D. (6, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

1040

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 6)

GRADING NOTE 2 + 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 5x, y is the dependent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

1041

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y depends on the value chosen for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (2, 8) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

1042

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3(2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 2

Understand a function's graph as its input-output pairs • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

The function f pairs each x with 2x − 1.

What output does it pair with x = 4?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

1044

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is {(1, 2), (2, 3), (1, 5)} a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

1045

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 1 is paired with two outputs, 2 and 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = x^2$?

A. (9, −3)

B. (−3, −9)

C. (−3, 9)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

1046

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 9)

GRADING NOTE $(-3)^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

1047

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The vertical line x = 2 is the graph of a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

1048

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 would have infinitely many outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 1

Derive a parabola from its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A parabola is the set of points equally far from the focus (0, 2) and the directrix $y = -2$.

Derive its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

1050

QUESTION ID

S30-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$x^2 = 8y$

GRADING NOTE REQUIRED WORK: The distance condition $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$ written, both sides squared, and simplified to $x^2 = 8y$.

Form required: the derivation and the equation (any equivalent form, such as $y = x^2 \div 8$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

Is (4, 2) on the parabola with focus (0, 2) and directrix $y = -2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

1051

QUESTION ID
S30-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes. It is 4 units from the focus and 4 units from the directrix.

GRADING NOTE Any equivalent explanation (or $4^2 = 8 \cdot 2$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

Where is the focus of the parabola $y = x^2 \div 4$?

A. (0, 4)

B. $(0, \frac{1}{4})$

C. (1, 0)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5

1052

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 1)

GRADING NOTE $x^2 = 4y = 4py$ with $p = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q3**

**High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5**

True or false?

Every point on a parabola is the same distance from its focus as from its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

1053

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a parabola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The vertex of a parabola lies on its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

1054

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The vertex is halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 2

Use the unit circle as the set of points (cos θ, sin θ) • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Explain why the point $(\cos\theta, \sin\theta)$ lies on the circle $x^2 + y^2 = 1$ for every angle θ.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

1056

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\cos^2\theta + \sin^2\theta = 1$, so the point satisfies the equation. (It is 1 unit from the origin.)

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

Which point on the unit circle corresponds to θ = π ÷ 2?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

1057

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. cos(π ÷ 2) = 0 and sin(π ÷ 2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

Which point is on the unit circle $x^2 + y^2 = 1$?

A. (0.5, 0.5)

B. (1, 1)

C. (0.6, 0.8)

D. (0.8, 0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

1058

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (0.6, 0.8)

GRADING NOTE 0.36 + 0.64 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

The point $(-1, 0)$ on the unit circle corresponds to an angle of π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

1059

QUESTION ID
S30-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\cos \pi = -1$ and $\sin \pi = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (cos 0, sin 0) is (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

1060

QUESTION ID
S30-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE cos 0 = 1 and sin 0 = 0, so the point is (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approximate solutions with graphs and tables

MAIN	Algebra I main question	**HSA-REI.D.11**
BACKWARD 1	Check whether points are on a graph	**HSA-REI.D.10**
BACKWARD 2	Read the solution of a system from the graphs of two lines	**8.EE.C.8.a**
FORWARD 1	Solve exponential equations exactly with logarithms	**HSF-LE.A.4**
FORWARD 2	Solve trigonometric equations with graphs and technology	**HSF-TF.B.7**

SET 31 • HSA-REI.D.11

MAIN

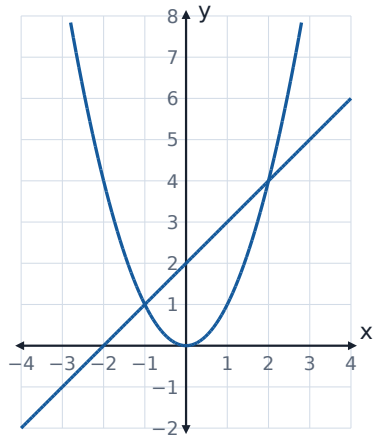
Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ are shown.

Use the graphs to solve $x^2 = x + 2$.

Solutions: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

1063

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = -1$ or $x = 2$

GRADING NOTE Form required: both solutions. The graphs meet at $(-1, 1)$ and $(2, 4)$. Check: $(-1)^2 = 1 = -1 + 2$ and $2^2 = 4 = 2 + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a table of values to find the solution of $2^x = 3x$ between $x = 3$ and $x = 4$, to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

1064

QUESTION ID

S31-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

3.3

GRADING NOTE

REQUIRED WORK: A table of 2^x and 3x at x-values that bracket the solution closely enough to decide the tenth (for example x = 3.3 and x = 3.35), showing where $2^x - 3x$ changes sign.

Form required: rounded to the nearest tenth, with the table. At 3.3, $2^x \approx 9.85 < 9.9 = 3x$; at 3.35, $2^x \approx 10.20 > 10.05 = 3x$, so the solution (about 3.31) rounds to 3.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows f(x) = |x − 1| and g(x) = 3 − x.

For which x is f(x) = g(x)?

x	0	1	2	3	4
f(x)	1	0	1	2	3
g(x)	3	2	1	0	−1

A. x = 2 **B.** x = 3 **C.** x = 0 **D.** x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

1065

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 2

GRADING NOTE At x = 2 both equal 1. In every other column the outputs differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why the x-coordinates of the points where the graphs of y = f(x) and y = g(x) meet are the solutions of f(x) = g(x).

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

1066

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

At a meeting point (a, b), b = f(a) and b = g(a), so f(a) = g(a). If f(a) = g(a), the point (a, f(a)) is on both graphs.

GRADING NOTE Any equivalent explanation that uses the coordinates of a point on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

True or false?

The graphs of $y = 3x + 1$ and $y = 3x - 4$ never meet, so $3x + 1 = 3x - 4$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

1067

QUESTION ID
S31-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The lines are parallel, and $1 = -4$ is false for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.D.11

BACKWARD 1

Check whether points are on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 4) on both $y = x^2$ and $y = x + 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

1069

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $2^2 = 4$ and $2 + 2 = 4$.

GRADING NOTE Any equivalent explanation that checks both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point $(-1, y)$ is on the graph of $y = x + 2$.

Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

1070

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = |x - 1|$?

A. $(-2, -3)$

B. $(-2, 3)$

C. $(3, -2)$

D. $(1, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

1071

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, 3)

GRADING NOTE $|-2 - 1| = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If (a, b) is on the graph of y = f(x), then b = f(a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

1072

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is the set of points (x, f(x)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 1) is on the graph of $y = 2^x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

1073

QUESTION ID
S31-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2^0 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

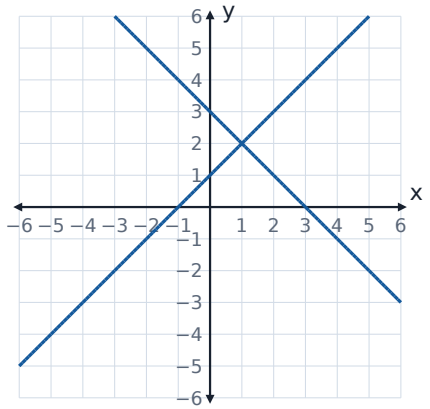
BACKWARD 2

Read the solution of a system from the graphs of two lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Use the graph to solve the system $y = x + 1$ and $y = -x + 3$.

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

1075

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Any equivalent form. The lines cross at (1, 2). Check: 1 + 1 = 2 and −1 + 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At what x-value do the graphs of $y = 2x - 3$ and $y = 5$ meet?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

1076

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2x - 3 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

The graphs of two linear equations are the same line.

How many solutions does the system have?

A. One

B. Infinitely many

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

1077

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Infinitely many

GRADING NOTE Every point of the line is on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

True or false?

Two lines with different slopes meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

1078

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lines with different slopes are not parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

True or false?

Two lines with the same slope always meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

1079

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Distinct lines with the same slope are parallel and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 1

Solve exponential equations exactly with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2^x$ and $y = 10$ meet at one point.

Find its x-coordinate exactly using logarithms, and to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

1081

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = log 10 ÷ log 2 = 1 ÷ log 2 ≈ 3.32

GRADING NOTE Form required: the exact expression (such as $\log_2 10$) and the rounded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5e^x = 20$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

1082

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 4

GRADING NOTE Form required: exact. $e^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph shows $y = 3^x$ meeting $y = 15$ at about $x = 2.5$.

Which expression is the exact solution of $3^x = 15$? (log means base 10.)

A. $15 \div 3$

B. log 5

C. log 15 ÷ log 3

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

1083

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. log 15 ÷ log 3

GRADING NOTE Take the logarithm of both sides: x log 3 = log 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_2 8 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

1084

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solution of $10^x = 0.01$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

1085

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.01 = 10^{-2}$, so $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 2

Solve trigonometric equations with graphs and technology • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = \frac{1}{2}$ for $0 \le x \le \pi$. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

1087

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = π ÷ 6 or

x = 5π ÷ 6

GRADING NOTE Form required: exact, both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

Use a calculator to solve $\cos x = 0.3$ for $0 \le x \le \pi$, to the nearest hundredth of a radian.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

1088

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.27

GRADING NOTE Form required: rounded to the nearest hundredth. $x = \cos^{-1}(0.3) \approx 1.266$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many solutions does sin x = 2 have?

A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

1089

QUESTION ID
S31-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. None

GRADING NOTE The values of sin x are between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graphs of y = sin x and y = 0.5 meet at infinitely many points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

1090

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sine graph repeats every 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

True or false?

For $0 \le x \le \pi$, the equation $\cos x = 0$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

1091

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = \pi \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

Solve linear-quadratic systems by graphing and check the solutions algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Read solutions of f(x) = g(x) from intersections	**HSA-REI.D.11**
BACKWARD 2	Check whether a point is on a graph	**HSA-REI.D.10**
FORWARD 1	Find where a line meets a circle with coordinates	**HSG-GPE.B.4**
FORWARD 2	Find where a line meets an ellipse	**HSG-GPE.A.3**

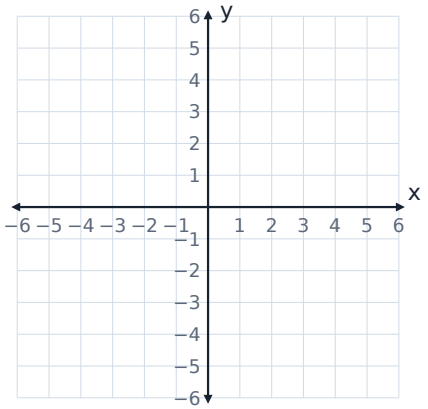
SET 73 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2 - 2$ and $y = x$ on the grid. Read the solutions of the system from the intersections, then check them algebraically.



Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

1094

QUESTION ID

S73-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The parabola $y = x^2 - 2$ (vertex $(0, -2)$) and the line $y = x$, crossing at $(-1, -1)$ and $(2, 2)$

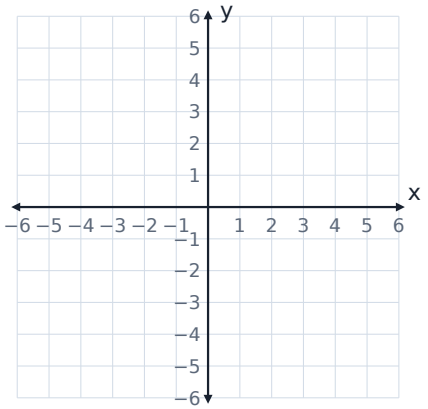
Written: $(-1, -1)$ and $(2, 2)$

GRADING NOTE Form required: both points with an algebraic check: $x^2 - 2 = x$ gives $(x - 2)(x + 1) = 0$. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q1/5

Graph the circle $x^2 + y^2 = 25$ and the line $y = x + 1$. Read the intersection points, then check one of them algebraically.



Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

1095

QUESTION ID

S73-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A circle of radius 5 centered at the origin and the line through (0, 1) with slope 1, crossing at (3, 4) and (−4, −3)

Written: (3, 4) and (−4, −3)

GRADING NOTE Form required: both points and one algebraic check, such as 9 + 16 = 25 and 4 = 3 + 1.

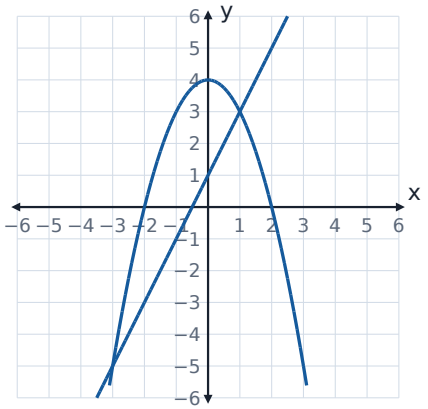
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = -x^2 + 4$ and $y = 2x + 1$ are shown.

Read the solutions of the system from the graph, then verify them algebraically.



Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

1096

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 3) and (−3, −5)

GRADING NOTE Form required: both points with the check $-x^2 + 4 = 2x + 1$, so $x^2 + 2x - 3 = (x + 3)(x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q3/5

The graphs of $y = x^2$ and $y = 2x - 1$ are shown.

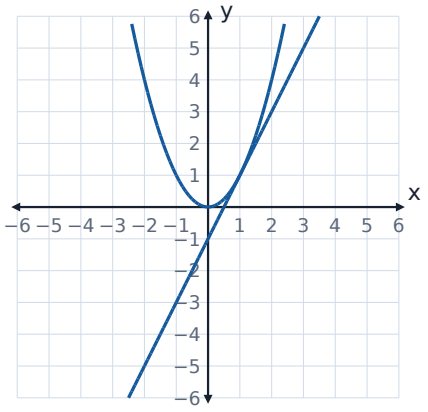
How many solutions does the system have?

A. One

B. Two

C. Infinitely many

D. None



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

1097

QUESTION ID

S73-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One

GRADING NOTE $x^2 = 2x - 1$ gives $(x - 1)^2 = 0$, so the line touches the parabola only at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q4/5

True or false?

The line $y = 5$ and the circle $x^2 + y^2 = 9$ intersect at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

1098

QUESTION ID

S73-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The circle has radius 3, so it reaches only up to $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 73 • MAIN • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 1

Read solutions of f(x) = g(x) from intersections • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = x^2$ and $y = 4$ meet at two points.
What are the solutions of $x^2 = 4$?

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

1100

QUESTION ID
S73-B1-Q1

TYPE
Short answer

CORRECT ANSWER
x = 2 or x = −2

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2x + 1$ and $y = x^2 + 1$ meet at (0, 1) and (2, 5). Which x-values solve $2x + 1 = x^2 + 1$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

1101

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0 and x = 2: at those x-values the two expressions have the same output.

GRADING NOTE Form required: both values and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of y = f(x) and y = g(x) meet only at (3, 7). Which statement is true?

A. f(x) = g(x) for every x

B. f(7) = g(7) = 3

C. f(3) = 3

D. f(3) = g(3) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

1102

QUESTION ID

S73-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(3) = g(3) = 7

GRADING NOTE An intersection point (a, b) means f(a) = g(a) = b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) never meet, then f(x) = g(x) has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

1103

QUESTION ID

S73-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A solution would give a common point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q4/5

True or false?

The solutions of f(x) = g(x) are the y-coordinates of the intersection points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

1104

QUESTION ID

S73-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are the x-coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

BACKWARD 2

Check whether a point is on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (3, 4) on the circle $x^2 + y^2 = 25$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

1106

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 9 + 16 = 25.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the parabola $y = x^2 - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

1107

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. $2^2 - 2 = 2$, not 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on both $y = x$ and $y = x^2 - 2$?

A. (2, 2)

B. (−2, −2)

C. (1, 1)

D. (0, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

1108

QUESTION ID

S73-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 2)

GRADING NOTE $2 = 2$ and $2^2 - 2 = 2$. Each other point fails one of the equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q3/5

True or false?

A point is a solution of a system exactly when it lies on the graphs of both equations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

1109

QUESTION ID
S73-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each graph is the set of solutions of its equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 5) is on both the circle $x^2 + y^2 = 25$ and the line $y = x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

1110

QUESTION ID
S73-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is on the circle, but 0 + 1 = 1, not 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 1

Find where a line meets a circle with coordinates • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the points where the line $y = 7 - x$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

1112

QUESTION ID
S73-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(3, 4) and (4, 3)

GRADING NOTE Form required: both points. $x^2 + (7 - x)^2 = 25$ gives $x^2 - 7x + 12 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q1/5

Show that the line $y = 5$ touches the circle $x^2 + y^2 = 25$ at exactly one point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S73-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

1113

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 25 = 25$ gives $x^2 = 0$, so the only point is (0, 5).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q2/5

How many points do the line $y = 2x$ and the circle $x^2 + y^2 = 20$ share?

A. One

B. Four

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

1114

QUESTION ID
S73-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Two

GRADING NOTE $x^2 + 4x^2 = 20$ gives $x = 2$ or $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q3/5

True or false?

The line $x = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

1115

QUESTION ID
S73-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The circle reaches only to $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q4/5

True or false?

The point (1, 5) lies on the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

1116

QUESTION ID
S73-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1 + 25 = 26$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSA-REI.C.7

FORWARD 2

Find where a line meets an ellipse • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find where the line $y = 2$ meets the ellipse $x^2 \div 16 + y^2 \div 4 = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

1118

QUESTION ID
S73-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Only at (0, 2): the line touches the ellipse there.

GRADING NOTE Form required: the point. $x^2 \div 16 + 1 = 1$ gives $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find where the x-axis meets the ellipse $x^2 \div 9 + y^2 \div 16 = 1$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

1119

QUESTION ID
S73-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(3, 0) and (−3, 0)

GRADING NOTE Form required: both points. $y = 0$ gives $x^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q2/5

Where does the line x = 0 meet the ellipse $x^2 \div 25 + y^2 \div 9 = 1$?

A. (0, 5) and (0, −5)

B. (5, 0) and (−5, 0)

C. Nowhere

D. (0, 3) and (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

1120

QUESTION ID
S73-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. (0, 3) and (0, −3)

GRADING NOTE x = 0 gives $y^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q3/5

True or false?

A line can meet an ellipse in at most two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

1121

QUESTION ID
S73-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Substituting a line into the ellipse gives an equation of degree at most 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q4/5

True or false?

The line $y = 4$ meets the ellipse $x^2 \div 25 + y^2 \div 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

1122

QUESTION ID
S73-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ellipse reaches only up to $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 73 • FORWARD 2 • Q5/5

Solve systems of linear equations by graphing, including approximate solutions

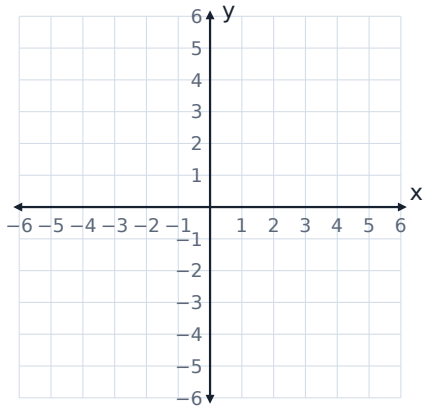
MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Find the solution of a system where two lines cross	**8.EE.C.8.a**
BACKWARD 2	Decide whether points are on a line	**HSA-REI.D.10**
FORWARD 1	Graph parallel and perpendicular lines and find where they meet	**HSG-GPE.B.5**
FORWARD 2	Check a graphical estimate with an inverse matrix	**HSA-REI.C.9**

SET 84 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Graph $y = 2x - 1$ and $y = -x + 5$ on the grid. What is the solution of the system?



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

1125

QUESTION ID

S84-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line through (0, −1) with slope 2 and the line through (0, 5) with slope −1, crossing at (2, 3)

Written: (2, 3)

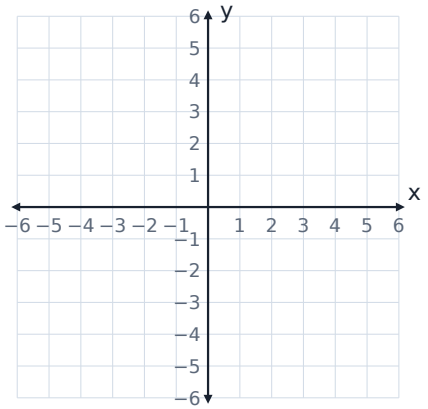
GRADING NOTE Form required: the solution read from the graph (it may be written x = 2, y = 3). Check: 4 − 1 = 3 and −2 + 5 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = 0.5x + 2 and y = −2x + 4. Estimate the solution to the nearest half unit, then solve algebraically to check.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

1126

QUESTION ID

S84-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The two lines crossing just left of x = 1 and just below y = 2.5

Written: Estimate about (1, 2.5); exact solution (0.8, 2.4)

GRADING NOTE Form required: an estimate within 0.5 of (0.8, 2.4) and the exact solution. 0.5x + 2 = −2x + 4 gives 2.5x = 2.

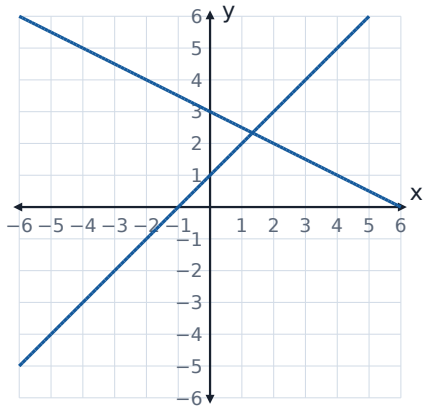
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = x + 1$ and $y = -0.5x + 3$ are shown.

Estimate the solution of the system to the nearest tenth, and explain how you made the estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

1127

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

About (1.3, 2.3): the crossing is about one third of the way from x = 1 to x = 2 and from y = 2 to y = 3.

GRADING NOTE Form required: an estimate within 0.2 of the exact (4/3, 7/3) and a reason from the graph.

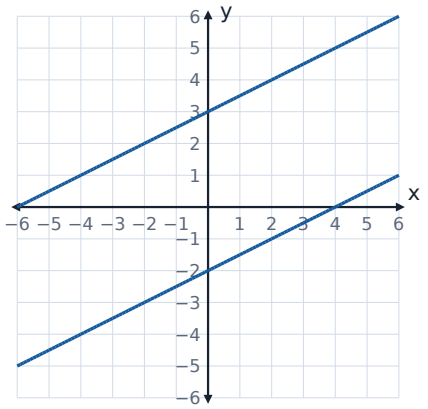
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of the two equations of a system are shown.

How many solutions does the system have?



A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

1128

QUESTION ID

S84-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE The lines are parallel (same slope, different intercepts), so they never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q4/5

True or false?

If the graphs of two linear equations lie on top of each other, the system has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

1129

QUESTION ID

S84-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every point of the line is on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 84 • MAIN • Q5/5

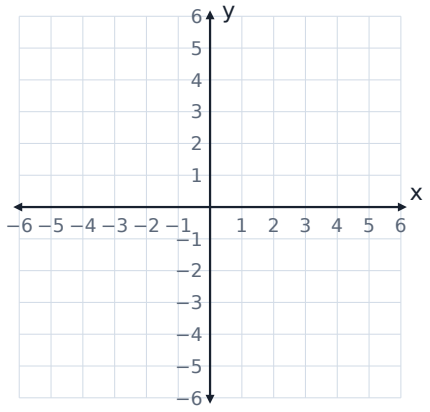
SET 84 • HSA-REI.C.6

BACKWARD 1

Find the solution of a system where two lines cross • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x and y = 4 − x, and mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

1131

QUESTION ID

S84-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Two lines crossing at (2, 2), marked

GRADING NOTE Any correct graph with (2, 2) marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines meet at (−1, 3). What is the solution of their system?

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

1132

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q2/5

The graphs of two lines cross at (4, −2). Which equations could they be?

A. y = x − 6 and y = x + 2

B. y = x − 6 and y = −x + 2

C. y = x + 6 and y = −x + 2

D. y = 2x and y = −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

1133

QUESTION ID

S84-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = x − 6 and y = −x + 2

GRADING NOTE 4 − 6 = −2 and −4 + 2 = −2. In each other pair, (4, −2) fails at least one equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q3/5

True or false?

A point that lies on both lines is a solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

1134

QUESTION ID

S84-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It makes both equations true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q4/5

True or false?

Two lines with different slopes can have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

1135

QUESTION ID

S84-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines with different slopes always cross exactly once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

BACKWARD 2

Decide whether points are on a line • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the line y = 2x − 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

1137

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(2) − 1 = 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 3) on the line y = −x + 4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

1138

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. −2 + 4 = 2, not 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the line y = 0.5x + 2?

A. (2, 3)

B. (3, 2)

C. (0, 0.5)

D. (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

1139

QUESTION ID

S84-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 3)

GRADING NOTE 0.5(2) + 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q3/5

True or false?

The graph of a linear equation in two variables is the set of all its solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

1140

QUESTION ID

S84-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point on the line is a solution, and each solution is on the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(0, 4) is on the graph of $y = -2x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q5/5

1141

QUESTION ID

S84-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

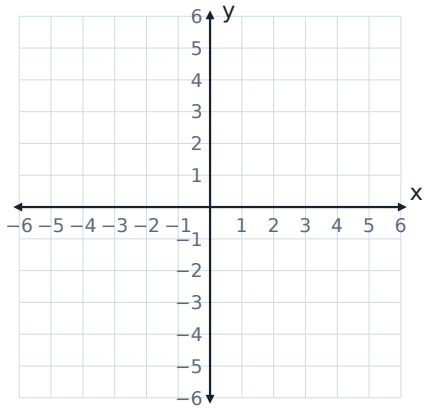
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 1

Graph parallel and perpendicular lines and find where they meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Graph y = 2x + 1 and the line through (4, 1) that is perpendicular to it. Mark the point where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

1143

QUESTION ID

S84-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line y = −0.5x + 3 through (4, 1), crossing y = 2x + 1 at (0.8, 2.6)

GRADING NOTE Any correct graph. The perpendicular slope is $-\frac{1}{2}$; 2x + 1 = −0.5x + 3 gives x = 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S84-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q1/5

Two lines have slopes 3 and $-\frac{1}{3}$. What angle do they form?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

1144

QUESTION ID
S84-F1-Q2

TYPE
Short answer

CORRECT ANSWER
90°

GRADING NOTE Any equivalent form. The slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q2/5

Which system has no solution?

A. $y = 3x + 1$ and $y = -3x + 1$

B. $y = 3x + 1$ and $y = x + 1$

C. $y = 3x + 1$ and $y = 3x - 4$

D. $y = 3x$ and $y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

1145

QUESTION ID
S84-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $y = 3x + 1$ and $y = 3x - 4$

GRADING NOTE Same slope, different intercepts: parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q3/5

True or false?

The lines $y = -x + 2$ and $y = x - 2$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

1146

QUESTION ID
S84-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(-1)(1) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q4/5

True or false?

The lines y = 2x + 5 and y = 2x + 5 have no points in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

1147

QUESTION ID
S84-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are the same line, so they share every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSA-REI.C.6

FORWARD 2

Check a graphical estimate with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

The system y = 0.5x + 2 and y = −2x + 4 is written −0.5x + y = 2 and 2x + y = 4. Find the determinant of the matrix A with rows [−0.5 1] and [2 1].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

1149

QUESTION ID
S84-F2-Q1

TYPE
Short answer

CORRECT ANSWER
−2.5

GRADING NOTE Any equivalent form. (−0.5)(1) − (1)(2) = −2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q1/5

A^{-1} is $(1 \div (-2.5))$ times the matrix with rows $[1\ -1]$ and $[-2\ -0.5]$. Use $X = A^{-1}B$, with B the column $[2, 4]$, to find the exact solution.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

1150

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = 0.8, y = 2.4

GRADING NOTE Form required: both values. $(1 \div (-2.5))$ times the column $[2 - 4, -4 - 2]$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q2/5

A graph shows two lines crossing near (1.3, 2.3). The inverse-matrix solution is (4/3, 7/3). Which statement is true?

A. The two results contradict each other.

B. The system has two solutions.

C. The graph is exact; the matrix method is approximate.

D. The graph gives an estimate; the matrix method gives the exact solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

1151

QUESTION ID

S84-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The graph gives an estimate; the matrix method gives the exact solution.

GRADING NOTE 4/3 ≈ 1.33 and 7/3 ≈ 2.33, which match the estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a 2-by-2 linear system has a nonzero determinant, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

1152

QUESTION ID

S84-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then A has an inverse and $X = A^{-1}B$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q4/5

True or false?

If the graphs of two lines coincide, the determinant of the coefficient matrix is not 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

1153

QUESTION ID

S84-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Coinciding lines give a matrix with determinant 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 84 • FORWARD 2 • Q5/5

Graph the solutions of linear inequalities and systems of linear inequalities in two variables as half-planes

MAIN	Algebra I main question	**HSA-REI.D.12**
BACKWARD 1	Solve an inequality in two variables for y	**HSA-REI.B.3**
BACKWARD 2	Plot points and find distances in the coordinate plane	**6.NS.C.8**
FORWARD 1	Find the area of a region described by inequalities	**HSG-GPE.B.7**
FORWARD 2	Use zeros to find where a polynomial is positive or negative	**HSA-APR.B.3**

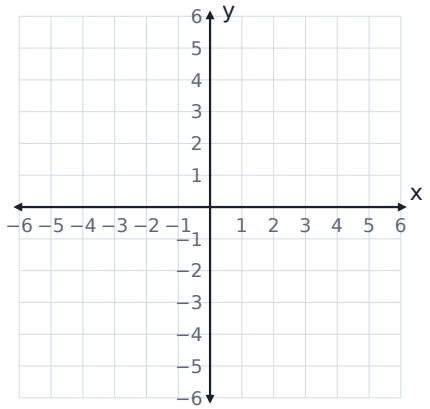
SET 32 • HSA-REI.D.12

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y < 2x - 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

1156

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, −1) and (2, 3), with the half-plane below and to the right of it shaded

GRADING NOTE Any correct graph. The line is dashed because points on it are not solutions; (0, −2) is a solution and (0, 0) is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

Is (4, 1) a solution of the system $y \leq -x + 6$ and $y > \frac{1}{2}x - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

1157

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. $1 \leq -4 + 6 = 2$, and $1 > 2 - 2 = 0$.

GRADING NOTE Any equivalent explanation that checks both inequalities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

A graph has a solid boundary line through (0, 2) and (3, 0), and the side that contains the origin is shaded.

Which inequality does it show?

A. $2x + 3y < 6$

B. $2x + 3y \geq 6$

C. $3x + 2y \leq 6$

D. $2x + 3y \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

1158

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x + 3y \leq 6$

GRADING NOTE The line is $2x + 3y = 6$; solid means "or equal to"; (0, 0) gives $0 \leq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

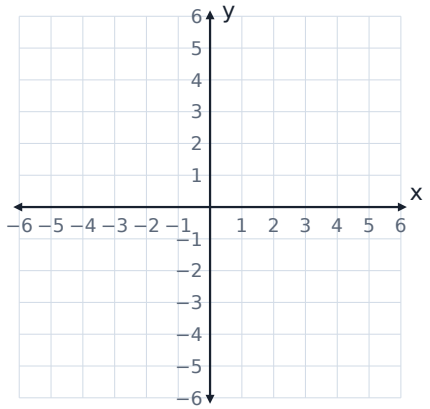
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $y \geq x - 2$ and $x + y < 4$ on the coordinate plane.

Name one point that is a solution of the system.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

1159

QUESTION ID

S32-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A solid line $y = x - 2$ with the side above it shaded, a dashed line $x + y = 4$ with the side below it shaded, and the overlap marked

Written: Any point in the overlap, such as (0, 0)

GRADING NOTE Any equivalent form: the point must satisfy both inequalities. (0, 0): $0 \geq -2$ and $0 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

True or false?

Points on the boundary line y = 3x are solutions of y > 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

1160

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE On the line, y = 3x, which is not greater than 3x. The boundary is dashed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 1

Solve an inequality in two variables for y • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 3y ≤ 6 for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

1162

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$y \leq -\frac{2}{3}x + 2$

GRADING NOTE Any equivalent form with y alone. 3y ≤ −2x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x - 2y > 4$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

1163

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$y < \frac{1}{2}x - 2$

GRADING NOTE Any equivalent form with y alone. Dividing by −2 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-y \geq x - 5$ for y.

A. $y \geq x + 5$

B. $y \geq -x + 5$

C. $y \leq -x + 5$

D. $y \leq x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

1164

QUESTION ID
S32-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $y \leq -x + 5$

GRADING NOTE Multiply by −1 and reverse the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Solving $4x + y < 3$ for y gives $y < -4x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

1165

QUESTION ID
S32-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Subtract 4x from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

True or false?

Solving $-3y < 9$ for y gives $y < -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

1166

QUESTION ID
S32-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Dividing by −3 reverses the sign: $y > -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 2

Plot points and find distances in the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (2, −3) and (2, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

1168

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

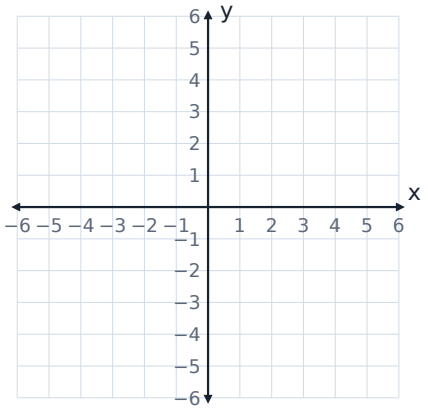
GRADING NOTE Any equivalent form. |5 − (−3)| = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot the points (−3, 2) and (4, −1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

1169

QUESTION ID

S32-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point at (−3, 2) and a point at (4, −1)

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which point is in Quadrant III?

A. (2, −5)

B. (−2, −5)

C. (2, 5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

1170

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, −5)

GRADING NOTE Quadrant III has both coordinates negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

The points (3, 4) and (3, −4) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

1171

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Only the sign of y changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The distance between (−4, 1) and (6, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

1172

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $|6 - (-4)| = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 1

Find the area of a region described by inequalities • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

The system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$ describes a triangle.

Find its vertices and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

1174

QUESTION ID
S32-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 0), (6, 0), and (0, 6); area 18 square units

GRADING NOTE Form required: the vertices and the area. $\frac{1}{2}(6)(6) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

Find the perimeter of the rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

1175

QUESTION ID
S32-F1-Q2

TYPE
Short answer

CORRECT ANSWER
14 units

GRADING NOTE Any equivalent form. 2(4 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

The region $0 \le x \le 4$ and $0 \le y \le 2x$ is a triangle.

What is its area?

A. 32

B. 16

C. 8

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

1176

QUESTION ID
S32-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 16

GRADING NOTE Vertices (0, 0), (4, 0), and (4, 8): $\frac{1}{2}(4)(8) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (3, 0), and (0, 4) has perimeter 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

1177

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 + 4 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

True or false?

The square with vertices (0, 0), (2, 0), (2, 2), and (0, 2) has area 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

1178

QUESTION ID
S32-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 2

Use zeros to find where a polynomial is positive or negative • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = (x - 1)(x - 4)$.

For which x is $p(x) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

1180

QUESTION ID
S32-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$1 < x < 4$

GRADING NOTE Form required: an inequality. Between the zeros, one factor is positive and one is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

$p(x) = x(x + 2)(x - 3)$.

Is $p(1)$ positive or negative? Explain using its factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

1181

QUESTION ID
S32-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Negative: $1 \cdot 3 \cdot (-2) = -6$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $(x + 3)(x - 2) > 0$?

A. $x < -3$ only

B. $x > 2$ only

C. $x < -3$ or $x > 2$

D. $-3 < x < 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

1182

QUESTION ID
S32-F2-Q3

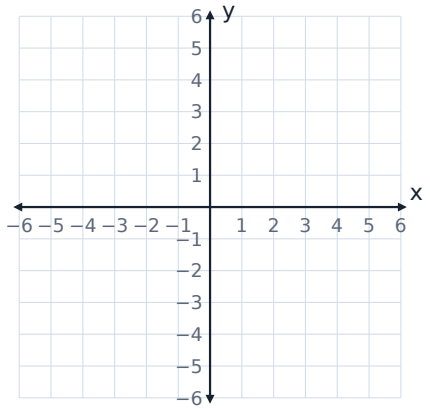
TYPE
Multiple choice

CORRECT ANSWER
C. $x < -3$ or $x > 2$

GRADING NOTE Both factors are positive when $x > 2$, and both are negative when $x < -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

Sketch a rough graph of $y = x(x + 2)(x - 3)$, and mark the x-values where $y < 0$. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

1183

QUESTION ID

S32-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 0, and 3, falling to the left and rising to the right; $y < 0$ for $x < -2$ and for $0 < x < 3$

GRADING NOTE Any correct sketch with both negative intervals marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)(x + 1) \geq 0$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

1184

QUESTION ID
S32-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At x = 0 it equals −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

Create equations in two or more variables and graph them on labeled axes

MAIN	Algebra I main question	**HSA-CED.A.2**
BACKWARD 1	Write and solve an equation in one variable	**HSA-CED.A.1**
BACKWARD 2	Write the equation of a linear function	**8.F.B.4**
FORWARD 1	Write the equation of a circle from a situation	**HSG-GPE.A.1**
FORWARD 2	Write and interpret polynomial models	**HSF-IF.C.7.c**

SET 33 • HSA-CED.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A savings account has $150 and grows by $25 each week.

Write an equation for the balance B after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

1187

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

B = 150 + 25w

GRADING NOTE Any equivalent form, such as B = 25w + 150.

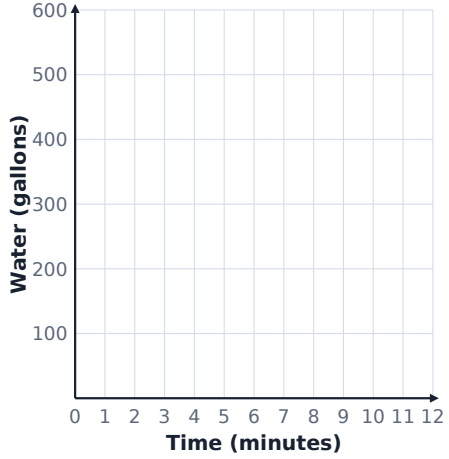
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

A pool holds 600 gallons and drains at 50 gallons per minute.

Write an equation for the gallons G left after t minutes, and graph it for 0 ≤ t ≤ 12.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

1188

QUESTION ID

S33-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line from (0, 600) to (12, 0)

Written:
G = 600 − 50t

GRADING NOTE Any equivalent form. Both the equation and the graph are graded. The pool is empty at t = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

A rectangle's length is 3 more than twice its width w.

Write an equation for its area A.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

1189

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

A = w(2w + 3)

GRADING NOTE Any equivalent form, such as $A = 2w^2 + 3w$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has 8,000 people, and its population grows 3% each year.

Which equation gives the population P after t years?

A. $P = 8{,}000(0.03)^t$

B. $P = 8{,}000 + 0.03t$

C. $P = 8{,}000(1.03)^t$

D. $P = 8{,}000(1.3)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

1190

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $P = 8{,}000(1.03)^t$

GRADING NOTE Each year the population is multiplied by 1.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

True or false?

Student tickets cost $4 and adult tickets cost $9. The equation 4s + 9a = 300 relates the numbers of tickets when sales total $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

1191

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 4s dollars from s student tickets plus 9a dollars from a adult tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 1

Write and solve an equation in one variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 150 + 25w = 400 for w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

1193

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. 25w = 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

A pool has 600 gallons and drains at 50 gallons per minute.

Write and solve an equation for when it is empty.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

1194

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

600 − 50t = 0; t = 12 minutes

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A plan costs $12 per month plus a $30 fee.

Which equation gives the number of months m for a total of $186?

A. 12m − 30 = 186

B. 30m + 12 = 186

C. 12(m + 30) = 186

D. 12m + 30 = 186

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

1195

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12m + 30 = 186

GRADING NOTE 12 per month for m months, plus 30 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

True or false?

w = 2.5 is a solution of w(2w + 3) = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

1196

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2.5(5 + 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Solving 4s + 9(20) = 300 gives s = 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

1197

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4s = 120, so s = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 2

Write the equation of a linear function • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11).

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

1199

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x + 5

GRADING NOTE Any equivalent form. The slope is 6 ÷ 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges $50 plus $30 per hour.

Write a function for the cost C(h).

C(h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

1200

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30h + 50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

A table has x-values 0, 1, 2 and y-values 4, 7, 10.

Which equation fits the table?

A. y = 4x + 3

B. y = x + 4

C. y = 3x

D. y = 3x + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

1201

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 4

GRADING NOTE y starts at 4 and increases by 3 for each 1 in x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

True or false?

The rate of change of $y = -2x + 7$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

1202

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rate of change is the slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

True or false?

The initial value of $y = 6x - 1$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

1203

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value (at $x = 0$) is -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 1

Write the equation of a circle from a situation • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A sprinkler at (3, −2) waters everything within 4 meters.

Write an equation of the boundary of the watered region in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

1205

QUESTION ID
S33-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y + 2)^2 = 16$

GRADING NOTE Form required: the standard form of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cell tower at the origin reaches 10 km. Is a phone at (6, 7) within range? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

1206

QUESTION ID
S33-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes.
$6^2 + 7^2 = 85 < 100$, so it is about 9.2 km away.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

Which equation describes all the points 5 units from (0, 0)?

A. $x^2 - y^2 = 25$

B. $x^2 + y^2 = 25$

C. $x + y = 5$

D. $x^2 + y^2 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

1207

QUESTION ID
S33-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. $x^2 + y^2 = 25$

GRADING NOTE The radius squared is 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 4$ passes through $(3, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

1208

QUESTION ID
S33-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $2^2 + 0^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The circle $x^2 + y^2 = 9$ passes through (3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

1209

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 + 9 = 18, not 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 2

Write and interpret polynomial models • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner.

Write an equation for its volume V.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

1211

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

V = x(10 − 2x)(8 − 2x)

GRADING NOTE Any equivalent form. The base is (10 − 2x) by (8 − 2x), and the height is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner and folding up the sides. Its volume is $V = x(10 - 2x)(8 - 2x)$.

Which values of x make sense in the situation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

1212

QUESTION ID
S33-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$0 < x < 4$

GRADING NOTE Form required: an inequality. Each side length must be positive, and $8 - 2x > 0$ means $x < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the zeros of $V = x(10 - 2x)(8 - 2x)$?

A. 5 and 4 only

B. 0, 10, and 8

C. 0, −5, and −4

D. 0, 5, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

1213

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0, 5, and 4

GRADING NOTE Set each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

True or false?

The graph of $y = x^3 - x$ crosses the x-axis three times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

1214

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

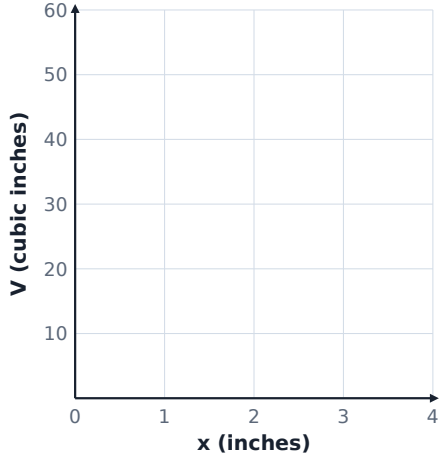
A. True

GRADING NOTE $x(x - 1)(x + 1) = 0$ at $x = -1$, 0, and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner. Its volume is $V = x(10 - 2x)(8 - 2x)$. Graph V for $0 \le x \le 4$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

1215

QUESTION ID
S33-F2-Q5

TYPE
Drawing / plotting

CORRECT ANSWER
A smooth curve through (0, 0), (1, 48), (2, 48), (3, 24), and (4, 0), highest near x ≈ 1.5 (V ≈ 52.5)

GRADING NOTE Any correct graph through these points with one maximum between x = 1 and x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

Represent constraints with equations and inequalities and decide whether solutions are viable

MAIN	Algebra I main question	**HSA-CED.A.3**
BACKWARD 1	Test points in a system of inequalities	**HSA-REI.D.12**
BACKWARD 2	Solve a system of linear equations	**HSA-REI.C.6**
FORWARD 1	Use constraints to optimize a design	**HSG-MG.A.3**
FORWARD 2	Use expected values to compare decisions — nearest related	**HSS-MD.B.7**

SET 34 • HSA-CED.A.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Write the system of constraints.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

1218

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

m + 3b ≤ 60, 2m + 5b ≤ 140, m ≥ 0, b ≥ 0 (m and b whole numbers)

GRADING NOTE Form required: both resource inequalities; the conditions m ≥ 0 and b ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Is making 40 muffins and 10 loaves possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

1219

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. It needs 40 + 30 = 70 cups of flour, more than 60 (the 130 minutes would be enough).

GRADING NOTE Any equivalent explanation that checks the constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A club needs at least $500 from selling shirts at $15 and hats at $10, and it can sell at most 45 items.

Write the constraints for s shirts and h hats.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

1220

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

15s + 10h ≥ 500, s + h ≤ 45, s ≥ 0, h ≥ 0 (whole numbers)

GRADING NOTE Form required: both inequalities; s, h ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

A club sells shirts for $15 and hats for $10. It needs at least $500 and can sell at most 45 items.

Which combination meets both constraints?

A. 15 shirts and 25 hats

B. 25 shirts and 15 hats

C. 30 shirts and 20 hats

D. 20 shirts and 10 hats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

1221

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 25 shirts and 15 hats

GRADING NOTE 375 + 150 = 525 ≥ 500 and 40 ≤ 45. 30 shirts and 20 hats is 50 items; 15 shirts and 25 hats brings $475; 20 shirts and 10 hats brings $400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

True or false?

If x counts people, x = 12.5 can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

1222

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A number of people must be a whole number, so 12.5 is not viable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 1

Test points in a system of inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Is (10, 10) a solution of $x + 3y \leq 60$ and $2x + 5y \leq 140$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

1224

QUESTION ID
S34-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 40 ≤ 60 and 70 ≤ 140.

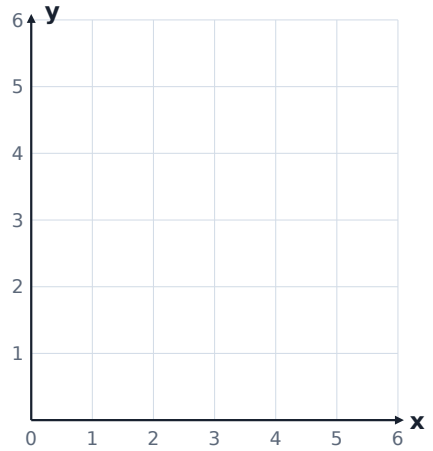
GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $x + y \leq 5$, $x \geq 0$, and $y \geq 0$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

1225

QUESTION ID

S34-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices (0, 0), (5, 0), and (0, 5), shaded, with solid sides

GRADING NOTE Any correct graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

Which point is in the solution region of $y \geq 2x$ and $y \leq 6$?

A. (4, 1)

B. (1, 7)

C. (3, 5)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

1226

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

GRADING NOTE $4 \geq 2$ and $4 \leq 6$. (3, 5) fails $5 \geq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

True or false?

The point (0, 0) satisfies $x \geq 0$ and $y \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

1227

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0 \geq 0$ is true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (2, 5) satisfies $x + y < 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

1228

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 + 5 = 7, which is not less than 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 2

Solve a system of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 45$

$15x + 10y = 500$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

1230

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(10, 35)

GRADING NOTE Any equivalent form. $15x + 450 - 10x = 500$, so $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 12$

$x + y = 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

1231

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 4)

GRADING NOTE Any equivalent form. Subtract: x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + 3y = 60$

$x + y = 40$

A. (30, 10)

B. (15, 15)

C. (10, 30)

D. (20, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

1232

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 10)

GRADING NOTE Subtract: 2y = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(10, 35) is a solution of x + y = 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

1233

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 + 35 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The system x + y = 10 and x + y = 12 has one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

1234

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pair can have both sums: no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-CED.A.3

FORWARD 1

Use constraints to optimize a design • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A feasible region has vertices (0, 0), (60, 0), and (0, 20).

Find the greatest value of P = 2m + 7b on the region and where it occurs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

1236

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

140, at (0, 20)

GRADING NOTE Form required: the value and the point. P = 0, 120, and 140 at the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangular pen uses at most 40 m of fence.

What dimensions give the greatest area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

1237

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 m by 10 m (area 100 m²)

GRADING NOTE Form required: the dimensions. A = w(20 − w) is greatest at w = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

The cost C = 3x + 2y is checked at the vertices (0, 6), (2, 2), and (6, 0) of a region.

Where is the cost least?

A. (2, 2)

B. (0, 0)

C. (6, 0)

D. (0, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

1238

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 2)

GRADING NOTE C = 12, 10, and 18 at the vertices. (0, 0) is not in the region.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

True or false?

For a linear expression on a polygonal region, the greatest value occurs at a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

1239

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the basis of linear programming.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with area 36, a square has the smallest perimeter, so its side is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

1240

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The side is $\sqrt{36} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

FORWARD 2

Use expected values to compare decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Choosing viable options under constraints extends to choosing between options by expected value.

ANSWER KEY Reduced copy of the question page, with its answer

Plan A earns $100 with probability 0.6 and $0 otherwise.
Plan B earns $70 for sure.

Which plan has the greater expected value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

1242

QUESTION ID
S34-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Plan B: $70 compared with $60 for Plan A.

GRADING NOTE Form required: the plan and both expected values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Guessing on a 4-choice question earns 1 point if right and loses 0.25 point if wrong.

What is the expected score from guessing?

Expected score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

1243

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.0625 point

GRADING NOTE Any equivalent form, such as $\frac{1}{16}$. $0.25(1) + 0.75(-0.25) = 0.0625$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5 • NEAREST RELATED

A $5 warranty pays $100 if the item breaks (probability 0.04) and $0 otherwise.

What is the expected net value of buying it?

A. −$5

B. $1

C. $4

D. −$1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

1244

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1

GRADING NOTE 100(0.04) − 5 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A choice with the higher expected value can still lose money on a single trial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

1245

QUESTION ID
S34-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected value describes the long-run average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a game has an expected value of 0, a player cannot lose money playing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

1246

QUESTION ID
S34-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Individual plays can lose; only the long-run average is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S34-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 35 • HSF-IF.A.1 • INTERPRETING FUNCTIONS

Understand a function as a rule that assigns exactly one output to each input, with a domain and a range

MAIN	Algebra I main question	**HSF-IF.A.1**
BACKWARD 1	Decide whether a relation is a function	**8.F.A.1**
BACKWARD 2	Check whether points are on the graph of an equation	**HSA-REI.D.10**
FORWARD 1	Describe transformations of the plane as functions of points	**HSG-CO.A.2**
FORWARD 2	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**

SET 35 • HSF-IF.A.1

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Does the relation {(−2, 4), (0, 1), (2, 4), (3, −1)} define y as a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

1249

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each input has exactly one output. (Two inputs may share the output 4.)

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and the range of $f(x) = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

1250

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

Domain: all real numbers. Range: $y \geq 1$.

GRADING NOTE Form required: both the domain and the range. $x^2 \geq 0$, so $x^2 + 1 \geq 1$, and every value 1 or more is reached.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation does NOT define y as a function of x?

A. $y = x^2$

B. $x = y^2$

C. $y = 2^x$

D. $y = |x|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

1251

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x = y^2$

GRADING NOTE For $x = y^2$, the input $x = 4$ has two outputs, $y = 2$ and $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √(x − 3). Over the real numbers, what is the domain of f?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

1252

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

x ≥ 3

GRADING NOTE Form required: an inequality (or interval [3, ∞)). The expression under the square root must not be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

True or false?

If f is a function and f(2) = 5, then f(2) can also equal 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

1253

QUESTION ID
S35-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A function assigns exactly one output to each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 1

Decide whether a relation is a function • Grade 8 • Functions • 8.F.A.1

A table pairs x = 1, 2, 3, 2 with y = 4, 5, 6, 7. Is y a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

1255

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The input 2 has two outputs, 5 and 7.

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each button on a vending machine gives exactly one kind of snack. Is the snack a function of the button? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

1256

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Each input (button) gives exactly one output (snack).

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

Which set of ordered pairs is a function?

A. {(5, 1), (5, 2), (6, 3)}

B. {(0, 0), (0, 1)}

C. {(1, 3), (2, 3), (3, 3)}

D. {(1, 2), (1, 3), (2, 4)}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

1257

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. {(1, 3), (2, 3), (3, 3)}

GRADING NOTE In {(1, 3), (2, 3), (3, 3)} every input appears once. Each of the other sets repeats an input with a different output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

True or false?

Two different inputs of a function can have the same output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

1258

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, $f(x) = x^2$ gives $f(2) = f(-2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A function can pair the input 4 with two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

1259

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each input has exactly one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 2

Check whether points are on the graph of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (−2, 5) on the graph of $y = x^2 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

1261

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $(-2)^2 + 1 = 5$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (a, 10) is on the graph of y = 2x + 4.

Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

1262

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 2a + 4 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = \sqrt{x}$?

A. (9, −3)

B. (−9, 3)

C. (9, 3)

D. (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

1263

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (9, 3)

GRADING NOTE $\sqrt{9} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of y = 4 is the set of all points with y-coordinate 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

1264

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every such point makes the equation true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(1, 3) is on the graph of $y = x^2 + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

1265

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1^2 + 3 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 1

Describe transformations of the plane as functions of points • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2

The translation T maps (x, y) to (x + 3, y − 2).

Find T(4, 1).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

1267

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, −1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflection across the y-axis maps (x, y) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

1268

QUESTION ID
S35-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(−x, y)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

Which transformation does NOT preserve distance?

A. (x, y) → (x + 5, y)

B. (x, y) → (−x, y)

C. (x, y) → (y, x)

D. (x, y) → (2x, y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

1269

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (2x, y)

GRADING NOTE (x, y) → (2x, y) stretches horizontal distances. (x, y) → (x + 5, y) is a translation, and (x, y) → (−x, y) and (x, y) → (y, x) are reflections, which keep distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

True or false?

A transformation of the plane is a function: each input point has exactly one image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

1270

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is what makes it a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

True or false?

A rotation of 90° counterclockwise about the origin maps (1, 0) to (0, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

1271

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps (1, 0) to (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 2

Restrict a domain to make a function invertible • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $f(x) = x^2$, with all real numbers as its domain, has no inverse function.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

1273

QUESTION ID
S35-F2-Q1

TYPE
Short answer

CORRECT ANSWER
f(2) = f(−2) = 4, so an inverse would have to send 4 to two different outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^2$. Restrict f to the largest interval that contains $x = 3$ on which f is one-to-one, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

1274

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$; $f^{-1}(x) = 1 + \sqrt{x}$, for $x \geq 0$

GRADING NOTE Form required: the interval and the inverse. On $x \geq 1$, f is increasing, and solving $x = (y - 1)^2$ with $y \geq 1$ gives $y = 1 + \sqrt{x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

Which function is one-to-one on all real numbers?

A. $f(x) = x^2 - 4$

B. $f(x) = x^3$

C. $f(x) = |x|$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

1275

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3$

GRADING NOTE x^3 takes each output once. x^2, $|x|$, and $x^2 - 4$ give the same output for x and $-x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ with domain $x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

1276

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE On $x \geq 0$, $\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

1277

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only one-to-one functions do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-IF.A.2 • INTERPRETING FUNCTIONS

Use function notation, evaluate functions, and interpret statements in function notation in context

MAIN	Algebra I main question	**HSF-IF.A.2**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve linear equations	**HSA-REI.B.3**
FORWARD 1	Evaluate compositions of functions	**HSF-BF.A.1.c**
FORWARD 2	Read values of an inverse function	**HSF-BF.B.4.c**

SET 36 • HSF-IF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x^2 - 3x + 1.$

Find $f(-2)$.

$f(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

1280

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 2(4) + 6 + 1 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet t seconds after it is hit.

Find h(1), and explain what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

1281

QUESTION ID
S36-M-Q2

TYPE
Short answer

CORRECT ANSWER
h(1) = 48: one second after it is hit, the ball is 48 feet high.

GRADING NOTE Form required: the value and its meaning (with units).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 5x - 7$.

Find the value of x for which $g(x) = 18$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

1282

QUESTION ID

S36-M-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5x - 7 = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

C(n) is the cost in dollars of n tickets.

What does C(4) = 60 mean?

A. 4 tickets cost $60.

B. C equals 4 times 60.

C. Each ticket costs $60.

D. 60 tickets cost $4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

1283

QUESTION ID

S36-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 tickets cost $60.

GRADING NOTE The input is the number of tickets and the output is the cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = x^2$, then $f(a + 1) = a^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

1284

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(a + 1) = (a + 1)^2 = a^2 + 2a + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2x^2 - 3x + 1$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

1286

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $18 - 9 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-16t^2 + 64t$ when $t = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

1287

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

64

GRADING NOTE Any equivalent form. $-64 + 128$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $3(a - 4)^2$ when $a = 6$.

A. 144

B. 12

C. 6

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

1288

QUESTION ID
S36-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 12

GRADING NOTE $3(2)^2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $x = -1$, $x^2 - x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

1289

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1 + 1 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $n = 5$, $2n^2 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

1290

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2(25) = 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 2

Solve linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5x − 7 = 18.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

1292

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3(x + 2) = 2x − 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

1293

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10

GRADING NOTE Any equivalent form. 3x + 6 = 2x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q2**

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.5x + 3 = 11.

A. x = 7

B. x = 28

C. x = 16

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

1294

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 16

GRADING NOTE 0.5x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = -2$ is the solution of $4x + 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

1295

QUESTION ID
S36-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $-8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

True or false?

The solution of $7 - x = 10$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

1296

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-x = 3$, so $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 1

Evaluate compositions of functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 4$ and $g(x) = 3x$.

Find $f(g(2))$.

$f(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

1298

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. g(2) = 6 and f(6) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x - 3$.

Write $f(g(x))$.

$f(g(x)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

1299

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Any equivalent form, such as $x^2 - 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ and $g(x) = x + 1$.

Find $g(f(5))$.

A. 11

B. 10

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

1300

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 11

GRADING NOTE $f(5) = 10$ and $g(10) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x - 1$ and $g(x) = x + 1$, then $f(g(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

1301

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1) - 1 = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = 2x$ and $g(x) = x^2$, then $f(g(3)) = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

1302

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $g(3) = 9$ and $f(9) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 2

Read values of an inverse function • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

The function f has domain {1, 2, 3, 4}, with f(1) = 3, f(2) = 5, f(3) = 7, and f(4) = 9. Its outputs are all different, so f has an inverse.

Find $f^{-1}(7)$.

$f^{-1}(7)$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

1304

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 7, so $f^{-1}(7) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x + 1$ is invertible.

Find $f^{-1}(11)$.

$f^{-1}(11) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

1305

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $2x + 1 = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

f is a one-to-one function with f(4) = 10.

Which statement is true?

A. $f^{-1}(10) = 4$

B. $f^{-1}(4) = 10$

C. $f(10) = 4$

D. $f^{-1}(10) = \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

1306

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(10) = 4$

GRADING NOTE The inverse reverses the pair (4, 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

True or false?

If f is a one-to-one function, the graph of f^{-1} is the reflection of the graph of f across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

1307

QUESTION ID
S36-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each point (a, b) on f becomes (b, a) on f^{-1}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f^{-1}(x)$ means $1 \div f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

1308

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f^{-1} is the inverse function, not the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-IF.A.3 • INTERPRETING FUNCTIONS

Recognize that sequences are functions whose domain is a set of integers

MAIN	Algebra I main question	**HSF-IF.A.3**
BACKWARD 1	Evaluate functions in function notation	**HSF-IF.A.2**
BACKWARD 2	Generate two numerical patterns and relate their terms	**5.OA.B.3**
FORWARD 1	Repeat a dilation and describe the lengths — nearest related	**HSG-SRT.A.1**
FORWARD 2	Find the sum of a finite geometric series	**HSA-SSE.B.4**

SET 37 • HSF-IF.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(1) = 3$ and $f(n) = f(n - 1) + 4$ for $n \geq 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

1311

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. 3, 7, 11, 15, 19.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(1) = 2$, $f(2) = 3$, and $f(n) = f(n - 1) + 2f(n - 2)$ for $n \geq 3$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

1312

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

27

GRADING NOTE Any equivalent form. $f(3) = 3 + 2(2) = 7$, $f(4) = 7 + 2(3) = 13$, $f(5) = 13 + 2(7) = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for the sequence 5, 10, 20, 40, ..., with f(1) = 5.

f(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

1313

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

$5 \cdot 2^{n-1}$

GRADING NOTE Any equivalent form, such as $2.5 \cdot 2^{n}$. Each term is 2 times the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

A sequence f(n) starts with f(1).

What is its domain?

A. The positive integers 1, 2, 3, ...

B. The numbers from 0 to 1

C. All real numbers

D. All integers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

1314

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The positive integers 1, 2, 3, ...

GRADING NOTE A sequence is defined only at the term numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

True or false?

The graph of the sequence f(n) = 2n + 1 for n = 1, 2, 3, ... is a solid line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

1315

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The domain is the positive integers, so the graph is separate points: (1, 3), (2, 5), (3, 7), ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 1

Evaluate functions in function notation • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 3n - 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

1317

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2^n$.

Find f(5).

f(5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

1318

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = n^2 + 1$.

Find f(3).

A. 7

B. 10

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

1319

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

GRADING NOTE $9 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

True or false?

If f(n) = 10 − n, then f(10) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

1320

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 − 10 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

True or false?

If f(n) = 4n, then f(3) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

1321

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \cdot 3 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 2

Generate two numerical patterns and relate their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first 5 terms of each pattern.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

1323

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A: 0, 3, 6, 9, 12; B: 0, 6, 12, 18, 24

GRADING NOTE Form required: both patterns in order. Each B term is 2 times the matching A term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Write ordered pairs (A term, B term) for the first 4 terms. What relationship do you see?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

1324

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 0), (4, 8), (8, 16), (12, 24); each B term is 2 times the matching A term.

GRADING NOTE Form required: the four pairs and the relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

How do their third terms compare?

A. A's is 3 times B's.

B. B's is 4 more than A's.

C. B's is 3 times A's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

1325

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. B's is 3 times A's.

GRADING NOTE A: 0, 2, 4. B: 0, 6, 12. 12 = 3 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is twice the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

1326

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A: 0, 5, 10, 15. B: 0, 10, 20, 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 2 and adds 4. Each term of B is 4 times the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

1327

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A: 1, 3, 5, 7. B: 2, 6, 10, 14. Each B term is 2 times the matching A term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 1

Repeat a dilation and describe the lengths • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Repeated dilations produce a geometric sequence of lengths, a function defined on the positive integers.

ANSWER KEY Reduced copy of the question page, with its answer

A segment of length 3 is dilated by scale factor 2, then the image is dilated by 2 again, and so on.

Write the length after n dilations.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

1329

QUESTION ID
S37-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$3 \cdot 2^n$

GRADING NOTE Any equivalent form. Each dilation multiplies the length by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A dilation with center (0, 0) and scale factor 3 maps (2, −1) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

1330

QUESTION ID
S37-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(6, −3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5 • NEAREST RELATED

A dilation with scale factor $\frac{1}{2}$ maps a segment of length 10 to a segment of what length?

A. 20

B. 9.5

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

1331

QUESTION ID
S37-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 5

GRADING NOTE $10 \cdot \frac{1}{2} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

1332

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of dilations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A dilation with scale factor 2 doubles the area of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

1333

QUESTION ID
S37-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Areas are multiplied by $2^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 37 • HSF-IF.A.3

FORWARD 2

Find the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 5 terms of 2, 6, 18, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

1335

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

242

GRADING NOTE Any equivalent form. $2(3^5 - 1) \div (3 - 1) = 242$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a formula for the sum of the first n terms of 5, 10, 20, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

1336

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$5(2^n - 1)$

GRADING NOTE Any equivalent form. a = 5 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 1 + 4 + 16 + 64?

A. 256

B. 85

C. 84

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

1337

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 85

GRADING NOTE $(4^4 - 1) \div 3 = 85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

True or false?

The sum of the first 4 terms of 1, 2, 4, 8, ... is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

1338

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1 + 2 + 4 + 8 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

True or false?

A geometric series with first term 4, ratio 1, and 5 terms has sum 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

1339

QUESTION ID
S37-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 + 4 + 4 + 4 + 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of graphs and tables of functions in context

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Describe a function qualitatively from a situation	**8.F.B.5**
BACKWARD 2	Use vertex form to find a maximum or minimum	**HSA-SSE.B.3.b**
FORWARD 1	Identify key features of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Identify amplitude, midline, and period of a periodic model	**HSF-TF.B.5**

SET 38 • HSF-IF.B.4

MAIN

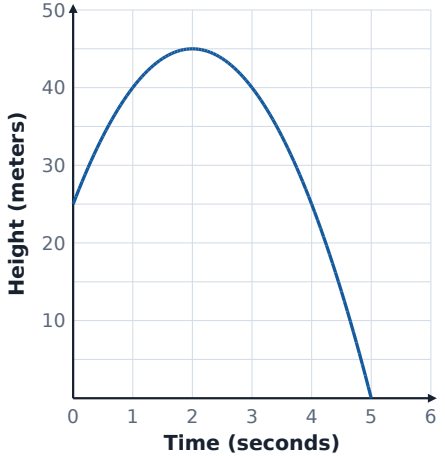
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

What is the maximum height, and when does it occur?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

1342

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

45 meters, at t = 2 seconds

GRADING NOTE Form required: the height and the time, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

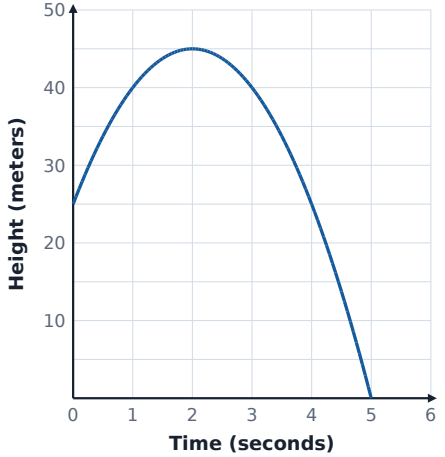
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony (t = 0 is the moment it is thrown).

What is the h-intercept of the graph, and what does it mean?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

1343

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

25: the ball was thrown from 25 meters above the ground.

GRADING NOTE Form required: the value and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

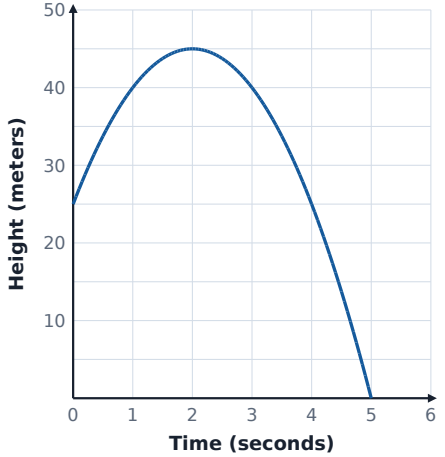
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

When does the ball reach the ground? Over what interval is its height increasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

1344

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

At t = 5 seconds; the height increases for 0 < t < 2.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

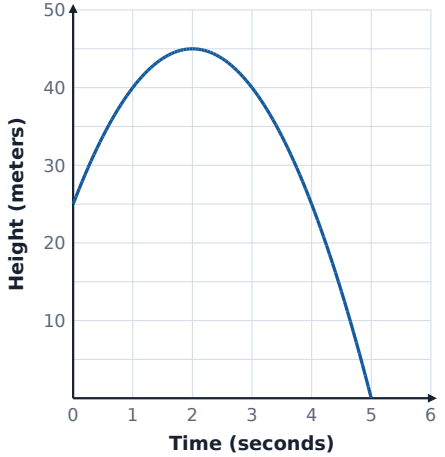
ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

Over what interval is the ball's height decreasing?

A. $0 < t < 2$ **B.** $0 < t < 5$

C. $2 < t < 5$ **D.** $25 < t < 45$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

1345

QUESTION ID

S38-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $2 < t < 5$

GRADING NOTE After the highest point at $t = 2$, the height falls until the ball lands at $t = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

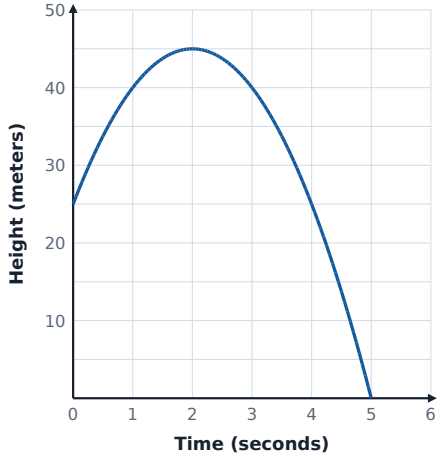
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony. The ball is 25 meters high twice: at t = 0 and at t = 4.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

1346

QUESTION ID
S38-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The graph is symmetric about t = 2, and 0 and 4 are each 2 seconds from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 1

Describe a function qualitatively from a situation • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A car speeds up, drives at a constant speed, then slows to a stop.

Describe the graph of its speed over time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

1348

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

It rises, then stays level, then falls to 0.

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

A graph of water level is horizontal from t = 2 to t = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

1349

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The water level did not change from t = 2 to t = 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation has a graph that first increases and then decreases?

A. Savings with a steady deposit each week

B. The height of a ball thrown up that comes back down

C. The distance a parked car travels

D. The height of a burning candle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

1350

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The height of a ball thrown up that comes back down

GRADING NOTE The ball rises and then falls. A burning candle only gets shorter, a parked car's distance stays the same, and steady deposits only increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

On a distance-time graph, a steeper segment means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

1351

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More distance in the same time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A decreasing graph always means the quantity is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

1352

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A quantity can decrease while staying positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 2

Use vertex form to find a maximum or minimum • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $-5t^2 + 20t + 25$ in the form $a(t - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

1354

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-5(t - 2)^2 + 45$

GRADING NOTE Form required: $a(t - h)^2 + k$. $-5(t^2 - 4t) + 25 = -5(t - 2)^2 + 20 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the maximum value of $-(x - 3)^2 + 7$, and for what x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

1355

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7, at x = 3

GRADING NOTE Form required: the value and the x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

What is the minimum value of $2(x + 1)^2 - 8$?

A. -8

B. -1

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

1356

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE $2(x + 1)^2 \geq 0$, and it is 0 at $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

$-5(t-2)^2 + 45$ is greatest when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

1357

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The squared term is 0 there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

The minimum value of $(x - 4)^2 + 1$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

1358

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum value is 1, at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 1

Identify key features of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Describe the end behavior of $f(x) = -2x^3 + x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

1360

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$.

GRADING NOTE Any equivalent description. The leading term $-2x^3$ decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

Find the zeros of f(x) = x(x − 2)(x + 3).

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

1361

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 2, and −3

GRADING NOTE Form required: all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function rises on the right and falls on the left?

A. $f(x) = -x^4$

B. $f(x) = x^3 - 4x$

C. $f(x) = x^2$

D. $f(x) = -x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

1362

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3 - 4x$

GRADING NOTE An odd degree with a positive leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A polynomial function of degree 4 has at most 4 real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

1363

QUESTION ID
S38-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

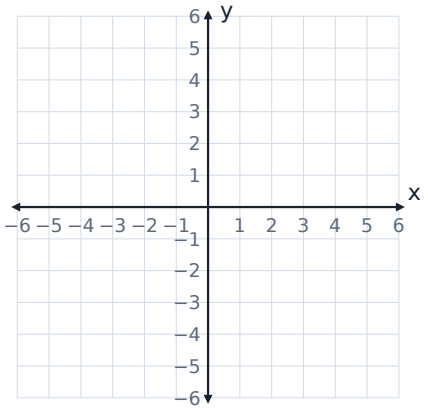
GRADING NOTE A degree-n polynomial has at most n zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = -x^2(x - 3)$. Mark the zeros and show what the graph does at each zero and at the ends. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

1364

QUESTION ID

S38-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve that rises to the left, touches the x-axis at 0 and turns (double zero), rises to a peak between 0 and 3, crosses the x-axis at 3, and falls to the right

GRADING NOTE Any correct graph: touch at 0, cross at 3, end behavior of $-x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 2

Identify amplitude, midline, and period of a periodic model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel seat's height is $h(t) = 30 - 25\cos(\pi t \div 5)$ meters after t minutes.

What are the greatest and least heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

1366

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

55 meters and 5 meters

GRADING NOTE Form required: both heights. cos is between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

What is the period of $h(t) = 30 - 25\cos(\pi t \div 5)$?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

1367

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 minutes

GRADING NOTE Any equivalent form. $2\pi \div (\pi \div 5) = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

What is the midline of y = 4sin x + 2?

A. y = 4

B. y = 2

C. y = 6

D. y = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

1368

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2

GRADING NOTE The graph goes from −2 to 6, centered on y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

The amplitude of $y = 3\sin x$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

1369

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph goes 3 units above and below its midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

True or false?

The function y = sin x has period π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

1370

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

SET 78 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Sketch graphs that show key features given in a verbal description

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Sketch a graph from a description	**8.F.B.5**
BACKWARD 2	Write and graph equations on labeled axes	**HSA-CED.A.2**
FORWARD 1	Sketch polynomial graphs from a description of their features	**HSF-IF.C.7.c**
FORWARD 2	Sketch periodic graphs from a description	**HSF-TF.B.5**

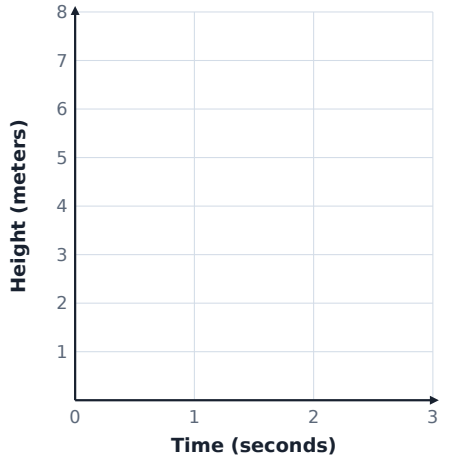
SET 78 • HSF-IF.B.4

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball is thrown up from a 2-m platform. It reaches its highest point, 7 m, after 1 second, and it lands on the ground after about 2.2 seconds.

Sketch its height against time. Label the starting point, the maximum, and the landing time.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

1373

QUESTION ID

S78-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A curve starting at (0, 2), rising to a peak at (1, 7), and falling to the t-axis near (2.2, 0), with those points labeled

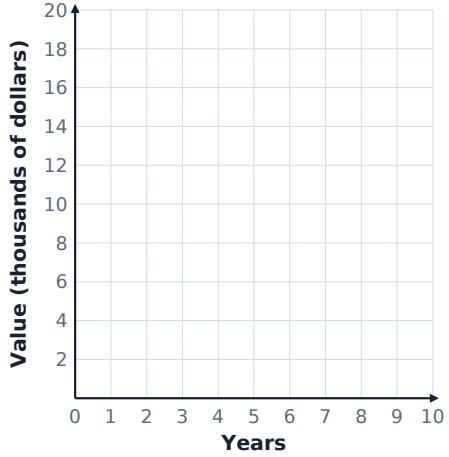
GRADING NOTE Any correct sketch: the intercept 2, the maximum (1, 7), increasing then decreasing, and the landing time near 2.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q1/5

A car is worth $20,000 when new, and it loses 15% of its value each year, never reaching $0.

Sketch its value for 0 to 10 years. Show the starting value and what happens as time goes on.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

1374

QUESTION ID

S78-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A decreasing curve from (0, 20) that falls more slowly each year and stays above 0 (about 4 at 10 years)

GRADING NOTE Any correct sketch: intercept 20, always decreasing, curving toward but never reaching 0. $20(0.85)^{10} \approx 3.9$.

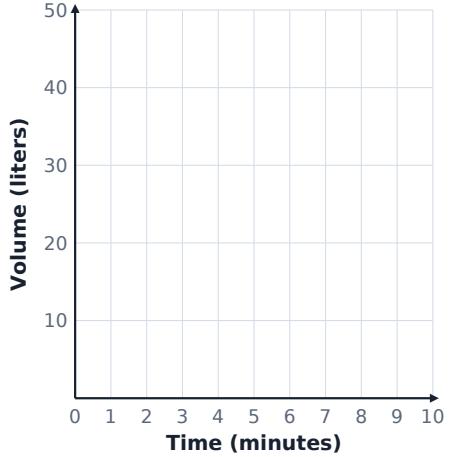
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 50-liter tank drains at a constant rate until it is empty after 5 minutes, stays empty for 2 minutes, and then refills at a constant rate to 30 liters in 3 minutes.

Sketch the volume over the 10 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

1375

QUESTION ID

S78-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

A segment from (0, 50) to (5, 0), a horizontal segment from (5, 0) to (7, 0), and a segment from (7, 0) to (10, 30)

GRADING NOTE Any correct sketch with the three pieces and their intervals of decrease, constancy, and increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q3/5

A town's population grew slowly at first and then faster and faster.

Which graph shape fits?

A. Increasing and leveling off

B. Increasing and curving upward

C. A horizontal line

D. Decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

1376

QUESTION ID

S78-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Increasing and curving upward

GRADING NOTE Growth that keeps speeding up gives an increasing graph that gets steeper.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q4/5

True or false?

A graph of the temperature of a cup of coffee left on a table falls at a constant rate until it reaches 0 °C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

1377

QUESTION ID

S78-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The coffee cools quickly at first and then more slowly, leveling off near room temperature, not 0 °C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 78 • MAIN • Q5/5

SET 78 • HSF-IF.B.4

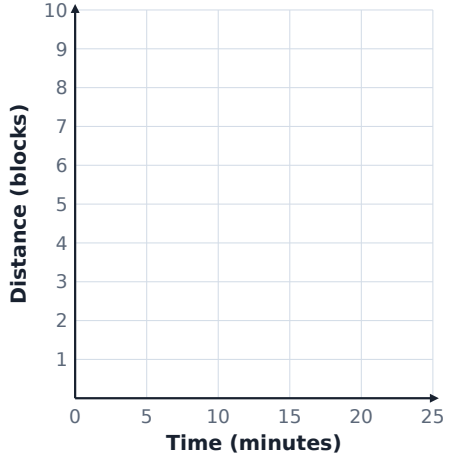
BACKWARD 1

Sketch a graph from a description • Grade 8 • Functions • 8.F.B.5

ANSWER KEY Reduced copy of the question page, with its answer

You walk away from home at a steady pace for 10 minutes, rest for 5 minutes, then walk home faster in 10 minutes.

Sketch your distance from home over 25 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

1379

GRADING NOTE Any correct qualitative sketch with the three pieces in order.

QUESTION ID

S78-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A rising segment for 0 to 10 minutes, a horizontal segment for 10 to 15, and a steeper falling segment back to 0 at 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of a plant's height rises steeply and then levels off. Describe what happened to the plant.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

1380

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

It grew quickly at first, then its growth slowed until its height stayed about the same.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q2/5

Which situation has a graph that stays constant (horizontal) for a while?

A. A rising balloon's height

B. A burning candle's height

C. A parked car's distance from home

D. A falling ball's height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

1381

QUESTION ID

S78-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A parked car's distance from home

GRADING NOTE A parked car does not move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q3/5

True or false?

A graph that rises and then falls has a highest point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

1382

QUESTION ID

S78-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It changes from increasing to decreasing at a maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q4/5

True or false?

A horizontal segment on a speed-time graph means the car is stopped.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

1383

QUESTION ID

S78-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means the speed is constant, which may not be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-IF.B.4

BACKWARD 2

Write and graph equations on labeled axes • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for the area A of a square with side s. What kind of graph does it have for s ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

1385

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

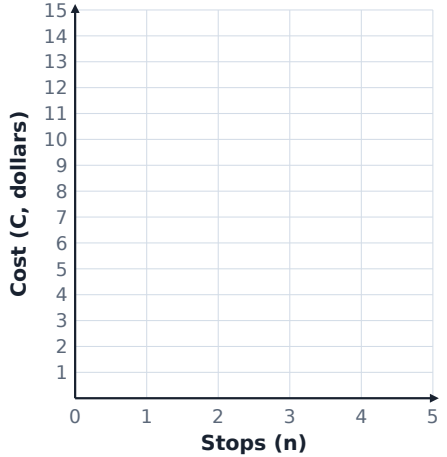
$A = s^2$; half of a parabola, starting at (0, 0) and curving upward

GRADING NOTE Form required: the equation and the shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q1/5

Graph C = 5 + 2n, the cost in dollars of a ride with n stops, for n = 0 to 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

1386

QUESTION ID

S78-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points (or a line) from (0, 5) to (5, 15), rising by 2 for each stop

GRADING NOTE Any correct graph; separate points are expected because n is a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A $1,200 laptop loses $200 of value each year. Which equation gives its value V after t years?

A. V = 1,200 − 200t

B. V = 200 − 1,200t

C. V = 1,200 + 200t

D. $V = 1{,}200(0.8)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

1387

QUESTION ID
S78-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. V = 1,200 − 200t

GRADING NOTE Start at 1,200 and subtract 200 per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q3/5

True or false?

An equation in two variables can be graphed on a coordinate plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

1388

QUESTION ID

S78-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each solution is a point (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of d = 50t is a curve.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

1389

QUESTION ID

S78-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a straight line through the origin with slope 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 78 • BACKWARD 2 • Q5/5

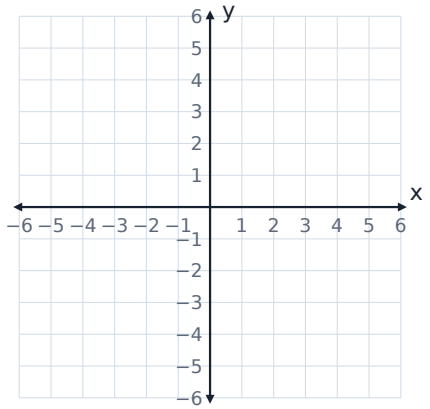
SET 78 • HSF-IF.B.4

FORWARD 1

Sketch polynomial graphs from a description of their features • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Sketch a cubic function that falls to the left, rises to the right, and has zeros at −2, 1, and 4. (The y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

1391

QUESTION ID

S78-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, 1, and 4, falling to the left and rising to the right, with one turn between −2 and 1 and one between 1 and 4

GRADING NOTE Any correct sketch, for example of y = (x + 2)(x − 1)(x − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A polynomial graph touches the x-axis at x = 3 and turns around there. What can you say about the factor x − 3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

1392

QUESTION ID

S78-F1-Q2

TYPE

Short answer

CORRECT ANSWER

It appears an even number of times, for example $(x - 3)^2$.

GRADING NOTE Any equivalent explanation (even multiplicity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which polynomial function has a graph that falls on both ends?

A. $y = -x^4 + 3x$

B. $y = x^3$

C. $y = -x^3$

D. $y = x^4 - 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

1393

QUESTION ID

S78-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = -x^4 + 3x$

GRADING NOTE Even degree with a negative leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q3/5

True or false?

The graph of a degree-5 polynomial with a positive leading coefficient rises to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

1394

QUESTION ID
S78-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Odd degree, positive leading coefficient: falls left, rises right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q4/5

True or false?

The graph of a cubic function can rise on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

1395

QUESTION ID
S78-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A cubic rises on one end and falls on the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 78 • FORWARD 1 • Q5/5

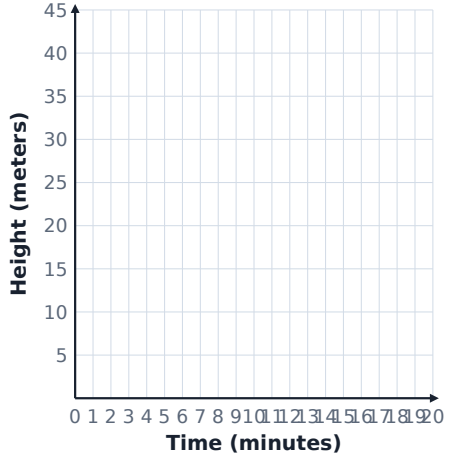
SET 78 • HSF-IF.B.4

FORWARD 2

Sketch periodic graphs from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat starts at its lowest height, 2 m, reaches its highest height, 42 m, after 5 minutes, and repeats every 10 minutes.

Sketch its height over 20 minutes.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

1397

QUESTION ID

S78-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A smooth wave starting at (0, 2), up to (5, 42), down to (10, 2), up to (15, 42), down to (20, 2), with midline 22

GRADING NOTE Any correct sketch with the minimum 2, maximum 42, period 10, and starting point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q1/5

Daylight lasts 15 hours in late June and 9 hours in late December. Describe a graph of daylight hours over a year.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

1398

QUESTION ID

S78-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A wave with maximum 15 (June), minimum 9 (December), midline 12, and period 12 months.

GRADING NOTE Form required: the maximum, minimum, midline, and period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q2/5

Which feature of a periodic graph is the time for one full cycle?

A. The maximum

B. The midline

C. The amplitude

D. The period

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

1399

QUESTION ID

S78-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The period

GRADING NOTE The period is the length of one cycle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q3/5

True or false?

The midline of a periodic graph is halfway between its maximum and minimum values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

1400

QUESTION ID
S78-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Midline = (maximum + minimum) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q4/5

True or false?

A tide graph that repeats every 12.4 hours has amplitude 12.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

1401

QUESTION ID
S78-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 12.4 hours is the period; the amplitude is a height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 78 • FORWARD 2 • Q5/5

SET 39 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

Relate the domain of a function to the quantity it describes

MAIN	Algebra I main question	**HSF-IF.B.5**
BACKWARD 1	Decide whether a solution is viable	**HSA-CED.A.3**
BACKWARD 2	Find where a quadratic model equals 0	**HSA-REI.B.4.b**
FORWARD 1	Use trigonometric ratios as functions of acute angles — nearest r..	**HSG-SRT.C.6**
FORWARD 2	Extend trigonometric functions to all real numbers with the unit ci…	**HSF-TF.A.2**

SET 39 • HSF-IF.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

n(k) = 12k gives the number of eggs in k full cartons.
The store has 30 cartons.

What is a reasonable domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

1404

QUESTION ID
S39-M-Q1

TYPE
Short answer

CORRECT ANSWER
The whole numbers 0, 1, 2, ..., 30

GRADING NOTE Form required: the domain, stated as whole numbers from 0 to 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet from the moment it is hit until it lands.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

1405

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \le t \le 4$

GRADING NOTE Form required: an inequality (or interval). $h(t) = -16t(t - 4) = 0$ at $t = 0$ and $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A taxi fare is C(m) = 3 + 2.5m dollars for m miles, with a limit of 40 miles per ride.

What is the domain?

A. All real numbers

B. m ≥ 3

C. 3 ≤ m ≤ 103

D. 0 ≤ m ≤ 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

1406

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0 ≤ m ≤ 40

GRADING NOTE The input is miles, from 0 to the 40-mile limit. 3 ≤ C ≤ 103 is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has perimeter 20. Its area is $A(w) = w(10 - w)$, where w is the width.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

1407

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

$0 < w < 10$

GRADING NOTE Form required: an inequality. Both w and the length $10 - w$ must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

True or false?

P(n) = 50n gives the number of seats on n buses.
n = 2.5 is in a reasonable domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

1408

QUESTION ID
S39-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The number of buses must be a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 1

Decide whether a solution is viable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Solving a problem gives 7.4 buses needed to carry every student.

What is the viable answer? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

1410

QUESTION ID
S39-B1-Q1

TYPE
Short answer

CORRECT ANSWER
8 buses: the number must be a whole number, and 7 buses are not enough.

GRADING NOTE Form required: the number and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

The number of tickets t must satisfy $5t \leq 60$ and $t \geq 0$.

What whole numbers of tickets are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

1411

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, …, 12

GRADING NOTE Any equivalent form, such as the whole numbers from 0 to 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is NOT viable for the number of people in a room?

A. 40

B. 0

C. 12

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

1412

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE A count cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A length of 0.25 meter can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

1413

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lengths can be fractions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

$w(10 - w) = 24$ has solutions $w = 4$ and $w = 6$, but only one of them is a viable width.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

1414

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both are viable: width 4 with length 6, or width 6 with length 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 2

Find where a quadratic model equals 0 • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

1416

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 or t = 4

GRADING NOTE Form required: both solutions. $-16t(t - 4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $w(10 - w) = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

1417

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$w = 0$ or $w = 10$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

Solve $-16t^2 + 64t + 80 = 0$.

A. $t = 4$ or $t = -2$

B. $t = 5$ or $t = -1$

C. $t = 5$ only

D. $t = -5$ or $t = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

1418

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $t = 5$ or $t = -1$

GRADING NOTE $t^2 - 4t - 5 = (t - 5)(t + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

True or false?

$t = 4$ is a solution of $t^2 - 4t = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

1419

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $16 - 16 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

1420

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = -9$ has no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 1

Use trigonometric ratios as functions of acute angles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: A trigonometric ratio defined by a right triangle has a domain limited to acute angles, a domain set by the situation.

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, with AC = 4, BC = 3, and AB = 5.

Find sin A.

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

1422

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form, such as 0.6. sin A = opposite ÷ hypotenuse = BC ÷ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

For which angle measures θ is sin θ defined as a ratio of sides of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

1423

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0° < θ < 90°

GRADING NOTE Form required: an inequality. Only the acute angles of a right triangle are used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5 • NEAREST RELATED

In a right triangle, $\cos A = \frac{12}{13}$.

What is sin A?

A. $\frac{5}{13}$

B. $\frac{12}{5}$

C. $\frac{1}{13}$

D. $\frac{13}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

1424

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/13

GRADING NOTE The sides are 5, 12, 13, so the opposite side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In a right triangle, sin A = cos B when A and B are the two acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

1425

QUESTION ID
S39-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The side opposite A is adjacent to B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For an acute angle θ, sin θ can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

1426

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The opposite side is shorter than the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry
HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 39 • HSF-IF.B.5

FORWARD 2

Extend trigonometric functions to all real numbers with the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Find cos π.

cos π = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

1428

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point at angle π is (−1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

Find sin(3π ÷ 2).

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

1429

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point is (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

With the unit circle, what is the domain of f(θ) = sin θ?

A. All real numbers

B. $0 < \theta < \pi \div 2$

C. $\theta \geq 0$

D. $0 \leq \theta \leq 2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

1430

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

GRADING NOTE Any real number of radians, positive or negative, gives a point on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

True or false?

A negative radian measure gives a clockwise rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

1431

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive angles turn counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

True or false?

sin θ is defined only for acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

1432

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit circle defines it for every real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate, interpret, and estimate the average rate of change of a function over an interval

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find the rate of change of a linear function	**8.F.B.4**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Build a periodic model from amplitude, period, and midline	**HSF-TF.B.5**
FORWARD 2	Analyze rational functions: rates of change and asymptotes	**HSF-IF.C.7.d**

SET 40 • HSF-IF.B.6

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2 + 2x$.

Find the average rate of change from $x = 1$ to $x = 4$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

1435

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $(f(4) - f(1)) \div (4 - 1) = (24 - 3) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town had 12,000 people in 2010 and 15,600 in 2022.

Find the average rate of change, with units.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

1436

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

300 people per year

GRADING NOTE Form required: the rate with its units. 3,600 ÷ 12 = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

The table gives the distance d a cart has traveled after t seconds.

Over which 2-second interval was its average speed greatest? Give that speed.

t (seconds)	0	2	4	6
d (meters)	0	30	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

1437

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

From t = 2 to t = 4, at 35 meters per second

GRADING NOTE Form required: the interval and the speed. The speeds are 15, 35, and 25 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

What is the average rate of change of $g(x) = 3^x$ from $x = 0$ to $x = 2$?

A. 4

B. 3

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

1438

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $(9 - 1) \div 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

True or false?

A linear function has the same average rate of change over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

1439

QUESTION ID
S40-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its average rate of change is always its slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 1

Find the rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the rate of change of the line through (2, 5) and (6, 17).

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

1441

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tank had 50 liters at t = 0 and 20 liters at t = 10 minutes, draining steadily.

What is the rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

1442

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3 liters per minute

GRADING NOTE Any equivalent form, with units. (20 − 50) ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the rate of change of $y = 7 - 4x$?

A. 7

B. -4

C. $-\frac{4}{7}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

1443

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE The coefficient of x is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A negative rate of change means the quantity is decreasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

1444

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outputs fall as inputs rise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

The rate of change from (0, 3) to (2, 3) is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

1445

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (3 − 3) ÷ 2 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A runner goes $\frac{3}{4}$ mile in $\frac{1}{8}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

1447

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 miles per hour

GRADING NOTE Any equivalent form. $\frac{3}{4} \div \frac{1}{8} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

A car uses 12 gallons to go 300 miles.

How many miles per gallon?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

1448

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

A recipe uses $\frac{2}{3}$ cup of sugar for $\frac{1}{3}$ of a batch.

How much sugar is used per batch?

A. $\frac{1}{2}$ cup

B. 1 cup

C. 2 cups

D. $\frac{2}{9}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

1449

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 cups

GRADING NOTE $\frac{2}{3} \div \frac{1}{3} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

120 km in 1.5 hours is 80 km per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

1450

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120 ÷ 1.5 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$18 for 4 pounds is $4 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

1451

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 18 ÷ 4 = $4.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 1

Build a periodic model from amplitude, period, and midline • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 8 minutes. A rider starts at the lowest point.

Write a model h(t) = A cos(Bt) + D for the rider's height after t minutes.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

1453

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−20cos(πt ÷ 4) + 25

GRADING NOTE Any equivalent form. Amplitude 20, midline 25, B = 2π ÷ 8 = π ÷ 4; A is negative because the ride starts at the lowest point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

$h(t) = -20\cos(\pi t \div 4) + 25$ gives a rider's height in meters after t minutes.

Give the amplitude, the period, and the midline.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

1454

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Amplitude 20 m; period 8 minutes; midline h = 25

GRADING NOTE Form required: all three. $2\pi \div (\pi \div 4) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

Daily high temperatures in a town vary from a low of 4 °C to a high of 28 °C over a 365-day year.

Which values fit a sinusoidal model?

A. Amplitude 24, midline 16, period 365 days

B. Amplitude 16, midline 12, period 365 days

C. Amplitude 12, midline 14, period 182.5 days

D. Amplitude 12, midline 16, period 365 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

1455

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Amplitude 12, midline 16, period 365 days

GRADING NOTE Amplitude (28 − 4) ÷ 2 = 12; midline (28 + 4) ÷ 2 = 16; one cycle per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

True or false?

A model $y = A\cos(Bt) + D$ with period 12 hours has $B = \pi \div 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

1456

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2\pi \div B = 12$ gives $B = \pi \div 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

True or false?

A tide that rises from 1 m to 5 m and falls back again has amplitude 4 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

1457

QUESTION ID
S40-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is half the distance from low to high: (5 − 1) ÷ 2 = 2 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 2

Analyze rational functions: rates of change and asymptotes • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 12 \div x$.

Find the average rate of change from $x = 2$ to $x = 6$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

1459

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. $(2 - 6) \div (6 - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $f(x) = 1 \div (x - 3) + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

1460

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Vertical x = 3; horizontal y = 2

GRADING NOTE Form required: both asymptotes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

Where is f(x) = (x + 1) ÷ (x − 4) undefined?

A. x = −4

B. x = 1

C. x = −1

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

1461

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 4

GRADING NOTE The denominator is 0 at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 1 \div x$ gets closer to the x-axis as x gets large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

1462

QUESTION ID
S40-F2-Q4

TYPE
True or false

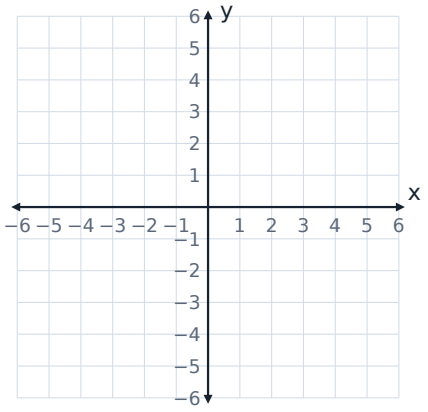
CORRECT ANSWER
A. True

GRADING NOTE $1 \div x$ approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

Graph f(x) = 1 ÷ (x − 2) + 1. Draw both asymptotes as lines and mark the intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

1463

QUESTION ID

S40-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Vertical asymptote x = 2 and horizontal asymptote y = 1; the curve passes through the x-intercept (1, 0) and the y-intercept (0, 0.5), with one branch below and left of (2, 1) and one above and right

GRADING NOTE Any correct graph. 1 ÷ (x − 2) = −1 gives x = 1; f(0) = −0.5 + 1 = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

SET 79 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Estimate the average rate of change of a function from its graph

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find a rate of change from two points	**8.F.B.4**
BACKWARD 2	Read a unit rate from a graph	**7.RP.A.2.b**
FORWARD 1	Estimate average rates of change from a polynomial graph	**HSF-IF.C.7.c**
FORWARD 2	Estimate average rates of change from a rational-function graph	**HSF-IF.C.7.d**

SET 79 • HSF-IF.B.6

MAIN

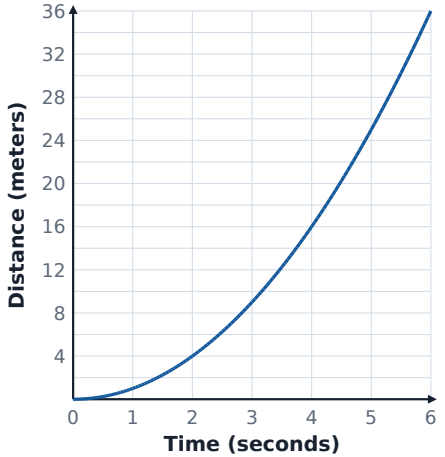
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 1 to t = 3, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

1466

QUESTION ID
S79-M-Q1

TYPE
Short answer

CORRECT ANSWER
About 4 meters per second

GRADING NOTE Form required: an estimate with units. Accept 3.5 to 4.5 m/s: the graph passes about (1, 1) and (3, 9), and (9 − 1) ÷ 2 = 4.

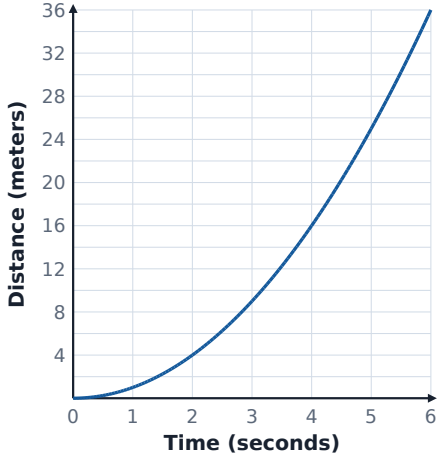
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q1/5

The graph shows the distance d, in meters, that a cart has rolled after t seconds.

Estimate the average rate of change from t = 3 to t = 5. Is the cart speeding up?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

1467

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 8 meters per second; yes, the rate is greater than on earlier intervals.

GRADING NOTE Form required: an estimate (7 to 9 m/s) and the conclusion. (25 − 9) ÷ 2 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

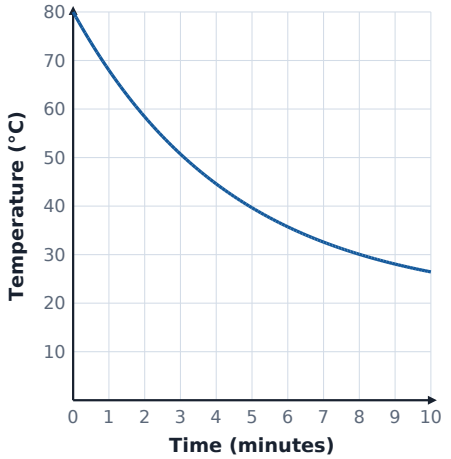
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows the temperature of a cup of coffee.

Estimate the average rate of change of the temperature over the first 5 minutes, with units.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

1468

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

About −8 °C per minute

GRADING NOTE Form required: an estimate with units. Accept −7 to −9 °C per minute: from about 80 °C to about 40 °C in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q3/5

The graph shows the temperature of a cup of coffee.

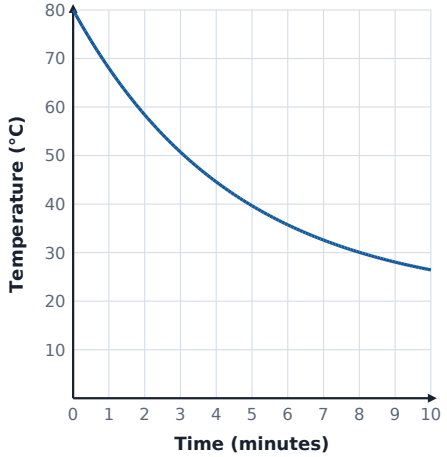
Over which 2-minute interval is the coffee cooling fastest on average?

A. 4 to 6 minutes

B. 8 to 10 minutes

C. 0 to 2 minutes

D. 6 to 8 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

1469

QUESTION ID

S79-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0 to 2 minutes

GRADING NOTE The graph is steepest at the start; it flattens as the coffee nears room temperature.

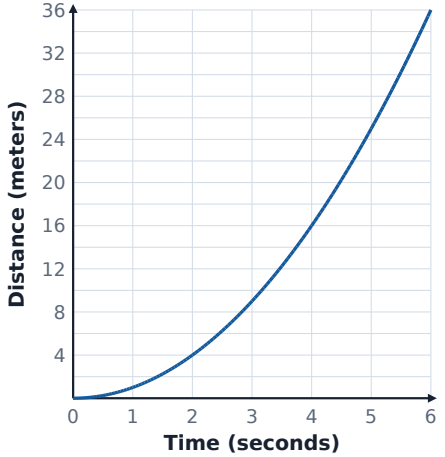
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q4/5

True or false?

The average rate of change of d from t = 0 to t = 6 equals the slope of the segment joining (0, 0) and (6, 36) on the graph.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

1470

QUESTION ID
S79-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The average rate of change is the slope of the secant line: 36 ÷ 6 = 6 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 79 • MAIN • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 1

Find a rate of change from two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (2, 10) and (6, 30). Find its rate of change.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

1472

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 20 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q1/5

A straight-line graph of a hike shows 3 km at 1 hour and 9 km at 3 hours. What is the rate of change, with units?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

1473

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 km per hour

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q2/5

What is the slope of the line through (0, 50) and (5, 25)?

A. 5

B. −5

C. 0.2

D. −25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

1474

QUESTION ID

S79-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −5

GRADING NOTE $(25 - 50) \div 5 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q3/5

True or false?

The slope between two points on a line is (change in y) ÷ (change in x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

1475

QUESTION ID

S79-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 4) and (3, 4) has rate of change 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

1476

QUESTION ID

S79-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4 - 4) \div 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

BACKWARD 2

Read a unit rate from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

The graph of a proportional relationship passes through (4, 12). What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

1478

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q1/5

A proportional cost graph passes through (0, 0) and (5, 7.5). What is the cost per item?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

1479

QUESTION ID

S79-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$1.50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A proportional graph of miles against gallons passes through (3, 90). What is the unit rate?

A. 3 miles per gallon

B. 270 miles per gallon

C. 90 miles per gallon

D. 30 miles per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

1480

QUESTION ID
S79-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 30 miles per gallon

GRADING NOTE 90 ÷ 3 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q3/5

True or false?

On a proportional graph, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

1481

QUESTION ID

S79-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE At 1 unit of x, y equals the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q4/5

True or false?

A proportional graph passes through (2, 5), so its unit rate is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

1482

QUESTION ID

S79-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5 \div 2 = 2.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-IF.B.6

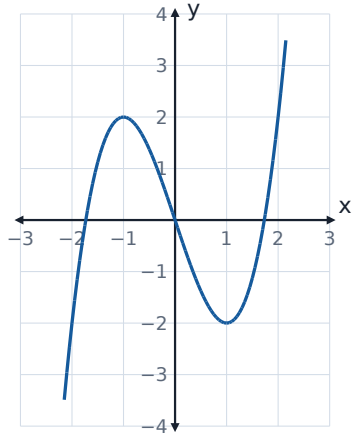
FORWARD 1

Estimate average rates of change from a polynomial graph • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = -1$ to $x = 1$.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

1484

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form. $(f(1) - f(-1)) \div 2 = (-2 - 2) \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

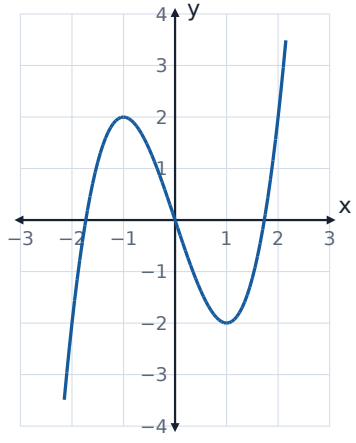
High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

Estimate the average rate of change from $x = 1$ to $x = 2$.

Rate: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

1485

QUESTION ID

S79-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $(2 - (-2)) \div 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $f(x) = x^3 - 3x$.

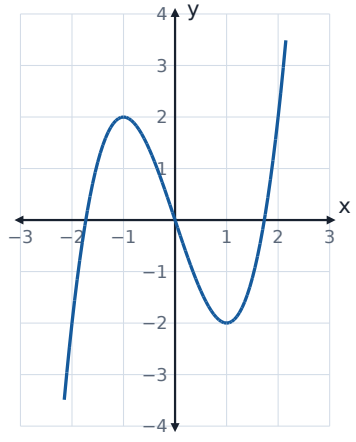
On which interval is f increasing?

A. $-1 < x < 1$

B. $1 < x < 2$

C. $-1 < x < 0$

D. $0 < x < 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

1486

QUESTION ID
S79-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 1 < x < 2

GRADING NOTE f falls between its turning points at $x = -1$ and $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q3/5

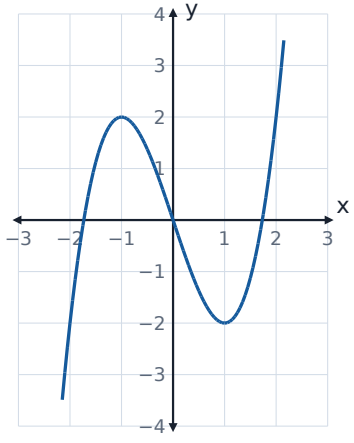
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The average rate of change of $f(x) = x^3 - 3x$ from $x = -2$ to $x = 2$ is 1.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

1487

QUESTION ID
S79-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(2 - (-2)) \div 4 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q4/5

True or false?

A positive average rate of change on an interval means the function increases at every point of the interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

1488

QUESTION ID
S79-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For $f(x) = x^3 - 3x$ on $-2 \le x \le 2$ the average rate is positive, but f decreases between -1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-IF.B.6

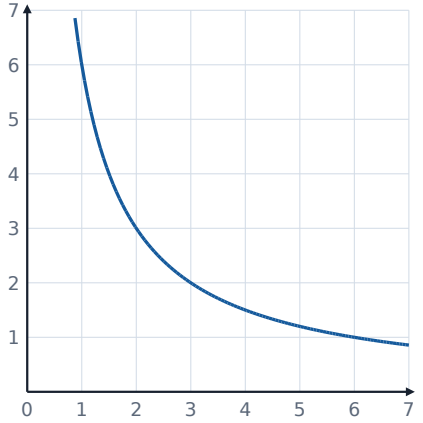
FORWARD 2

Estimate average rates of change from a rational-function graph • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 1 to x = 3.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

1490

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−2

GRADING NOTE Any equivalent form. (2 − 6) ÷ 2.

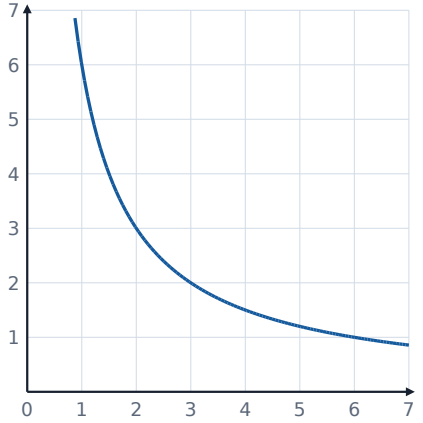
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

Estimate the average rate of change from x = 3 to x = 6.



Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

1491

QUESTION ID

S79-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/3

GRADING NOTE Any equivalent form (about −0.33). (1 − 2) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows f(x) = 6 ÷ x.

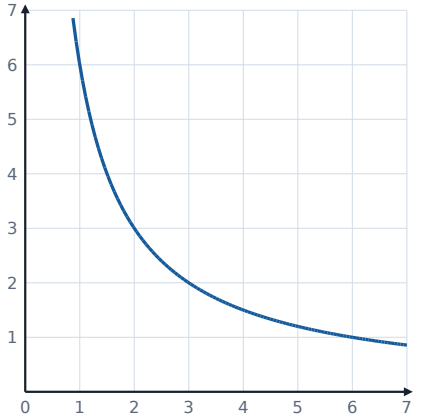
What happens to its average rate of change on intervals farther to the right?

A. It gets closer to 0.

B. It becomes positive.

C. It stays −2.

D. It gets more negative.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

1492

QUESTION ID

S79-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It gets closer to 0.

GRADING NOTE The graph flattens toward its horizontal asymptote.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 6 \div x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

1493

QUESTION ID

S79-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $6 \div x$ approaches 0 as x grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 6 \div x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

1494

QUESTION ID
S79-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 6 ÷ 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 79 • FORWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph linear functions and show their intercepts

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Find slope and write y = mx + b	**8.EE.B.6**
BACKWARD 2	Write linear equations in two variables from a situation	**HSA-CED.A.2**
FORWARD 1	Find a point that partitions a segment in a given ratio	**HSG-GPE.B.6**
FORWARD 2	Find and verify inverses of linear functions	**HSF-BF.B.4.b**

SET 41 • HSF-IF.C.7.a

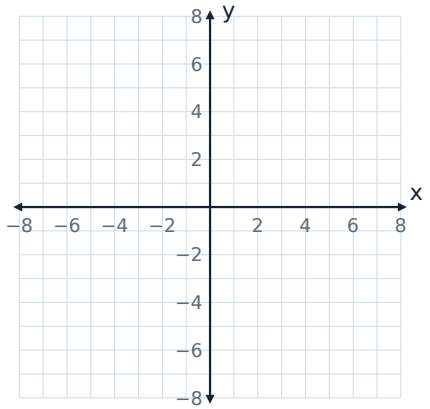
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = -\frac{2}{3}x + 4$ on the coordinate plane.

Give the x-intercept and the y-intercept.



Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

1497

QUESTION ID

S41-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 4), (3, 2), and (6, 0)

Written: x-intercept (6, 0); y-intercept (0, 4)

GRADING NOTE Form required: both intercepts. Both the graph and the intercepts are graded. $-\frac{2}{3}x + 4 = 0$ gives $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x- and y-intercepts of 3x − 5y = 15.

Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

1498

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 0) and (0, −3)

GRADING NOTE Form required: both intercepts. Set y = 0, then x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (−2, 7) and (4, −5) in slope-intercept form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

1499

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = −2x + 3

GRADING NOTE Form required: slope-intercept form, y = mx + b. The slope is −12 ÷ 6 = −2, and 7 = −2(−2) + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

Which describes the graph of y = −3?

A. A line with slope 1 through (−3, −3)

B. A vertical line through (−3, 0)

C. A line with slope −3 through the origin

D. A horizontal line through (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

1500

QUESTION ID

S41-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A horizontal line through (0, −3)

GRADING NOTE y is −3 for every x, and the slope is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

True or false?

The graph of $2x + y = 6$ has x-intercept 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

1501

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At $y = 0$, $2x = 6$, so the x-intercept is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 1

Find slope and write y = mx + b • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (3, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

1503

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line has slope 2 and y-intercept −1.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

1504

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

What is the y-intercept of y = 5x − 4?

A. $\frac{4}{5}$

B. 4

C. 5

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

1505

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

GRADING NOTE y = mx + b with b = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Any two distinct points on a nonvertical line give the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

1506

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar slope triangles have equal ratios. (A vertical line has no slope.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The line y = 4x passes through (1, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

1507

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4(1) = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 2

Write linear equations in two variables from a situation • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $25 to join plus $15 per month.

Write an equation for the cost C after m months.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

1509

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 25 + 15m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A candle 20 cm tall burns 2 cm per hour.

Write an equation for its height h after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

1510

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

h = 20 − 2t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ana earns $8 per hour plus $5 in tips.

Which equation gives her earnings E for h hours?

A. E = 8h + 5

B. E = 5h + 8

C. E = 13h

D. E = 8(h + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

1511

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. E = 8h + 5

GRADING NOTE 8 per hour, plus 5 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

True or false?

d = 60t gives the distance traveled at 60 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

1512

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance = rate × time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

True or false?

y = 100 − 4x describes a quantity that grows by 4 each step.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

1513

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 4 each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 1

Find a point that partitions a segment in a given ratio • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (−2, 3) to (6, 7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

1515

QUESTION ID
S41-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(2, 5)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q1**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the point $\frac{1}{4}$ of the way from A(0, 0) to B(8, 12).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

1516

QUESTION ID
S41-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(2, 3)

GRADING NOTE Any equivalent form. $\frac{1}{4}$ of (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

Point P divides the segment from A(1, 2) to B(7, 11) in the ratio 1 : 2 (AP : PB).

Find P.

A. (5, 8)

B. (3, 5)

C. (4, 6.5)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

1517

QUESTION ID
S41-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (3, 5)

GRADING NOTE P is $\frac{1}{3}$ of the way: (1 + 2, 2 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

True or false?

The midpoint of (0, 0) and (10, 4) is (5, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

1518

QUESTION ID
S41-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

True or false?

The point $\frac{1}{3}$ of the way from (0, 0) to (9, 6) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

1519

QUESTION ID
S41-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 2); (6, 4) is $\frac{2}{3}$ of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S41-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 2

Find and verify inverses of linear functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Find the inverse of f(x) = 3x − 6.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

1521

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6) ÷ 3

GRADING NOTE Any equivalent form, such as $\frac{1}{3}x + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

Show that f(x) = 2x + 1 and g(x) = (x − 1) ÷ 2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

1522

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = 2 · (x − 1) ÷ 2 + 1 = x, and g(f(x)) = (2x + 1 − 1) ÷ 2 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of f(x) = x ÷ 4 + 1?

A. $f^{-1}(x) = 4x + 1$

B. $f^{-1}(x) = 4(x + 1)$

C. $f^{-1}(x) = 4x - 4$

D. $f^{-1}(x) = x \div 4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

1523

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f^{-1}(x) = 4x - 4$

GRADING NOTE Solve x = y ÷ 4 + 1 for y: y = 4(x − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

True or false?

If f(g(x)) = x and g(f(x)) = x for every x, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

1524

QUESTION ID
S41-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = x + 5$ is $f^{-1}(x) = x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

1525

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The inverse is $x - 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph quadratic functions and show intercepts, maxima, and minima

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Factor to find the zeros of a quadratic	**HSA-SSE.B.3.a**
BACKWARD 2	Interpret key features of a function	**HSF-IF.B.4**
FORWARD 1	Relate a parabola's equation to its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use factors to describe the graph of a polynomial	**HSF-IF.C.7.c**

SET 42 • HSF-IF.C.7.a

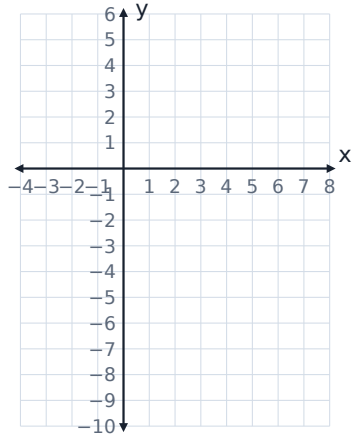
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = x^2 - 4x - 5$.

Give the vertex, the x-intercepts, and the y-intercept.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

1528

QUESTION ID

S42-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A parabola opening up with vertex (2, −9), through (−1, 0), (0, −5), (4, −5), and (5, 0)

Written: Vertex (2, −9); x-intercepts (−1, 0) and (5, 0); y-intercept (0, −5)

GRADING NOTE Form required: all four points. Both the graph and the points are graded. $f(x) = (x - 5)(x + 1) = (x - 2)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

Find the vertex and the axis of symmetry of $y = 2x^2 + 8x + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

1529

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (−2, −5); axis x = −2

GRADING NOTE Form required: both. $x = -b \div (2a) = -8 \div 4 = -2$, and $y = 8 - 16 + 3 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = -3x^2 + 6x - 1$ have a maximum or a minimum? Find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

1530

QUESTION ID

S42-M-Q3

TYPE

Short answer

CORRECT ANSWER

A maximum of 2, at x = 1

GRADING NOTE Form required: maximum or minimum, its value, and where. $a = -3 < 0$, so the parabola opens down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

What are the x-intercepts of $y = (x + 3)(x - 1)$?

A. (3, 0) and (−1, 0)

B. (−3, 0) and (1, 0)

C. (−3, 0) and (−1, 0)

D. (0, −3) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

1531

QUESTION ID

S42-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−3, 0) and (1, 0)

GRADING NOTE $y = 0$ when $x + 3 = 0$ or $x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

True or false?

The graph of $y = x^2 + 4$ has no x-intercepts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

1532

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 + 4 \geq 4$ for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 1

Factor to find the zeros of a quadratic • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 4x - 5$, and give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

1534

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 5)(x + 1); zeros 5 and −1

GRADING NOTE Form required: the factored form and both zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

Find the zeros of $2x^2 - 8$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

1535

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 and −2

GRADING NOTE Form required: both zeros. $2(x - 2)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

What are the zeros of $x^2 + 6x + 8$?

A. 2 and −4

B. −2 and −4

C. 2 and 4

D. −2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

1536

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2 and −4

GRADING NOTE (x + 2)(x + 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

True or false?

The zeros of $(x - 7)(x + 2)$ are 7 and -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

1537

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each factor is 0 at one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

True or false?

The zeros of $x^2 - 9x$ are 0 and −9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

1538

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x(x − 9): the zeros are 0 and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 2

Interpret key features of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A table has x = −2, −1, 0, 1, 2 and y = 5, 2, 1, 2, 5.

Describe its symmetry and its minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

1540

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric about x = 0; minimum 1 at x = 0.

GRADING NOTE Form required: both features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

A profit function P(x) is positive for $10 < x < 50$, where x is the number of items sold.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

1541

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The company makes a profit when it sells between 10 and 50 items.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

The highest point of a graph is (3, 40).

Which statement describes it?

A. A maximum of 40 at x = 3

B. A maximum of 3 at x = 40

C. A minimum of 3 at x = 40

D. A y-intercept of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

1542

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 40 at x = 3

GRADING NOTE The maximum value is the output, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

True or false?

A function is increasing on an interval when its outputs get larger as its inputs get larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

1543

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

The x-intercepts of a graph are where the output is greatest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

1544

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The output is 0 at an x-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 1

Relate a parabola's equation to its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = \frac{1}{8}x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

1546

QUESTION ID
S42-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Focus (0, 2); directrix y = −2

GRADING NOTE Form required: both. $x^2 = 8y = 4py$ with $p = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Derive the equation of the parabola with focus $(0, -1)$ and directrix $y = 1$ from the distance definition.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

1547

QUESTION ID
S42-F1-Q2

TYPE
Answer with required work

CORRECT ANSWER
$x^2 = -4y$

GRADING NOTE REQUIRED WORK: $\sqrt{(x^2 + (y + 1)^2)} = |y - 1|$ written, both sides squared, and simplified: $x^2 + 2y = -2y$, so $x^2 = -4y$.

Form required: the derivation and the equation (any equivalent form, such as $y = -\frac{1}{4}x^2$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parabola has focus (0, 3) and directrix y = −3.

Where is its vertex?

A. (0, 0)

B. (0, 3)

C. (0, −3)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

1548

QUESTION ID
S42-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. (0, 0)

GRADING NOTE Halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

True or false?

The parabola $x^2 = 12y$ opens upward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

1549

QUESTION ID
S42-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $p = 3 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

True or false?

The parabola with focus (0, 1) and directrix $y = -1$ is $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

1550

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 4y$, so $y = \frac{1}{4}x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 2

Use factors to describe the graph of a polynomial • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$f(x) = (x + 1)(x - 2)(x - 4)$.

Find the zeros and the y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

1552

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Zeros −1, 2, and 4; y-intercept 8

GRADING NOTE Form required: all three zeros and the y-intercept. f(0) = (1)(−2)(−4) = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At most how many turning points can the graph of a degree-3 polynomial have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

1553

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. At most n − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

$f(x) = (x - 3)^2(x + 1)$.

What does the graph do at $x = 3$?

A. It touches the x-axis and turns around.

B. It is undefined.

C. It has a vertical asymptote.

D. It crosses the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

1554

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It touches the x-axis and turns around.

GRADING NOTE The factor $(x - 3)$ is squared, so f does not change sign at 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^3 - 8$ has exactly one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

1555

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

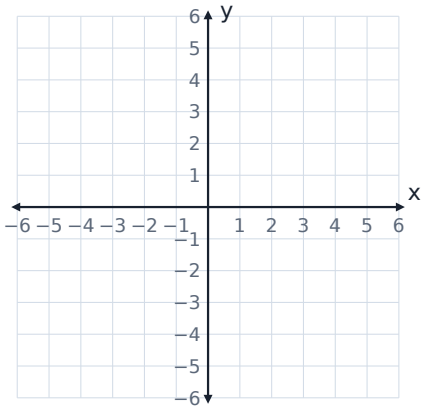
A. True

GRADING NOTE $x^3 = 8$ gives only $x = 2$ among the real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

Graph $f(x) = (x + 2)(x - 1)^2$. Mark the zeros and the y-intercept, and show what the graph does at each zero. (Use the grid for x; the y-scale is up to you.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

1556

QUESTION ID

S42-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A curve crossing the x-axis at −2, touching it at 1 and turning back up, through (0, 2), falling to the left and rising to the right

GRADING NOTE Any correct graph. f(0) = (2)(1) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph absolute value functions and show their key features

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find the domain and range of a function	**HSF-IF.A.1**
BACKWARD 2	Understand absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Solve radical equations and check for extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Read values of inverses of power and root functions	**HSF-BF.B.4.c**

SET 43 • HSF-IF.C.7.b

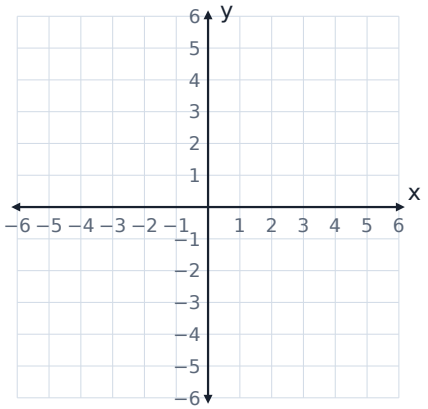
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph f(x) = |x − 2| − 3.

Give the vertex and the x-intercepts.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

1559

QUESTION ID

S43-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A V-shape opening up with vertex (2, −3), through (−1, 0), (0, −1), (4, −1), and (5, 0)

Written: Vertex (2, −3); x-intercepts (−1, 0) and (5, 0)

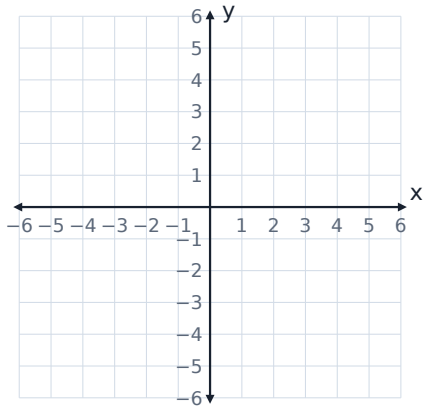
GRADING NOTE Form required: the vertex and both intercepts. Both the graph and the answer are graded. |x − 2| = 3 gives x = 5 or x = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph g(x) = −2|x + 1| + 4, and label its vertex and its x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

1560

QUESTION ID

S43-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A V-shape opening down with vertex (−1, 4), slopes 2 and −2, crossing the x-axis at (−3, 0) and (1, 0)

GRADING NOTE Any correct graph. |x + 1| = 2 gives x = 1 or x = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = |x + 3| + 1$ is related to the graph of $y = |x|$, and give its range.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

1561

QUESTION ID
S43-M-Q3

TYPE
Short answer

CORRECT ANSWER
It is the graph of $y = |x|$ shifted 3 units left and 1 unit up (vertex $(-3, 1)$); range $y \geq 1$.

GRADING NOTE Form required: the shifts and the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

Which function's graph is a V opening downward with vertex (2, 5)?

A. $y = |x - 2| + 5$

B. $y = -|x + 2| + 5$

C. $y = -|x - 5| + 2$

D. $y = -|x - 2| + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

1562

QUESTION ID

S43-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = -|x - 2| + 5$

GRADING NOTE $y = -|x - h| + k$ opens downward with vertex $(h, k) = (2, 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

True or false?

The graph of $y = 3|x|$ is narrower than the graph of $y = |x|$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

1563

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each output is tripled, so the sides are steeper (slopes 3 and −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 1

Find the domain and range of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $f(x) = \sqrt{(x - 5)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

1565

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 5$

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of f(x) = |x|?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

1566

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

Which value is NOT in the domain of $g(x) = \sqrt{(6 - x)}$?

A. 7

B. −3

C. 0

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

1567

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE $6 - 7 = -1$ is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

Every real number is in the domain of $f(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

1568

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every real number has a real cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

The range of $f(x) = x^2$ is all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

1569

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range is $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 2

Understand absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

What is |−9|?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

1571

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. −9 is 9 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is farther from 0, −7 or 5? Explain with absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

1572

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−7, because |−7| = 7 and |5| = 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

Which value equals |3 − 8|?

A. 5

B. −5

C. 11

D. −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

1573

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE |−5| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

True or false?

$|-4| = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

1574

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −4 is 4 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|0| = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

1575

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0 is 0 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check for extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x − 2) = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

1577

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. x − 2 = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt{(x + 6)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

1578

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3; x = −2 is extraneous.

GRADING NOTE Form required: the solution and the extraneous value. $x^2 - x - 6 = 0$ gives 3 and −2, but $\sqrt{4} = 2 \neq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{(x + 1)} = 2$.

A. x = 7

B. x = 3

C. x = 9

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

1579

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 7

GRADING NOTE x + 1 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{x} + 3 = 0$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

1580

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{x} \geq 0$, so $\sqrt{x} + 3 \geq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 16$ is a solution of $\sqrt{x} = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

1581

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{16} = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 2

Read values of inverses of power and root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$g(x) = x^3$.

Find $g^{-1}(27)$.

$g^{-1}(27) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

1583

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (4, 2) is on the graph of $f(x) = \sqrt{x}$.

Which point must be on the graph of f^{-1}?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

1584

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

$f(x) = x^2$ with domain $x \geq 0$.

What is $f^{-1}(49)$?

A. -7

B. 24.5

C. 2,401

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

1585

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE $f(7) = 49$, and 7 is in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

True or false?

The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

1586

QUESTION ID
S43-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cubing and taking the cube root undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

True or false?

f is a one-to-one function. If (3, 10) is on the graph of f, then (3, 10) must be on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

1587

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (10, 3) must be on the graph of f^{-1}, but (3, 10) need not be: for f(x) = x + 7, $f^{-1}(3) = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find domains and ranges of root functions	**HSF-IF.A.1**
BACKWARD 2	Evaluate square roots and cube roots	**8.EE.A.2**
FORWARD 1	Solve radical equations and check the solutions	**HSA-REI.A.2**
FORWARD 2	Find inverses of square root and cube root functions	**HSF-BF.B.4.c**

SET 74 • HSF-IF.C.7.b

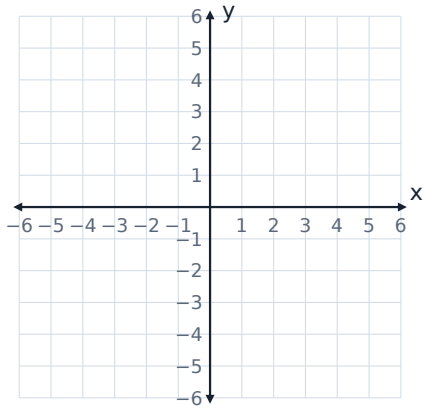
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Graph f(x) = √(x + 4).

Give its domain and range.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

1590

QUESTION ID

S74-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve starting at (−4, 0) and rising to the right through (−3, 1), (0, 2), and (5, 3)

Written: Domain x ≥ −4; range y ≥ 0

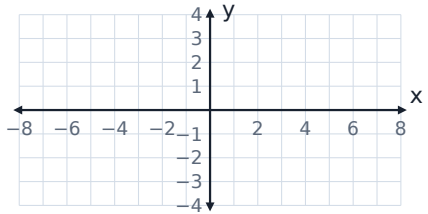
GRADING NOTE Form required: the domain and the range. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $g(x) = \sqrt[3]{x}$ for $-8 \le x \le 8$. Plot the points where x is a perfect cube, then draw the curve.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

1591

QUESTION ID

S74-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A curve through (−8, −2), (−1, −1), (0, 0), (1, 1), and (8, 2), increasing everywhere and flattest away from the origin

GRADING NOTE Any correct graph through these five points.

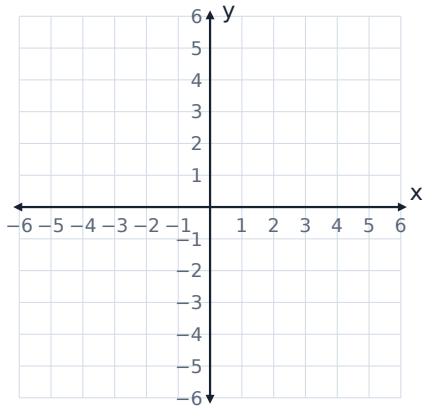
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph h(x) = −√x + 3.

Give its starting point and its range.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q3/5

1592

QUESTION ID

S74-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve starting at (0, 3) and falling to the right through (1, 2) and (4, 1)

Written: Starting point (0, 3); range y ≤ 3

GRADING NOTE Form required: the starting point and the range. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q3/5

Which describes the graph of $y = \sqrt[3]{x}$?

A. It starts at (0, 0) and goes only to the right.

B. It has the horizontal asymptote y = 0.

C. It passes through (−8, −2), (0, 0), and (8, 2).

D. It is V-shaped with vertex (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

1593

QUESTION ID

S74-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. It passes through (−8, −2), (0, 0), and (8, 2).

GRADING NOTE Cube roots of negative numbers are negative, so the graph extends in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = \sqrt{(x - 1)}$ starts at $(-1, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q5/5

1594

QUESTION ID

S74-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x - 1)}$ is defined for $x \geq 1$, so the graph starts at $(1, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.b

BACKWARD 1

Find domains and ranges of root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $g(x) = \sqrt{(2x - 6)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

1596

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 3$

GRADING NOTE Form required: an inequality (or interval). $2x - 6 \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of h(x) = √x + 2?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

1597

QUESTION ID

S74-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y ≥ 2

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q2/5

What is the domain of $y = \sqrt[3]{(x - 5)}$?

A. All real numbers

B. $x \leq 5$

C. $x > 5$

D. $x \geq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

1598

QUESTION ID

S74-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

GRADING NOTE Every real number has a real cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range of $y = -\sqrt{x}$ is $y \leq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

1599

QUESTION ID

S74-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{x} \geq 0$, so $-\sqrt{x} \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q4/5

1599

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = -1$ is in the domain of $y = \sqrt{x}$ over the real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q5/5

1600

QUESTION ID

S74-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(-1)}$ is not a real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 74 • BACKWARD 1 • Q5/5

1600

SET 74 • HSF-IF.C.7.b

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt{81}$ and $\sqrt[3]{(-27)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

1602

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9 and −3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 125$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

1603

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\sqrt[3]{8} + \sqrt{4}$?

A. 6

B. 12

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

1604

QUESTION ID

S74-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE 2 + 2 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{0.25} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

1605

QUESTION ID

S74-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5^2 = 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt[3]{(-8)}$ is not a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

1606

QUESTION ID

S74-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt[3]{(-8)} = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check the solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x + 4) = 3.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

1608

QUESTION ID

S74-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. x + 4 = 9. Check: √9 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{(2x)} = -2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

1609

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form. 2x = −8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

Solve $\sqrt{(x + 2)} = x$.

A. $x = -1$ only

B. $x = 2$ or $x = -1$

C. No solution

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

1610

QUESTION ID

S74-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 only

GRADING NOTE $x^2 - x - 2 = 0$ gives 2 or −1; $x = -1$ is extraneous because $\sqrt{1} = 1$, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(x - 3)} = -2$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q4/5

1611

QUESTION ID

S74-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A principal square root is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q4/5

True or false?

Cubing both sides of $\sqrt[3]{x} = 4$ gives $x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q5/5

1612

QUESTION ID

S74-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 2

Find inverses of square root and cube root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$ for $x \geq 0$. Find $f^{-1}(5)$.

$f^{-1}(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

1614

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form. $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = \sqrt[3]{x}$. Find $g^{-1}(-2)$.

$g^{-1}(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

1615

QUESTION ID

S74-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $\sqrt[3]{(-8)} = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $f(x) = \sqrt{(x - 1)}$, for $x \geq 1$?

A. $f^{-1}(x) = \sqrt{(x + 1)}$

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

C. $f^{-1}(x) = x^2 - 1$

D. $f^{-1}(x) = (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

1616

QUESTION ID

S74-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

GRADING NOTE $y = \sqrt{(x - 1)}$ gives $x = y^2 + 1$ with $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt{x}$ $(x \geq 0)$ is the reflection of the graph of $y = x^2$ $(x \geq 0)$ across the line $y = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q4/5

1617

QUESTION ID

S74-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$g(x) = \sqrt[3]{x}$ has no inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q5/5

1618

QUESTION ID
S74-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its inverse is $g^{-1}(x) = x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and step functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate piecewise-defined functions	**HSF-IF.A.2**
BACKWARD 2	Order numbers and describe intervals on a number line	**6.NS.C.7.a**
FORWARD 1	Graph the probability distribution of a discrete random variable —..	**HSS-MD.A.1**
FORWARD 2	Read inverse values of piecewise and step functions	**HSF-BF.B.4.c**

SET 75 • HSF-IF.C.7.b

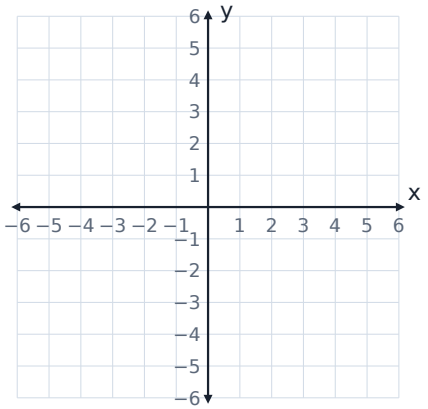
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 1$ for $x < 2$, and
$f(x) = 5 - x$ for $x \geq 2$.

Graph f. Then find f(0), f(2), and f(4).



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

1621

QUESTION ID

S75-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A line rising to (2, 3) from the left, joined at (2, 3) to a line falling to the right through (4, 1); (2, 3) is a filled point

Written: f(0) = 1, f(2) = 3, f(4) = 1

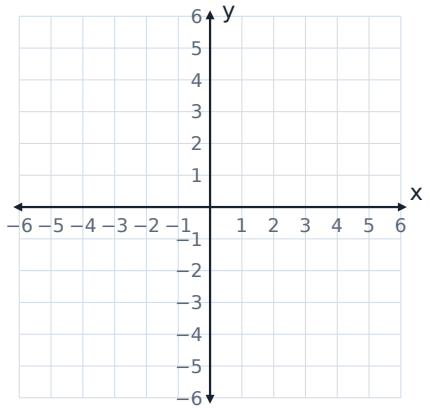
GRADING NOTE Form required: all three values. Both the graph and the values are graded. x = 2 belongs to the second piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

$g(x) = -1$ for $x < 0$, $g(x) = 2$ for $0 \le x < 3$, and $g(x) = x - 1$ for $x \ge 3$.

Graph g. Use open and closed dots at the breaks.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

1622

QUESTION ID

S75-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A horizontal ray at y = −1 for x < 0 with an open dot at (0, −1); a segment at y = 2 from a closed dot at (0, 2) to an open dot at (3, 2); a ray from a closed dot at (3, 2) rising with slope 1

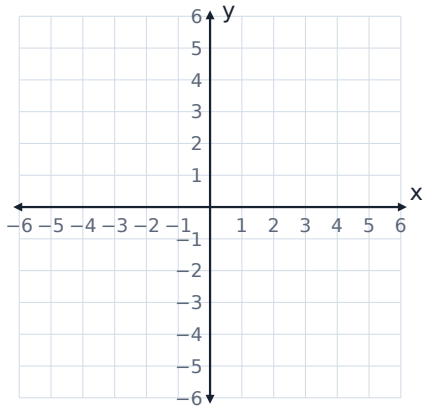
GRADING NOTE Any correct graph with the correct open and closed dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the greatest integer function y = ⌊x⌋ (the greatest integer less than or equal to x) for −2 ≤ x < 3.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

1623

QUESTION ID

S75-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Horizontal steps of length 1: y = −2 for −2 ≤ x < −1, y = −1 for −1 ≤ x < 0, y = 0 for 0 ≤ x < 1, y = 1 for 1 ≤ x < 2, y = 2 for 2 ≤ x < 3; each step has a closed left end and an open right end

GRADING NOTE Any correct graph with the closed and open ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour.

What is the charge for 2.25 hours?

A. $6

B. $6.75

C. $7.50

D. $9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

1624

QUESTION ID

S75-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $9

GRADING NOTE 2.25 hours is 3 started hours: 3 × $3 = $9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

True or false?

For the greatest integer function, ⌊2.7⌋ = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q5/5

1625

QUESTION ID

S75-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⌊2.7⌋ is the greatest integer less than or equal to 2.7, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.b

BACKWARD 1

Evaluate piecewise-defined functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x - 1$ for $x < 3$, and $f(x) = 8$ for $x \geq 3$.

Find $f(3)$ and $f(1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

1627

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

f(3) = 8 and f(1) = 1

GRADING NOTE Form required: both values. x = 3 uses the second piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = |x|$ for $x \le 0$, and $g(x) = x^2$ for $x > 0$.

Find $g(-4)$ and $g(4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

1628

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$g(-4) = 4$ and $g(4) = 16$

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(x) = 3$ for $x < 1$, and $h(x) = 2x$ for $x \geq 1$. What is $h(1)$?

A. 2

B. 5

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

1629

QUESTION ID

S75-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE $x = 1$ uses the second piece: $2(1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$p(x) = 5$ for $x \leq 2$, and $p(x) = x$ for $x > 2$. Then $p(2) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q4/5

1630

QUESTION ID

S75-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 2$ uses the first piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\lfloor -1.5 \rfloor = -1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q5/5

1631

QUESTION ID

S75-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The greatest integer less than or equal to −1.5 is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.b

BACKWARD 2

Order numbers and describe intervals on a number line • Grade 6 • The Number System • 6.NS.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Is −3 to the left or to the right of −1 on a number line? Which number is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

1633

QUESTION ID

S75-B2-Q1

TYPE

Short answer

CORRECT ANSWER

To the left; −1 is greater.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an inequality for "x is at least 0 and less than 3".

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

1634

QUESTION ID

S75-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq x < 3$

GRADING NOTE Form required: a compound inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement is true?

A. $-4 > -2$

B. $-4 < -2$

C. $0 < -1$

D. $-2 < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

1635

QUESTION ID

S75-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-4 < -2$

GRADING NOTE -4 is to the left of -2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

−5 is less than −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

1636

QUESTION ID

S75-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −5 is to the left of −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

True or false?

On a number line, 2 is to the left of −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

1637

QUESTION ID

S75-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 is to the right of −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

FORWARD 1

Graph the probability distribution of a discrete random variable • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A probability distribution assigns a value to each separate outcome, like a function defined piece by piece, and its bar graph is read the same way.

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice, and X is the number of heads. Give P(X = 0), P(X = 1), and P(X = 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

1639

QUESTION ID
S75-F1-Q1

TYPE
Short answer

CORRECT ANSWER
1/4, 1/2, 1/4

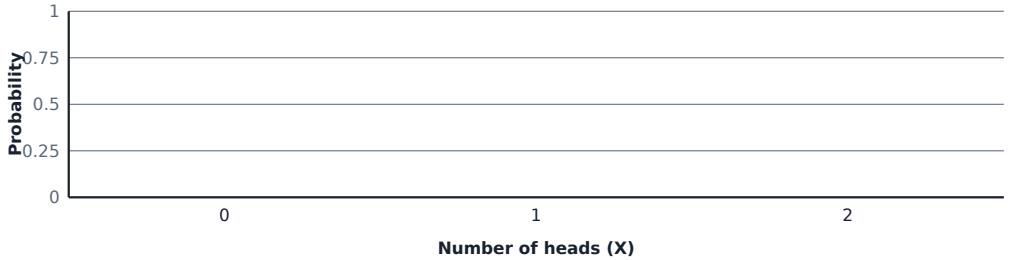
GRADING NOTE Form required: all three. The outcomes HH, HT, TH, TT are equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice, and X is the number of heads. Graph the probability distribution of X as a bar graph.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

1640

QUESTION ID
S75-F1-Q2

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of height 0.25 at X = 0, 0.5 at X = 1, and 0.25 at X = 2

GRADING NOTE Any correct bar graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

X is the result of one roll of a fair number cube numbered 1 to 6. What is P(X ≥ 5)?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

1641

QUESTION ID

S75-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

GRADING NOTE 5 and 6: 2 of 6 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

The probabilities of a probability distribution add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

1642

QUESTION ID
S75-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The outcomes cover every possibility.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For X = the number of heads in 2 fair coin flips, $P(X = 1) = \frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

1643

QUESTION ID
S75-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE HT and TH are 2 of 4 equally likely outcomes, so $P(X = 1) = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 75 • HSF-IF.C.7.b

FORWARD 2

Read inverse values of piecewise and step functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$. This f is one-to-one. Find $f^{-1}(9)$ and $f^{-1}(-4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

1645

QUESTION ID
S75-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(9) = 3$ and $f^{-1}(-4) = -2$

GRADING NOTE Form required: both values. $3^2 = 9$ with $3 \geq 0$, and $2(-2) = -4$ with $-2 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

$g(x) = x + 1$ for $x < 2$, and $g(x) = 2x - 1$ for $x \geq 2$. This g is one-to-one. Find $g^{-1}(7)$.

$g^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

1646

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $x + 1 = 7$ gives 6, which is not less than 2; $2x - 1 = 7$ gives 4, which is at least 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

Why does the greatest integer function $y = \lfloor x \rfloor$ have no inverse function?

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

B. It is a linear function.

C. It is not defined for negative x.

D. Its outputs are not integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

1647

QUESTION ID

S75-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

GRADING NOTE A function needs to be one-to-one to have an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

True or false?

A piecewise function whose pieces are increasing and whose output ranges do not overlap can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

1648

QUESTION ID

S75-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then every output comes from exactly one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

True or false?

If f is one-to-one and $f(3) = 5$, then $f^{-1}(3) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

1649

QUESTION ID
S75-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $f^{-1}(5) = 3$; $f^{-1}(3)$ need not be 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions and show intercepts and end behavior

MAIN	Algebra I main question	**HSF-IF.C.7.e**
BACKWARD 1	Evaluate integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate exponential functions	**HSF-IF.A.2**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Understand logarithms as inverses of exponential functions	**HSF-BF.B.5**

SET 44 • HSF-IF.C.7.e

MAIN

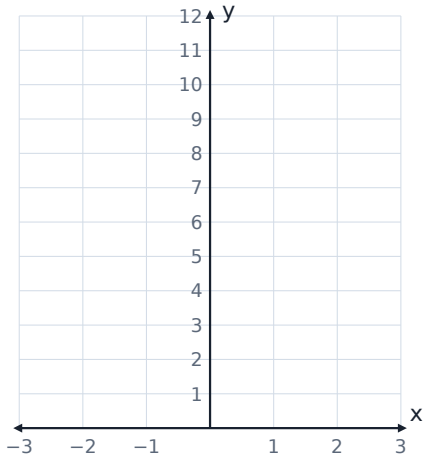
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = 3 \cdot 2^x$ for $-2 \le x \le 2$.

Give its y-intercept and its horizontal asymptote.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

1652

QUESTION ID

S44-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve through (−2, 0.75), (−1, 1.5), (0, 3), (1, 6), and (2, 12), rising more steeply to the right

Written: y-intercept (0, 3); asymptote y = 0

GRADING NOTE Form required: both. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

$g(x) = 50(0.8)^x$.

Is this growth or decay? Give its y-intercept, and say what happens to g(x) as x gets large.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

1653

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

Decay; y-intercept 50; g(x) gets closer and closer to 0.

GRADING NOTE Form required: all three parts. 0.8 < 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

$h(x) = 2^x - 4$.

Find the y-intercept, the x-intercept, and the horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

1654

QUESTION ID
S44-M-Q3

TYPE
Short answer

CORRECT ANSWER
y-intercept (0, −3); x-intercept (2, 0); asymptote y = −4

GRADING NOTE Form required: all three. $2^0 - 4 = -3$, and $2^x = 4$ at $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function's graph rises from left to right and passes through (0, 5)?

A. $y = 1.2(5)^x$

B. $y = 0.8(5)^x$

C. $y = 5(0.8)^x$

D. $y = 5(1.2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

1655

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 5(1.2)^x$

GRADING NOTE A base greater than 1 gives growth, and the coefficient is the y-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = 4(0.5)^x$ crosses the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

1656

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4(0.5)^x$ is always positive; the x-axis is an asymptote.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 1

Evaluate integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{-2}.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

1658

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form, such as 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $3 \cdot 2^{-1}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

1659

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE Any equivalent form, such as 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which number equals $(0.5)^{-1}$?

A. -0.5

B. 0.5

C. 2

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

1660

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $1 \div 0.5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

$5^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

1661

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any nonzero number to the 0 power is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^{-1} = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

1662

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^{-1} = 1 \div 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 2

Evaluate exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 50(0.8)^x$.

Find f(2).

f(2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

1664

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form. 50(0.64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2^x - 4$.

Find g(3).

g(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

1665

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 8 − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

$h(x) = 3 \cdot 2^x$.

Find $h(-1)$.

A. -1.5

B. 6

C. -6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

1666

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1.5

GRADING NOTE $3 \cdot \frac{1}{2} = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

If $f(x) = 10^x$, then $f(3) = 1{,}000$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

1667

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

If $f(x) = 3^x$, then $f(2) = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

1668

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 2^x = 48$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

1670

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2^x = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $50(0.8)^x = 20$. Round to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

1671

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.1

GRADING NOTE Form required: rounded to the nearest tenth. $x = \log 0.4 \div \log 0.8 \approx 4.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 12$. (log means base 10.)

A. x = log 6

B. x = log 12 ÷ log 2 ≈ 3.58

C. x = 3.5

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

1672

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = log 12 ÷ log 2 ≈ 3.58

GRADING NOTE Take the logarithm of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

log 1 = 0 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

1673

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

log 100 = 2 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

1674

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 2

Understand logarithms as inverses of exponential functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

1676

QUESTION ID
S44-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$\log_2 x$

GRADING NOTE Any equivalent form, such as log x ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\log_3 81 = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

1677

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3^4 = 81$, and $\log_3 81$ is the exponent on 3 that gives 81.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

The graph of $y = \log_2 x$ is the reflection of the graph of $y = 2^x$ across which line?

A. $x = 0$

B. $y = 0$

C. $y = x$

D. $y = -x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

1678

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = x

GRADING NOTE Inverse functions reflect across y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

True or false?

The domain of y = log x is x > 0. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

1679

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the range of $y = 10^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

True or false?

$\log_2(2^5) = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

1680

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log_2(2^5) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a • INTERPRETING FUNCTIONS

Factor and complete the square to show zeros, extreme values, and symmetry, and interpret them

MAIN	Algebra I main question	**HSF-IF.C.8.a**
BACKWARD 1	Complete the square	**HSA-SSE.B.3.b**
BACKWARD 2	Use the greatest common factor	**6.NS.B.4**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Complete the square to find the center of a hyperbola	**HSG-GPE.A.3**

SET 45 • HSF-IF.C.8.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

$f(x) = x^2 - 8x + 12$.

Write f in factored form and in vertex form. Give the zeros, the vertex, and the axis of symmetry.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

1683

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x − 6) and (x − 4)² − 4; zeros 2 and 6; vertex (4, −4); axis x = 4

GRADING NOTE Form required: both forms and all three features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company's profit is $P(x) = -2x^2 + 120x - 1{,}000$ dollars when it charges x dollars.

Complete the square to find the price that gives the most profit, and the greatest profit.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

1684

QUESTION ID

S45-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$P(x) = -2(x - 30)^2 + 800$; a price of $30 gives the greatest profit, $800.

GRADING NOTE REQUIRED WORK: The completed square shown:
$-2(x^2 - 60x) - 1{,}000 = -2(x^2 - 60x + 900) + 1{,}800 - 1{,}000 = -2(x - 30)^2 + 800$.

Form required: the completed square with its steps, the price, and the profit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A rock's height is $h(t) = -16t^2 + 32t + 48$ feet after t seconds.

Factor to find when it hits the ground.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

1685

QUESTION ID
S45-M-Q3

TYPE
Short answer

CORRECT ANSWER
$h(t) = -16(t - 3)(t + 1)$; at t = 3 seconds

GRADING NOTE Form required: the factored form and the time. $t = -1$ is not in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which form of $f(x) = x^2 + 6x + 5$ shows the vertex directly?

A. $(x + 1)(x + 5)$

B. $x^2 + 6x + 5$

C. $(x + 3)^2 - 4$

D. $x(x + 6) + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

1686

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(x + 3)^2 - 4$

GRADING NOTE Vertex form $a(x - h)^2 + k$ shows the vertex $(-3, -4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

True or false?

The axis of symmetry of $y = (x - 1)(x - 7)$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

1687

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The axis is halfway between the zeros 1 and 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 1

Complete the square • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 - 8x + 12$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

1689

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 4)^2 - 4$

GRADING NOTE Form required: $(x - h)^2 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 + 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

1690

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1.5)^2 - 2.25$

GRADING NOTE Form required: $(x - h)^2 + k$, with fractions or decimals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

1690

ANSWER KEY Reduced copy of the question page, with its answer

Which expression equals $-(x^2 - 6x)$?

A. $-(x - 3)^2 - 9$

B. $-(x - 3)^2 + 9$

C. $(x - 3)^2 - 9$

D. $-(x + 3)^2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

1691

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-(x - 3)^2 + 9$

GRADING NOTE $x^2 - 6x = (x - 3)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 2x + 10 = (x + 1)^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

1692

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1)^2 = x^2 + 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)^2 - 3$ has its minimum at $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

1693

QUESTION ID
S45-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its minimum is at $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 2

Use the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Use the greatest common factor to write 16 + 40 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

1695

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8(2 + 5)

GRADING NOTE Form required: the GCF times a sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the greatest common factor of 32 and 48.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

1696

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

Which expression equals 36 + 27 and uses the greatest common factor?

A. 9(4 + 3)

B. 3(12 + 6)

C. 6(6 + 27)

D. 9(4 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

1697

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(4 + 3)

GRADING NOTE $9 \cdot 4 = 36$ and $9 \cdot 3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

True or false?

The greatest common factor of 12 and 18 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

1698

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6 divides both, and no larger number does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$20 + 50 = 5(4 + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

1699

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5(4 + 5) = 45$. $20 + 50 = 10(2 + 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Show that $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ by expanding both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

1701

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Left: $x^4 + 2x^2y^2 + y^4$. Right: $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The two sides are equal.

GRADING NOTE Form required: both sides expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $x = 2$ and $y = 1$ in $x^2 - y^2$, $2xy$, and $x^2 + y^2$ to make a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

1702

QUESTION ID
S45-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3, 4, 5

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

Which equation is an identity (true for all numbers a and b)?

A. $a^3 + b^3 = (a + b)^3$

B. $a^3 - b^3 = (a - b)^3$

C. $a^2 + b^2 = (a + b)^2$

D. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

1703

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D.

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

GRADING NOTE Expanding $(a - b)(a^2 + ab + b^2)$ gives $a^3 - b^3$. Each other choice fails for $a = 2$, $b = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

True or false?

$(a + b)^2 - (a - b)^2 = 4ab$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

1704

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

1705

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 2

Complete the square to find the center of a hyperbola • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center of the hyperbola $x^2 - 4y^2 - 6x - 8y - 11 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

1707

QUESTION ID
S45-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(3, −1)

GRADING NOTE Any equivalent form. $(x - 3)^2 - 4(y + 1)^2 = 11 + 9 - 4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

The hyperbola $x^2 \div 16 - y^2 \div 9 = 1$ has foci (5, 0) and (−5, 0).

Show that (5, 2.25) is on the hyperbola and that the absolute difference of its distances to the foci is 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

1708

QUESTION ID

S45-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

It satisfies the equation, and $|10.25 - 2.25| = 8$.

GRADING NOTE

REQUIRED WORK: $25 \div 16 - 5.0625 \div 9 = 1.5625 - 0.5625 = 1$ shown; the distances 2.25 and $\sqrt{(100 + 5.0625)} = 10.25$ computed and subtracted.

Form required: the substitution and both distances. $8 = 2a$ with $a = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation is a hyperbola?

A. $y = x^2 \div 4$

B. $x^2 \div 4 + y^2 \div 9 = 1$

C. $x^2 \div 4 - y^2 \div 9 = 1$

D. $x^2 + y^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

1709

QUESTION ID
S45-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $x^2 \div 4 - y^2 \div 9 = 1$

GRADING NOTE A hyperbola has a difference of squared terms; $x^2 \div 4 + y^2 \div 9 = 1$ is an ellipse, $y = x^2 \div 4$ a parabola, and $x^2 + y^2 = 9$ a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

True or false?

For a hyperbola, the absolute value of the difference of the distances from any of its points to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

1710

QUESTION ID
S45-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the definition of a hyperbola. The signed difference has opposite signs on the two branches, so the absolute value is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

True or false?

The hyperbola $x^2 - y^2 = 1$ passes through (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

1711

QUESTION ID
S45-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $0 - 1 = -1$, not 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Use the properties of exponents to interpret growth and decay rates

MAIN	Algebra I main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth and decay from an exponential function	**HSF-IF.C.7.e**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Sum a geometric series in a savings context	**HSA-SSE.B.4**
FORWARD 2	Find times in exponential models with logarithms	**HSF-LE.A.4**

SET 46 • HSF-IF.C.8.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

A savings balance is $y = 800(1.035)^t$ dollars after t years.

Is this growth or decay, and what is the percent rate of change per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

1714

QUESTION ID
S46-M-Q1

TYPE
Short answer

CORRECT ANSWER
Growth, at 3.5% per year

GRADING NOTE Form required: growth or decay and the percent. 1.035 = 1 + 0.035.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A loan balance grows by 1.5% each month:

$B = 2{,}000(1.015)^{12t}$, where t is in years.

Rewrite it to find the yearly growth factor (to four decimal places) and the yearly percent rate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

1715

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

$B = 2{,}000(1.015^{12})^{t} \approx 2{,}000(1.1956)^{t}$; about 19.56% per year

GRADING NOTE Form required: the yearly factor and the percent. $(1.015)^{12t} = ((1.015)^{12})^{t}$, and $1.015^{12} \approx 1.19562$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

The value of a machine is $V = 45{,}000(0.88)^t$ dollars after t years.

What is the yearly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

1716

QUESTION ID

S46-M-Q3

TYPE

Short answer

CORRECT ANSWER

12% per year

GRADING NOTE Any equivalent form. $0.88 = 1 - 0.12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A forest's area is $A = 600(1.3)^{t/20}$ hectares after t years.

By what percent does A grow every 20 years?

A. 1.3%

B. 13%

C. 3%

D. 30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

1717

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 30%

GRADING NOTE When t increases by 20, the exponent increases by 1, so A is multiplied by 1.3: a 30% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 100(0.5)^{t/8}$ halves every 8 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

1718

QUESTION ID
S46-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each increase of 8 in t multiplies y by 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 1

Read growth and decay from an exponential function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 30(1.5)^x$ increasing or decreasing? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

1720

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Increasing, because the base 1.5 is greater than 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 7(0.9)^x$?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

1721

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form, such as (0, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

Which function is decreasing?

A. $y = 2(1.6)^x$

B. $y = 0.6(1.2)^x$

C. $y = 2(0.6)^x$

D. $y = 0.6(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

1722

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2(0.6)^x$

GRADING NOTE $y = 2(0.6)^x$ is the only one whose base, 0.6, is between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

True or false?

The graph of $y = 3^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

1723

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3^x approaches 0 as x decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

True or false?

$y = 5(1.1)^x$ has y-intercept 1.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

1724

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $40 shirt is marked up 15%.

What is the new price?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

1726

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$46

GRADING NOTE Any equivalent form. 40 × 1.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town grows from 2,000 to 2,300 people.

What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

1727

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15%

GRADING NOTE Any equivalent form. 300 ÷ 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A price drops from $80 to $60.

What is the percent decrease?

A. 25%

B. 20%

C. 33%

D. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

1728

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 25%

GRADING NOTE 20 ÷ 80 = 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying by 1.06 increases a number by 6%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

1729

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.06 = 1 + 0.06.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying by 0.85 decreases a number by 85%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

1730

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases the number by 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 1

Sum a geometric series in a savings context • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

You save $100 the first month and 2% more each month after that.

Write the total for 12 months as a sum formula, and find it to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

1732

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.02^{12} - 1) \div 0.02 \approx \$1{,}341$

GRADING NOTE Form required: the formula and the rounded total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $4 + 4(1.5) + 4(1.5)^2 + 4(1.5)^3$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

1733

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

32.5

GRADING NOTE Any equivalent form. 4 + 6 + 9 + 13.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is the sum of the first n terms of $a + ar + ar^2 + \ldots$ $(r \neq 1)$?

A. $a \div (1 - r)$

B. $a(1 - r) \div (1 - r^n)$

C. $a(1 - r^n) \div (1 - r)$

D. nar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

1734

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $a(1 - r^n) \div (1 - r)$

GRADING NOTE The finite geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1 + 0.5 + 0.25 + 0.125 = 1.875$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

1735

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add the terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

True or false?

The sum of 10 terms of $3 + 3 + 3 + \ldots$ is 3^{10}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

1736

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot 3 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 2

Find times in exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(0.5)^{t/8}$.

Find t when y = 10. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

1738

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26.6

GRADING NOTE Form required: rounded to the nearest tenth. t = 8 · log 0.1 ÷ log 0.5 ≈ 26.58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the continuous rate r for which $e^{10r} = 2$. Round to the nearest thousandth.

r ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

1739

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.069

GRADING NOTE Form required: rounded to the nearest thousandth. r = ln 2 ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

To the nearest tenth, how long does $y = (1.05)^t$ take to double?

A. 20

B. 7.2

C. 10.5

D. 14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

1740

QUESTION ID
S46-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 14.2

GRADING NOTE $t = \log 2 \div \log 1.05 \approx 14.21$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

True or false?

If $y = 2^{t/3}$, then y doubles every 3 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

1741

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Adding 3 to t adds 1 to the exponent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

True or false?

$\log(2^{10}) = 20$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

1742

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(2^{10}) = 10 \log 2 \approx 3.01$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of two functions represented in different ways

MAIN	Algebra I main question	**HSF-IF.C.9**
BACKWARD 1	Interpret key features in context	**HSF-IF.B.4**
BACKWARD 2	Compute average rates of change	**HSF-IF.B.6**
FORWARD 1	Compare periodic functions given in different forms	**HSF-TF.B.5**
FORWARD 2	Compare normal distributions by mean and standard deviation —…	**HSS-ID.A.4**

SET 47 • HSF-IF.C.9

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 3x + 2. The table shows g.

Which function has the greater value at x = 3? Which has the greater average rate of change from x = 0 to x = 3?

x	0	1	2	3
g(x)	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

1745

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

g at x = 3 (16 compared with 11); g has the greater rate (5 compared with 3).

GRADING NOTE Form required: both comparisons with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

Function A is $y = -2(x - 3)^2 + 10$. Function B is a parabola that opens down with vertex (1, 12).

Which has the greater maximum value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

1746

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

B: its maximum is 12, and A's is 10.

GRADING NOTE Form required: the function and both maxima.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Function f has y-intercept 4 and increases by 5 for each 1-unit increase in x. g(x) = 5x − 1.

Compare their slopes and y-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

1747

QUESTION ID

S47-M-Q3

TYPE

Short answer

CORRECT ANSWER

Same slope, 5; f has the greater y-intercept (4 compared with −1), so their graphs are parallel lines.

GRADING NOTE Form required: both comparisons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 1.08^x$ grows 8% per year. The table shows Q.

Which grows faster?

x	0	1	2
Q(x)	1	1.1	1.21

A. P: its graph is a line.

B. Q: it grows 10% per year.

C. P: it grows 8% per year and Q grows 1%.

D. They grow at the same rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

1748

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Q: it grows 10% per year.

GRADING NOTE Q is multiplied by 1.1 each year: 1, 1.1, 1.21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(x) = x^2$, and g is a parabola opening up with vertex (0, −4). The two functions have the same minimum value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

1749

QUESTION ID
S47-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The minimum of f is 0 and the minimum of g is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 1

Interpret key features in context • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

T(h) is the temperature h hours after midnight, and its minimum is T(6) = 4.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

1751

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The lowest temperature, 4 degrees, happened 6 hours after midnight (6 a.m.).

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of savings S(m) is horizontal from m = 3 to m = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

1752

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The savings stayed the same from month 3 to month 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A function's graph is an unbroken curve. It is positive for $0 < x < 8$ and negative for $x > 8$.

What is true at $x = 8$?

A. The output is greatest.

B. The output is 8.

C. The output is 0.

D. The function is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

1753

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The output is 0.

GRADING NOTE An unbroken curve that changes sign passes through the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A y-intercept gives the starting value of a quantity that changes over time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

1754

QUESTION ID
S47-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is the output at time 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

An x-intercept tells the starting value of a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

1755

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An x-intercept is where the output is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 2

Compute average rates of change • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$.

Find the average rate of change from $x = 1$ to $x = 3$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

1757

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (8 − 2) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A savings account had $100 at week 0 and $180 at week 4.

What is the average rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

1758

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$20 per week

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = x^2 - 1$.

What is the average rate of change from $x = -1$ to $x = 2$?

A. 1

B. -1

C. 3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

1759

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1

GRADING NOTE $(3 - 0) \div 3 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

True or false?

The average rate of change of $y = 5$ is 0 on every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

1760

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The output never changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

1760

True or false?

The average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

1761

QUESTION ID
S47-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 1

Compare periodic functions given in different forms • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3sin(2x). Function g has amplitude 4 and period π.

Which has the greater amplitude? Do they have the same period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

1763

QUESTION ID
S47-F1-Q1

TYPE
Short answer

CORRECT ANSWER
g has the greater amplitude (4 compared with 3); yes, both periods are π.

GRADING NOTE Form required: both parts. The period of f is 2π ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tide A rises to 12 ft and falls to 2 ft. Tide B is $h(t) = 6 + 4\sin t$.

Which has the greater amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

1764

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Tide A: 5 ft compared with 4 ft.

GRADING NOTE Form required: the tide and both amplitudes. (12 − 2) ÷ 2 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

What is the period of y = sin(4x)?

A. π ÷ 2

B. π ÷ 4

C. 2π

D. 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

1765

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. π ÷ 2

GRADING NOTE 2π ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

True or false?

y = 2cos x and y = 2sin x have the same amplitude and the same period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

1766

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both have amplitude 2 and period 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

True or false?

y = 5sin x has a maximum value of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

1767

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its maximum value is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 2

Compare normal distributions by mean and standard deviation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Comparing two functions by their key features leads to comparing two normal curves by their center and spread.

ANSWER KEY Reduced copy of the question page, with its answer

Test A scores are normal with mean 70 and standard deviation 5. Test B scores are normal with mean 70 and standard deviation 10.

Which test's scores are more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

1769

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Test B: its standard deviation is larger.

GRADING NOTE Form required: the test and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Heights are normal with mean 64 in and standard deviation 3 in.

About what percent are between 61 in and 67 in?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

1770

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 68%

GRADING NOTE Any equivalent form. That interval is within 1 standard deviation of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Scores are normal with mean 100 and standard deviation 15.

About what percent of scores are above 130?

A. About 16%

B. About 34%

C. About 2.5%

D. About 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

1771

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. About 2.5%

GRADING NOTE 130 is 2 standard deviations above the mean; about 95% lie within 2, so about 2.5% lie above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A normal distribution is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

1772

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its curve is a symmetric bell shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal distribution, about 95% of the data lie within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

1773

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About 68% lie within 1; about 95% lie within 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

Write an explicit expression or a recursive process for a function from a context

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Evaluate a sequence from a recursive or explicit rule	**HSF-IF.A.3**
BACKWARD 2	Write equations in two or more variables	**HSA-CED.A.2**
FORWARD 1	Build a composite function from a context	**HSF-BF.A.1.c**
FORWARD 2	Find and interpret the inverse of a model	**HSF-BF.B.4.b**

SET 48 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A phone plan has a $60 activation fee and costs $35 per month.

Write an explicit expression for the total cost C(n) after n months, and a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

1776

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

C(n) = 60 + 35n; C(0) = 60 and C(n) = C(n − 1) + 35 for n ≥ 1

GRADING NOTE Form required: both the explicit expression and the recursive rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tree is 4 ft tall and grows 1.5 ft per year.

Write a function for its height h(t) after t years.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

1777

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 + 1.5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball is dropped from 10 m. Each bounce reaches 60% of the height of the bounce before.

Write a function for the height h(n) of the nth bounce.

h(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

1778

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

$10(0.6)^n$

GRADING NOTE Any equivalent form. The first bounce reaches 10(0.6) = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows n = 1, 2, 3, 4 and a(n) = 7, 11, 15, 19.

Which recursive process fits?

A. a(1) = 4, a(n) = a(n − 1) + 7

B. a(1) = 7, a(n) = 4a(n − 1)

C. a(1) = 7, a(n) = a(n − 1) + 4

D. a(n) = 7n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

1779

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a(1) = 7, a(n) = a(n − 1) + 4

GRADING NOTE Start at 7 and add 4 each time. (7n + 4 gives 11 for n = 1.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The steps "start at 500 and multiply by 1.03 each year" give f(t) = 500 + 1.03t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

1780

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Repeated multiplication gives $f(t) = 500(1.03)^t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 1

Evaluate a sequence from a recursive or explicit rule • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 2$ and $f(n) = 3f(n - 1)$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

1782

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form. 2, 6, 18, 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 10 - 2n$ for $n \geq 1$.

List the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

1783

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8, 6, 4, 2

GRADING NOTE Form required: the four terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 5$ and $f(n) = f(n - 1) - 3$.

What is $f(3)$?

A. -1

B. 11

C. 2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

1784

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −1

GRADING NOTE 5, 2, −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

True or false?

The terms f(1), f(2), f(3) of a sequence are its outputs for the inputs 1, 2, 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

1785

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A sequence is a function on the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(n) = n^2$ gives the terms 1, 2, 3, 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

1786

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives 1, 4, 9, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 2

Write equations in two or more variables • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A car rental costs $45 per day plus $0.20 per mile.

Write an equation for the cost C for d days and m miles.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

1788

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 45d + 0.20m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for the perimeter P of a rectangle with length l and width w.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

1789

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

P = 2l + 2w

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A $900 laptop loses $150 of value each year.

Which equation gives its value V after t years?

A. $V = 900 - 150t$

B. $V = 150 - 900t$

C. $V = 900(150)^t$

D. $V = 900 + 150t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

1790

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. V = 900 − 150t

GRADING NOTE Start at 900 and subtract 150 per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

True or false?

$A = s^2$ gives the area of a square with side s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

1791

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Side times side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 12x gives the number of feet in x inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

1792

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 12x gives the number of inches in x feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 1

Build a composite function from a context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

An oil spill's radius is $r(t) = 2t$ meters after t hours, and the area of a circle is $A(r) = \pi r^2$.

Write $A(r(t))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

1794

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$A(r(t)) = 4\pi t^2$: the area of the spill after t hours.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store takes 20% off and then $5 off: d(x) = 0.8x and c(x) = x − 5.

Find c(d(50)).

c(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

1795

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35

GRADING NOTE Any equivalent form. d(50) = 40, and c(40) = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

With d(x) = 0.8x and c(x) = x − 5, which expression is d(c(x))?

A. 0.8x − 4

B. 0.8x + 4

C. 0.8x − 5

D. x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

1796

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.8x − 4

GRADING NOTE 0.8(x − 5) = 0.8x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

True or false?

A store takes 20% off and then $5 off: d(x) = 0.8x and c(x) = x − 5. For these functions, c(d(50)) and d(c(50)) are different.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

1797

QUESTION ID
S48-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE c(d(50)) = 35 and d(c(50)) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

True or false?

If f(x) = 3x and g(x) = x + 2, then f(g(x)) = 3x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

1798

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(x)) = 3(x + 2) = 3x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 2

Find and interpret the inverse of a model • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

C(n) = 60 + 35n is the total cost after n months.

Find the inverse and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

1800

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

n = (C − 60) ÷ 35: the number of months for a total cost of C dollars.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that f(x) = 5x − 2 and g(x) = (x + 2) ÷ 5 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

1801

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = (x + 2) − 2 = x and g(f(x)) = (5x − 2 + 2) ÷ 5 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

F = 1.8C + 32 converts Celsius to Fahrenheit.

Which formula is its inverse?

A. C = F ÷ 1.8 − 32

B. C = 1.8F + 32

C. C = (F + 32) ÷ 1.8

D. C = (F − 32) ÷ 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

1802

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. C = (F − 32) ÷ 1.8

GRADING NOTE Subtract 32, then divide by 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

True or false?

If f is invertible and $f(2) = 9$, then $f^{-1}(9) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

1803

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The inverse reverses each input-output pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = 2x$ is $f^{-1}(x) = -2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

1804

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f^{-1}(x) = x \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

Build quadratic functions that model situations

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Write equations in two variables for areas and perimeters	**HSA-CED.A.2**
BACKWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
FORWARD 1	Use quadratic models to make the best design choice	**HSG-MG.A.3**
FORWARD 2	Compose functions to build quadratic models	**HSF-BF.A.1.c**

SET 80 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A farmer has 100 m of fence for three sides of a rectangular pen; a barn wall forms the fourth side. Each side perpendicular to the wall is x m long.

Write the area A(x) of the pen.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

1807

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x(100 - 2x)$

GRADING NOTE Any equivalent form, such as $100x - 2x^2$. The side parallel to the wall uses the remaining $100 - 2x$ meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A theater sells 300 tickets at $20 each. For each $1 increase in price, it sells 10 fewer tickets.

Write the revenue R(n) after n price increases of $1.

R(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

1808

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

(20 + n)(300 − 10n)

GRADING NOTE Any equivalent form, such as $-10n^2 + 100n + 6{,}000$. Revenue = price × tickets sold.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A stone is thrown upward at 20 m/s from the top of a 5-m wall. Using $h(t) = -4.9t^2 + v_0t + h_0$, where v_0 is the starting upward speed and h_0 is the starting height, write the stone's height h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

1809

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-4.9t^2 + 20t + 5$

GRADING NOTE Any equivalent form. $v_0 = 20$ and $h_0 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table has x = 0, 1, 2, 3 and y = 3, 5, 11, 21.

Which function fits the table?

A. $y = 3(2)^x$

B. $y = 2x + 3$

C. $y = 2x^2 + 3$

D. $y = x^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

1810

QUESTION ID

S80-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2x^2 + 3$

GRADING NOTE The second differences are all 4, so the x^2 coefficient is 2; y(0) = 3. Check: 2(9) + 3 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If each of n people shakes hands once with every other person, the number of handshakes is $h(n) = n(n - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

1811

QUESTION ID

S80-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each handshake is counted twice in $n(n - 1)$, so $h(n) = n(n - 1) \div 2$ (still a quadratic function).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 1

Write equations in two variables for areas and perimeters • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has length 2w and width w. Write an equation for its perimeter P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

1813

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6w

GRADING NOTE Any equivalent form, such as 2(2w) + 2w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has base b and height 2b. Write an equation for its area A.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

1814

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

b^2

GRADING NOTE Any equivalent form, such as $\frac{1}{2}(b)(2b)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

A rectangle has perimeter 40 and width w. Which equation gives its area A?

A. $A = w^2 + 20$

B. $A = w(40 - w)$

C. $A = 20w$

D. $A = w(20 - w)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

1815

QUESTION ID

S80-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A = w(20 − w)

GRADING NOTE The length is 20 − w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

True or false?

The area of a circle as a function of its radius is $A = \pi r^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

1816

QUESTION ID

S80-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the area formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

True or false?

If a rectangle's length is its width w plus 2, then its area is w + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

1817

QUESTION ID

S80-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is w(w + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 2

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

In R = (20 + n)(300 − 10n), what does the factor 300 − 10n represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

1819

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of tickets sold after n price increases of $1.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In A = x(100 − 2x) for a pen against a wall, what does 100 − 2x represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

1820

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The length of the side parallel to the wall, in meters.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In $h = -4.9t^2 + 20t + 5$, what does 5 represent?

A. The starting upward speed

B. The starting height

C. The effect of gravity

D. The time in the air

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

1821

QUESTION ID

S80-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The starting height

GRADING NOTE At t = 0, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $-4.9t^2 + 20t + 5$, the coefficient of t is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

1822

QUESTION ID

S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The term is 20t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

1822

True or false?

The constant term of $2w^2 + 3w$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

1823

QUESTION ID

S80-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There is no constant term (it is 0); 3 is the coefficient of w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 1

Use quadratic models to make the best design choice • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

For a pen with area A(x) = x(100 − 2x), which x gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

1825

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 25 m; 1,250 m²

GRADING NOTE Form required: x and the area. The zeros are 0 and 50, so the vertex is at x = 25; A(25) = 25 · 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For R(n) = (20 + n)(300 − 10n), which number of $1 increases gives the most revenue? What are the ticket price and the revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

1826

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

n = 5: a price of $25 and revenue of $6,250

GRADING NOTE Form required: n, the price, and the revenue. The zeros are −20 and 30, so the vertex is at n = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 24-m rope is laid out as a rectangle. Which dimensions give the greatest area?

A. 2 m by 10 m

B. 12 m by 12 m

C. 6 m by 6 m

D. 4 m by 8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

1827

QUESTION ID

S80-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 m by 6 m

GRADING NOTE Area = w(12 − w) is greatest at w = 6 (36 m²). A 12 m by 12 m rectangle would need 48 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

True or false?

Of all rectangles with perimeter 40, the square has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

1828

QUESTION ID

S80-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A = w(20 − w) is greatest at w = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

True or false?

For A(x) = x(100 − 2x), the area is greatest when x = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

1829

QUESTION ID
S80-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A(50) = 0; the greatest area is at x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 2

Compose functions to build quadratic models • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Ticket demand is $q(p) = 500 - 10p$ tickets at a price of p dollars, and the price after n increases is $p(n) = 20 + n$.

Write $q(p(n))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

1831

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$q(p(n)) = 300 - 10n$: the number of tickets sold after n price increases.

GRADING NOTE Form required: the simplified composite and its meaning. $500 - 10(20 + n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square's area is $A(s) = s^2$, and its side grows with time: $s(t) = 2 + 0.5t$.

Write $A(s(t))$ in standard form.

$A(s(t)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

1832

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$0.25t^2 + 2t + 4$

GRADING NOTE Form required: standard form. $(2 + 0.5t)^2 = 4 + 2t + 0.25t^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x + 3$. Which is $f(g(x))$ in standard form?

A. $x^2 + 3x$

B. $x^2 + 9$

C. $x^2 + 3$

D. $x^2 + 6x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

1833

QUESTION ID

S80-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 + 6x + 9$

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $g(f(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

1834

QUESTION ID

S80-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(3) = 9$ and $g(9) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $f(g(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

1835

QUESTION ID

S80-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $g(3) = 6$ and $f(6) = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

Combine standard function types using arithmetic operations

MAIN	Algebra I main question	**HSF-BF.A.1.b**
BACKWARD 1	Add, subtract, and multiply polynomials	**HSA-APR.A.1**
BACKWARD 2	Evaluate and interpret functions	**HSF-IF.A.2**
FORWARD 1	Rewrite rational expressions such as average cost	**HSA-APR.D.6**
FORWARD 2	Graph and interpret an average-cost rational function	**HSF-IF.C.7.d**

SET 49 • HSF-BF.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Revenue is R(x) = 25x and cost is C(x) = 400 + 9x dollars for x items.

Write the profit function P(x) in simplest form and find P(50).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

1838

QUESTION ID
S49-M-Q1

TYPE
Short answer

CORRECT ANSWER
P(x) = 16x − 400; P(50) = 400

GRADING NOTE Form required: P(x) simplified and the value. P = R − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -4.9t^2 + 30t + 2$ meters after t seconds.

Name the three simple functions added together, and what each one describes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

1839

QUESTION ID
S49-M-Q2

TYPE
Short answer

CORRECT ANSWER
The constant 2 (the starting height), the linear 30t (height gained from the upward speed), and the quadratic $-4.9t^2$ (the pull of gravity).

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2 + 1$ and $g(x) = 2x - 3$.

Find $(f - g)(x)$ in simplest form, and find $(f \cdot g)(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

1840

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(f - g)(x) = x^2 - 2x + 4$; $(f \cdot g)(2) = 5$

GRADING NOTE Form required: the simplified difference and the value. $f(2) = 5$ and $g(2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

The temperature of tea is $T(t) = 20 + 70(0.9)^t$ °C after t minutes.

Which description fits this combination of functions?

A. A linear function with slope 0.9

B. An exponential function that grows 90% per minute

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

D. A constant temperature of 90 °C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

1841

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

GRADING NOTE The constant 20 is added to an exponential decay term that starts at 70.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x^2$, then $(f + g)(2) = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

1842

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(2) + g(2) = 6 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 1

Add, subtract, and multiply polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Subtract.

$25x - (400 + 9x)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

1844

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 400

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$(x^2 + 1) + (2x - 3)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

1845

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 2$

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x^2 + 1)(2x - 3)$

A. $2x^3 + 2x - 3$

B. $2x^2 + 2x - 3$

C. $2x^3 - 3x^2 - 3$

D. $2x^3 - 3x^2 + 2x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

1846

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x^3 - 3x^2 + 2x - 3$

GRADING NOTE Multiply each term of the first by each term of the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

The product of two polynomials is a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

1847

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Polynomials are closed under multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

$(x + 2) - (x - 2) = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

1848

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 2) - (x - 2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 2

Evaluate and interpret functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 16x - 400.$

Find x so that $P(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

1850

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 20 + 70(0.9)^t$.

Find T(0).

T(0) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

1851

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90

GRADING NOTE Any equivalent form. $(0.9)^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

R(x) is the revenue in dollars from selling x items.

What does R(10) = 250 mean?

A. Selling 250 items brings in $10.

B. Each item costs $10.

C. The profit is $250.

D. Selling 10 items brings in $250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

1852

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Selling 10 items brings in $250.

GRADING NOTE Input: items. Output: dollars of revenue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

True or false?

If C(x) = 400 + 9x, then C(0) = 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

1853

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9 \cdot 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

If $h(t) = -4.9t^2 + 30t + 2$, then $h(0) = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

1854

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $h(0) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 1

Rewrite rational expressions such as average cost • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Write the average cost A(x) = (400 + 9x) ÷ x as a constant plus a fraction.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

1856

QUESTION ID
S49-F1-Q1

TYPE
Short answer

CORRECT ANSWER
9 + 400 ÷ x

GRADING NOTE Form required: a constant plus a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

Assume $x \neq -1$. Write $(x^2 + 3x + 5) \div (x + 1)$ in the form $q(x) + r \div (x + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

1857

QUESTION ID
S49-F1-Q2

TYPE
Short answer

CORRECT ANSWER
x + 2 + 3 ÷ (x + 1)

GRADING NOTE Form required: $q(x) + r \div (x + 1)$. $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 0$. Which expression equals $(6x^2 + 4x) \div (2x)$?

A. $3x + 4x$

B. $3x^2 + 2$

C. $6x + 2$

D. $3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

1858

QUESTION ID
S49-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 3x + 2

GRADING NOTE Divide each term by 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

True or false?

$(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

1859

QUESTION ID
S49-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 8) \div x = 9$ for $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

1860

QUESTION ID
S49-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 8) \div x = 1 + 8 \div x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 2

Graph and interpret an average-cost rational function • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

A(x) = 9 + 400 ÷ x is the average cost per item, in dollars, for x items.

Give its horizontal asymptote and what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

1862

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 9: for very many items, the average cost gets close to $9.

GRADING NOTE Form required: the asymptote and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Find A(100).

A(100) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

1863

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 9 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

What is the vertical asymptote of y = 400 ÷ x + 9?

A. x = 400

B. x = 9

C. y = 9

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

1864

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 0

GRADING NOTE The function is undefined at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

True or false?

As x increases, 9 + 400 ÷ x decreases toward 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

1865

QUESTION ID
S49-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

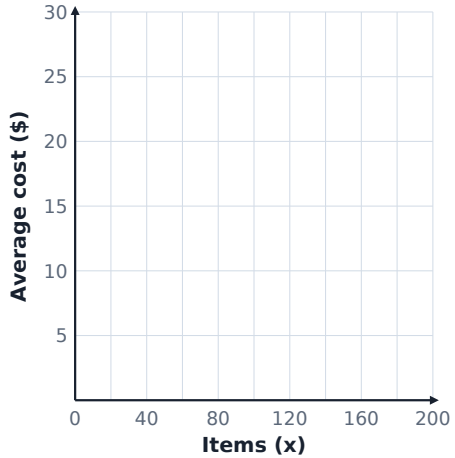
GRADING NOTE 400 ÷ x shrinks toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A(x) = 9 + 400 ÷ x is the average cost per item, in dollars, for x items. Graph A for 20 ≤ x ≤ 200, and draw its horizontal asymptote.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

1866

QUESTION ID

S49-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A decreasing curve through (20, 29), (40, 19), (50, 17), (100, 13), and (200, 11), approaching the line y = 9 from above

GRADING NOTE Any correct graph with the asymptote y = 9 drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

Write arithmetic and geometric sequences recursively and with an explicit formula

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate recursively defined sequences	**HSF-IF.A.3**
BACKWARD 2	Generate a pattern from a rule and describe its features	**4.OA.C.5**
FORWARD 1	Sum geometric sequences	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find a term number	**HSF-LE.A.4**

SET 50 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula and a recursive formula for 7, 4, 1, −2, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

1869

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

a(n) = 10 − 3n;
a(1) = 7 and
a(n) = a(n − 1) − 3

GRADING NOTE Form required: both formulas. The explicit formula may be written 7 − 3(n − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula and a recursive formula for 3, 12, 48, 192, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

1870

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

$a(n) = 3 \cdot 4^{n-1}$;
$a(1) = 3$ and
$a(n) = 4a(n - 1)$

GRADING NOTE Form required: both formulas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An arithmetic sequence has a(3) = 14 and a(8) = 39.

Find a(1) and the common difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

1871

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

a(1) = 4; d = 5

GRADING NOTE Form required: both. 5 steps add 25, so d = 5, and a(1) = 14 − 2(5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A geometric sequence of real numbers has a(1) = 2 and a(4) = 54.

What is the common ratio?

A. 3

B. 18

C. 27

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

1872

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $2r^3 = 54$, so $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The sequence 2, 4, 8, 14, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

1873

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 2, 4, 6, which are not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 1

Evaluate recursively defined sequences • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 4$ and $f(n) = f(n - 1) + 6$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

1875

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. 4, 10, 16, 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 1$ and $f(n) = 2f(n - 1) + 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

1876

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 1, 3, 7, 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2n - 1$.

What is $f(10)$?

A. 20

B. 21

C. 19

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

1877

GRADING NOTE $2(10) - 1$.

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A recursive formula defines each term after the starting term(s) using one or more earlier terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

1878

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(n) = f(n − 1) + 6 uses one earlier term; the rule f(n + 1) = f(n) + f(n − 1) uses two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

True or false?

f(1) = 3 and f(n) = 2f(n − 1) give 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

1879

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They give 3, 6, 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 2

Generate a pattern from a rule and describe its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 5 and add 5.

Write the first 6 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

1881

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5, 10, 15, 20, 25, 30

GRADING NOTE Form required: the six terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 1 and add 4.

Explain why every term is odd.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

1882

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The first term is odd, and adding 4 (an even number) to an odd number gives an odd number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 2 and multiply by 2.

What is the 5th term?

A. 10

B. 16

C. 64

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

1883

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 32

GRADING NOTE 2, 4, 8, 16, 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

1883

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

With the rule "start at 3 and add 3," every term is a multiple of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

1884

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3, 6, 9, 12, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

With the rule "start at 10 and add 10," the 4th term is 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

1885

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10, 20, 30, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 6 terms of 3, 12, 48, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

1887

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4,095

GRADING NOTE Any equivalent form. $3(4^6 - 1) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A job pays $1 on day 1, and the pay doubles each day.

What is the total pay for 10 days?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

1888

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$1,023

GRADING NOTE Any equivalent form. $2^{10} - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 5 + 10 + 20 + 40 + 80?

A. 160

B. 80

C. 155

D. 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

1889

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 155

GRADING NOTE $5(2^5 - 1) = 155$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ works for $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

1890

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It works for every $r \neq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

True or false?

$2 + 6 + 18 + 54 = 2(3^4 - 1)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

1891

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is $80 = 2(3^4 - 1) \div 2$, not 160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 2

Use logarithms to find a term number • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$a(n) = 3 \cdot 4^{n-1}$.

What is the smallest n for which a(n) > 10,000?

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

1893

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $4^{n-1} > 3{,}333.3...$ gives $n - 1 > \log 3{,}333.3 \div \log 4 \approx 5.85$. Check: a(6) = 3,072 and a(7) = 12,288.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence starts at 5 and grows by 20% per term: $a(n) = 5(1.2)^{n-1}$. Which term is the first to exceed 100? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

1894

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. $1.2^{n-1} > 20$ gives $n - 1 > \log 20 \div \log 1.2 \approx 16.4$. Check: $a(17) \approx 92.4$ and $a(18) \approx 110.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^n = 600$. (log means base 10.)

A. n = 7

B. n = log 600 ÷ log 2 ≈ 9.23

C. n = log 120 ÷ log 2 ≈ 6.91

D. n = 120 ÷ 2 = 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

1895

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = log 120 ÷ log 2 ≈ 6.91

GRADING NOTE $2^n = 120$, so n = log 120 ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

If $4^k = 64$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

1896

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

If $3^n = 30$, then n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

1897

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3^{10} is much larger than 30; n = log 30 ÷ log 3 ≈ 3.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

Model situations with arithmetic and geometric sequences and translate between recursive and explicit forms

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate sequences from recursive and explicit rules	**HSF-IF.A.3**
BACKWARD 2	Write a function or a recursive process from a situation	**HSF-BF.A.1.a**
FORWARD 1	Sum geometric sequences that model situations	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find when a sequence reaches a value	**HSF-LE.A.4**

SET 81 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $50 to join and $30 each month. a(n) is the total paid after n months, with a(1) = 80.

Write a recursive rule and an explicit rule for a(n).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

1900

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

a(1) = 80 and
a(n) = a(n − 1) + 30
for n ≥ 2;
a(n) = 50 + 30n

GRADING NOTE Form required: both rules. The explicit rule may be written 80 + 30(n − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball bounces to 80% of its previous height. Its first bounce reaches 5 m.

Write an explicit rule and a recursive rule for the height h(n) of the nth bounce.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

1901

QUESTION ID

S81-M-Q2

TYPE

Short answer

CORRECT ANSWER

$h(n) = 5(0.8)^{n-1}$; $h(1) = 5$ and $h(n) = 0.8h(n - 1)$ for $n \geq 2$

GRADING NOTE Form required: both rules.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Translate the explicit rule a(n) = 7 + 4(n − 1), for n ≥ 1, into a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

1902

QUESTION ID

S81-M-Q3

TYPE

Short answer

CORRECT ANSWER

a(1) = 7 and
a(n) = a(n − 1) + 4
for n ≥ 2

GRADING NOTE Form required: the starting value and the recursion with its condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which explicit rule matches $a(1) = 3$ and $a(n) = 2a(n - 1)$ for $n \geq 2$?

A. $a(n) = 3 \cdot 2^{n-1}$

B. $a(n) = 2 \cdot 3^{n-1}$

C. $a(n) = 3 \cdot 2^{n}$

D. $a(n) = 3 + 2(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

1903

QUESTION ID

S81-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $a(n) = 3 \cdot 2^{n-1}$

GRADING NOTE Each term doubles, starting from 3: 3, 6, 12, ... $3 \cdot 2^{n}$ gives 6 for $n = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

True or false?

The recursive rule a(n) = a(n − 1) − 6 needs a starting value, such as a(1) = 50, to define a sequence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

1904

QUESTION ID

S81-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Without a starting value, many sequences fit the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 1

Evaluate sequences from recursive and explicit rules • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 50$ and $f(n) = f(n - 1) - 6$ for $n \geq 2$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

1906

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form. 50, 44, 38, 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 5(0.8)^{n-1}$.

Find f(3).

f(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

1907

QUESTION ID

S81-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.2

GRADING NOTE Any equivalent form. 5(0.64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

$f(1) = 2$ and $f(n) = 3f(n - 1) + 1$ for $n \geq 2$.

What is $f(3)$?

A. 7

B. 22

C. 19

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

1908

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 22

GRADING NOTE $f(2) = 7$ and $f(3) = 22$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The domain of the sequence a(n) = 2n + 1, n ≥ 1, is the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

1909

QUESTION ID

S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A sequence is defined at term numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the sequence 1, 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

1910

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives 1, 4, 9, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 2

Write a function or a recursive process from a situation • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A savings jar starts with $20 and gains $5 each week.

Write a function for the amount A after w weeks.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

1912

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20 + 5w

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

A culture of 100 cells triples every day.

Write a function for the number of cells N after d days.

N = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

1913

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$100 \cdot 3^d$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which recursive process means "start at 10 and multiply by 1.1 each year"?

A. $a(0) = 10, a(n) = a(n - 1) + 1.1$

B. $a(n) = 10 + 1.1n$

C. $a(0) = 1.1, a(n) = 10a(n - 1)$

D. $a(0) = 10, a(n) = 1.1a(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

1914

QUESTION ID

S81-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $a(0) = 10$, $a(n) = 1.1a(n - 1)$

GRADING NOTE Each year's amount is 1.1 times the year before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

True or false?

Adding the same amount at each step gives a linear function of the step number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

1915

QUESTION ID

S81-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal differences mean linear growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 2 at each step, starting at 1, gives f(n) = 2n.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

1916

QUESTION ID

S81-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives f(n) = 2^n: 1, 2, 4, 8, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences that model situations • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A ball's bounce heights are 5, 4, 3.2, ... meters (each 80% of the one before). Find the total of the first 5 bounce heights, to the nearest hundredth.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

1918

QUESTION ID

S81-F1-Q1

TYPE

Short answer

CORRECT ANSWER

16.81 m

GRADING NOTE Form required: rounded to the nearest hundredth. $5(1 - 0.8^5) \div (1 - 0.8) = 25(0.67232) = 16.808$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You deposit $100 the first month and 3% more each month after that. Find the total deposited in 12 months, to the nearest cent.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

1919

QUESTION ID

S81-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$1,419.20

GRADING NOTE Form required: to the nearest cent. $100(1.03^{12} - 1) \div 0.03 \approx 1{,}419.20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

Which expression is the sum of the first 10 terms of $5(0.8)^{n-1}$?

A. $5(1 - 0.8^{10}) \div 0.2$

B. $5(0.8)^{10}$

C. $10(5)(0.8)$

D. $5 \div 0.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

1920

QUESTION ID

S81-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $5(1 - 0.8^{10}) \div 0.2$

GRADING NOTE $a(1 - r^n) \div (1 - r)$ with $a = 5$ and $r = 0.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

True or false?

The sum of the first n bounce heights 5, 4, 3.2, ... is always less than 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

1921

QUESTION ID

S81-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sum is $25(1 - 0.8^n)$, which is less than 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

True or false?

For a geometric series with ratio 2, adding one more term always doubles the sum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

1922

QUESTION ID

S81-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 + 2 = 3, but 1 + 2 + 4 = 7, not 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 2

Use logarithms to find when a sequence reaches a value • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$h(n) = 5(0.8)^{n-1}$ is the height of the nth bounce, in meters. Which bounce is the first below 1 m? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

1924

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $0.8^{n-1} < 0.2$ gives $n - 1 > \log 0.2 \div \log 0.8 \approx 7.2$. Check: $h(8) \approx 1.05$ and $h(9) \approx 0.84$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$a(n) = 1{,}000(1.05)^n$ is a savings balance after n years. After how many whole years does it first exceed $2,000? (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

1925

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. $n > \log 2 \div \log 1.05 \approx 14.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3^n = 50$. (log means base 10.)

A. n = log 3 ÷ log 50

B. n = 50 ÷ 3

C. n = log 50 ÷ log 3 ≈ 3.56

D. n = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

1926

QUESTION ID

S81-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = log 50 ÷ log 3 ≈ 3.56

GRADING NOTE Take the logarithm of both sides: n log 3 = log 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

True or false?

$\log(0.8^n) = n \log 0.8$, and $\log 0.8$ is negative. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

1927

QUESTION ID

S81-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.8 < 1$, so its logarithm is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $2^n = 10$, then n is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

1928

QUESTION ID

S81-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 = 8$ and $2^4 = 16$, so $n \approx 3.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recognize even and odd functions

MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph absolute value and square root functions	**HSF-IF.C.7.b**
BACKWARD 2	Read key features of quadratic graphs	**HSF-IF.C.7.a**
FORWARD 1	Describe a sequence of rigid transformations	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs	**HSF-TF.B.5**

SET 51 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

$g(x) = (x - 4)^2 + 1$.

Describe how to get the graph of g from the graph of $y = x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

1931

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

Shift it 4 units right and 1 unit up.

GRADING NOTE Any equivalent description of both shifts.

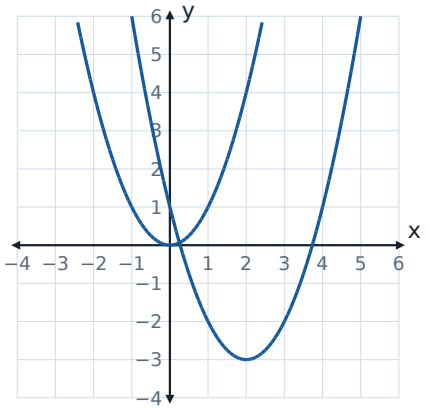
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $y = x^2$ and a translated graph $y = g(x)$.

Write g(x).



g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

1932

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 - 3$

GRADING NOTE Any equivalent form, such as $x^2 - 4x + 1$. The vertex moved from (0, 0) to (2, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$. Compare $g(x) = 3f(x)$ and $h(x) = f(3x)$.

Which is a vertical stretch and which is a horizontal compression? Are g and h the same function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

1933

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

$g(x) = 3x^2$ is a vertical stretch by 3; $h(x) = 9x^2$ is a horizontal compression by a factor of $\frac{1}{3}$. They are not the same.

GRADING NOTE Form required: both descriptions and the comparison. $(3x)^2 = 9x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

$f(x) = x^3$.

Is f even, odd, or neither?

A. Odd: $f(-x) = -f(x)$

B. Even: $f(-x) = f(x)$

C. Both even and odd

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

1934

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Odd: $f(-x) = -f(x)$

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

True or false?

The graph of y = f(x + 2) is the graph of y = f(x) shifted 2 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

1935

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 2 units left: the output at x is f's output at x + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 1

Graph absolute value and square root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = |x + 3| - 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

1937

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give two points on the graph of $y = \sqrt{(x - 1)}$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

1938

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

For example (1, 0) and (5, 2)

GRADING NOTE Any two points with $y = \sqrt{(x - 1)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

What is the vertex of $y = -|x - 1| + 4$?

A. (1, 4)

B. (1, −4)

C. (4, 1)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

1939

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 4)

GRADING NOTE The expression inside is 0 at x = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

True or false?

y = |x| has its minimum at (0, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

1940

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE |x| ≥ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

True or false?

The domain of $y = \sqrt{(x + 2)}$ is $x \geq 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

1941

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 2 \geq 0$ gives $x \geq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 2

Read key features of quadratic graphs • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Give the vertex of $y = (x - 2)^2 - 3$, and say whether it is a maximum or a minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

1943

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, −3), a minimum

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q1**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x-intercepts of $y = x^2 - 9$.

x-intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

1944

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 0) and (−3, 0)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q2**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 2x^2 - 5x + 7$?

A. 7

B. 0

C. −5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

1945

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE Substitute x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = -x^2 + 4$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

1946

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient of x^2 is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

True or false?

The vertex of $y = x^2 + 6$ is (6, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

1947

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The vertex is (0, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 1

Describe a sequence of rigid transformations • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Describe a transformation that maps the triangle with vertices (0, 0), (2, 0), (0, 1) to the triangle with vertices (3, 2), (5, 2), (3, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

1949

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A translation by (x, y) → (x + 3, y + 2).

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

The point (2, 5) is reflected across the x-axis and then translated 4 units left.

Where does it end up?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

1950

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

GRADING NOTE Any equivalent form. (2, 5) → (2, −5) → (−2, −5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which single transformation maps every (x, y) to (−x, −y)?

A. A translation

B. A rotation of 180° about the origin

C. A reflection across the y-axis

D. A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

1951

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A rotation of 180° about the origin

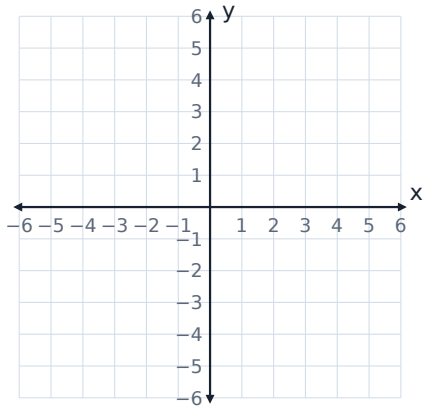
GRADING NOTE Both coordinates change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(1, 1), B(4, 1), and C(1, 3). Reflect it across the y-axis, then translate the image 2 units down. Draw the final image and label its vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

1952

QUESTION ID

S51-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with A'(−1, −1), B'(−4, −1), and C'(−1, 1)

GRADING NOTE Any correct drawing. Reflecting across the y-axis gives (−1, 1), (−4, 1), (−1, 3); then subtract 2 from each y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

True or false?

Translating and then reflecting always gives the same result as reflecting and then translating.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

1953

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For (1, 0), translating right 1 then reflecting across the y-axis gives (−2, 0); reflecting first then translating gives (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of y = sin x + 3 related to the graph of y = sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

1955

QUESTION ID
S51-F2-Q1

TYPE
Short answer

CORRECT ANSWER
It is shifted 3 units up; its midline is y = 3.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a sine function with amplitude 2, midline y = 1, and period 2π whose graph passes through (0, 1) and is rising there.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

1956

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2sin x + 1

GRADING NOTE Any equivalent form, such as 2sin(x + 2π) + 1. The conditions at (0, 1) fix the phase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How is y = 4sin x obtained from y = sin x?

A. A shift 4 units right

B. A horizontal stretch by a factor of 4

C. A vertical stretch by a factor of 4

D. A shift 4 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

1957

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A vertical stretch by a factor of 4

GRADING NOTE Every output is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = -\sin x$ is the reflection of $y = \sin x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

1958

QUESTION ID
S51-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every output changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = \sin x - 2$ has midline $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

1959

QUESTION ID
S51-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its midline is $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

Experiment with transformations of graphs and recognize even and odd functions from graphs and equations

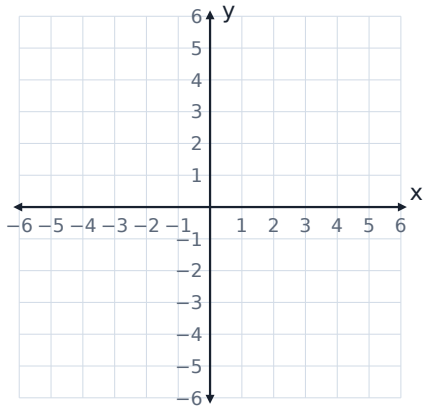
MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph quadratic functions	**HSF-IF.C.7.a**
BACKWARD 2	Graph absolute value and root functions	**HSF-IF.C.7.b**
FORWARD 1	Draw rigid transformations of figures	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs and recognize their symmetry	**HSF-TF.B.5**

SET 82 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Graph $f(x) = x^2$ and $g(x) = (x - 3)^2 - 2$ on the same grid. Label both, and describe how g is related to f.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

1962

QUESTION ID

S82-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Both parabolas; g has vertex (3, −2): it is f shifted 3 units right and 2 units down

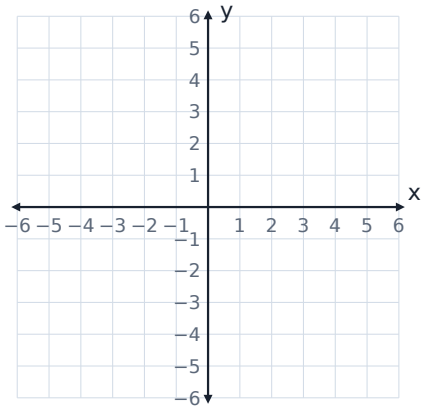
GRADING NOTE Any correct graphs with the shift described.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x|$. Graph $g(x) = 2f(x)$ and $h(x) = f(2x)$ for $-3 \le x \le 3$, using a table or graphing technology. Are g and h the same function?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

1963

QUESTION ID

S82-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: One V-shape with vertex (0, 0) and slopes 2 and −2 (both g and h)

Written: Yes. $|2x| = 2|x|$ for every x, so g and h have the same graph.

GRADING NOTE Form required: the answer with a reason or table. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a table or graphing technology to compare $f(x) = \sqrt{x}$ with $g(x) = \sqrt{(x - 4)} + 1$ at $x = 4$, 5, and 8. Describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

1964

QUESTION ID

S82-M-Q3

TYPE

Short answer

CORRECT ANSWER

g(4) = 1, g(5) = 2, g(8) = 3, which match f(0) = 0, f(1) = 1, f(4) = 2 moved 4 right and 1 up: g is f shifted 4 units right and 1 unit up.

GRADING NOTE Form required: the values and the transformation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of f is shown.

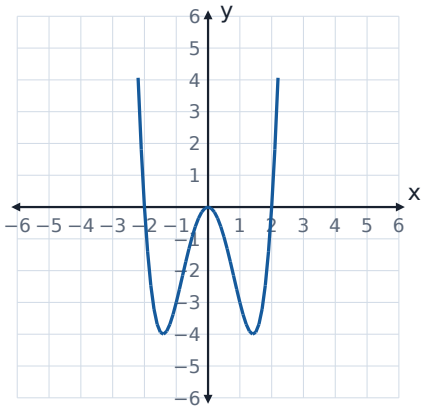
Is f even, odd, or neither?

A. Both even and odd

B. Even

C. Odd

D. Neither



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

1965

QUESTION ID

S82-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Even

GRADING NOTE The graph is symmetric about the y-axis, so f(−x) = f(x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

True or false?

$f(x) = x^3 + x$ is an odd function, because $f(-x) = -f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

1966

QUESTION ID

S82-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-x)^3 + (-x) = -(x^3 + x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

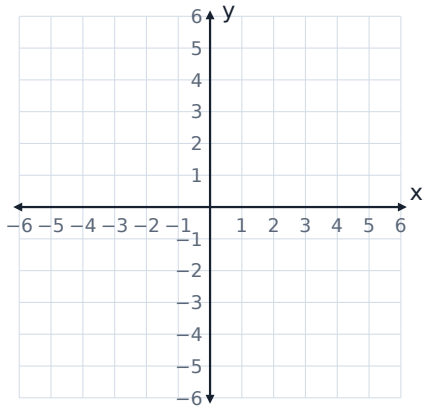
SET 82 • HSF-BF.B.3

BACKWARD 1

Graph quadratic functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2 - 4$, and mark its vertex and x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

1968

QUESTION ID

S82-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A parabola opening up with vertex (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

GRADING NOTE Any correct graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = (x + 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

1969

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

What is the axis of symmetry of $y = x^2 - 6x + 1$?

A. $x = 6$

B. $x = 3$

C. $x = -3$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

1970

QUESTION ID

S82-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3

GRADING NOTE $x = -b \div (2a) = 6 \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = -x^2 + 1$ has a maximum value of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

1971

QUESTION ID

S82-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-x^2 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

True or false?

The x-intercepts of $y = x^2 + 1$ are 1 and −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

1972

QUESTION ID

S82-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 1$ is never 0, so there are no x-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 2

Graph absolute value and root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = |x - 4| + 1$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

1974

QUESTION ID

S82-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does the graph of $y = \sqrt{(x + 2)} - 3$ start?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

1975

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function's graph is V-shaped?

A. $y = \sqrt[3]{(x + 1)}$

B. $y = |x + 1|$

C. $y = \sqrt{(x + 1)}$

D. $y = (x + 1)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

1976

QUESTION ID

S82-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = |x + 1|

GRADING NOTE Absolute value graphs are V-shaped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt[3]{x}$ passes through $(-1, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

1977

QUESTION ID

S82-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-1)^3 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

True or false?

The graph of $y = |x| - 2$ has vertex (2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

1978

QUESTION ID

S82-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its vertex is (0, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

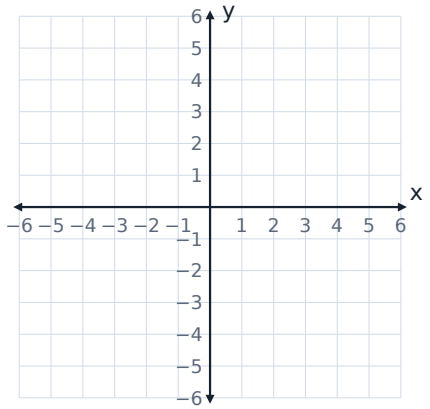
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 1

Draw rigid transformations of figures • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Translate triangle A(0, 0), B(2, 0), C(0, 3) by (x, y) → (x − 3, y + 1). Draw and label the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

1980

QUESTION ID

S82-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle A'(−3, 1), B'(−1, 1), C'(−3, 4)

GRADING NOTE Any correct drawing with labels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

The point (4, −1) is rotated 90° counterclockwise about the origin. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

1981

QUESTION ID
S82-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(1, 4)

GRADING NOTE Any equivalent form. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sequence of transformations maps (x, y) to (−x, y − 2)?

A. Reflect across the x-axis, then translate 2 units down

B. Rotate 180°, then translate 2 units up

C. Reflect across the y-axis, then translate 2 units down

D. Translate 2 units left, then reflect across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

1982

QUESTION ID
S82-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Reflect across the y-axis, then translate 2 units down

GRADING NOTE −x comes from the reflection, and y − 2 from the translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

True or false?

A translation followed by another translation is a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

1983

QUESTION ID

S82-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The two shifts add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

True or false?

Reflecting a figure across the same line twice moves it to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

1984

QUESTION ID

S82-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second reflection undoes the first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

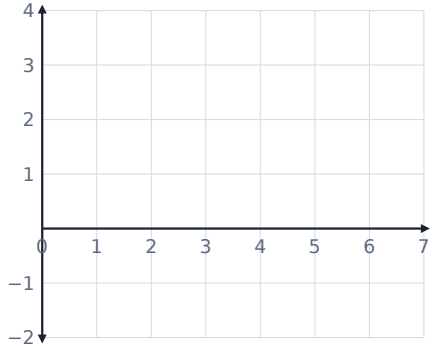
SET 82 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs and recognize their symmetry • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch $y = 2\sin x + 1$ for $0 \le x \le 2\pi$ (about 6.28).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

1986

QUESTION ID

S82-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A wave starting at (0, 1), rising to 3 near x = 1.57, back to 1 near 3.14, down to −1 near 4.71, and back to 1 near 6.28

GRADING NOTE Any correct sketch with midline 1, amplitude 2, and period 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = \sin(x - \pi \div 2)$ is related to the graph of $y = \sin x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

1987

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is the graph of $y = \sin x$ shifted $\pi \div 2$ units to the right.

GRADING NOTE Any equivalent description (it is also the graph of $y = -\cos x$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has period π?

A. y = sin x + 2

B. y = sin(x + 2)

C. y = 2sin x

D. y = sin(2x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

1988

QUESTION ID

S82-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = sin(2x)

GRADING NOTE 2π ÷ 2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

True or false?

y = −cos x is the reflection of y = cos x across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

1989

QUESTION ID
S82-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every output changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = cos x is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

1990

QUESTION ID

S82-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos(−x) = cos x, so cos is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

Solve f(x) = c for a simple function f and write an expression for the inverse

MAIN	Algebra I main question	**HSF-BF.B.4.a**
BACKWARD 1	Rearrange a formula for another variable	**HSA-CED.A.4**
BACKWARD 2	Use cube roots to solve $x^3 = p$	**8.EE.A.2**
FORWARD 1	Verify inverses by composition	**HSF-BF.B.4.b**
FORWARD 2	Restrict a domain to produce an inverse	**HSF-BF.B.4.d**

SET 52 • HSF-BF.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3x^3 - 1$. Solve $f(x) = 80$, and write an expression for the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

1993

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 3$; $f^{-1}(x) = \sqrt[3]{((x + 1) \div 3)}$

GRADING NOTE Form required: the solution and the inverse (any equivalent expression). $3x^3 = 81$, so $x^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (2x + 3) \div (x - 4)$, for $x \neq 4$. Write an expression for the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

1994

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(4x + 3) \div (x - 2)$, for $x \neq 2$

GRADING NOTE Any equivalent form. Solve $y = (2x + 3) \div (x - 4)$ for x: $xy - 4y = 2x + 3$, so $x(y - 2) = 4y + 3$. Check: $f(0) = -0.75$ and $f^{-1}(-0.75) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

F(C) = 1.8C + 32 gives degrees Fahrenheit.

Write the inverse C(F), and find C when F = 77.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

1995

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

C(F) = (F − 32) ÷ 1.8; 25 °C

GRADING NOTE Form required: the inverse (any equivalent form) and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 4x - 7$.

What is $f^{-1}(x)$?

A. $(x - 7) \div 4$

B. $(x + 7) \div 4$

C. $x \div 4 + 7$

D. $4x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

1996

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (x + 7) ÷ 4

GRADING NOTE Solve $y = 4x - 7$ for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

True or false?

If $f(x) = x \div 3 + 2$, then $f^{-1}(x) = 3x - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

1997

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = y \div 3 + 2$ gives $y = 3(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 1

Rearrange a formula for another variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = 4x - 7$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

1999

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(y + 7) \div 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

2000

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(P − 2w) ÷ 2

GRADING NOTE Any equivalent form, such as P ÷ 2 − w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

2000

ANSWER KEY Reduced copy of the question page, with its answer

Solve d = rt for t. (r ≠ 0)

A. t = r ÷ d

B. t = d − r

C. t = d ÷ r

D. t = dr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

2001

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = d ÷ r

GRADING NOTE Divide both sides by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

2001

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $y = x^3$ for x gives $x = \sqrt[3]{y}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

2002

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take the cube root of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

2002

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving y = 5x for x gives x = 5y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

2003

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = y ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

2003

SET 52 • HSF-BF.B.4.a

BACKWARD 2

Use cube roots to solve $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 27$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

2005

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt[3]{125}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

2006

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

2006

ANSWER KEY Reduced copy of the question page, with its answer

What is $\sqrt[3]{64}$?

A. 21.3

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

2007

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

2007

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt[3]{(-27)} = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

2008

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

2008

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{16} = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

2009

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^2 = 16$, and $\sqrt{}$ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 1

Verify inverses by composition • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Show that $f(x) = 2x^3$ and $g(x) = \sqrt[3]{(x \div 2)}$ are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

2011

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = 2 \cdot (x \div 2) = x$ and $g(f(x)) = \sqrt[3]{(x^3)} = x$.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = (x + 1) ÷ (x − 1).

Find f(f(3)).

f(f(3)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

2012

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 2 and f(2) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

Which function is its own inverse?

A. $f(x) = 1 \div x$

B. $f(x) = 2x$

C. $f(x) = x + 1$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

2013

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f(x) = 1 \div x$

GRADING NOTE $1 \div (1 \div x) = x$ for $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

True or false?

If f(g(x)) = x on the domain of g and g(f(x)) = x on the domain of f, then g is the inverse of f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

2014

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the composition test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

f(x) = x + 3 and g(x) = x − 3 are not inverses, because f(g(1)) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

2015

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(1)) = f(−2) = 1. They are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

2015

SET 52 • HSF-BF.B.4.a

FORWARD 2

Restrict a domain to produce an inverse • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2 + 2$. Restrict f to the largest interval that contains $x = 1$ on which f is one-to-one, and give the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

2017

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$; $f^{-1}(x) = \sqrt{(x - 2)}$, for $x \geq 2$

GRADING NOTE Form required: the interval and the inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why f(x) = |x| has no inverse on all real numbers.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

2018

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
f(3) = f(−3) = 3, so the input of 3 for an inverse would need two outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

$f(x) = (x + 4)^2$.

What is the largest interval containing $x = 0$ on which f is one-to-one?

A. All real numbers

B. $-8 \le x \le 0$

C. $x \ge 0$

D. $x \ge -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

2019

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x \ge -4$

GRADING NOTE f is increasing for $x \ge -4$. On $-8 \le x \le 0$, $f(-8) = f(0) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ restricted to $x \le 0$ has the inverse $f^{-1}(x) = -\sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

2020

QUESTION ID
S52-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE For $x \le 0$, $-\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(x) = x^2 + 1$ is one-to-one on all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

2021

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(1) = f(-1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

Prove that linear functions grow by equal differences and exponential functions grow by equal factors over equal intervals

MAIN	Algebra I main question	**HSF-LE.A.1.a**
BACKWARD 1	Tell linear from nonlinear functions	**8.F.A.3**
BACKWARD 2	Compare average rates of change over intervals	**HSF-IF.B.6**
FORWARD 1	Derive the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to turn equal factors into equal differences	**HSF-BF.B.5**

SET 53 • HSF-LE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3x + 5. Show that f grows by equal differences over equal intervals: find f(x + h) − f(x) in simplest form, and explain why it does not depend on x.

f(x + h) − f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

2024

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

3h

GRADING NOTE REQUIRED WORK: Algebra showing f(x + h) − f(x) = 3(x + h) + 5 − (3x + 5) = 3h, and the statement that 3h depends only on the interval length h.

Form required: the simplified difference and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

2024

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$. Show that g grows by equal factors over equal intervals: find $g(x + h) \div g(x)$ in simplest form, and explain why it does not depend on x.

$g(x + h) \div g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

2025

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

2^h

GRADING NOTE REQUIRED WORK: Algebra showing $4 \cdot 2^{x+h} \div (4 \cdot 2^x) = 2^h$, and the statement that 2^h depends only on h.

Form required: the simplified ratio and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 2, 4, 6 and y = 5, 11, 17, 23.

Is the relationship linear or exponential? Explain using equal intervals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

2026

QUESTION ID
S53-M-Q3

TYPE
Short answer

CORRECT ANSWER
Linear: over each interval of 2, y increases by the same amount, 6.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 80, 40, 20, 10.

Which description is correct?

A. Linear: y decreases by 40 each time

B. Exponential: y is multiplied by $\frac{1}{2}$ each time x increases by 1

C. Linear: y decreases by 20 each time

D. Neither linear nor exponential

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

2027

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Exponential: y is multiplied by 1/2 each time x increases by 1

GRADING NOTE The differences are −40, −20, −10 (not equal), but the ratios are all $\frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

2027

True or false?

A table with y-values 2, 5, 10, 17 for x = 0, 1, 2, 3 is linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

2028

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences 3, 5, 7 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 1

Tell linear from nonlinear functions • Grade 8 • Functions • 8.F.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = x^2 + 1$ linear? Use the points (0, 1), (1, 2), and (2, 5) to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

2030

QUESTION ID
S53-B1-Q1

TYPE
Short answer

CORRECT ANSWER
No. y increases by 1 and then by 3, so the rate is not constant.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 4 − 2x linear? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

2031

GRADING NOTE Form required: the answer and the reason.

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. It has the form y = mx + b with slope −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function is linear?

A. $y = 2^x$

B. $y = x^2$

C. $y = 0.5x - 3$

D. $y = 1 \div x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

2032

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 0.5x - 3$

GRADING NOTE Only $y = 0.5x - 3$ has the form $y = mx + b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

True or false?

The graph of a linear function is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

2033

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is why it is called linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The area of a square, $A = s^2$, is a linear function of s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

2034

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its graph curves; the area grows by 1, 3, 5, ... for s = 0, 1, 2, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 2

Compare average rates of change over intervals • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3x + 5$.

Find the average rate of change from $x = 0$ to $x = 10$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

2036

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Find the average rate of change from $x = 0$ to $x = 1$, and from $x = 1$ to $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

2037

QUESTION ID
S53-B2-Q2

TYPE
Short answer

CORRECT ANSWER
4 and 8

GRADING NOTE Form required: both rates. (8 − 4) ÷ 1 and (16 − 8) ÷ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the average rate of change of $5 \cdot 3^x$ from $x = 0$ to $x = 2$?

A. 15

B. 20

C. 40

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

2038

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

GRADING NOTE $(45 - 5) \div 2 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An exponential growth function has a larger average rate of change on later intervals of the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

2039

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its outputs, and their changes, keep multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of 2^x from $x = 0$ to $x = 3$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

2040

QUESTION ID
S53-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(8 - 1) \div 3 = \frac{7}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 1

Derive the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Let $S = 1 + 2 + 4 + \ldots + 2^9$. Show that $2S - S = 2^{10} - 1$, and find S.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

2042

QUESTION ID

S53-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

1,023

GRADING NOTE REQUIRED WORK: 2S written as $2 + 4 + \ldots + 2^{10}$, then subtracting S so that every term except 2^{10} and 1 cancels.

Form required: the subtraction and the value of S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find the sum of the first 6 terms of 1, 3, 9, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

2043

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

364

GRADING NOTE Any equivalent form. $(1 - 729) \div (1 - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

When deriving the sum formula, why do we multiply S by r?

A. To double the sum

B. So that S − rS cancels all but two terms

C. To make every term equal

D. To make r equal 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

2044

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. So that S − rS cancels all but two terms

GRADING NOTE S and rS share every term except the first term of S and the last term of rS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $r \neq 1$, $a + ar + \ldots + ar^{n-1} = a(1 - r^n) \div (1 - r)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

2045

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

True or false?

$1 + 3 + 9 + 27 = 3^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

2046

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is 40, and $3^4 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 2

Use logarithms to turn equal factors into equal differences • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Show that $\log_2 g(x) = x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

2048

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2(4 \cdot 2^x) = \log_2$

$4 + \log_2 2^x = 2 + x.$

GRADING NOTE Form required: the steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 10^x$.

What is log y? (log means base 10.)

log y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

2049

QUESTION ID
S53-F2-Q2

TYPE
Short answer

CORRECT ANSWER

x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The values y = 1, 10, 100, 1,000 occur at x = 0, 1, 2, 3.

What are the values of log y? (log means base 10.)

A. 0, 1, 2, 3

B. 0, 10, 20, 30

C. 1, 2, 3, 4

D. 1, 10, 100, 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

2050

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 1, 2, 3

GRADING NOTE $\log 10^k = k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $a > 0$ and $b > 0$, taking the logarithm of $a \cdot b^x$ gives a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

2051

QUESTION ID
S53-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\log(a \cdot b^x) = \log a + x \log b$, a linear function of x. The logarithm needs a positive input, so $a > 0$ and $b > 0$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(100 \cdot 10^x) = 100x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

2052

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(100 \cdot 10^x) = 2 + x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

Recognize situations in which a quantity changes at a constant rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.b**
BACKWARD 1	Find the constant of proportionality	**7.RP.A.2.b**
BACKWARD 2	Write a linear function from a rate and an initial value	**8.F.B.4**
FORWARD 1	Relate arc length and sector area to the central angle — nearest r…	**HSG-C.B.5**
FORWARD 2	Use radian measure as arc length on a circle — nearest related	**HSF-TF.A.1**

SET 54 • HSF-LE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A pool fills at a constant rate. At 2 p.m. it holds 1,500 gallons; at 5 p.m. it holds 3,300 gallons.

What is the rate, and how much water was in it at noon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

2055

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

600 gallons per hour; 300 gallons at noon

GRADING NOTE Form required: the rate with units and the amount. 1,800 ÷ 3 = 600, and 1,500 − 2(600) = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plant is 6 cm tall on day 0 and grows 0.8 cm each day.

How tall is it on day 15?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

2056

QUESTION ID
S54-M-Q2

TYPE
Short answer

CORRECT ANSWER
18 cm

GRADING NOTE Any equivalent form. 6 + 0.8(15).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

A taxi charges the same amount for every mile.

Explain why the fare for the miles driven is a linear function of the miles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

2057

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

Each mile adds the same amount, so the fare grows by equal differences over equal intervals, which is linear growth.

GRADING NOTE Any equivalent explanation that uses the constant rate per mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation is modeled by a linear function?

A. A cell population doubles each hour.

B. A tank loses 4 liters each minute.

C. A loan grows 4% each month.

D. A car loses 15% of its value each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

2058

QUESTION ID

S54-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A tank loses 4 liters each minute.

GRADING NOTE Only the tank changes by the same amount each unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

True or false?

A savings account that earns the same dollar amount of interest each year grows linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

2059

QUESTION ID
S54-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A constant amount per year is a constant rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 1

Find the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

y = 2.5x gives the cost of x pounds of apples.

What is the constant of proportionality?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

2061

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A proportional table has x = 2, 4, 6 and y = 7, 14, 21.

What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

2062

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE Any equivalent form. 7 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cyclist rides 12 miles in 3 hours at a constant rate.

What is the constant of proportionality in miles per hour?

A. 15

B. 36

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

2063

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 12 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 9x, y increases by 9 whenever x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

2064

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A proportional relationship through (5, 20) has constant of proportionality 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

2065

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 \div 5 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 2

Write a linear function from a rate and an initial value • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (2, 1,500) and (5, 3,300).

Find its slope and its y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

2067

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Slope 600;
y-intercept 300

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A phone battery is at 80% and drops 5 percentage points per hour.

Write a function B(t) for the charge after t hours.

B(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

2068

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

80 − 5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has rate of change 0.8 and initial value 6?

A. $y = 6x + 0.8$

B. $y = 0.8x + 6$

C. $y = 6.8x$

D. $y = 6(0.8)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

2069

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 0.8x + 6$

GRADING NOTE $y = mx + b$ with $m = 0.8$ and $b = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

True or false?

A function with initial value 10 and rate of change −2 is y = 10 − 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

2070

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 10 and m = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (2, 10) has rate of change 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

2071

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(10 - 4) \div 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 1

Relate arc length and sector area to the central angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Arc length grows by a constant amount (the radius) for each radian, a constant-rate situation.

ANSWER KEY Reduced copy of the question page, with its answer

In a circle of radius 5 cm, by how much does the arc length increase for each 1-radian increase in the central angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

2073

QUESTION ID
S54-F1-Q1

TYPE
Short answer

CORRECT ANSWER
5 cm

GRADING NOTE Any equivalent form. $s = 5\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 10.

Find the arc lengths for central angles of 0.5, 1, and 1.5 radians.

Arc lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

2074

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5, 10, and 15

GRADING NOTE Form required: all three, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The area of a sector is $A = \frac{1}{2}r^2\theta$, and $r = 4$.

By how much does A increase when θ increases by 1 radian?

A. 4

B. 2

C. 16

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

2075

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

GRADING NOTE $\frac{1}{2}(16)(1) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For a fixed radius, arc length is a linear function of the central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

2076

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE s = rθ has the constant rate r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Doubling the radius while keeping the angle the same doubles the area of a sector.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

2077

QUESTION ID
S54-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The area depends on r^2, so it is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • HSF-LE.A.1.b

FORWARD 2

Use radian measure as arc length on a circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Algebra II standard. Connection: Radian measure is arc length per unit of radius, continuing the constant-rate arc length of FORWARD 1.

A wheel of radius 0.3 m turns through 4π radians.

How far does a point on its rim travel? Give the exact distance and the distance to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

2079

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.2π m, about 3.77 m

GRADING NOTE Form required: exact and rounded. Arc length = rθ = 0.3 × 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A central angle cuts off an arc of length 6 on a circle of radius 2.

What is its radian measure?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

2080

QUESTION ID
S54-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

A wheel turns at 3 radians per second.

Through about how many degrees does it turn in 2 seconds?

A. 6°

B. 172°

C. 1,080°

D. 344°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

2081

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 344°

GRADING NOTE 6 radians × (180 ÷ π) ≈ 343.8°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

One radian is the angle that cuts off an arc as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

2082

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rotation of $-\pi \div 2$ radians is a quarter turn counterclockwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

2083

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative angles turn clockwise; $-\pi \div 2$ is a quarter turn clockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.c**
BACKWARD 1	Read growth and decay rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Solve percent problems	**7.RP.A.3**
FORWARD 1	Find times in percent-rate models with logarithms	**HSF-LE.A.4**
FORWARD 2	Use natural logarithms with continuous growth	**HSF-BF.B.5**

SET 55 • HSF-LE.A.1.c

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

A car worth $24,000 loses 15% of its value each year.

Write a function V(t), and find the value after 3 years to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

2086

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

$V(t) = 24{,}000(0.85)^t$; $14,739

GRADING NOTE Form required: the function and the rounded value. $0.85^3 = 0.614125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a quantity that grows by 6% each year grows by equal factors over equal intervals.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

2087

QUESTION ID
S55-M-Q2

TYPE
Short answer

CORRECT ANSWER
Each year the quantity is multiplied by the same factor, 1.06, so every 1-year interval multiplies it by 1.06.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town's population increases by 2.5% per year.

By what factor does it grow in 2 years? What is the percent increase over 2 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

2088

QUESTION ID
S55-M-Q3

TYPE
Short answer

CORRECT ANSWER
1.050625; about 5.06%

GRADING NOTE Form required: the factor and the percent. $1.025^2 = 1.050625$, slightly more than 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation is exponential?

A. A worker earns $15 each hour.

B. A medicine dose decreases by 20% each hour.

C. A tank gains 3 liters each second.

D. A runner covers 200 meters each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

2089

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A medicine dose decreases by 20% each hour.

GRADING NOTE Only the dose changes by a constant percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

True or false?

A 10% increase followed by a 10% decrease returns a quantity to its original value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

2090

QUESTION ID
S55-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1.1 \times 0.9 = 0.99$, a 1% loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 1

Read growth and decay rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

$V = 24{,}000(0.85)^t$.

What is the yearly percent change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

2092

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A 15% decrease per year

GRADING NOTE Any equivalent form. $0.85 = 1 - 0.15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 500(1.025)^t$.

Is this growth or decay, and at what rate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

2093

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Growth, at 2.5% per unit of t

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function shows 4% decay per year?

A. $y = 300(1.04)^t$

B. $y = 300(0.04)^t$

C. $y = 300 - 0.04t$

D. $y = 300(0.96)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

2094

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 300(0.96)^t$

GRADING NOTE $1 - 0.04 = 0.96$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 50(1.5)^t$ grows by 50% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

2095

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.5 = 1 + 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 80(0.75)^t$ decreases by 75% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

2096

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps 75%, so it decreases by 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 2

Solve percent problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $24,000 car loses 15% of its value.

What is its new value?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

2098

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$20,400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 8% sales tax on $45?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

2099

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$3.60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the price of a $60 item after a 25% discount?

A. $15

B. $45

C. $75

D. $35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

2100

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $45

GRADING NOTE 60 × 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

20% of 150 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

2101

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.2 × 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A 50% increase of 40 is 60, and a 50% decrease of 60 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

2102

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 50% decrease of 60 is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 1

Find times in percent-rate models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$V(t) = 24{,}000(0.85)^t$.

When is V = 12,000? Round to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

2104

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4.3

GRADING NOTE Form required: rounded to the nearest tenth. t = log 0.5 ÷ log 0.85 ≈ 4.27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $1.025^t = 2$. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

2105

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

28.1

GRADING NOTE Form required: rounded to the nearest tenth. log 2 ÷ log 1.025 ≈ 28.07.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

Which expression gives the half-life of $y = (0.9)^t$? (log means base 10.)

A. 0.9 ÷ 2

B. 0.5 ÷ 0.9

C. log 0.9 ÷ log 0.5

D. log 0.5 ÷ log 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

2106

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. log 0.5 ÷ log 0.9

GRADING NOTE Solve $0.9^t = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(0.85^t) = t \log 0.85$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

2107

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The power property of logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $2^t = 5$, then $t = 2.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

2108

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^{2.5} \approx 5.66$. $t = \log 5 \div \log 2 \approx 2.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 2

Use natural logarithms with continuous growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $y = e^x$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

2110

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = \ln x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{0.05t} = 3$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

2111

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 ln 3

GRADING NOTE Form required: exact. 0.05t = ln 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\ln(e^4)$? (ln means base e.)

A. 4e

B. 4

C. 1

D. e

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

2112

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE ln and e^x are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

True or false?

e raised to the power ln 7 equals 7. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

2113

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE e^x and ln x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

True or false?

ln 1 = 1 (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

2114

QUESTION ID
S55-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $e^0 = 1$, so ln 1 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

Construct linear and exponential functions from a graph, a description, two points, or a table

MAIN	Algebra I main question	**HSF-LE.A.2**
BACKWARD 1	Write a linear function from two points or a table	**8.F.B.4**
BACKWARD 2	Write formulas for arithmetic and geometric sequences	**HSF-BF.A.2**
FORWARD 1	Build a sinusoidal model from a description	**HSF-TF.B.5**
FORWARD 2	Find a quadratic through three points with a matrix equation	**HSA-REI.C.8**

SET 56 • HSF-LE.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function whose graph passes through (2, 11) and (6, 23).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

2117

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 5

GRADING NOTE Any equivalent form. The slope is 12 ÷ 4 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function $y = ab^x$ $(b > 0)$ whose graph passes through (1, 6) and (3, 54).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

2118

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
$2(3)^x$

GRADING NOTE Any equivalent form. $ab^3 \div (ab) = b^2 = 9$, so $b = 3$ and $a = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

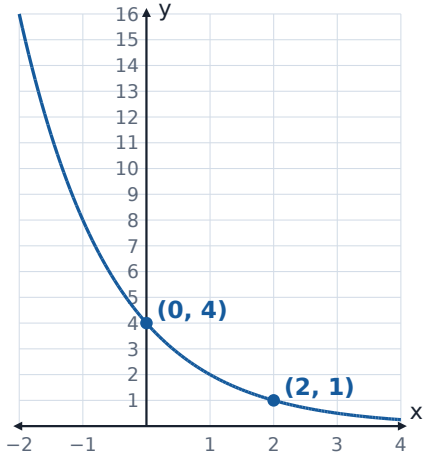
High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of an exponential function $y = ab^x$ passes through the marked points.

Write the function.

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

2119

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
$4(0.5)^x$

GRADING NOTE Any equivalent form, such as $4(1/2)^x$. a = 4, and $4b^2 = 1$ gives b = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 1.5, 3, 6, 12.

Which function fits?

A. $y = 2(1.5)^x$

B. $y = 1.5x + 1.5$

C. $y = 3x$

D. $y = 1.5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

2120

QUESTION ID

S56-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 1.5(2)^x$

GRADING NOTE y starts at 1.5 and doubles each time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

True or false?

The linear function with f(0) = 10 that decreases by 2 for each unit increase in x is $f(x) = 10(0.8)^x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

2121

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal differences give f(x) = 10 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 1

Write a linear function from two points or a table • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, −3) and (4, 5).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

2123

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2x − 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 1, 2, 3 and y = 10, 7, 4.

Write the linear function.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

2124

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

Which line passes through (−1, 2) and (1, 6)?

A. y = −2x + 4

B. y = 2x + 4

C. y = 2x + 2

D. y = 4x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

2125

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 4

GRADING NOTE The slope is 4 ÷ 2 = 2; 6 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

True or false?

The line through (0, 0) and (3, 6) is y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

2126

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The slope is 2 and the y-intercept is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 1) and (2, 3) is y = 2x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

2127

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 2x − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 2

Write formulas for arithmetic and geometric sequences • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A geometric sequence has a(1) = 2 and ratio 3.

Write an explicit formula.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

2129

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$2 \cdot 3^{n-1}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for 10, 7, 4, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

2130

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a recursive formula for 6, 3, 1.5, ...?

A. $a(n) = 6 - 3n$

B. $a(1) = 6$, $a(n) = 0.5a(n - 1)$

C. $a(1) = 6$, $a(n) = 2a(n - 1)$

D. $a(1) = 6$, $a(n) = a(n - 1) - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

2131

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 6$, $a(n) = 0.5a(n - 1)$

GRADING NOTE Each term is half the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function on the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

2132

QUESTION ID
S56-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equal differences between consecutive terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The geometric sequence 5, 10, 20, ... has ratio 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

2133

QUESTION ID
S56-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ratio is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 1

Build a sinusoidal model from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A tide has high water 10 ft and low water 2 ft, with period 12 hours and high tide at t = 0.

Write a cosine model h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

2135

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4cos(πt ÷ 6) + 6

GRADING NOTE Any equivalent form. Amplitude 4, midline 6, and 2π ÷ 12 = π ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sinusoid has maximum 10 and minimum 2.

What are its amplitude and midline?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

2136

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Amplitude 4; midline y = 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

A model y = cos(Bt) has period 12.

What is B?

A. 12

B. π ÷ 12

C. π ÷ 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

2137

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. π ÷ 6

GRADING NOTE 2π ÷ B = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

True or false?

A sinusoidal model with maximum 8 and minimum 0 has amplitude 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

2138

QUESTION ID
S56-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE (8 − 0) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 3cos(2t) has period 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

2139

QUESTION ID
S56-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The period is 2π ÷ 2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 2

Find a quadratic through three points with a matrix equation • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

ANSWER KEY Reduced copy of the question page, with its answer

$y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7).

Write the system for a, b, and c as a matrix equation $AX = B$, with X the column [a, b, c]. Give A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

2141

QUESTION ID
S56-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A has rows [0 0 1], [1 1 1], and [4 2 1]; B is the column [1, 2, 7].

GRADING NOTE Form required: A and B. Each point (x, y) gives the row $[x^2\ x\ 1]$ and the constant y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A has rows [0 0 1], [1 1 1], and [4 2 1], X is the column [a, b, c], and B is the column [1, 2, 7]. Solve AX = B to find the parabola $y = ax^2 + bx + c$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

2142

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1, c = 1, so $y = 2x^2 - x + 1$

GRADING NOTE Form required: all three values (or the equation). c = 1, a + b = 1, and 4a + 2b = 6, so 2a = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

A parabola $y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7). It is written as $AX = B$ with X the column [a, b, c].

Which row of A comes from the point (2, 7)?

A. [4 2 1]

B. [2 4 1]

C. [4 2 7]

D. [2 2 1]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

2143

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. [4 2 1]

GRADING NOTE $4a + 2b + c = 7$, so the row is [4 2 1] and the matching entry of B is 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

True or false?

Three points with different x-values that are not on one line determine exactly one parabola $y = ax^2 + bx + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

2144

QUESTION ID
S56-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The 3-by-3 system has exactly one solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

True or false?

A parabola $y = ax^2 + bx + c$ through (0, 1), (1, 2), and (2, 7) gives a matrix equation $AX = B$ in which A has 2 rows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

2145

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each of the 3 points gives one equation, so A has 3 rows (and 3 columns).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

Observe that exponential growth eventually exceeds linear, quadratic, and polynomial growth

MAIN	Algebra I main question	**HSF-LE.A.3**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Compare two functions given in different forms	**8.F.A.2**
FORWARD 1	Compare the growth of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Compare logarithmic growth with other growth	**HSF-BF.B.5**

SET 57 • HSF-LE.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 100x$ and $g(x) = 2^x$.

What is the smallest whole number $x \geq 1$ for which $g(x) > f(x)$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

2148

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $g(9) = 512 < 900 = f(9)$, and $g(10) = 1{,}024 > 1{,}000 = f(10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

$f(x) = x^2$ and $g(x) = 1.5^x$.

Which is larger at x = 10? At x = 20? What happens for larger x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

2149

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

f at x = 10 (100 compared with about 57.7); g at x = 20 (about 3,325 compared with 400); g stays larger from then on.

GRADING NOTE Form required: all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A pays $1,000 per day. Plan B pays 1 cent on day 1, and its daily pay doubles each day.

On which day does Plan B's daily pay first exceed Plan A's?

Day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

2150

QUESTION ID
S57-M-Q3

TYPE
Short answer

CORRECT ANSWER
Day 18

GRADING NOTE Any equivalent form. Day n pays $0.01 \cdot 2^{n-1}$ dollars: day 17 pays $655.36 and day 18 pays $1,310.72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function eventually has the greatest values as x increases?

A. $y = 50x^2$

B. $y = 1.01^x$

C. $y = x^3 + 500$

D. $y = 1{,}000x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

2151

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 1.01^x$

GRADING NOTE Any exponential function with base greater than 1 eventually exceeds every polynomial function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

True or false?

Because $3x^2 > 2^x$ at $x = 5$, $3x^2$ stays greater than 2^x for all $x > 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

2152

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At $x = 10$, $3x^2 = 300$ and $2^x = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 12, 24.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

2154

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: y is multiplied by 2 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 9, 12.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

2155

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Linear: y increases by 3 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function grows by equal factors over equal intervals?

A. $y = 7 + 1.1x$

B. $y = 7(1.1)^x$

C. $y = 7x$

D. $y = 1.1x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

2156

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 7(1.1)^x$

GRADING NOTE Only the exponential function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

Linear functions grow by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

2157

QUESTION ID
S57-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The difference is the slope times the interval length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

$y = 5x^2$ grows by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

2158

QUESTION ID
S57-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $5(2)^2 \div 5(1)^2 = 4$ but $5(3)^2 \div 5(2)^2 = 2.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 2

Compare two functions given in different forms • Grade 8 • Functions • 8.F.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Function A is y = 4x + 1. Function B has x = 0, 1, 2 and y = 3, 5, 7.

Which has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

2160

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A: 4 compared with 2.

GRADING NOTE Form required: the function and both rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

Function A is y = −x + 6. Function B is the line through (0, 2) and (2, 8).

Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

2161

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A: 6 compared with 2.

GRADING NOTE Form required: the function and both intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

$f(x) = 3x - 2$. Function g has $g(0) = 1$ and $g(1) = 5$, and it is linear.

Which statement is true?

A. f has the greater rate of change.

B. f has the greater initial value.

C. g has the greater rate of change and the greater initial value.

D. f and g have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

2162

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g has the greater rate of change and the greater initial value.

GRADING NOTE g has rate 4 and initial value 1; f has rate 3 and initial value −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

True or false?

A linear function with slope 5 increases faster than one with slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

2163

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It gains 5 per unit instead of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 2x + 10 has greater values than y = 3x for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

2164

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For x > 10, 3x is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 1

Compare the growth of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

For which $x > 0$ is $x^3 > 100x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

2166

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x > 100

GRADING NOTE Form required: an inequality. Divide by $x^2 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which grows faster for large x, x^4 or $1{,}000x^3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

2167

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x^4: $x^4 \div (1{,}000x^3) = x \div 1{,}000$, which is greater than 1 when $x > 1{,}000$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As $x \to \infty$, what does $f(x) = x^4 - 50x^3$ do?

A. $f(x) \to -\infty$

B. $f(x) \to -50$

C. $f(x) \to \infty$

D. $f(x) \to 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

2168

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f(x) \to \infty$

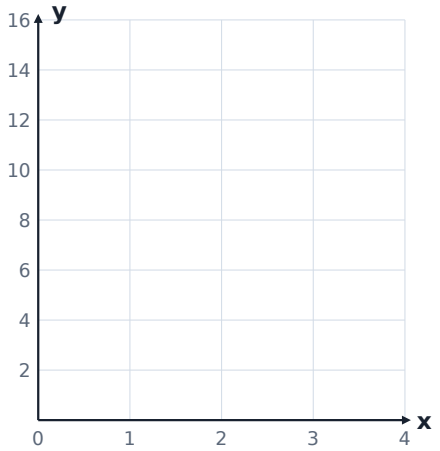
GRADING NOTE The leading term x^4 decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2$ and $y = 0.5x^3$ for $0 \le x \le 4$ on the same grid. Mark where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

2169

QUESTION ID

S57-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Both curves through (0, 0) and meeting again at (2, 4); $y = 0.5x^3$ is below $y = x^2$ for $0 < x < 2$ and above it for $x > 2$ (it leaves the grid near $x = 3.2$)

GRADING NOTE Any correct graph with both meeting points. $x^2 = 0.5x^3$ gives $x = 0$ or $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For every large x, $0.001x^3$ is less than $1{,}000x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

2170

QUESTION ID
S57-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For $x > 1{,}000{,}000$, $0.001x^3$ is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 2

Compare logarithmic growth with other growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find $\log_2 1{,}024$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

2172

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $2^{10} = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

By what factor must x be multiplied for log x to increase by 1? (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

2173

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. log(10x) = 1 + log x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

Which is greatest when x = 1,000,000? (log means base 10; ln means base e.)

A. $\sqrt{x}$

B. $\log_2 x$

C. ln x

D. log x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

2174

QUESTION ID
S57-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $\sqrt{x}$

GRADING NOTE $\sqrt{x} = 1{,}000$; $\log x = 6$; $\ln x \approx 13.8$; $\log_2 x \approx 19.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

Eventually, log x grows more slowly than any linear function with positive slope. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

2175

QUESTION ID
S57-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE log x is the inverse of the fast-growing 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

$\log_2 x = 100$ when $x = 200$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

2176

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 2^{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

Interpret the parameters of a linear or exponential function in context

MAIN	Algebra I main question	**HSF-LE.B.5**
BACKWARD 1	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
BACKWARD 2	Interpret the rate and initial value of a linear function	**8.F.B.4**
FORWARD 1	Interpret the parameters of a sinusoidal model	**HSF-TF.B.5**
FORWARD 2	Interpret the mean and standard deviation of a normal model — n…	**HSS-ID.A.4**

SET 58 • HSF-LE.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$C(m) = 0.35m + 40$ is the cost in dollars of renting a car for one day and driving m miles.

What do 0.35 and 40 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

2179

QUESTION ID
S58-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.35: the cost per mile ($0.35); 40: the daily fee ($40) charged even for 0 miles.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

$P(t) = 1{,}200(1.04)^t$ is a town's population t years after 2020.

What do 1,200 and 1.04 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

2180

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
1,200: the population in 2020; 1.04: the population grows 4% each year.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

$A(t) = 80(0.5)^{t/6}$ is the milligrams of a medicine left in the body after t hours.

What do 80, 0.5, and 6 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

2181

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

80 mg is the starting dose; half of the medicine remains after every 6 hours (a half-life of 6 hours).

GRADING NOTE Form required: all three meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

V(t) = 18,000 − 1,500t is a machine's value in dollars after t years.

What does −1,500 mean?

A. The machine loses $1,500 of value each year.

B. The machine loses 1,500% of its value each year.

C. The machine was bought for $1,500.

D. The machine is worth $18,000 after 1,500 years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

2182

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The machine loses $1,500 of value each year.

GRADING NOTE The slope is the change in value per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

True or false?

In $y = 250(0.92)^t$, the value decreases by 92% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

2183

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 92% remains each year, so it decreases by 8%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 1

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges 75 + 60h dollars for h hours of work.

What does 75 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

2185

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A $75 fee charged for every visit, before any hours of work.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3{,}000(1.05)^t$, what does the factor 1.05 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

2186

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The amount grows 5% for each unit of t.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

A box with a square base of side s and height h has surface area $2s^2 + 4sh$.

What does 4sh represent?

A. The volume

B. The area of the four side faces

C. The perimeter of the base

D. The area of the top and bottom

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

2187

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The area of the four side faces

GRADING NOTE Each side face is s by h, and there are four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

In 500 − 20w, the term −20w shows a decrease of 20 for each unit of w.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

2188

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient −20 is the change per unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In 1.5x + 7, the coefficient of x is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

2189

QUESTION ID
S58-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The coefficient of x is 1.5; 7 is the constant term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 2

Interpret the rate and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

Cost y = 12x + 30 for x hours of bike rental.

What is the initial value, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

2191

GRADING NOTE Form required: the value and its meaning.

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30: a $30 fee before any hours are counted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

The water in a tank is $y = 500 - 25t$ gallons after t minutes.

What does -25 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

2192

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The tank loses 25 gallons each minute.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

A plant's height is a linear function of time. It is 8 cm on day 0 and 20 cm on day 4.

What is the rate of change?

A. 5 cm per day

B. 20 cm per day

C. 8 cm per day

D. 3 cm per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

2193

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 cm per day

GRADING NOTE (20 − 8) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

True or false?

In y = 40x + 100, 100 is the value of y when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

2194

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In d = 65t, 65 is the starting distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

2195

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 65 is the rate (distance per unit of time); the starting distance is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

2195

SET 58 • HSF-LE.B.5

FORWARD 1

Interpret the parameters of a sinusoidal model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What do 15 and 10 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

2197

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 °C is the average (midline) temperature; the temperature rises and falls 10 °C above and below it.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What is the period of T, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

2198

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24 hours: the temperature pattern repeats every day.

GRADING NOTE Form required: the period and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

$h(t) = 1.5\cos(\pi t \div 6) + 3$ is the depth of water in meters.

What is the greatest depth?

A. 4.5 m

B. 3 m

C. 6 m

D. 1.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

2199

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.5 m

GRADING NOTE 3 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = A sin(Bx) + D, D gives the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

2200

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is centered on y = D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

The amplitude of $y = 1.5\cos(\pi t \div 6) + 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

2201

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The amplitude is 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 2

Interpret the mean and standard deviation of a normal model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Interpreting the parameters of a function in context leads to interpreting the parameters of a normal model.

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is normal with mean 400 hours and standard deviation 20 hours.

What does the 20 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

2203

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It measures spread: battery lives typically differ from the mean by roughly 20 hours (about 68% are within 20 hours of 400).

GRADING NOTE Any equivalent description of the standard deviation as a typical (root-mean-square) distance from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

2203

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is normal with mean 400 hours and standard deviation 20 hours.

About what percent of batteries last between 360 and 440 hours?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

2204

QUESTION ID
S58-F2-Q2

TYPE
Short answer

CORRECT ANSWER
About 95%

GRADING NOTE Any equivalent form. That is within 2 standard deviations of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Which change makes a normal curve wider?

A. Increasing the standard deviation

B. Increasing the mean

C. Decreasing the standard deviation

D. Decreasing the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

2205

QUESTION ID
S58-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Increasing the standard deviation

GRADING NOTE The standard deviation measures spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

2205

True or false?

Changing the mean of a normal model shifts the curve left or right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

2206

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The mean is its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a normal model with mean 50 and standard deviation 5, the value 60 is 10 standard deviations above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

2207

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(60 - 50) \div 5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 59 • HSS-ID.A.1 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Represent data with dot plots, histograms, and box plots

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display numerical data in plots	**6.SP.B.4**
BACKWARD 2	Compute measures of center and variation	**6.SP.B.5.c**
FORWARD 1	Decide whether a normal model fits a distribution	**HSS-ID.A.4**
FORWARD 2	Decide whether results are consistent with a model by simulation…	**HSS-IC.A.2**

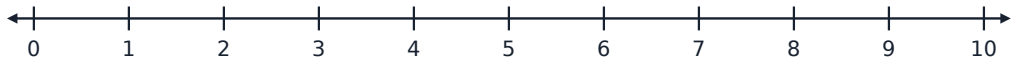
SET 59 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a dot plot of the numbers of books read by 10 students:

2, 3, 3, 4, 4, 4, 5, 5, 6, 9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

2210

QUESTION ID
S59-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
One dot above 2, two above 3, three above 4, two above 5, one above 6, and one above 9

GRADING NOTE Any correct dot plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the five-number summary of 2, 3, 3, 4, 4, 4, 5, 5, 6, 9. Use the median of each half (lower five and upper five values) as Q1 and Q3.

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

2211

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

Minimum 2, Q1 3, median 4, Q3 5, maximum 9

GRADING NOTE Form required: all five values. The median is the mean of the 5th and 6th values; Q1 and Q3 are the medians of the lower and upper five values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

For the data 2, 3, 3, 4, 4, 4, 5, 5, 6, 9 (Q1 and Q3 are the medians of the lower five and upper five values), use the 1.5 × IQR rule to decide whether 9 is an outlier.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

2212

QUESTION ID

S59-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. IQR = 5 − 3 = 2, and Q3 + 1.5(2) = 8. Since 9 > 8, it is an outlier.

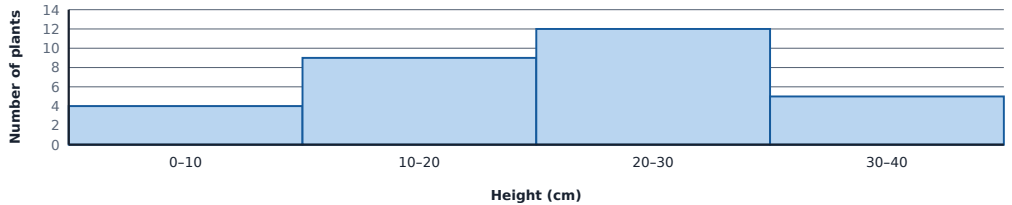
GRADING NOTE Form required: the answer with the fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

The histogram shows the heights of 30 plants. Each bar includes its lower edge but not its upper edge.

How many plants are at least 20 cm tall?



A. 21 **B.** 17 **C.** 5 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

2213

QUESTION ID

S59-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

GRADING NOTE The bars for 20–30 and 30–40 have 12 + 5 = 17 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

True or false?

A box plot shows the mean of a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

2214

QUESTION ID
S59-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A box plot shows the five-number summary; the mean is not shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 1

Display numerical data in plots • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Which display shows every data value, a dot plot or a box plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

2216

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A dot plot

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Make a dot plot of 1, 2, 2, 3, 3, 3, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

2217

GRADING NOTE Any correct dot plot.

QUESTION ID

S59-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

One dot above 1, two above 2, three above 3, and one above 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which display groups data values into intervals?

A. Box plot

B. Dot plot

C. Histogram

D. Ordered list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

2218

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Histogram

GRADING NOTE A histogram counts the values in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

True or false?

A box plot is made from the five-number summary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

2219

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Minimum, Q1, median, Q3, and maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A histogram shows each individual data value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

2220

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows only how many values fall in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 2

Compute measures of center and variation • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median of 3, 8, 5, 10, 4.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

2222

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. In order: 3, 4, 5, 8, 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 3, 8, 5, 10, 4.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

2223

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 30 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Q1 = 12 and Q3 = 20.

What is the interquartile range?

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

2224

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 20 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range of 2, 7, 9, 15 is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

2225

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 15 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 2, 3, 100 is 26.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

2226

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is (2 + 3) ÷ 2 = 2.5; 26.5 is the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 1

Decide whether a normal model fits a distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A data set is strongly skewed to the right.

Is a normal model appropriate? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

2228

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No. A normal distribution is symmetric, and this one is not.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Scores are normal with mean 75 and standard deviation 8.

What score is 1 standard deviation above the mean?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

2229

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

83

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Values are normal with mean 20 and standard deviation 4.

About what percent are between 12 and 28?

A. 99.7%

B. 50%

C. 68%

D. 95%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

2230

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 95%

GRADING NOTE 12 and 28 are 2 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

True or false?

In a normal distribution, the mean and the median are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

2231

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The distribution is symmetric about its mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

About 50% of a normal distribution lies within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

2232

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About 68% does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 2

Decide whether results are consistent with a model by simulation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Simulation results are summarized with the data displays of the MAIN set and then compared with a model.

ANSWER KEY Reduced copy of the question page, with its answer

In 100 simulated sets of 10 fair coin flips, 8 or more heads happened in 5 sets.

Estimate the probability of 8 or more heads in 10 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

2234

QUESTION ID
S59-F2-Q1

TYPE
Short answer

CORRECT ANSWER
0.05

GRADING NOTE Any equivalent form, such as 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

A spinner model says red comes up 25% of the time. In 200 spins, red came up 52 times.

What proportion is that? Is it consistent with the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

2235

QUESTION ID
S59-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.26; yes, it is close to 0.25.

GRADING NOTE Form required: the proportion and a judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A model says a coin lands heads 50% of the time.

Which result most strongly questions the model?

A. 52 heads in 100 flips

B. 6 heads in 10 flips

C. 95 heads in 100 flips

D. 48 heads in 100 flips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

2236

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 95 heads in 100 flips

GRADING NOTE 95 of 100 is very far from 50 and almost never happens with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A result that rarely happens in simulations of a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

2237

QUESTION ID
S59-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It suggests the model may be wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a coin lands heads 6 times in 10 flips, the coin must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

2238

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 of 10 happens often with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

Make histograms from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display data in a histogram	**6.SP.B.4**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Use a histogram's shape to decide whether a normal model fits	**HSS-ID.A.4**
FORWARD 2	Use a histogram of simulated results to judge a model	**HSS-IC.A.2**

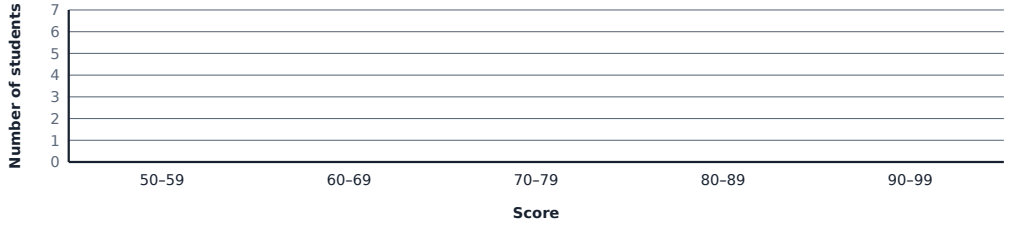
SET 76 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a histogram of these 20 test scores. Use the intervals 50–59, 60–69, 70–79, 80–89, and 90–99.

52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

2241

QUESTION ID
S76-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of heights 2, 4, 6, 5, and 3

GRADING NOTE Any correct histogram; the bar heights must add to 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the intervals 50–59, 60–69, 70–79, 80–89, and 90–99 for the scores 52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97:

how many scores are in the 70–79 interval, and which interval has the tallest bar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

2242

QUESTION ID
S76-M-Q2

TYPE
Short answer

CORRECT ANSWER
6 scores; the 70–79 interval is tallest.

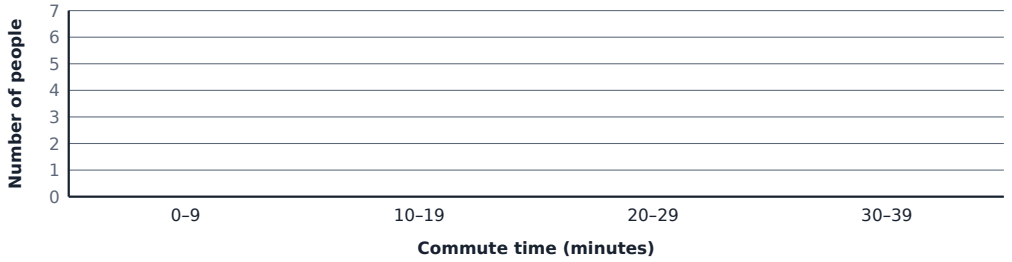
GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

Make a histogram of these 16 commute times, in minutes. Use the intervals 0–9, 10–19, 20–29, and 30–39.

4, 8, 12, 14, 15, 18, 19, 21, 22, 22, 25, 28, 29, 33, 36, 38



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

2243

QUESTION ID
S76-M-Q3

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of heights 2, 5, 6, and 3

GRADING NOTE Any correct histogram; the bar heights must add to 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram uses the intervals 10–19 and 20–29 for whole-number data.

In which bar is the value 20 counted?

A. 20–29

B. 10–19

C. Both

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q4/5

2244

QUESTION ID

S76-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20–29

GRADING NOTE Each value belongs to exactly one interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q4/5

True or false?

Using wider intervals for the same data gives a histogram with more bars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

2245

QUESTION ID

S76-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each wider interval covers more values, so fewer bars are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

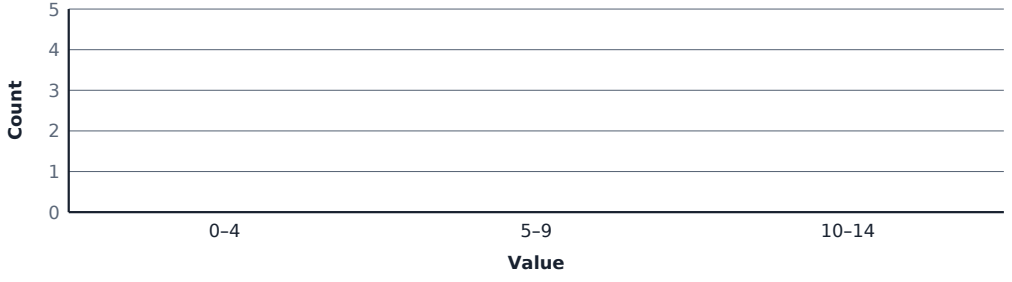
SET 76 • HSS-ID.A.1

BACKWARD 1

Display data in a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Make a histogram of 1, 3, 4, 6, 7, 8, 9, 12 with the intervals 0–4, 5–9, and 10–14.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

2247

QUESTION ID

S76-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Bars of heights 3, 4, and 1

GRADING NOTE Any correct histogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram has bars of heights 3, 5, and 2. How many data values does it show?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

2248

QUESTION ID

S76-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

2248

Which data are best shown in a histogram?

A. The heights of 200 students

B. Each student's favorite color

C. One test score

D. The names of 5 cities

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

2249

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The heights of 200 students

GRADING NOTE A histogram shows numerical data grouped into intervals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a histogram, the height of a bar shows how many values are in that interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

2250

QUESTION ID

S76-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Bar heights are counts (frequencies).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A histogram is the best display for categories such as favorite colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

2251

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Categories are shown with a bar graph, which has gaps between bars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many observations are in the data set 4, 7, 7, 9, 10, 12?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

2253

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram has bars of heights 2, 4, 6, 5, and 3. How many observations does it show?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

2254

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A survey recorded the number of pets owned by each of 25 students. How many observations are there?

A. 1

B. 25

C. 100

D. The total number of pets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

2255

QUESTION ID

S76-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25

GRADING NOTE One observation per student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

True or false?

The number of observations is the number of data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

2256

QUESTION ID

S76-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each value is one observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

True or false?

If 8 students each report 2 siblings, there are 16 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

2257

QUESTION ID

S76-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are 8 observations (each is the value 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 1

Use a histogram's shape to decide whether a normal model fits • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A histogram of 1,000 adult heights is single-peaked and symmetric, with mean 170 cm and standard deviation 8 cm. About how many heights are between 162 cm and 178 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

2259

QUESTION ID
S76-F1-Q1

TYPE
Short answer

CORRECT ANSWER
About 680

GRADING NOTE Any equivalent form. That interval is within 1 standard deviation of the mean: about 68% of 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram of household incomes is strongly skewed right. Would a normal model give good estimates of percents? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

2260

QUESTION ID
S76-F1-Q2

TYPE
Short answer

CORRECT ANSWER
No. A normal model is symmetric, and the data are strongly skewed.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

Which histogram shape suggests that a normal model may fit?

A. Two separate peaks

B. Flat (uniform)

C. Single-peaked and symmetric

D. Strongly skewed to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

2261

QUESTION ID

S76-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Single-peaked and symmetric

GRADING NOTE A normal curve is one symmetric bell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

True or false?

In a normal model, about 95% of values are within 2 standard deviations of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

2262

QUESTION ID

S76-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is part of the 68-95-99.7 rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any histogram with one peak is normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

2263

QUESTION ID
S76-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A single peak can still be skewed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 2

Use a histogram of simulated results to judge a model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

100 simulated trials of 20 fair coin flips give a histogram of the number of heads centered near 10. What does the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

2265

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The distribution of the number of heads when the coin is fair: which counts are typical and which are rare.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

In that simulation, 2 of the 100 trials had 15 or more heads. Estimate the probability of 15 or more heads in 20 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

2266

QUESTION ID
S76-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE Any equivalent form, such as 2%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner is claimed to land on blue 50% of the time. In 50 spins it landed on blue 12 times. In 200 simulated sets of 50 spins with a 50% chance, 12 or fewer blue never happened.

What do you conclude?

A. Nothing can be concluded.

B. The result supports the 50% claim.

C. The spinner lands on blue exactly 24% of the time.

D. The result is not consistent with the 50% claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

2267

QUESTION ID
S76-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. The result is not consistent with the 50% claim.

GRADING NOTE The observed result is far outside what the model usually produces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

True or false?

A simulation repeats a random process many times to show which results are typical under a model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

2268

QUESTION ID
S76-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is the purpose of the simulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

True or false?

If a result never happens in 200 simulated trials, it is impossible under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

2269

QUESTION ID
S76-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is rare, not impossible; more trials might produce it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

Make box plots from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Draw and read box plots	**6.SP.B.4**
BACKWARD 2	Find medians, quartiles, and ranges	**6.SP.B.5.c**
FORWARD 1	Estimate the quartiles of a normal distribution	**HSS-ID.A.4**
FORWARD 2	Compare two treatments with box plots and a randomization test	**HSS-IC.B.5**

SET 77 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a box plot of 3, 5, 6, 7, 8, 9, 10, 12, 15. (Use the median of each half, not counting the middle value, as Q1 and Q3.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

2272

QUESTION ID
S77-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
Whiskers from 3 to 5.5 and from 11 to 15; a box from Q1 = 5.5 to Q3 = 11 with the median line at 8

GRADING NOTE Any correct box plot. Lower half 3, 5, 6, 7 gives Q1 = 5.5; upper half 9, 10, 12, 15 gives Q3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the five-number summary of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

2273

QUESTION ID

S77-M-Q2

TYPE

Short answer

CORRECT ANSWER

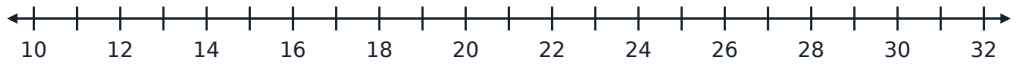
Minimum 12, Q1 15, median 19, Q3 23.5, maximum 30

GRADING NOTE Form required: all five values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

Make a box plot of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

2274

QUESTION ID
S77-M-Q3

TYPE
Drawing / plotting

CORRECT ANSWER

Whiskers from 12 to 15 and from 23.5 to 30; a box from 15 to 23.5 with the median line at 19

GRADING NOTE Any correct box plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

In a box plot, what does the length of the box show?

A. The interquartile range

B. The mean

C. The range

D. The median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

2275

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The interquartile range

GRADING NOTE The box runs from Q1 to Q3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

True or false?

In a box plot whose whiskers reach the minimum and maximum, each whisker covers about a quarter of the data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

2276

QUESTION ID

S77-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The quartiles split the ordered data into four parts of about equal size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 1

Draw and read box plots • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Draw a box plot with minimum 2, Q1 4, median 5, Q3 8, and maximum 10.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

2278

QUESTION ID

S77-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Whiskers from 2 to 4 and from 8 to 10; a box from 4 to 8 with a line at 5

GRADING NOTE Any correct box plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A box plot has Q1 = 20 and Q3 = 35. What is the interquartile range?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

2279

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which part of a box plot marks the median?

A. The line inside the box

B. The left end of the box

C. The right whisker

D. The left whisker

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

2280

QUESTION ID

S77-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The line inside the box

GRADING NOTE The median divides the box.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

2280

True or false?

A box plot shows the minimum and the maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

2281

QUESTION ID

S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are the ends of the whiskers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A box plot shows how many data values there are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

2282

QUESTION ID

S77-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows the five-number summary, not the count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 2

Find medians, quartiles, and ranges • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median of 4, 9, 2, 7, 5, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

2284

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. In order: 2, 4, 5, 7, 8, 9; (5 + 7) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The ordered data are 2, 4, 5, 7, 8, 9. Find Q1, the median of the lower half 2, 4, 5.

Q1 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

2285

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of 14, 3, 9, 21?

A. 21

B. 18

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

2286

QUESTION ID

S77-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18

GRADING NOTE 21 − 3 = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The interquartile range is Q3 − Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

2287

QUESTION ID

S77-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The median of 1, 3, 100 is 34.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

2288

QUESTION ID

S77-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is the middle value, 3; 34 is the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 1

Estimate the quartiles of a normal distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

For a normal distribution, the quartiles are about $\mu - 0.67\sigma$ and $\mu + 0.67\sigma$. Estimate Q1 and Q3 for mean 100 and standard deviation 15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

2290

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Q1 ≈ 90 and Q3 ≈ 110

GRADING NOTE Form required: both values (about 89.95 and 110.05 are also accepted). 0.67 × 15 ≈ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using those quartiles, estimate the interquartile range of that normal distribution.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

2291

QUESTION ID

S77-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 20

GRADING NOTE Any equivalent form. 110 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a normal distribution, about what percent of values are between Q1 and Q3?

A. 25%

B. 95%

C. 50%

D. 68%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

2292

QUESTION ID
S77-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 50%

GRADING NOTE Q1 to Q3 always holds the middle half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

2292

True or false?

A box plot of data from a normal distribution is roughly symmetric about the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

2293

QUESTION ID
S77-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A normal distribution is symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For normal data, the median is far from the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

2294

QUESTION ID

S77-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For a normal distribution they are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 2

Compare two treatments with box plots and a randomization test • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Students were randomly assigned to a new study method (group A) or the usual method (group B). Box plots show a median of 78 for A and 70 for B, with similar spreads. In 1,000 random regroupings, a difference in means at least as large as observed happened 30 times.

Using a 5% threshold, is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

2296

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Yes.
30 ÷ 1,000 = 3%, which is below 5%.

GRADING NOTE Form required: the decision with the proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

Why is it useful to draw side-by-side box plots of the two groups before running the test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

2297

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
They show the centers, spreads, overlap, and any outliers of the two groups at a glance.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

In another randomized comparison, random regroupings matched or beat the observed difference 22% of the time.

Using a 5% threshold, what is the conclusion?

A. The difference is significant.

B. The methods are proved to have the same effect.

C. The new method is proved to work.

D. The difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

2298

QUESTION ID

S77-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The difference is not significant.

GRADING NOTE 22% is above 5%, so chance could easily explain the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

True or false?

In a randomized comparison, a difference that rarely appears in random regroupings is evidence of a treatment effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

2299

QUESTION ID
S77-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Chance alone rarely produces it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

True or false?

Overlapping box plots prove that two treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

2300

QUESTION ID
S77-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Overlap shows variation; it does not prove the effects are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

2300

SET 60 • HSS-ID.A.2 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Compare the center and spread of two or more data sets

MAIN	Algebra I main question	**HSS-ID.A.2**
BACKWARD 1	Tell measures of center from measures of variation	**6.SP.A.3**
BACKWARD 2	Compare two distributions using center and variability	**7.SP.B.3**
FORWARD 1	Use simulation to decide whether a difference between treatment…	**HSS-IC.B.5**
FORWARD 2	Evaluate reports based on data	**HSS-IC.B.6**

SET 60 • HSS-ID.A.2

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Class A scores: 70, 75, 80, 85, 90. Class B scores: 60, 70, 80, 90, 100.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

2303

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

Both have mean and median 80. Class B is more spread out: its range is 40 compared with 20 (and its standard deviation is larger).

GRADING NOTE Form required: a comparison of center and of spread, with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the population standard deviation (divide by n) of 2, 4, 4, 4, 5, 5, 7, 9.

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

2304

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. The mean is 5; the squared deviations add to 32; 32 ÷ 8 = 4, and √4 = 2. (Dividing by n − 1 gives about 2.14.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

Store X's daily sales have median 50 and IQR 10. Store Y's have median 45 and IQR 25.

Which store typically sells more? Which is more consistent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

2305

QUESTION ID

S60-M-Q3

TYPE

Short answer

CORRECT ANSWER

Store X on both: its median is higher, and its IQR is smaller.

GRADING NOTE Form required: both answers with reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

Which measure of spread is best for skewed data with outliers?

A. Range

B. Standard deviation

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

2306

QUESTION ID

S60-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The IQR uses the middle half of the data, so outliers barely affect it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

2307

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterexample: 70, 75, 80, 85, 90 and 60, 70, 80, 90, 100 both have mean 80, but the second set is more spread out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 1

Tell measures of center from measures of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Which describes a typical value of a data set, the median or the IQR?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

2309

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The median

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of the data set 4, 4, 4, 4?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

2310

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. There is no variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

Which is a measure of variation?

A. Median

B. Mode

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

2311

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The median, the mean, and the mode are measures of center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

True or false?

A measure of center summarizes all of the values with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

2312

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, the mean or the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range is a measure of center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

2313

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range measures variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 2

Compare two distributions using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Two classes have mean scores 78 and 84, and each has MAD 3.

How many MADs apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

2315

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (84 − 78) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Team A's mean height is 170 cm and Team B's is 180 cm; each has MAD 5 cm.

Express the difference of the means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

2316

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 MADs

GRADING NOTE Any equivalent form. 10 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

Two means differ by 12, and each data set has MAD 4.

How many MADs is the difference?

A. 48

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

2317

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

When two distributions have little variability, a small difference in their centers is easier to notice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

2318

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The data overlap less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If two data sets overlap, their means must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

2319

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Overlapping sets can have different means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 1

Use simulation to decide whether a difference between treatments is significant • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Plants were randomly assigned to a fertilizer group or a control group with no fertilizer. The fertilizer group averaged 24 cm and the control group 20 cm. To test chance alone, the same heights were randomly regrouped 1,000 times, and a difference of 4 cm or more (first group minus second) happened 12 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

2321

QUESTION ID
S60-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. Only about 1.2% of the regroupings gave a difference this large, which is below 5%, so chance alone is an unlikely explanation.

GRADING NOTE Form required: the decision with the simulated proportion (12 ÷ 1,000 = 1.2%) compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why are subjects randomly assigned to treatments in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

2322

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Random assignment tends to balance other variables between the groups, so a large difference can be credited to the treatment rather than to how the groups were formed.

GRADING NOTE Any equivalent explanation. Random assignment balances other variables on average; it does not make the groups identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a regrouping simulation, a difference as large as the observed one happened 400 times out of 1,000.

What does that suggest?

A. The treatment definitely works.

B. The difference could easily be due to chance.

C. The treatment definitely does not work.

D. The experiment had no control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

2323

QUESTION ID
S60-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. The difference could easily be due to chance.

GRADING NOTE 40% is not rare.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

True or false?

In a randomized experiment, a small simulated probability for the observed difference is evidence that the treatment had an effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

2324

QUESTION ID
S60-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Random assignment makes chance the main other explanation, and chance alone rarely produces the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

True or false?

In an experiment, the control group receives the new treatment being tested.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

2325

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The control group is the comparison group: it gets no treatment, a placebo, or a standard treatment, not the new one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 2

Evaluate reports based on data • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "9 out of 10 dentists recommend Brand X," based on 10 dentists chosen by Brand X.

What is the flaw?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

2327

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The sample is tiny and was chosen by the company, so it is likely biased.

GRADING NOTE Any equivalent explanation that names size or bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An online poll of a website's visitors finds 80% support a policy.

Can the result be applied to all adults? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

2328

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER

No. The respondents chose themselves and are only that site's visitors, so they do not represent all adults.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is most likely to represent a school's students?

A. The first 50 students to arrive

B. The 50 students in the chess club

C. A random sample of 50 students from the school roster

D. 50 volunteers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

2329

QUESTION ID
S60-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. A random sample of 50 students from the school roster

GRADING NOTE Only random selection gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

True or false?

A larger random sample generally gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

2330

QUESTION ID
S60-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Estimates from larger random samples vary less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A survey with many volunteers is free of bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

2331

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers may differ from the population, whatever their number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.3 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret differences in shape, center, and spread, accounting for outliers

MAIN	Algebra I main question	**HSS-ID.A.3**
BACKWARD 1	Describe a distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Choose measures of center and spread that fit the data	**6.SP.B.5.d**
FORWARD 1	Use z-scores to judge how unusual a value is	**HSS-ID.A.4**
FORWARD 2	Distinguish a population parameter from a sample statistic — nea…	**HSS-IC.A.1**

SET 61 • HSS-ID.A.3

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

Find the mean and the median. Which better describes a typical salary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

2334

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

Mean ≈ 74.29 thousand; median 45 thousand. The median is better: the one salary of 250 pulls the mean up.

GRADING NOTE Form required: both values (the mean to the nearest hundredth or as 520/7) and the choice with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

If the salary of 250 is removed, what happens to the mean and to the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

2335

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

The mean drops from about 74.29 to 45; the median stays 45.

GRADING NOTE Form required: both changes. 270 ÷ 6 = 45, and the middle values are 45 and 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

A histogram of wait times has a long tail on the right.

Describe its shape and the usual relationship between its mean and median.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

2336

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

Skewed right; the mean is usually greater than the median, because the long right tail pulls the mean up.

GRADING NOTE Form required: the shape and the usual relationship. This is a typical tendency, not a rule that holds for every data set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which data set has the largest standard deviation?

A. 28, 29, 30, 31, 32

B. 25, 30, 30, 30, 35

C. 10, 20, 30, 40, 50

D. 30, 30, 30, 30, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

2337

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10, 20, 30, 40, 50

GRADING NOTE All four have mean 30; the values 10, 20, 30, 40, 50 are spread farthest from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

True or false?

An outlier affects the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

2338

QUESTION ID
S61-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The mean uses every value; the median depends only on the middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 1

Describe a distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5.

Describe the shape of the distribution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

2340

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, centered at 3

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value in 2, 3, 3, 3, 4, 15 is unusual?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

2341

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A dot plot has most values on the left and a few values far to the right.

What is its shape?

A. Skewed right

B. Symmetric

C. Uniform

D. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

2342

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Skewed right

GRADING NOTE The tail points to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A distribution can be described by its center, its spread, and its shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

2343

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE These are the three main features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A symmetric distribution must have an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

2344

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many symmetric distributions have no outliers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 2

Choose measures of center and spread that fit the data • Grade 6 • Statistics & Probability • 6.SP.B.5.d

ANSWER KEY Reduced copy of the question page, with its answer

For data with an extreme outlier, should you report the mean or the median as the center? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

2346

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The median, because one extreme value barely changes it.

GRADING NOTE Form required: the choice and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For symmetric data with no outliers, which measures of center and spread are usually reported?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

2347

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The mean and the mean absolute deviation

GRADING NOTE Any equivalent form. (The mean and the standard deviation are also accepted.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Home prices in a town include one mansion.

Which measure best describes a typical price?

A. The mean

B. The median

C. The maximum

D. The range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

2348

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The median

GRADING NOTE The mansion pulls the mean up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

The median is not changed much by a single extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

2349

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It depends on the middle of the ordered data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

The IQR changes greatly when the largest value is made even larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

2350

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The IQR depends on Q1 and Q3, which do not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 1

Use z-scores to judge how unusual a value is • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A distribution has mean 50 and standard deviation 4.

What is the z-score of 58?

z = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

2352

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (58 − 50) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A score of 92 on a test with mean 80 and standard deviation 6, or a score of 85 on a test with mean 70 and standard deviation 10: which is better relative to its test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

2353

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

92: its z-score is 2, compared with 1.5.

GRADING NOTE Form required: the score and both z-scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

In a normal distribution, which z-score is the most unusual?

A. −3.1

B. −0.5

C. 1.2

D. 2.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

2354

QUESTION ID
S61-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. −3.1

GRADING NOTE It is the farthest from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A z-score of 0 means the value equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

2355

QUESTION ID
S61-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is 0 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A z-score of −2 is above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

2356

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative z-scores are below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

FORWARD 2

Distinguish a population parameter from a sample statistic • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Centers and spreads computed from data are sample statistics that estimate population parameters.

ANSWER KEY Reduced copy of the question page, with its answer

What is the difference between a population and a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

2358

QUESTION ID
S61-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A population is the whole group of interest; a sample is part of it that is actually studied.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 200 students has a mean of 7.1 hours of sleep.

What does 7.1 estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

2359

QUESTION ID
S61-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The mean hours of sleep of all the students in the population.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A parameter describes

A. a graph

B. one individual

C. a population

D. a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

2360

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a population

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Different random samples from the same population can give different sample means.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

2361

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A sample statistic always equals the population parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

2362

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is only an estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 62 • HSS-ID.B.5 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Summarize categorical data in two-way frequency tables and interpret relative frequencies

MAIN	Algebra I main question	**HSS-ID.B.5**
BACKWARD 1	Read two-way tables for association	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a percent	**7.RP.A.3**
FORWARD 1	Use a two-way table to find conditional probabilities and test inde…	**HSS-CP.A.4**
FORWARD 2	Explain conditional probability and independence in everyday lang..	**HSS-CP.A.5**

SET 62 • HSS-ID.B.5

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a survey of 200 students.

What percent of all the students play a sport?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

2365

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

52.5%

GRADING NOTE Any equivalent form. 105 ÷ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

What percent of Grade 10 students play a sport? What percent of Grade 9 students do?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

2366

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

60% of Grade 10; 45% of Grade 9

GRADING NOTE Form required: both conditional relative frequencies. 60 ÷ 100 and 45 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

Is there an association between grade and playing a sport? Explain.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

2367

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. 60% of Grade 10 students play a sport compared with 45% of Grade 9 students.

GRADING NOTE Form required: the answer and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the joint relative frequency of "Grade 9 and does not play a sport"?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. 50% **B.** 57.9% **C.** 55% **D.** 27.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

2368

QUESTION ID

S62-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 27.5%

GRADING NOTE 55 out of all 200 students. (55% is the share of Grade 9 students; about 57.9% is the share of non-players.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

True or false?

Of the students who play a sport, more than half are in Grade 10.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

2369

QUESTION ID

S62-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 ÷ 105 ≈ 57%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 1

Read two-way tables for association • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

12 of 20 boys and 9 of 15 girls in a class like math.

How many students in all like math?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

2371

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

12 of 20 boys and 9 of 15 girls in a class like math.

Is there an association between gender and liking math? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

2372

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 ÷ 20 = 60% and 9 ÷ 15 = 60%: the same share.

GRADING NOTE Form required: the answer with both relative frequencies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the row and column totals of a two-way table called?

A. Joint frequencies

B. Conditional frequencies

C. Marginal frequencies

D. Outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

2373

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Marginal frequencies

GRADING NOTE They are written in the margins of the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

True or false?

A two-way table shows data on two categorical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

2374

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One variable for the rows and one for the columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a two-way table, every row must have the same total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

2375

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Row totals can be different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a percent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write 45 out of 100 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

2377

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 60 out of 105 as a percent, rounded to the nearest tenth.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

2378

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

57.1%

GRADING NOTE Form required: rounded to the nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

27 is what percent of 60?

A. 45%

B. 60%

C. 33%

D. 27%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

2379

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45%

GRADING NOTE 27 ÷ 60 = 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

55 out of 200 is 27.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

2380

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 55 ÷ 200 = 0.275.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

40 out of 95 is 40%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

2381

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40 ÷ 95 ≈ 42.1%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 1

Use a two-way table to find conditional probabilities and test independence • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student from the table is chosen at random.

Find P(plays a sport | Grade 10).

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

2383

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form. 60 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are "Grade 10" and "plays a sport" independent? Explain with probabilities.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

2384

QUESTION ID
S62-F1-Q2

TYPE
Short answer

CORRECT ANSWER
No. P(plays a sport) = 0.525, but P(plays a sport | Grade 10) = 0.6.

GRADING NOTE Form required: the answer with both probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is P(Grade 9 | plays a sport)?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. About 0.43

B. 0.225

C. 0.525

D. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

2385

QUESTION ID
S62-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. About 0.43

GRADING NOTE 45 ÷ 105 ≈ 0.43.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

True or false?

If P(B) > 0, events A and B are independent exactly when P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

2386

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Knowing B does not change the probability of A. P(B) > 0 is needed for P(A | B) to be defined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A | B) always equals P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

2387

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE In the table, P(sport | Grade 10) = 0.6 but P(Grade 10 | sport) ≈ 0.57.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 2

Explain conditional probability and independence in everyday language • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Explain in everyday language what P(late | rain) > P(late) means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

2389

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A person is more likely to be late on a rainy day than on a typical day.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent, P(A) = 0.3, and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

2390

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.15

GRADING NOTE Any equivalent form. 0.3 × 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of events is most likely independent?

A. Height and shoe size

B. Studying and passing a test

C. Flipping heads and rolling a 6

D. Rain and carrying an umbrella

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

2391

QUESTION ID
S62-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Flipping heads and rolling a 6

GRADING NOTE A coin and a number cube do not affect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

2392

QUESTION ID
S62-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the multiplication rule for independent events.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If P(A | B) = 0.8 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

2393

QUESTION ID
S62-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Knowing B doubles the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear, quadratic, or exponential function to data and use it to solve problems

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Describe patterns in scatter plots	**8.SP.A.1**
BACKWARD 2	Write linear and exponential functions through given points	**HSF-LE.A.2**
FORWARD 1	Use logarithms with an exponential model fitted to data	**HSF-LE.A.4**
FORWARD 2	Fit a sinusoidal model to periodic data	**HSF-TF.B.5**

SET 63 • HSS-ID.B.6.a

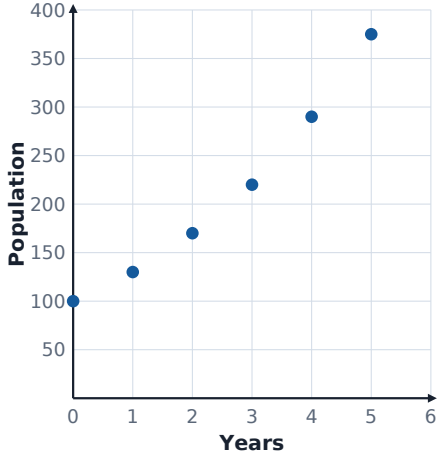
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows a population over 5 years.

Which type of function fits best? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

2396

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: the population is multiplied by about 1.3 each year (130 ÷ 100, 170 ÷ 130, ...).

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An exponential model for the data is $P(t) = 100(1.3)^t$.

Use it to predict the population at t = 8, to the nearest whole number.

P(8) ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

2397

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

816

GRADING NOTE Form required: rounded to the nearest whole number. $100(1.3)^8 \approx 815.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Data: x = 0, 1, 2, 3, 4 and y = 2, 5, 10, 17, 26.

Which quadratic function fits these data exactly? Write it.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

2398

QUESTION ID

S63-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 2$

GRADING NOTE Any equivalent form, such as $(x + 1)^2 + 1$. The second differences are all 2, so the x^2 coefficient is 1; y(0) = 2 gives the constant 2, and y(1) = 5 gives the x coefficient 2. Check: y(4) = 16 + 8 + 2 = 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

A scatter plot rises and then falls in a symmetric curve.

Which model fits best?

A. Quadratic

B. Exponential growth

C. Exponential decay

D. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

2399

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Quadratic

GRADING NOTE A parabola rises and then falls symmetrically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

True or false?

Using a model to predict far outside the range of the data is always reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

2400

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Extrapolation assumes the pattern continues, which it may not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 1

Describe patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

2402

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is an outlier in a scatter plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

2403

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A point far from the overall pattern of the others

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Points fall from left to right in a curved pattern.

Which describes the association?

A. Positive and nonlinear

B. Positive and linear

C. Negative and nonlinear

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

2404

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Negative and nonlinear

GRADING NOTE Falling means negative; curved means nonlinear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

True or false?

A scatter plot shows the relationship between two quantitative variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

2405

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point is a pair of values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

True or false?

If the points show no pattern, there is a strong association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

2406

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pattern means little or no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 2

Write linear and exponential functions through given points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function through (0, 100) and (1, 130).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

2408

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.3)^x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, 2) and (4, 26).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

2409

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

Which exponential function $y = ab^x$ $(b > 0)$ passes through $(0, 50)$ and $(2, 12.5)$?

A. $y = 50(0.25)^x$

B. $y = 12.5(2)^x$

C. $y = 50(0.5)^x$

D. $y = 50 - 18.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

2410

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 50(0.5)^x$

GRADING NOTE $a = 50$ and $50b^2 = 12.5$, so $b = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 3(2)^x$ passes through (2, 12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

2411

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $3 \cdot 4 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q4/5

True or false?

$y = 5(1.2)^x$ passes through (0, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

2412

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 1

Use logarithms with an exponential model fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$P(t) = 100(1.3)^t$.

When does P reach 1,000? Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

2414

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.8

GRADING NOTE Form required: rounded to the nearest tenth. t = log 10 ÷ log 1.3 ≈ 8.78.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2(1.5)^x = 30$. Round to the nearest hundredth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

2415

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6.68

GRADING NOTE Form required: rounded to the nearest hundredth. $1.5^x = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

Which is the exact solution of $100(1.3)^t = 500$? (log means base 10.)

A. t = log 1.3 ÷ log 5

B. t = 500 ÷ 130

C. t = log 5 ÷ log 1.3

D. t = 5 ÷ 1.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

2416

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = log 5 ÷ log 1.3

GRADING NOTE $1.3^t = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

True or false?

If $1.3^t = 1.69$, then t = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

2417

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.3^2 = 1.69$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S63-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $10^t = 50$, then $t = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

2418

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = \log 50 \approx 1.70$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

FORWARD 2

Fit a sinusoidal model to periodic data • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Monthly average temperatures peak at 30 °C and bottom out at 10 °C.

What amplitude and midline fit a sinusoidal model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

2420

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Amplitude 10; midline y = 20

GRADING NOTE Form required: both. (30 − 10) ÷ 2 and (30 + 10) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What period fits monthly temperatures that repeat every year?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

2421

QUESTION ID
S63-F2-Q2

TYPE
Short answer

CORRECT ANSWER
12 months

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

Daylight hours vary from 9 to 15 over a year.

Which model fits?

A. A linear model

B. A sinusoidal model with amplitude 3 and period 12 months

C. A quadratic model with vertex at month 12

D. An exponential model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

2422

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sinusoidal model with amplitude 3 and period 12 months

GRADING NOTE The pattern repeats each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

True or false?

A repeating pattern in data suggests a sinusoidal model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

2423

QUESTION ID
S63-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Sinusoids are periodic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

True or false?

Data that vary between 9 and 15 have amplitude 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

2424

QUESTION ID
S63-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The amplitude is half the distance: 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Construct scatter plots from paired data and describe how the variables are related

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Make and describe scatter plots	**8.SP.A.1**
BACKWARD 2	Plot points and find distances on the coordinate plane	**6.NS.C.8**
FORWARD 1	Choose an exponential model from a scatter plot and solve it with…	**HSF-LE.A.4**
FORWARD 2	Evaluate claims based on scatter plots	**HSS-IC.B.6**

SET 83 • HSS-ID.B.6.a

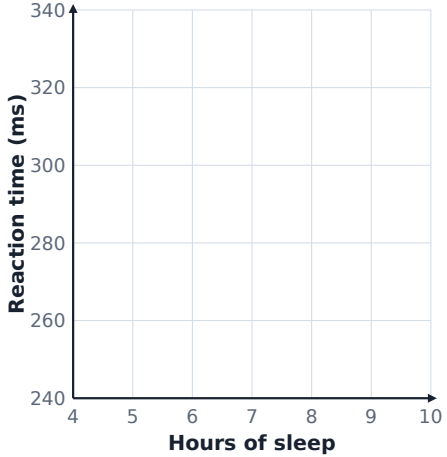
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

Make a scatter plot of hours of sleep (x) and reaction time in milliseconds (y):

(5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

2427

QUESTION ID

S83-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Six points plotted at the given pairs, falling from left to right

GRADING NOTE Any correct scatter plot with all six points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe the association between hours of sleep and reaction time in the data (5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255): its direction, form, and strength.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

2428

QUESTION ID
S83-M-Q2

TYPE
Short answer

CORRECT ANSWER
Negative, roughly linear, and strong: more sleep goes with shorter reaction times.

GRADING NOTE Form required: direction, form, and strength.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

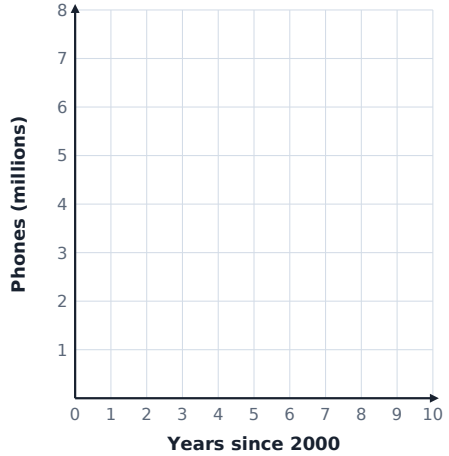
ANSWER KEY Reduced copy of the question page, with its answer

Make a scatter plot of years since 2000 (x) and phones in millions (y):

(0, 1), (2, 1.5), (4, 2.3), (6, 3.4), (8, 5.1), (10, 7.6)

Which type of model does the plot suggest?

Model: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

2429

QUESTION ID

S83-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Six points rising from left to right and curving upward

Written: Exponential: the points curve upward, growing by about 1.5 times every 2 years.

GRADING NOTE Form required: the model type with a reason. Both the plot and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

When you make a scatter plot to see whether one variable helps explain another, which variable usually goes on the horizontal axis?

A. Whichever variable has larger numbers

B. Whichever variable has fewer values

C. The explanatory variable (the one that may explain the other)

D. The response variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

2430

QUESTION ID

S83-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. The explanatory variable (the one that may explain the other)

GRADING NOTE By convention the explanatory variable is x and the response variable is y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

True or false?

Both axes of a scatter plot must start at 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

2431

QUESTION ID

S83-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Axes can start near the data, as long as the scales are labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

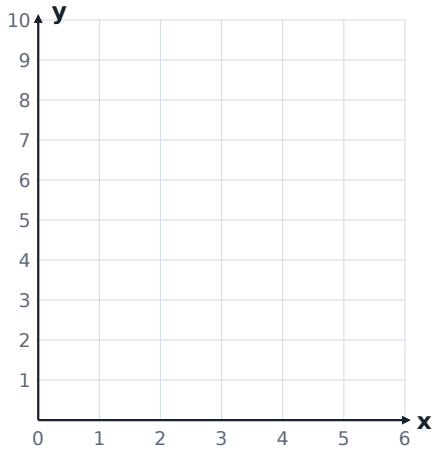
SET 83 • HSS-ID.B.6.a

BACKWARD 1

Make and describe scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Plot (1, 2), (2, 3), (3, 5), (4, 6), and (5, 8) as a scatter plot.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

2433

QUESTION ID

S83-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Five points rising from left to right

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right but curve upward. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

2434

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive and nonlinear

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes a scatter plot whose points show no trend?

A. Linear association

B. Strong positive association

C. Strong negative association

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

2435

QUESTION ID

S83-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. No association

GRADING NOTE No trend means no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

True or false?

Each point in a scatter plot represents one pair of values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

2436

QUESTION ID

S83-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One point per pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A cluster in a scatter plot is the same thing as an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

2437

QUESTION ID

S83-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A cluster is a group of points; an outlier is a single point far from the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

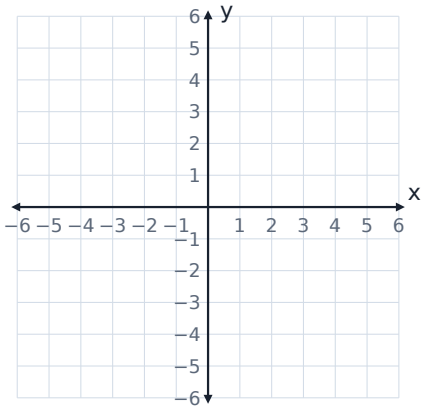
SET 83 • HSS-ID.B.6.a

BACKWARD 2

Plot points and find distances on the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Plot the points (2, 5), (−3, 1), and (0, −4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

2439

QUESTION ID

S83-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Three points at (2, 5), (−3, 1), and (0, −4)

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (−3, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

2440

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

GRADING NOTE Any equivalent form. |5 − (−3)| = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

Which point lies on the y-axis?

A. (1, −1)

B. (0, −4)

C. (−4, 0)

D. (4, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

2441

QUESTION ID

S83-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (0, −4)

GRADING NOTE Points on the y-axis have x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

True or false?

The point (−2, 6) is in Quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

2442

QUESTION ID

S83-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Quadrant II has $x < 0$ and $y > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The distance between (1, −3) and (1, 4) is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

2443

QUESTION ID

S83-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $|4 - (-3)| = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 1

Choose an exponential model from a scatter plot and solve it with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A model for the phone data is $y = 1(1.22)^x$, where x is years since 2000 and y is millions of phones. When does it predict 20 million? Round to the nearest tenth. (log means base 10.)

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

2445

QUESTION ID
S83-F1-Q1

TYPE
Short answer

CORRECT ANSWER
15.1 years after 2000

GRADING NOTE Form required: rounded to the nearest tenth. x = log 20 ÷ log 1.22 ≈ 15.07.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using $y = 1(1.22)^x$, how many years does it take the number of phones to double? Round to the nearest tenth. (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

2446

QUESTION ID
S83-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3.5

GRADING NOTE Form required: rounded to the nearest tenth. log 2 ÷ log 1.22 ≈ 3.49.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

Solve $4(1.25)^x = 20$. (log means base 10.)

A. $x = \log 5 \div \log 1.25 \approx 7.21$

B. $x = 5 \div 1.25$

C. $x = 4$

D. $x = \log 20 \div \log 1.25 \approx 13.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

2447

QUESTION ID
S83-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $x = \log 5 \div \log 1.25 \approx 7.21$

GRADING NOTE $1.25^x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

True or false?

If $y = a \cdot b^x$ with $a > 0$ and $b > 1$, then y doubles every log 2 ÷ log b units of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

2448

QUESTION ID

S83-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $b^d = 2$ gives $d = \log 2 \div \log b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The model $y = 1(1.22)^x$ predicts 0 phones in 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

2449

QUESTION ID
S83-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At x = 0, y = 1 (million).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 2

Evaluate claims based on scatter plots • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A website plots shoe size against reading score for students in Grades 1 to 8, finds a strong positive association, and claims that bigger feet improve reading. What is wrong?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

2451

QUESTION ID
S83-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Age (grade) is a lurking variable: older students have bigger feet and read better. Observational data do not show cause.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A scatter plot of 6 students from one class shows a strong trend. Why should you be careful applying it to all students?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

2452

QUESTION ID
S83-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The sample is small and was not randomly selected, so it may not represent all students.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

Which detail would make a claim based on a scatter plot more trustworthy?

A. Both axes start at 0.

B. The plot uses bright colors.

C. There are only 5 points.

D. The data come from a large random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

2453

QUESTION ID
S83-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. The data come from a large random sample.

GRADING NOTE Large random samples represent the population better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q3**
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

True or false?

Extrapolating a trend far beyond the plotted data can give misleading predictions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

2454

QUESTION ID

S83-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The pattern may not continue outside the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

True or false?

If a scatter plot shows a strong trend, one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

2455

QUESTION ID
S83-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Association does not show causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Assess the fit of a function by plotting and analyzing residuals

MAIN	Algebra I main question	**HSS-ID.B.6.b**
BACKWARD 1	Use a fitted function to make predictions	**HSS-ID.B.6.a**
BACKWARD 2	Subtract rational numbers	**7.NS.A.1.c**
FORWARD 1	Use logarithms to solve exponential models fitted to data	**HSF-LE.A.4**
FORWARD 2	Evaluate reports that use fitted models	**HSS-IC.B.6**

SET 64 • HSS-ID.B.6.b

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

ANSWER KEY Reduced copy of the question page, with its answer

The model $\hat{y} = 2.2x + 0.8$ is fitted to the data (1, 3), (2, 5), (3, 8), (4, 9), (5, 12).

Find the residual for each point.

Residuals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

2458

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

0, −0.2, 0.6, −0.6, 0.2

GRADING NOTE Form required: all five, in order. Residual = actual − predicted; the predictions are 3, 5.2, 7.4, 9.6, 11.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

For that model, what does the residual −0.6 at x = 4 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

2459

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

The actual y-value is 0.6 less than the model predicts.

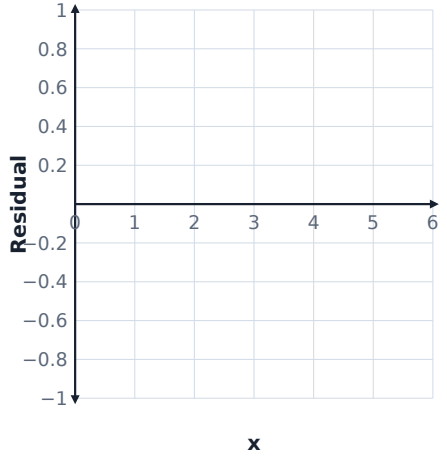
GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Make a residual plot for the residuals 0, −0.2, 0.6, −0.6, 0.2 at x = 1, 2, 3, 4, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

2460

QUESTION ID

S64-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The points (1, 0), (2, −0.2), (3, 0.6), (4, −0.6), and (5, 0.2), scattered above and below 0 with no pattern

GRADING NOTE Any correct plot. The lack of a pattern supports a linear form; the small size of the residuals (all within 0.6) shows the errors are small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

2460

A residual plot shows a clear U-shaped pattern.

What does that suggest?

A. The correlation is exactly 1.

B. A linear model is not a good fit; a curved model may fit better.

C. The linear model fits well.

D. There are no residuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

2461

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A linear model is not a good fit; a curved model may fit better.

GRADING NOTE A pattern in the residuals means the model misses part of the relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

True or false?

A good linear fit has residuals scattered randomly around 0, with no pattern.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

2462

QUESTION ID
S64-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The model then captures the pattern of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 1

Use a fitted function to make predictions • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The model is $\hat{y} = 2.2x + 0.8$.

Predict y when x = 6.

$\hat{y}$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

2464

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. 13.2 + 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The model is $y = 3x^2 - 2$.

Predict y when x = 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

2465

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

46

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-B1-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The model is $y = 50(0.9)^x$.

Predict y when x = 2.

A. 50

B. 40.5

C. 45

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

2466

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40.5

GRADING NOTE 50(0.81) = 40.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

True or false?

A predicted value comes from the model, not from the data table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

2467

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the model's output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

True or false?

If a model predicts 14 and the actual value is 14, the residual is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

2468

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The residual is $14 - 14 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 8 − 7.4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

2470

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 9 − 9.6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

2471

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find 5 − 5.2.

A. −10.2

B. 10.2

C. 0.2

D. −0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

2472

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.2

GRADING NOTE 5.2 is greater than 5, so the difference is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$3 - (-2) = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

2473

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtracting −2 adds 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-4 - 6 = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

2474

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4 - 6 = -10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 1

Use logarithms to solve exponential models fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A model fitted to bacteria counts is $N = 150(1.4)^t$, with t in hours. When does N reach 1,000? Give the exact expression and the value to the nearest tenth of an hour. (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

2476

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

t

= log(1,000 ÷ 150) ÷ log 1.4 ≈ 5.6 hours

GRADING NOTE Form required: the exact expression (any equivalent, such as log(20/3) ÷ log 1.4) and the rounded value. $1.4^t = 6.67$, $t \approx 5.64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line fitted to the logarithms of data is log y = 0.3x + 1 (log means base 10).

Write y as an exponential function of x.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

2477

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$10 \cdot 10^{0.3x}$

GRADING NOTE Any equivalent exact form, such as $10^{0.3x+1}$ or $10(10^{0.3})^x$. A rounded base such as $10(2)^x$ is not exact: $10^{0.3} \approx 1.995$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fitted model is $y = 20e^{0.08x}$. For which x is y = 60? (ln means base e.)

A. $x = 3 \div 0.08 = 37.5$

B. $x = \ln 60 \div 0.08 \approx 51.2$

C. $x = 0.08 \ln 3 \approx 0.09$

D. $x = \ln 3 \div 0.08 \approx 13.7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

2478

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \ln 3 \div 0.08 \approx 13.7$

GRADING NOTE $e^{0.08x} = 3$, so $0.08x = \ln 3 \approx 1.0986$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For the model $y = 5(1.5)^x$, the solution of $5(1.5)^x = 50$ is $x = \log 10 \div \log 1.5 \approx 5.68$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

2479

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $1.5^x = 10$, so $x = \log 10 \div \log 1.5 = 1 \div 0.1761 \approx 5.68$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

The doubling time of $N = 150(1.4)^t$ is log 2 ÷ log 1.4 ≈ 0.49 hour. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

2480

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE log 2 ÷ log 1.4 ≈ 0.301 ÷ 0.146 ≈ 2.06 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 2

Evaluate reports that use fitted models • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A report fits a line to the heights of children aged 5 to 15 and uses it to predict the height of a 40-year-old.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

2482

QUESTION ID
S64-F2-Q1

TYPE
Short answer

CORRECT ANSWER

It extrapolates far beyond the data; people stop growing, so the linear pattern does not continue.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report finds a strong correlation between ice cream sales and drownings and concludes that ice cream causes drowning.

Do these observational data establish that conclusion? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

2483

QUESTION ID
S64-F2-Q2

TYPE
Short answer

CORRECT ANSWER

No. A correlation in observational data does not establish causation; a lurking variable, such as hot weather, could increase both.

GRADING NOTE Any equivalent explanation that association alone does not establish causation. Naming a possible lurking variable is accepted but not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

A residual plot shows a strong curve, but a report says the linear model is excellent.

What should you conclude?

A. The report is correct.

B. The data must be wrong.

C. The residuals are all 0.

D. The linear model may not be appropriate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

2484

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The linear model may not be appropriate.

GRADING NOTE A curved residual pattern shows a poor linear fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

True or false?

A report should state how its data were collected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

2485

QUESTION ID
S64-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The method decides what conclusions are justified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

True or false?

A model with a high correlation coefficient is always the right type of model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

2486

QUESTION ID
S64-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Curved data can have a high r for a line and still be fit better by a curve.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear function to a scatter plot that suggests a linear association

MAIN	Algebra I main question	**HSS-ID.B.6.c**
BACKWARD 1	Judge a line of fit informally	**8.SP.A.2**
BACKWARD 2	Write a line through two points	**HSF-LE.A.2**
FORWARD 1	Evaluate conclusions drawn from a line of fit	**HSS-IC.B.6**
FORWARD 2	Use a randomized experiment and simulation to test whether mo…	**HSS-IC.B.5**

SET 65 • HSS-ID.B.6.c

MAIN

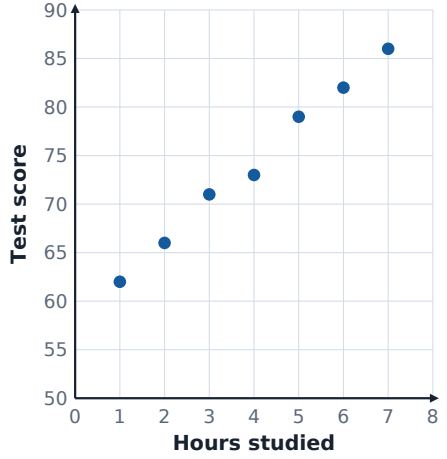
Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows hours studied and test scores.

Draw a line of fit and write its equation.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

2489

QUESTION ID

S65-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line that follows the trend of the points closely (balancing points above and below is a helpful check)

Written: About y = 4x + 58

GRADING NOTE Any equation close to the least-squares line $\hat{y} = 4x + 58.14$ (slope about 3.5 to 4.5). The equation should match the student's line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line for the data is $\hat{y} = 4x + 58.14$.

Predict the score for 4.5 hours, to the nearest whole number.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

2490

QUESTION ID

S65-M-Q2

TYPE

Short answer

CORRECT ANSWER

76

GRADING NOTE Form required: rounded to the nearest whole number. 18 + 58.14 = 76.14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through the data points (1, 62) and (7, 86), to use as a line of fit.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

2491

QUESTION ID

S65-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 4x + 58

GRADING NOTE Any equivalent form. The slope is 24 ÷ 6 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line fits the points (0, 1), (1, 3), (2, 5), (3, 7) exactly?

A. y = 3x + 1

B. y = 2x + 1

C. y = 2x

D. y = x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

2492

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 1

GRADING NOTE y increases by 2 for each 1 in x, starting at 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

True or false?

A line of fit should pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

2493

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It should be close to the points overall, not through all of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 1

Judge a line of fit informally • Grade 8 • Statistics & Probability • 8.SP.A.2

Is a straight line a good model for points that curve strongly upward? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

2495

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The points do not follow a straight pattern.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How can you judge whether a line of fit is good?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

2496

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The points are close to the line and follow its trend, with no pattern of points far above or below.

GRADING NOTE Any equivalent description. Balancing points above and below is a helpful check, not a definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

2496

ANSWER KEY Reduced copy of the question page, with its answer

Which is the best sign of a good line of fit?

A. All of the points are below it.

B. All of the points are above it.

C. The points are close to the line and follow its trend.

D. Exactly two points are on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

2497

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The points are close to the line and follow its trend.

GRADING NOTE A good line stays close to the points; balancing points above and below is a helpful check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q3/5

True or false?

Lines of fit are used for data with a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

2498

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Straight-line patterns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A line of fit must pass through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

2499

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its y-intercept can be any value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 2

Write a line through two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (2, 66) and (6, 82).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

2501

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 58

GRADING NOTE Any equivalent form. The slope is 16 ÷ 4 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (0, 7) and (3, 13).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

2502

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which line passes through (0, 10) and (5, 0)?

A. $y = 2x + 10$

B. $y = 10x - 2$

C. $y = -2x + 10$

D. $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

2503

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -2x + 10$

GRADING NOTE The slope is $-10 \div 5 = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The line through (0, 3) with slope 4 is y = 4x + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

2504

QUESTION ID
S65-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE y = mx + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The line through (2, 2) and (4, 8) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

2505

QUESTION ID
S65-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $6 \div 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 1

Evaluate conclusions drawn from a line of fit • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

In a study of 7 volunteer students, each extra hour of study went with about 4 more points.

Can a school claim this holds for all of its students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

2507

QUESTION ID

S65-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Not with confidence: 7 volunteers are a small sample that was not randomly selected, and an observational study shows association, not cause.

GRADING NOTE Any equivalent explanation that names the sampling problem (small, not random) or the association-causation limit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study of 7 students who studied from 1 to 7 hours found the line $\hat{y} = 4x + 58$, where x is hours of study and $\hat{y}$ is the test score out of 100.

A report uses this line to predict a score of 138 for 20 hours of study. Why is that unreasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

2508

QUESTION ID
S65-F1-Q2

TYPE
Short answer

CORRECT ANSWER
20 hours is far outside the data (1 to 7 hours), and the score cannot exceed 100.

GRADING NOTE Any equivalent explanation that names the extrapolation or the 100-point maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which detail makes a regression conclusion more trustworthy?

A. The prediction is far outside the data.

B. The line was drawn by hand.

C. The data came from a large random sample.

D. The sample was 5 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

2509

QUESTION ID
S65-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. The data came from a large random sample.

GRADING NOTE Random, large samples represent the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

True or false?

Observational data alone cannot prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

2510

QUESTION ID
S65-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Lurking variables may explain the association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

True or false?

A line of fit from 7 volunteer students proves that extra study raises the score of every student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

2511

QUESTION ID
S65-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It describes an association in that small, nonrandom sample; it does not prove an effect for everyone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 2

Use a randomized experiment and simulation to test whether more study raises scores • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

To test whether extra study raises scores, 40 volunteers were randomly assigned: 20 studied 3 extra hours and 20 did not. The extra-study group averaged 5 points higher. In 1,000 random regroupings of the 40 scores, a difference of 5 points or more (first group minus second) happened 18 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2513

QUESTION ID
S65-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 18 ÷ 1,000 = 1.8%, which is below 5%, so chance alone is unlikely to explain the difference; the extra study likely raised scores for volunteers like these.

GRADING NOTE Form required: the decision with the simulated proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a randomized experiment, a tutoring group averaged 6 points more than a control group. Random regroupings of the scores gave differences of 6 points or more 30% of the time.

Using a 5% threshold, what is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2514

QUESTION ID
S65-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The difference is not significant: 30% is above 5%, so it could easily be due to chance.

GRADING NOTE Form required: the conclusion with the comparison to 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the purpose of random assignment in an experiment?

A. To balance other variables between the groups, on average

B. To make the sample smaller

C. To guarantee a significant result

D. To remove the need for a control group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2515

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. To balance other variables between the groups, on average

GRADING NOTE Random assignment tends to balance other variables, so a large difference can be credited to the treatment. It does not make the groups identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

True or false?

A result is statistically significant when a difference that large would rarely happen by chance alone, using a threshold chosen in advance such as 5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2516

QUESTION ID
S65-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is the meaning of significance in a randomization test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

True or false?

Using a 5% threshold, a difference that happens in 40% of random regroupings is statistically significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2517

QUESTION ID
S65-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 40% is far above 5%, so such a difference is common by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

Interpret the slope and the intercept of a linear model in the context of the data

MAIN	Algebra I main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**
BACKWARD 2	Use a linear model to solve problems	**8.SP.A.3**
FORWARD 1	Interpret the parameters of continuous exponential models	**HSF-LE.A.4**
FORWARD 2	Understand a fitted slope as an estimate of a population value	**HSS-IC.A.1**

SET 66 • HSS-ID.C.7

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 4x + 58$ predicts a test score from x hours of study.

Interpret the slope and the intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

2520

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: each additional hour of study goes with about 4 more points (predicted). Intercept: the predicted score for 0 hours is 58.

GRADING NOTE Form required: both interpretations, in context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for used cars is $\hat{y} = 22{,}000 - 1{,}800x$, where x is the age in years and $\hat{y}$ is the price in dollars.

Interpret $-1{,}800$ and $22{,}000$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

2521

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

The predicted price drops about $1,800 for each year of age; the predicted price of a new (0-year-old) car is $22,000.

GRADING NOTE Form required: both interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$\hat{y} = 0.9x - 20$ predicts an adult's weight in pounds from height in inches.

Does the intercept make sense in context? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

2522

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. It predicts −20 pounds at a height of 0 inches, which is impossible and far outside the data.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

$\hat{y} = 2.5x + 10$ predicts the cost in dollars of a pizza with x toppings.

What does 2.5 mean?

A. Each additional topping adds about $2.50 to the predicted cost.

B. A pizza has 2.5 toppings.

C. A pizza costs $2.50.

D. A plain pizza costs $2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

2523

QUESTION ID

S66-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Each additional topping adds about $2.50 to the predicted cost.

GRADING NOTE The slope is the change in predicted cost per topping.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

True or false?

The slope of a fitted line gives the exact change in y for every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

2524

QUESTION ID

S66-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives the predicted (average) change; individuals vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 1

Interpret the parameters of linear and exponential functions • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A technician charges C = 50 + 20h dollars for h hours.

What does 20 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

2526

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$20 for each hour of work

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 3{,}000(0.9)^t$ is a population after t years.

What does 0.9 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

2527

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The population decreases 10% each year.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$d = 300 - 60t$ is a driver's distance from home, in miles, after t hours.

What does 300 mean?

A. The starting distance from home

B. The speed

C. The distance driven each hour

D. The time to arrive home

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

2528

QUESTION ID
S66-B1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. The starting distance from home

GRADING NOTE At $t = 0$, $d = 300$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

True or false?

In y = 1.5x + 4, increasing x by 2 increases y by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

2529

QUESTION ID
S66-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 1.5 × 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $y = 8(1.25)^x$, y increases by 1.25 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

2530

QUESTION ID
S66-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE y is multiplied by 1.25 for each unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

2530

SET 66 • HSS-ID.C.7

BACKWARD 2

Use a linear model to solve problems • Grade 8 • Statistics & Probability • 8.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

y = 2x + 5 models a plant's height in cm after x weeks.

Predict the height at 6 weeks.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

2532

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

17 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using y = 2x + 5, after how many weeks is the plant 21 cm tall?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

2533

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 2x + 5 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 2x + 5$ models a plant's height in cm after x weeks.

In this model, what does 5 mean?

A. The height at week 0

B. The weeks needed to grow 1 cm

C. The final height

D. The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

2534

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The height at week 0

GRADING NOTE $y = 5$ when $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The slope of a linear model is a rate of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

2535

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the change in y per unit of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 2x + 5, the plant grows 5 cm each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

2536

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It grows 2 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 1

Interpret the parameters of continuous exponential models • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A(t) = 500e^{0.04t}$.

What does 0.04 mean? When does A double, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

2538

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A continuous growth rate of 4% per unit of t; t ≈ 17.3

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.04.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$N(t) = 1{,}000e^{-0.1t}$.

What does −0.1 mean? When is N = 500, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

2539

QUESTION ID
S66-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Continuous decay at 10% per unit of t; t ≈ 6.9

GRADING NOTE Form required: the meaning and the rounded time. t = ln 2 ÷ 0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{2x} = 9$ exactly. (ln means base e.)

A. $x = \ln 3$

B. $x = 4.5$

C. $x = e^{4.5}$

D. $x = \ln 9 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

2540

QUESTION ID
S66-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. x = ln 3

GRADING NOTE $2x = \ln 9 = 2 \ln 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\ln(e^{0.04t}) = 0.04t$ (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

2541

QUESTION ID

S66-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ln and e^x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $e^k = 1$, then $k = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

2542

QUESTION ID

S66-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $e^0 = 1$, so $k = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 2

Understand a fitted slope as an estimate of a population value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 50 used cars of one model gives $\hat{y} = 22{,}000 - 1{,}800x$.

What population value does $-1{,}800$ estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

2544

QUESTION ID
S66-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The average yearly drop in price for all used cars of that model.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Would another random sample of 50 cars give exactly the same slope? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

2545

QUESTION ID
S66-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Probably not: different samples give different statistics (sampling variability).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which of these is a parameter?

A. The mean price of the 50 sampled cars

B. One car's price

C. The slope from the sample

D. The mean price of all used cars of this model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

2546

QUESTION ID

S66-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The mean price of all used cars of this model

GRADING NOTE A parameter describes the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

True or false?

Larger random samples tend to give statistics closer to the parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

2547

QUESTION ID
S66-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Their sampling variability is smaller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

True or false?

A statistic is a number that describes a whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

2548

QUESTION ID
S66-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

Compute (using technology) and interpret the correlation coefficient of a linear fit

MAIN	Algebra I main question	**HSS-ID.C.8**
BACKWARD 1	Describe the direction of an association	**8.SP.A.1**
BACKWARD 2	Fit and use a line for linear data	**HSS-ID.B.6.c**
FORWARD 1	Use a randomized experiment and simulation to test whether an a…	**HSS-IC.B.5**
FORWARD 2	Evaluate claims based on a correlation	**HSS-IC.B.6**

SET 67 • HSS-ID.C.8

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use technology to find the correlation coefficient r for the data

(1, 9), (2, 8), (3, 8), (4, 6), (5, 5), (6, 5), (7, 3), (8, 2),

to the nearest hundredth. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

2551

QUESTION ID

S67-M-Q1

TYPE

Short answer

CORRECT ANSWER

r ≈ −0.98: a strong negative linear association.

GRADING NOTE Form required: r to the nearest hundredth and a description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$r = -0.85$ for hours of TV and test score.

Interpret this value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

2552

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

A strong negative linear association: more TV goes with lower scores.

GRADING NOTE Any equivalent interpretation of the direction, strength, and form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value of r shows the strongest linear relationship?

A. 0.10

B. −0.92

C. −0.40

D. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

2553

QUESTION ID

S67-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −0.92

GRADING NOTE Strength depends on |r|, and |−0.92| is the largest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Can a correlation coefficient equal 1.3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

2554

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. r is always between −1 and 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

True or false?

r = 0 means there is no relationship of any kind between the variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

2555

QUESTION ID
S67-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means no linear relationship; there could be a curved one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 1

Describe the direction of an association • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Points rise from left to right, fairly close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

2557

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points for hours of sleep and number of errors fall from left to right.

Is the association positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

2558

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Negative

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of variables likely has a positive association?

A. Height and arm span

B. Outdoor temperature and heating cost

C. A car's age and its value

D. Speed and travel time for a fixed trip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

2559

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Height and arm span

GRADING NOTE Taller people tend to have longer arm spans.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a negative association, as one variable increases, the other tends to decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

2560

QUESTION ID

S67-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the meaning of negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Clusters in a scatter plot mean the data are linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

2561

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Clusters describe grouping, not form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 2

Fit and use a line for linear data • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Write a line of fit through (2, 5) and (8, 14).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

2563

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5x + 2

GRADING NOTE Any equivalent form. The slope is 9 ÷ 6 = 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The least-squares line is $\hat{y} = 1.5x + 2$.

Predict y when $x = 6$.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

2564

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What does the least-squares line minimize?

A. The y-intercept

B. The sum of the squared residuals

C. The number of points

D. The slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

2565

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The sum of the squared residuals

GRADING NOTE That is what "least squares" means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

2566

QUESTION ID

S67-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of the least-squares line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

True or false?

Two people drawing lines of fit by eye always get the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

2567

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines drawn by eye vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 1

Use a randomized experiment and simulation to test whether an association is causal • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Observational data show r = −0.6 between evening screen time and hours of sleep. To test cause, 50 volunteers were randomly assigned to no screens after 8 p.m. or to their usual habits. The no-screen group slept 0.5 hour more on average. In 1,000 random regroupings, a difference of at least 0.5 hour (no-screen minus usual) happened 2 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

2569

QUESTION ID
S67-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 2 ÷ 1,000 = 0.2%, which is below 5%, so chance alone is very unlikely to explain it; cutting evening screens likely increased sleep for volunteers like these.

GRADING NOTE Form required: the decision with the simulated proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can this randomized experiment, but not the observational correlation alone, support the claim that evening screens reduce sleep?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

2570

QUESTION ID

S67-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Random assignment tends to balance other variables (such as stress or age) between the groups, so a significant difference can be credited to the treatment; in observational data a lurking variable could explain the correlation.

GRADING NOTE Any equivalent explanation that contrasts random assignment with possible lurking variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a randomization test, the observed difference was matched or exceeded in 350 of 1,000 regroupings.

Using a 5% threshold, what is the conclusion?

A. The treatment definitely has no effect.

B. The difference is not significant; chance could easily explain it.

C. The difference is significant.

D. The experiment proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

2571

QUESTION ID

S67-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The difference is not significant; chance could easily explain it.

GRADING NOTE 350 ÷ 1,000 = 35%, far above 5%. A non-significant result does not prove there is no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

True or false?

In a randomization test, the regroupings show how large the difference between groups tends to be when the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

2572

QUESTION ID
S67-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Regrouping the same results at random imitates assignment with no treatment effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

True or false?

If an experiment's difference is not significant, that proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

2573

QUESTION ID

S67-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means the data do not show an effect beyond chance; a small effect could still exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 2

Evaluate claims based on a correlation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A news story says, "r = 0.6, so 60% of the change in sales is caused by advertising."

Name two errors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2575

QUESTION ID
S67-F2-Q1

TYPE
Short answer

CORRECT ANSWER
r is not a percent of change ($r^2 \approx 0.36$ describes the share of variation explained), and correlation does not show causation.

GRADING NOTE Form required: two errors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study of 8 cities finds r = 0.7 between the number of parks and happiness scores.

Why should you be cautious?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2576

QUESTION ID

S67-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The sample is small, the study is observational, and other variables (such as income) could explain both.

GRADING NOTE Any two valid reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement about r = −0.9 is correct?

A. There is a strong negative linear association.

B. The slope of the line is −0.9.

C. 90% of the points are on the line.

D. An increase in x causes y to fall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2577

QUESTION ID

S67-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. There is a strong negative linear association.

GRADING NOTE r measures the direction and strength of a linear association only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

True or false?

A correlation can be strong even when the relationship is not causal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2578

QUESTION ID
S67-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A lurking variable can create it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

If r = 0.8, the slope of the least-squares line is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2579

QUESTION ID
S67-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The slope depends on the units and spreads of x and y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-ID.C.9 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Distinguish between correlation and causation

MAIN	Algebra I main question	**HSS-ID.C.9**
BACKWARD 1	Interpret the correlation coefficient	**HSS-ID.C.8**
BACKWARD 2	Use random samples to make valid generalizations	**7.SP.A.1**
FORWARD 1	Distinguish surveys, experiments, and observational studies	**HSS-IC.B.3**
FORWARD 2	Use random methods to make fair decisions — nearest related	**HSS-MD.B.6**

SET 68 • HSS-ID.C.9

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Data show that cities with more firefighters have more fire damage.

Do these data establish that hiring firefighters causes damage? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

2582

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

No. The data show an association only. A lurking variable, such as the size of the city or of its fires, could explain both.

GRADING NOTE Any equivalent explanation that association does not establish causation. A plausible lurking variable is accepted but not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

Ice cream sales and sunburn rates are strongly correlated.

Name a lurking variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

2583

QUESTION ID

S68-M-Q2

TYPE

Short answer

CORRECT ANSWER

Hot, sunny weather

GRADING NOTE Any reasonable lurking variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How could you test whether a new study app causes higher test scores?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

2584

QUESTION ID
S68-M-Q3

TYPE
Short answer

CORRECT ANSWER
Run a randomized experiment: randomly assign students to use the app or not, then compare the groups' scores.

GRADING NOTE Any design that uses random assignment and a comparison group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

Which study design can show that one variable causes changes in another?

A. A survey of volunteers

B. A scatter plot of existing data

C. An observational study

D. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

2585

QUESTION ID

S68-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A randomized experiment

GRADING NOTE Random assignment balances other variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

True or false?

A strong correlation between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

2586

QUESTION ID

S68-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Correlation alone never proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 1

Interpret the correlation coefficient • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

ANSWER KEY Reduced copy of the question page, with its answer

r = 0.88.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

2588

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A strong positive linear association

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which shows the stronger linear association, r = −0.7 or r = 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

2589

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

r = −0.7

GRADING NOTE Any equivalent form. |−0.7| > |0.5|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is r for points that lie exactly on a line with negative slope?

A. 1

B. 0

C. −0.5

D. −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

2590

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −1

GRADING NOTE A perfect negative linear relationship.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

The correlation coefficient has no units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

2591

QUESTION ID

S68-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is computed from standardized values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

Switching the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

2592

QUESTION ID
S68-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE r is the same either way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 2

Use random samples to make valid generalizations • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why is a random sample more likely to represent a population than a convenience sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

2594

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Every member has the same chance of being chosen, so the method does not favor any group (no selection bias), although a random sample can still differ from the population by chance.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

A survey at a gym asks about the exercise habits of all town residents.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

2595

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

People at a gym probably exercise more than other residents, so the sample is biased.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a random sample of a school's students?

A. Drawing 40 names from a list of all students

B. Asking the first 40 students in line

C. Asking your friends

D. Asking one math class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

2596

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Drawing 40 names from a list of all students

GRADING NOTE Only the drawing gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

Generalizations about a population are valid only if the sample represents the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

2597

QUESTION ID

S68-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is why random sampling matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

2598

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers choose themselves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 1

Distinguish surveys, experiments, and observational studies • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Researchers record students' screen time and grades without changing anything.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

2600

QUESTION ID
S68-F1-Q1

TYPE
Short answer

CORRECT ANSWER
An observational study

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Researchers randomly assign plants to receive fertilizer or no fertilizer.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

2601

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

An experiment

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the main purpose of a survey?

A. To test whether a treatment causes an effect

B. To assign treatments to subjects

C. To control lurking variables

D. To estimate a characteristic of a population from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

2602

QUESTION ID
S68-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. To estimate a characteristic of a population from a sample

GRADING NOTE A survey collects information without assigning treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

True or false?

Random selection lets you generalize to a population; random assignment lets you conclude cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

2603

QUESTION ID
S68-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two kinds of randomization do different jobs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

True or false?

In an observational study, the researchers decide which treatment each subject gets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

2604

QUESTION ID
S68-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They only observe.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 2

Use random methods to make fair decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Random assignment, which is needed to show causation, is a fair random method.

ANSWER KEY Reduced copy of the question page, with its answer

Describe a fair way to split 20 volunteers into two groups of 10 using random numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

2606

QUESTION ID
S68-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Number the volunteers 1 to 20, randomly draw 10 different numbers for group A, and put the rest in group B.

GRADING NOTE Any method that gives every volunteer the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A coach picks a captain by drawing one name from a hat with 12 names.

What is the probability that a given player is chosen?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

2607

QUESTION ID

S68-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Which method is fair for deciding who goes first among 4 players?

A. The tallest goes first

B. Each rolls a number cube; ties roll again; the highest goes first

C. The coach's favorite goes first

D. Whoever asks first goes first

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

2608

QUESTION ID
S68-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. Each rolls a number cube; ties roll again; the highest goes first

GRADING NOTE Only the cube roll gives everyone the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A fair lottery gives every participant the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

2609

QUESTION ID
S68-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what fair means here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Flipping one fair coin once is a fair way to choose one person from 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED

2610

QUESTION ID
S68-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A coin has only 2 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED

2610